

Introduction to Applied Educational Learning

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Luke Springer



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Preface

Early on, adults taught children the knowledge and abilities essential to their community, thereby beginning education. This was accomplished orally and through imitation in preliterate communities. Through storytelling, knowledge, morals, and talents were passed down from one generation to the next. Cultures started to expand their knowledge beyond what could be easily gained by imitation as formal education started to emerge. In Egypt during the Middle Kingdom, there were schools. The first college in Europe was established by Plato when he established the Academy in Athens. After Athens ceased to be the intellectual center of Ancient Greece, Alexandria, an Egyptian city founded in 330 BCE, took its place. In the third century BCE, the enormous Alexandrian Library was erected. Following the fall of Rome in CE 476, European civilizations saw a collapse in literacy and structure.

The Catholic Church became the sole keeper of literate learning in Western Europe after Rome fell. In the Early Middle Ages, the church built cathedral schools as institutions of higher learning. Some of these institutions eventually developed into medieval universities, serving as the ancestors of numerous European modern universities. During the High Middle Ages, Chartres Cathedral ran the renowned and significant Chartres Cathedral School. Thomas Aquinas of the University of Naples, Robert Grosseteste of the University of Oxford, an early expositor of a systematic method of scientific experimentation, and Saint Albert the Great, a pioneer of biological field research, were just a few of the fine scholars and natural philosophers who came from the well-integrated medieval universities of Western Christendom who encouraged freedom of inquiry. The Institution of Bologna, established in 1088, is regarded as the first and the oldest continuously running the university. In most nations today, children up to a particular age must get full-time education, whether in school or another setting. UNESCO estimates that more individuals will get a formal education in the next 30 years than in all of human history due to the development of compulsory schooling and population increase.

The act of promoting learning or the development of knowledge, skills, values, beliefs, and habits is known as education. Narrative, conversation, instruction, training, and focused research are all forms of education. Educators are typically present when learning occurs, but students can also educate themselves. Any event that impacts a person's thoughts, feelings, or actions may be deemed educational. Education can occur in official or informal settings. Pedagogy is the name given to the teaching approach. The phases of education are typically broken down into preschool or kindergarten, elementary school, and secondary school, followed by college, university, or an apprenticeship. Several nations and the UN have acknowledged a right to education. Education is typically required till a particular age.

Author

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CHAPTER

Introduction

The education sector has become a major focus of attention amongst development partners in the last two years, driven primarily by the desire to assist India in achieving the United Nations (UN) Millennium Development Goals (MDGs) for education. The Government aims to improve national competitiveness and reduce poverty through its Medium Term Strategic Plan. It is committed to the achievement of the MDGs and the Education for All (EFA) goals, and if current efforts are sustained, these key education goals are set to be reached by 2010.

Key challenges for the Government include widening access to education without compromising quality, improving quality in a newly decentralised environment, and ensuring that the newly increased funds for education are well spent. To address this, the Ministry of Education is working through its strategic plan for 2005-2009, which has three pillars to its strategy, including:

- Improving access,
- Increasing quality, and
- Improving the governance and management of education.

Globally, the European Commission (EC) , along with a number of development partners, seeks where possible to provide support to partner governments through SWAPs and budget support. This approach is based on two key principles: one, that programmes are led by partner governments, and two, that they have the common goal of improving the efficiency and effectiveness with which internal and external resources are used. This reflects a mutual concern to improve results of government and donor spending both by focusing resources on the priorities stated in national planning documents and

by improving the quality of spending. A number of development partners are currently supporting programmes in basic education, including:

Australian Agency for International Development (Ausaid) , United States Agency for International Development (USAID) , Asian Development Bank (ADB) , Japan International Cooperation Agency (JICA) , UN Children's Fund (UNICEF) and UN Educational, Scientific and Cultural Organisation (UNESCO) . Large amounts of development assistance funds are being made available to support education reform and development.

There is growing interest in moving towards a SWAP among this group, which formalised inter-donor coordination in 2005 with the establishment of the Education Sector Working Group (ESWG) , currently co-chaired by the EC and the Netherlands for one year. This group is taking forward the Paris Declaration on Aid Effectiveness, and is seeking to work with the Government in support of achievement of MDGs. World Bank support to basic education is moving into a new phase towards teacher quality support and district level SWAPs, joining forces with the Netherlands and the EC, meanwhile both the ADB and Ausaid have developed major programmes aimed at working towards a SWAP in Indonesia.

For development assistance through a SWAP to become effective, the Government needs to take full ownership of the process, to take a lead in donor coordination and harmonisation, and establish a platform for policy dialogue at sector level which will allow development partners to monitor developments in education and to engage productively with the Government. Alongside this, a medium term expenditure framework will allow development partners' assistance to fill financing gaps, and will give the Government much clearer information than it has at present on how development assistance is being used and can be used in the future.

OBJECTIVES OF EDUCATIONAL ADMINISTRATION

The goals of education can be formulated only in the light of the needs of society for which the education is meant. Education constitutes the curriculum and the teaching techniques, or in simple words—'what we teach and how we teach'. Difference in social and political institutions will determine different purposes of education. The ends of education will be different with totalitarian, theocratic and democratic societies., Totalitarian governments use schools and colleges as they use all other institutions to create tools of the state rather than free citizens.

George S. Count and Nucia P. Lodge through translation of Russian writings revealed that the purpose of the elementary schools under Stalin was to promote love of the motherland, reverence for Stalin and hatred for enemies of both. In a free society the general purpose of schooling is to prepare the young for participation in a life of citizenship in which all may share equally—for duties and responsibilities of Self Government. The value structures of society with agricultural bias will be different from those depending on industrial economies. To make education really meaningful

the ends of education will change from rural to urban societies and vice-versa. The requirements of an industrial economy will significantly differ from those of an agricultural economy; likewise the goals of education with a rural bias will vary from those of urban bias.

The end values to be sought through education must be, therefore, identified, elaborated and given operational meaning by reference to social conditions and personal experiences, to the efforts and designs of society and to the aims, needs, and aspirations of those who are to be educated. Ours is a democratic society with a mixed economy. Our education should lead to the liberation of the mind and the training of the hand, heart and the head.

GENERAL ADMINISTRATION IN EDUCATION

It is being argued that “all administration is the same. Whether one looks at administration from the standpoint of its purpose, its task, its situational milieu or its process, he will find much that is common in the management of Business, Government, Army, Education and other organisations.”

Those who see no significant difference between the two say that human behaviour, cult of efficiency and bureaucracy are common to all administration.

Educational administration includes most of the general administrative functions such as appraisal of the organisational purposes and coordination of all the forces—staffing, training, public relations, community service, raising funds for institutions, maintenance of accounts and a score of other things to achieve the objectives.

On this account they say that educational administration very well fits into the pattern of administration in general:

- Despite the common elements, there are differences at least in degree between the administrative processes related to various social organisations and educational institutions. Educational administration is a “non-profit making enterprise”. It is chiefly welfare or service oriented. The purpose of an industrial administration is to produce goods, of course, to meet the needs of the people but their motive is to make profits. No worthwhile business administration can run even on ‘no profit, no loss’ basis whereas educational administration worth the name aims at the development of human personality without caring for the monetary losses. In fact, educational enterprise is a “welfare enterprise” first and last and more often than not runs in financial losses which—are met by donations, public contribution or by levies or taxes etc.
- Peculiarities of educational administration arise from the objectives of education in our society. Education is closely linked to life. It needs to be transferred into a powerful instrument of social change. National development can be brought about only through universal education. Even the survival of a democratic society is possible through education. “Destiny of India is now being shaped in her classrooms.” This, we believe is no mere rhetoric. In a world based on science

and technology it is education that determines the level of prosperity, welfare and security of the people. On the quality and number of persons coming out of our schools and colleges will depend our success in the great enterprise of our national reconstruction; whose principal objective is to raise the standard of living of our people.—Indian Education Commission (1964-66) .

- Good educational institutions free men to rise to the level of their natural abilities. Hope for personal advancement and the advancement of one's children is, of course, one of great wellsprings of human energy. The schools and colleges only sustain this hope. Public education has an inbuilt corrective of our society. Only through general public education can we succeed as the largest democracy of the world. The educational administrators have, therefore, a special responsibility.
- Education deals with people directly and intimately. "The most intimate relationship the school/college has, is with children and youth. The pupils represent the dearest possessions of their parents and these parents, by and large, are greatly concerned with how the school/college is treating their offspring. In other words, the school/college is confronted with the emotional attachment of parents to their children and these attachments are much stronger than those which most parents have for other public or private services." The educational administrator is thus concerned with a group of clients who may not be his by choice. But without whose cooperation the institution, he is responsible for, cannot function properly.
- The main concern of educational administration being, people and their welfare, the techniques of educational administration are mostly persuasive while those in other kinds of administration are mostly coercive, dictatorial and authoritative. An educational administrator has to look to the interaction of the persons involved, whereas in other types of administration one can afford to ignore the reactions of the persons affected by his decisions—though sometimes he too has to pay through the nose for the neglect.
- Another special feature of the educational enterprise is the nature of the learning process. Learning is a complicated affair. It comes more from informal situations than from the class room lessons. Much informal education goes on without any body's intending to learn—simply as a process of social interaction. Much of what a pupil learns is learnt as a consequence of social interaction among pupils. Thus learning is preceded by a number of factors. It does not merely mean acquiring some knowledge or understanding but it may provoke many emotional commitments. Shaheed-i-Azam Bhagat Singh who shook the foundations of the British Empire, Subhash Chandra Bose who alienated the heavenly born Indian Civil Service and Lala Lajpat Rai, the "Lion of the Punjab", achieved more than learning in their school and college days. The channels of contact and communication have to be always kept open and free with the

teachers, the students and their parents. The educational administrator has to be highly visible and accessible to the public in general too as the educational institutions are the public affair.

- Evaluation of results in education is another area which poses special problems to educational administrators. In business, one can use sales or production or the profits as evaluative measures. In the military, objectives are definite and it is not difficult to assess whether these goals have been achieved or not. Likewise the targets of the town architects and dam builders can be measured to a large extent as these are more or less concrete. But it is not so in the case of educational objectives and targets. The “purpose of all education at all levels varies from literacy to critical thinking, intellectual leisure and facilitation of social mobility.” Literacy could of course be taken as something tangible, *i.e.* reading, writing and numeration but critical thinking, intellectual leisure or facilitation of social mobility may almost mean anything to anybody. To say, therefore, about any institution as to how far these abstract goals have been achieved is difficult if not impossible. People may disagree even on their interpretations.
- Educational administration is different from other types of administrations in one more way—that is staffing. The teaching staff of a college is more or less equally qualified. All of them hold post-graduate degrees if not Ph.D. or M.Phil degrees and generally with peer merit. Many members of the staff may even be better trained and experienced than the principal. They may have even more of research work and teaching merit to their credit. The educational administrator is thus working with professionals who feel, and often rightly that they know more about teaching and learning than he does.
Gatzels and Cuba have pointed out that professional trained people, in contrast to day labourers or skilled people or production workers, are more inclined towards ‘idiographic’ than ‘nomothetic’ behaviour. A college organisation as contrasted to any business enterprise is more persuasive and idiographic. In other words, the training of teachers and the institutional settings in which the teachers work produce behaviour which stresses personality-factors more than role-factors.
- Authority, of course, serves as fuel on which all formal organisations are run. But authority is far more sophisticated than simple “command and obey” situation or relationship and the role of authority in the academic organisation is even more discreet than it is in other enterprises. The zone of acceptance in the case of an academic faculty is much narrower than in other enterprises.
- We often find that the governing bodies or the boards of control of educational institutes know very little about the complexities of learning and teaching. The unfortunate part of the whole situation is that they sit in judgement on the educational programmes they do not fully understand. But this thing seldom

happens in other organisations. The manager of a firm or a factory works for a board most of whom are thoroughly familiar with the factory affairs, its organisation and objectives.

- Lastly the educational administrator in a college has to perform two major functions. Firstly, he is principal of the college, and secondly, he needs to continuously reshape his organisation. The principal of a college has not only to be an administrator but an educational statesman too. He has to achieve the immediate objectives of the organisation and also to plan the long-range development and growth of the institution. He is not only to implement the educational policies framed by the powers that be, but is expected to suggest the changes in the policies. A general administrator is bound to plan and operate a programme designed to implement certain policies. He has to work within the given steel framework. If he ever trespasses the framework, he has to pay the price for it; whereas an educational administrator who is worth his salt spearheads new tracks. He may be, even ploughing a lonely furrow.

MANAGEMENT OF SCHOOL BUILDING

During The Construction Period: Liaison with the local authority and the contractor:

- Before construction activities commence on site, it is important for the school to be aware of the project communication arrangements between the local authority project manager and the contractor. Regular site meetings are likely to be held, and it may be appropriate for the school to be represented, or to be briefed on progress.
- In many projects, day-to-day communication should be encouraged between the school co-ordinator and the contractor's site manager regarding 'housekeeping' matters which will affect the running of the school.
- As well as having a clear understanding of the design and contract programme, the school should also be aware of what terms and conditions, if any, were included in the contract regarding the contractor's working restrictions, health and safety, security and so on. The school itself will have no authority to enforce the terms of the contract. It is therefore essential to establish a single point of contact with the local authority for all communication regarding the project.

KEEPING STAFF AND PUPILS INFORMED

- The flow of information to and from school users is very important throughout the project. This is usually an element of the school co-ordinator's role.
- Existing communication arrangements within the school such as regular departmental and whole staff meetings should be used wherever possible rather than setting up separate project meetings. However, staff should be able to raise individual issues directly with the school co-ordinator.

- Pupils are usually kept informed about the project through school assemblies and feedback is often provided through the pupil council or through ‘house structures’ where these operate in secondary schools.
- The project is also likely to feature heavily on the agenda of any School Board and PTA meetings. The HT and school co-ordinator, as well as the contractor, may well be expected to make regular progress reports at these meetings.
- One secondary school undergoing a major refurbishment set up a specific project users group which met monthly throughout the duration of the project. The group consisted of members representing staff, pupils, parents and others from the school community. Meetings were chaired by the HT and were attended by the local authority project manager and the contractor.

MANAGING DISRUPTIONS

- A certain degree of inconvenience is unavoidable during a major construction project. The level of disruption likely to be experienced by the school will depend upon a variety of factors such as the scale and type of building operations, their proximity to the school activities, the time at which the works are carried out, and the constraints of the existing buildings and site.
- The following examples are the most common disruptions reported by schools during construction projects:
 - *Dust and dirt:* Apart from being particularly uncomfortable, airborne dust and dirt can give rise to medical complaints leading to staff and pupil absences. Locating classrooms which require natural ventilation away from construction activities, and insisting on the constructing and maintaining of seals in affected areas can help to minimise the ingress of dust and dirt. In some cases, it may be appropriate to arrange additional cleaning for the school during the project.
 - *Noise:* Health and Safety regulations ensure that noise levels will not be hazardous to health, but they may still be extremely distracting. Where construction works are in close proximity to the school, the contractor may be excluded from undertaking certain noisy activities during particular periods such as exams. It may also be possible for the school to request the contractor, on an informal basis, to reduce noise levels for short periods from time to time.
 - *Distractions from increased traffic, both vehicular and personnel:* In order to avoid continuing distraction it may be possible to locate particularly sensitive school classes and activities away from the main site access and construction works. Ensure site activities are appropriately screened and avoid allowing contractor’s staff access through pupil areas.
 - *Frequent changes to access points and circulation routes:* The need to advise school users of continuing changes in access arrangements can be disruptive

and resource intensive. This should be considered when agreeing the sequence of phasing of the construction works.

- Planned and unplanned interruptions to water, gas, power, ICT services; prior consultation and contingency planning with staff about the consequences of a particular service failure or disconnection will allow the school to better manage these situations when they occur.
- *Reduced playground space, loss of playing fields and car parking:* The loss of amenities during the construction process is often an unavoidable source of inconvenience to school users. However, early consultation with those affected, provides the opportunity to investigate and implement alternatives.

MANAGING HAZARDS AND DISRUPTIONS

- The contractor will manage the health and safety and security aspects of the construction activities in any project. However, this in itself does not prevent the school from having to consider a number of health and safety matters or having to prepare at certain times for considerable disruption to the normal day-to-day running of the school.

HEALTH AND SAFETY/SECURITY

- The school will need to monitor its health and safety management procedures throughout the project and carry out additional risk assessments where appropriate. In particular, any alterations to fire escape routes and gathering points, access arrangements and site boundaries should be clearly identified. Arrangements should be clearly displayed in the school with signage amended as appropriate. Fire drills should be undertaken each time escape routes are changed.
- Prior to construction commencing on site, all school users should be briefed on the health and safety arrangements for the project. Presentations to school assemblies by the contractor, often incorporating protective clothing and some basic statistics about construction site safety, are generally considered to have more impact than if these were delivered by the school management team.
- As school holidays approach further advice should be issued to pupils regarding the hazards of building sites. Security monitoring will be provided by the contractor but in some instances it may also be appropriate to alert the local police who may patrol the site at high risk periods.
- Where work needs to be carried out in occupied buildings, contractor's staff may require to gain access to parts of the construction site through operational parts of the school. These situations generally place greater responsibilities on the contractor's staff and school users. These responsibilities need to be understood by all.
- It may be considered appropriate in these situations for the contractor's staff to wear agreed forms of identification and to be prepared to be challenged by

staff on school premises. It may also be considered necessary to caution contractor's staff to avoid initiating contact with pupils. It has also been known for school pupils to abuse construction workers, so the conduct of school users must also be considered and managed.

- Despite contractual agreements, breaches in health and safety procedures can still occur in these situations. For example contractor's staff may leave doors unlocked or materials and equipment unattended in circulation routes. School users should be particularly vigilant about possible hazards and procedures should be in place to report such incidents.

GOVERNMENT AND LOCAL AUTHORITY SCHOOLS

Some problems of schools under the management of government and local authorities deserve notice. Government institutions, for instance, have certain advantages, such as good financial support, good system of remuneration and retirement benefits and security of tenure for their teachers, and a fairly adequate provision of other physical facilities. In spite of all these assets, however, most government schools show, an average performance; and though some of them raise the average, very few qualify for the top places in the school system.

This is so because of several reasons. The average government school is isolated from its community, and sometimes even indifferent to it. The over-security of service creates an atmosphere of complacency and lethargy, especially because the conduct and discipline rules are such that it is difficult to reward merit, and even more so, to punish slackness. The teachers are recruited, not to individual institutions, but to a cadre and are frequently transferred from one institution to another. Consequently, they do not ordinarily develop loyalty to any individual institution.

They also have the minimum academic freedom and are hampered by rules and regulations at every step. The local authority schools also suffer from all these disadvantages. They have one compensating asset, namely, that they are more closely involved with their community. In practice, however, this generally proves to be not an asset, but a disadvantage, because their teachers are often harassed through postings and transfers and become involved in local politics and factions. A programme is, therefore, needed to overcome these weaknesses of government and local authority schools, so that the country can get an adequate return for the comparatively large investment it makes in these schools.

For this purpose we make the following recommendations:

- There should be a School Committee to look after every government or local authority school.

Such a committee will bring these schools closer to their local communities. Half the members of these committees should be elected by the local authority in charge of the area-village panchayat or municipality-and the remaining should be persons interested in education, nominated by the District School Boards.

The functions of these committees will include, among other things, the following:

- Responsibility for securing proper accommodation and construction and maintenance of school buildings, school gardens, children's parks and playgrounds;
- Provision of equipment;
- Distribution of books and writing materials to children;
- Grant of uniforms, scholarships and prizes;
- Enforcement of compulsory education within the area; Assisting in the organisation of extra-curricular activities and in building up a close relationship between the schools and the community;
- Provision of midday meals;
- Securing residential accommodation for teachers; and
- Generally taking all such measures as will help in improving school education within the area.

Each school committee should have a fund of its own for the proper discharge of its responsibilities. This fund, which may be designated as the School Fund, will consist of amounts placed at its disposal by the municipality or the village panchayat in the area; donations and contributions voluntarily made by the parents and local community from time to time; and a grant-in-aid given by the District School Board to stimulate local collections under and on some basis of equalisation, *i.e.*, a larger grant-in-aid being given to the poor areas while the richer ones may get a smaller one or none. The entire proceeds of the fund should be locally available for the development of such services in the schools as would supplement the effort made by the District School Board. It is evident that all school committees will not be equally efficient.

The system to be designed should, therefore, be elastic. School committees that are functioning well should be given more powers and more funds. The powers of those which are not working efficiently may be curtailed. If properly developed, this programme would make the local communities vie with one another in improving their schools, and their efforts should be a good supplement to those of the District School Board itself. Rational and appropriate policies have to be developed for transfers and postings of teachers which now cause considerable harassment, particularly to primary teachers under local bodies. As a rule, teachers should be allowed to remain in the same schools as long as possible and to develop loyalties to individual institutions. It is also necessary to give greater freedom to government and local authority schools and to reduce the existing red-tape to the minimum. Private schools already have a good deal of freedom, which is their main asset. There is no reason why similar freedom should not be given to government schools, where teachers and facilities are ordinarily of a better quality and where the freedom is likely to be better utilised.

PRIVATE SCHOOLS

The private educational institutions form a very heterogeneous group, falling into three main categories: recognised and aided institutions, recognised but unaided or

independent institutions, and unrecognised institutions. The magnitude of the last two of these categories is small and we shall deal with them separately later. But the recognised and aided institutions, in spite of their 'private' management, have to be treated as an integral part of the system of public education.

Most of their expenditure comes from government grants and fees; and where fees have been abolished, they depend almost exclusively on government funds. Their main assets are: strong ties with the local community on whom they depend for support; a fair measure of freedom, although this is disappearing rapidly wider increasing departmental controls; and the loyalty of teachers who are recruited, unlike in government or local authority service, to individual institutions.

These institutions have main weaknesses: a precarious financial position, due partly to the uncertainty of government grants and partly to their own increasing incapacity to raise funds; and very often, a bad and even unscrupulous management. From the point of view of quality and efficiency, these institutions fall into two broad groups: a small group of very efficient institutions and a large group of weak and even undesirable ones.

The institutions in the former have attracted-and continue to attract-competent and dedicated teachers who often form a self-perpetuating body of their own, and who remain virtually in charge of the management. Consequently, they maintain very good standards. The latter group includes a number of voluntary organisations which are dominated by sectarian considerations that affect the recruitment of teachers as well as their atmosphere. Several of them are run, not for purposes of education or social service, but for exploitation and patronage and are like commercial undertakings.

The conditions of service of teachers working under several of these organisations are far from satisfactory. They have little security of tenure and no pensionary benefits and sometimes not even a contributory provident fund; their remuneration is generally lower than that given to teachers of corresponding status under government or local bodies. In many cases, they do not even receive the amount which supposed to have been paid, to them because the managements, who are unable to raise popular contributions, often try to produce the matching contributions required of them under the grant-in-aid rules by an illegal and unacknowledged cut in teachers' salaries. It must be admitted that, by and large, these schools make a rather negative contribution to education and life, and they pose a major problem in school education.

In spite of all these limitations and deficiencies, however, these institutions will have to be treated as an integral part of the common school system of public education. It is the responsibility of government to see that they are improved through adequate support and proper management. This responsibility becomes all the greater because the bulk of the students in secondary schools are in these institutions. Unfortunately, the efforts of the Education Departments in dealing with this problem have not been very successful. As the existing grant-in-aid rules are based on egalitarian principles, all private schools are treated alike for purposes of financial assistance. This very often

cramps the progress of the good schools while funds are unduly wasted on the poor ones. The attempt to check malpractices which are often found in the second group of these institutions and to give security of tenure to their teachers has resulted in a large measure of detailed control which again is applied to all schools alike. This has not achieved its primary purpose, but has, on the contrary, weakened the discipline in these institutions and made things unnecessarily difficult even for good institutions who need, not greater control, but greater freedom.

Moreover, the grant-in-aid rules are generally complicated and difficult to administer and the amount of aid is inadequate. Consequently, most of these institutions have remained in a very unsatisfactory condition. What is really needed is a discriminating rather than a uniform policy in respect of assistance to and control of private aided institutions.

The good private schools which maintain high standards and which have been able to attract the services of dedicated and competent teachers will have to be identified and given more freedom and adequate financial assistance. These institutions even today are the quality schools in the system and set the pace for others. They can quickly and effectively be developed as the 'seed farms' in the common school system of public education.

At the same time, a sympathetic but firm policy will have to be adopted with the large group of private institutions which are substandard. They should be given time and assistance to put their houses in order. We expect that many of them will rise to the occasion and become good schools; but there will be many more that may not do so. These latter should be dealt with firmly and either closed down or taken over by the Government.

The position of private schools will be greatly affected by our recommendation that tuition fees should be abolished till the end of class X. When this recommendation is implemented, all fee-charging and aided private schools should be given the option either to abolish the fees and remain within the system or continue to charge fees and become independent. We anticipate, however, that most of the private schools will choose to remain within the common school system of public education.

2

CHAPTER

Formal and Informal Education

FORMAL EDUCATION

A formal education programme is the process of training and developing people in knowledge, skills, mind, and character in a structured and certified programme.

SYSTEMS OF FORMAL EDUCATION

Education is a concept, referring to the process in which students can learn something:

- Instruction refers to the facilitating of learning towards identified objectives, delivered either by an instructor or other forms.
- Teaching refers to the actions of a real live instructor designed to impart learning to the student.
- Learning refers to learning with a view towards preparing learners with specific knowledge, skills, or abilities that can be applied immediately upon completion.

Preschool Education

Preschool education or Infant education is the provision of education for children before the commencement of statutory and obligatory education, usually between the ages of zero or three and five, depending on the jurisdiction. In British English, nursery school or simply “nursery” or playgroup is the usual term for preschool education. In the United States preschool and Pre-K are used, while “nursery school” is an older term. Preschool work is organized within a framework that professional educators create. The framework includes structural, process and alignment components that are associated with each

individual unique child that has both social and academic outcomes. Arguably the first pre-school institution was opened in 1816 by Robert Owen in New Lanark, Scotland. The Hungarian countess Theresa Brunszvik followed in 1828. In 1837, Friedrich Fröbel opened one in Germany, coining the term “kindergarten”.

Developmental Areas

The areas of development which preschool education covers varies from country to country. However, the following main themes are represented in the majority of systems.

- Personal, social, economical, and emotional development
- Communication, including sign language, talking and listening
- Knowledge and understanding of the world
- Creative and aesthetic development
- Educational software
- Mathematical awareness and development
- Physical development
- Playing
- Self-help skills
- Social skills.

Allowing preschool aged children to discover and explore freely within each of these areas of development is the foundation for developmental learning. While the National Association for the Education of Young Children and the National Association of Child Care Professionals have made tremendous strides in publicizing and promoting the idea of developmentally appropriate practice, there is still much work to be done. It is widely recognized that although many preschool educators are aware of the guidelines for developmentally appropriate practice, putting this practice to work effectively in the classroom is more challenging. The NAEYC published that although 80% of Kindergarten classrooms claim to be developmentally appropriate, only 20% actually are.

Age and Importance

Preschool is generally considered appropriate for children between zero or three and five years of age, between the baby or toddler and school stages. During this stage of development, children learn and assimilate information rapidly, and express interest and fascination in each new discovery. It is well established that the most important years of learning are begun at birth.

A child’s brain at this age is making connections that will last the rest of their life. During these early years, a human being is capable of absorbing more information at a time than they will ever be able to again. The environment of the young child influences the development of cognitive skills and emotional skills due to the rapid brain growth that occurs in the early years. Studies have shown that high quality preschools have a short and long term effect in improving the outcomes of a child, especially a

disadvantaged child. However, some more recent studies dispute the accuracy of the earlier results which cited benefits to preschool education, and actually point at preschool being detrimental to a child's cognitive and social development.

A study by UC Berkeley and Stanford University on 14,000 Kindergarteners revealed that while there is a temporary cognitive boost in pre-reading and math, preschool holds detrimental effects on social development and cooperation. The Universal Preschool movement is an international effort to make access to preschool available to families in a similar way to compulsory primary education. Various jurisdictions and advocates have differing priorities for access, availability and funding sources. There has been a shift from preschools that operated primarily as controlled play groups to educational settings in which children learn specific, if basic, skills. It examines several different perspectives on teaching in kindergarten, including those of the developmentally appropriate practice, the academic approach, the child-centered approach, and the Montessori approach to the curriculum.

Gratuity

The gratuity of infant education has been established in some countries, as Spain, beginning in the second cycle but extending to the first cycle.

Methods of Preschool Education

Some preschools have adopted specialized methods of teaching, such as Montessori, Waldorf, Head Start, HighReach Learning, High Scope, The Creative Curriculum, Reggio Emilia approach, Bank Street, Forest kindergartens, and various other pedagogies which contribute to the foundation of education. Creative Curriculum has an interactive web site where parents and teachers can work together in evaluating preschool age children. The web site is very user friendly and prints off many reports that are helpful in evaluating children and the classroom itself.

The web site has a variety of activities that are targeted to each of the fifty goals on the continuum. The International Preschool Curriculum adopted a bilingual approach to teaching and offers a curriculum that embraces international standards and recognizes national requirements for preschool education. In the United States most preschool advocates support the National Association for the Education of Young Children's Developmentally Appropriate Practices. Family childcare can also be nationally accredited by the National Association of Family Childcare if the provider chooses to go through the process. National accreditation is only awarded to those programmes who demonstrate the quality standards set forth by the NAFCC.

Funding for Preschool Programmes

While a majority of American preschool programmes remain tuition-based, support for some public funding of early childhood education has grown over the years. As of 2008, 38 states and the District of Columbia invested in at least some pre-kindergarten

programmes, and many school districts were providing preschool services on their own, using local and federal funds. The benefits and challenges of a public preschool are closely tied to the amount of funding provided. Funding for a public preschool can come in a variety of sources. According to Levin and Schwartz funding can range from federal, state, local public allocations, private sources, and parental fees. The problem of funding a public preschool occurs not only from limited sources but from the cost per child. The average cost across the 48 states is \$6,582.

There are four categories that determine the costs of public preschools: personnel ratios, personnel qualifications, facilities and transportation, and health and nutrition services. According to Levin and Schwartz these structural elements depend heavily on the cost and quality of services provided. The main personnel factor related to cost is the qualifications each preschool require for a teacher. Another determinate of cost is the length of a preschool day. The longer the session, the more increase in cost.

Therefore, the quality of programme accounts presumably for a major component of cost. Collaboration has been a solution for funding issues in several districts. Wilma Kaplan, principal, turned to collaborating with the area Head Start and other private preschool to fund a public preschool in her district. “We’re very pleased with the interaction. It’s really added a dimension to our programme that’s been very positive”. The National Head Start Bureau has been looking for more opportunities to partner with public schools. Torn Schultz of the National Head Start Bureau states, “We’re turning to partnership as much as possible, either in funds or facilities to make sure children get everything necessary to be ready for school”. The goal for funding is to develop a variety of sources that provide for all children to benefit from early learning within a public preschool.

Special Education in Preschool

In the United States, students who may benefit from special education receive services in preschools. Since the inception of the Individuals with Disabilities Education Act Public Law 101-476 in 1975 and its amendments, PL 102-119 and PL 105-17 in 1997, the educational system has moved away from self-contained classrooms and progressed to inclusion. As a result, there has been a need for special education teachers to practice in various settings in order to assist children with special needs, particularly by working with regular classroom teachers when possible to strengthen the inclusion of children with special needs. As with other stages in the life of a child with special needs, the Individualized Education Plan or an Individual Family Service Plan is an important way for special education teachers, regular classroom teachers, administrators and parents to set guidelines for a partnership to help the child succeed in preschool.

PRIMARY EDUCATION

Primary education is the first stage of compulsory education. It is preceded by pre-school or nursery education and is followed by secondary education. In North America

this stage of education is usually known as elementary education and is generally followed by middle school. In most countries, it is compulsory for children to receive primary education, though in many jurisdictions it is permissible for parents to provide it. The transition to secondary school or high school is somewhat arbitrary, but it generally occurs at about eleven or twelve years of age. Some educational systems have separate middle schools with the transition to the final stage of education taking place at around the age of fourteen. The major goals of primary education are achieving basic literacy and numeracy amongst all pupils, as well as establishing foundations in science, mathematics, geography, history and other social sciences. The relative priority of various areas, and the methods used to teach them, are an area of considerable political debate. Typically, primary education is provided in schools, where the child will stay in steadily advancing classes until they complete it and move on to high school/secondary school.

Children are usually placed in classes with one teacher who will be primarily responsible for their education and welfare for that year. This teacher may be assisted to varying degrees by specialist teachers in certain subject areas, often music or physical education. The continuity with a single teacher and the opportunity to build up a close relationship with the class is a notable feature of the primary education system.

Traditionally, various forms of corporal punishment have been an integral part of early education. Recently this practice has come under attack, and in many cases been outlawed, especially in Western countries. The National Council of Educational Research and Training is the apex body for school education in India. The NCERT provides support and technical assistance to a number of schools in India and oversees many aspects of enforcement of education policies. In India, the various bodies governing school education system are:

- The state government boards, in which the majority of Indian children are enrolled.
- The Central Board of Secondary Education board.
- The Council for the Indian School Certificate Examinations board.
- The National Institute of Open Schooling.
- International schools affiliated to the International Baccalaureate Programme and/or the Cambridge International Examinations.
- Islamic Madrasah schools, whose boards are controlled by local state governments, or autonomous, or affiliated with Darul Uloom Deoband.
- Autonomous schools like Woodstock School, Auroville, Patha Bhavan and Ananda Marga Gurukula.

Primary school teaching in India consists of 12 grade levels. These are:

- Kindergarten: nursery - 3 years, Lower Kindergarten (LKG) -4 years, Upper Kindergarten (UKG) - 5 years
- 1st class: 6 years
- 2nd class: 7 years
- 3rd class: 8 years

- 4th class: 9 years
- 5th class: 10 years
- 6th class: 11 years
- 7th class: 12 years
- 8th class: 13 years
- 9th class: 14 years
- 10th class: 15 years
- 11th class: 16 years
- 12th class: 17 years.

SECONDARY EDUCATION

Secondary education is the stage of education following primary school. Secondary education is generally the final stage of compulsory education. However, secondary education in some countries includes a period of compulsory and a period of non-compulsory education. The next stage of education is usually college or university. Secondary education is characterized by transition from the typically compulsory, comprehensive primary education for minors to the optional, selective tertiary, “post-secondary”, or “higher” education for adults. Depending on the system, schools for this period or a part of it may be called secondary schools, high schools, gymnasias, lyceums, middle schools, colleges, vocational schools and preparatory schools, and the exact meaning of any of these varies between the systems.

NON-FORMAL EDUCATION

Non-formal education became part of the international discourse on education policy in the late 1960s and early 1970s. It can be seen as related to the concepts of recurrent and lifelong learning. Tight suggests that whereas the latter concepts have to do with the extension of education and learning throughout life, non-formal education is about ‘acknowledging the importance of education, learning and training which takes place outside recognized educational institutions’. Fordham suggests that in the 1970s, four characteristics came be associated with non-formal education:

- Relevance to the needs of disadvantaged groups.
- Concern with specific categories of person.
- A focus on clearly defined purposes.
- Flexibility in organization and methods.

In many northern countries the notion of non-formal education is not common in internal policy debates - preferred alternatives being community education and community learning, informal education and social pedagogy.

OPEN SCHOOL

The third instance of non-formal education corresponds to open systems or open learning, which have drifted much farther apart from the features of formal education, creating a

wide, deep rift. As remarked by Butts, “open learning systems are defined as those which offer students a measure of flexibility and autonomy, to study the programmes of their choice when and where they wish, and at a pace to suit their circumstances. “The features ascribed to open systems, by this author, necessarily set them up as non-formal education instances, jointly with correspondence learning and distance study. As Butts points out, “...distance learning is seen... as one type of open learning.” Correspondence learning can be deemed a type of distance learning and, as distance learning can be said to be an instance of open systems, we conclude that this latter is in the most widely encompassing class among non-formal education examples. Some authors also consider rather freely the concept of open education - as synonymous with open systems. As Yalli says, “the idea of openness may be twofold: open as to structures, that is, a rupture of the physical barriers of educative institutions, so as to provide free access to schools; or open as to methodology and learning resources.” And, he concludes: “The essential fact about open education is that it does not matter how knowledge is acquired, all means are valid. The open learning system aims at the formation of independent students who have capacity for self-discipline and a high capacity for synthesis and for analysis.” This author defines that in an open system, learning is the function of an interaction between the student and the actual world.

DISTANCE LEARNING

According to Holmberg “Distance study is learning supported by those teaching methods in which, because of the physical separateness of learners and teachers, the interactive, as well as the preactive phase of teaching is conducted through print, mechanical or electronic devices.” Distance learning is based on non-contiguous communication, that is, “the learner is at a distance from the teacher for much, most or even all the time during the teaching-learning process”. Based on this definition, we may infer that the concept of distance learning is wider than that of correspondence learning, with which it is sometimes confused.

Thus, Butts remarks that “the rapid adoption, over the past 10 years, of the phrase ‘distance learning’ to replace ‘correspondence courses’ would seem to reflect the incorporation of media other than print; the fresh impetus coming from research into individualized learning and self-instructional methods; the broadening of the social base for open learning systems; and the development of courses and qualifications designed specifically to meet the needs of distance learning students.” The concept of open learning systems used by Butts is wider than that of distance learning, as below analysed.

In sum, according to Holmberg, the three universally accepted features of distance learning are as follows:

- Typical of the whole distance study is that it is based on non-contiguous communication, *i.e.*, the learner is at a distance from the teacher for much, most or even all of the time during the teaching-learning process.

- A pre-produced course, as self-instructional as possible, printed and/or consisting of presentation brought about by other means than print guides the study.
- Organized non-contiguous two-way communication is a constitutive element of distance study. It is in most cases principally brought about by assignments for submission for the students to solve and answer and for the tutors to comment on but freer forms of communication also occur.

The organization and administration of distance learning significantly differs from those of formal education. Thus, for instance, no students attend classes at the institution, except for occasional visitors. There are no classrooms; instead there are places where multidisciplinary teams comprised of redactors, authors, audio-visual experts, and so on, plan and compose the materials that will be used. In distance learning we find no “academic semesters”. The students may at will discontinue studies whenever he needs or wants to do so.

As per Holmberg, distance learning is comprised of the following basic activities:

- The development and technical production of distance study courses;
- The distribution of course materials;
- The non-contiguous two-way communication between students and tutors/counselors; and
- Record-keeping.

Holmberg also reminds us that, in some case, other activities may be required, as for instance:

- Course certificate
- Examination and degrees
- Supplementary face-to-face contacts between students and tutors/counselors.

A large experience has been obtained over these years through application of distance study at various levels, for different target populations. Our main concern in this stage relates to the use of distance study for higher level education. A well-succeeded example of such use is the Open University. As pointed out by Oliveira, Open Universities are generally based on distance study through one or more communication media, such as radio, TV and the printed press.

They mostly formulate the instructional materials used in their courses, for the most part employing a distance tutoring system that contracts teachers to provide the required support to the performance of supplementary activities. The assessment and graduation requirements are not uniform and in some cases, the diplomas are on a par with those issued by regular universities, whereas in others we find that certain restrictions are made with regard to given courses. There are also open universities which are in no way concerned with the validation or equivalence of the offered courses and of their diplomas to those given in the existing formal universities. Open universities need their own organizational structure that differs widely from that of traditional universities. According to Oliveira, “The nature of their tasks and the modus operandi of open universities provide a mixture of academic culture and industrial activity”, requiring

the cooperation of professionals from varied backgrounds to act as redactors, educational planners, professors specializing in the different fields, audio-visual experts, and so on, thus displaying a multidisciplinary character.

The materials forwarded to the students, comprising printed texts, audio or videotapes, kits, etc., is usually validated prior to their utilization, so as to ensure a high degree of efficacy and efficiency. Oliveira also notes that “in countries with a shallower academic sedimentation, open universities seldom have their start on an academic basis and this results in their remaining for the most part on the fringes of the educational process” - an extremely significant aspect which will be relevant to the proposal that will be submitted below.

The British Open University may be mentioned as being the most successful among all open universities. As described by Grayson, the British Open University founded in 1969 was created in order to remain open to new people, methods and ideas. The traditional matriculation requirements were abolished and efforts were made to attract working students. Approximately 63.000 students enrolled in 1980, and its syllabus includes printed materials, audiotapes, reading, study guides, self-assessments and radio and TV programmes. Tutorial assistance and counseling are available in about 280 study centers throughout Great Britain. We must include here a mention to the high quality level of the produced instructional resources, as well as to the disposition shown by the planners of the British Open University always to remain receptive to non-conventional programmes. The courses encompass six areas, namely: education, mathematics, sciences, social sciences, and technology

Their duration is approximately one year. The success of the British Open University led to the creation of several open universities in France, in Germany and in the United States as from 1971, without even mentioning the several Latin-American efforts in this field. Oliveira mentions the pilot-experiment of the China Open University which is planning to enroll approximately two and a half million students. According to that author, “despite a variety of forms and contexts, Open Universities illustrate the many possibilities for widening the scope of higher level education.”

As to the expression “Open” he remarks that it may relate to:

- The moment in which the student enrolls on a course for which the required credits system has been extremely simplified;
- The educational process itself, as well as the range of options offered to the students as regards programmes and courses;
- The fact that the course is taught at a distance; and
- The fact that although not generally providing final degrees, they offer to students the possibility and the required flexibility to stay on or to leave the courses.

CORRESPONDENCE LEARNING

Correspondence Learning organized, structured correspondence schools date from more than one century. Several works and authors mention that in 1856, in Berlin, Toussaint

and Langenscheidt founded a correspondence languages course. In 1886, in England, a graduate studies correspondence course was introduced. A “Society to Encourage Study at Home” was organized in 1873, in Boston, and the first formal experience took place in 1883, in New York, the “Correspondence University”. In several countries similar efforts were made, named “enseignement par correspondance” in France, “fern Unterricht” and “fernStudium” in Germany, “home study”, “tuition mail” and “postal tuition” in England; “ensino por correspondência” in Portugal; “ensenanza por correo” and “ensenanza por correspondencia” in Spain. Nowadays, there is a large number of correspondence schools all over the World, encompassing studies that range from basic education to university studies, including a wide variety of subjects in the professional area.

Correspondence course participants are found in all age brackets and economic-social classes. But, which are the main features of correspondence learning? It is a planned and systematized activity, based on the preparation of printed educational materials which are forwarded to students who are physically separated from the teachers who can give but a limited assistance to them. Correspondence learning is an individualized learning system that allows students to proceed at their own pace, according to their interests. The institutional materials are for the most part printed and are generally prepared by a teacher who has not enough didactic and technical knowledge to prepare top quality educational material. Although a number of correspondence courses currently offer other types of instructional material - audio-tapes and videotapes, kits, etc. - we shall for classification purposes solely consider the printed materials offered by correspondence courses. We shall reserve the name “distance learning” to the courses prepared on a high technical level, by a multidisciplinary team, administered by a relatively large institution, comprising a wide variety of educational materials.

Correspondence courses generally establish a bi-directional communication by mail, supported by the teacher who corrects the paperwork, offers guidance and the requested explanations. A degree may or may not be obtained and there is no pressure - the student's motivation is the basic factor for the program's success. It is not difficult to see that correspondence courses do not incorporate several features of the formal education and are thus classified in the field of non-formal education.

NFE CENTERS

The Centrally Sponsored Scheme of Non Formal Education (NFE) was introduced in 1979-80 on a pilot basis with a view to support the formal system in providing education to all children upto the age of 14 years as enunciated in the Directive Principles of the Constitution. In subsequent years, the NFE scheme was expanded to cover 10 educationally backward states of Andhra Pradesh, Assam, Bihar, Himachal Pradesh, Jammu and Kashmir, Madhya Pradesh, Orissa, Rajasthan, Uttar Pradesh, and West Bengal. The National Policy on Education (NPE) , 1986 recognised that the school could not reach all children and a large and systematic programme of non formal education would be required for school dropouts, for children from habitations without

schools, working children and girls who could not attend whole day schools. Thus NFE became an important component of the overall strategy for achievement of Universalisation of Elementary Education. The NFE scheme was revised in 1987-88. While the focus continued to be on 10 educationally backward states, but it also included urban slums, hilly, tribal and desert areas and projects for working children in other states and Union Territories as well. A major portion of the NFE scheme is run by the state governments which set up NFE centres. One component of this scheme provides grants to Voluntary Agencies directly from the central government for running of NFE centres and a third is for projects of experimental innovative nature by VAs.

The Programme of Action 1992 outlined strategies for strengthening of the NFE scheme including:

- Setting up NFE centres based on a micro-planning exercise carried out for UEE.
- Central role for community by involving them in setting up of the centre, identification of the instructor and supervision of the NFE centre.
- Efforts to evolve different models of NFE programme for different target groups.
- Adequate training and orientation of NFE instructors. 30 days initial training of instructors and 20 days in subsequent years etc.
- Linkage with the formal school to facilitate lateral entry of the learners from the NFE stream.
- Efforts to link non-formal courses with formal schools.
- Adoption of learner-centered approach. The learning levels for the learners to be equivalent to the formal system.

Current Status of NFE Scheme

- Being implemented in 25 States/UTs by the state governments and by 826 VAs.
- 2.38 lakh primary and 6800 upper primary centres are presently sanctioned in the state sector.
- 58,000 primary and 1000 upper primary centres are run by VAs.
- 41 experimental and innovative education projects are being implemented by VAs.
- Total coverage of children under NFE scheme is about 74 lakh.

Review and Assessment of the NFE Scheme

Several evaluations and assessments by state governments, institutions and most notably the Programme Evaluation Organisation of the Planning Commission have indicated that the implementation of the Scheme has not been satisfactory.

The PEO's findings were:

- Insufficient involvement of the local community, the Village Education Committees and the Panchayati Raj Institutions.
- The absence of linkages for entry at different levels into formal schools and for tie-ups with the National Open School.

- The notion that the alternative system is inferior, second-rate and second-grade, both qualitatively and quantitatively.
 - Insufficient decentralisation of administrative and financial powers.
 - Insufficient flexibility. NFE needs to recognise that different children's groups have different educational needs and modify itself accordingly.
 - Lack of success with girls. The attendance at girls' centres and the number of women functionaries in the programme have been noticeably low.
 - Poor coordination of the work of VAs with state governments.
 - Low overall coverage of the scheme. It covers less than 10% of the out of school children.
 - Delay in release of funds at all levels.
 - Poor completion rates for the primary level by children studying NFE centres.
- Very low transition rates to the formal system

The NFE centres function for two hours daily at a time suitable for learners. But in many states the centres have functioned in the evening and night to accommodate children who are working during the day. The NFE Scheme did advocate flexibility in various aspects of running of the centre but the manner of its implementation resulted in a uniformity and rigidity almost across the country. Certain states like Andhra Pradesh and Madhya Pradesh could not implement modified approaches which had been worked out by them within the existing NFE scheme. A large number of NFE centres were set up in habitations which had formal schools and therefore the target of small and scattered SC and ST habitations without schools, did not receive a high priority. Clearly, in its present form the NFE Scheme could not ensure quality primary education for out of school children and the objectives and measures outlined in the NPE and POA could not be adequately met.

3

CHAPTER

Education and the Practice Environment

INTRODUCTION

According to an agreement between the International Association of Schools of Social Work and the International Federation of Social Workers, the social work profession “promotes social change, problem solving in human relationships and the empowerment and liberation of people to enhance well-being. Utilizing theories of human behaviour and social systems, social work intervenes at the points where people interact with their environments. Principles of human rights and social justice are fundamental to social work.” As such, social workers are a key component of modern social service provision. Among other things, they provide counseling, assist individuals in accessing social services and other benefits, lobby for the disenfranchised, and engage in actions designed to influence social policies.

During the Soviet era, social problems were either unrecognized or minimized in the Europe and Eurasia (E&E) region, which limited the availability of social and other support services. Former Communist regimes in the region relied heavily on financial allocations, many of them universal allowances, to solve social problems while services were severely underdeveloped. Social issues, including child abandonment, disabilities, domestic violence, or unemployment were hidden in homes or large institutions, which provided medical care and accommodation. Social workers (as well as many other types of service providers familiar to those who live in the West) did not exist. Thus, in

attempting to create modern systems of community-based social services, E&E countries have had to build the profession of social work from the ground up in many cases. While some progress is being made in this regard across the region, it is uneven, with some countries being further along than others. Building the social work profession is a complex undertaking that involves creating education programmes, developing and strengthening curricula, helping to nurture professional associations for social workers, developing licensing and practice standards, and raising awareness about the need for social workers, among other things. Moreover, building the profession must be closely coordinated with other programme activities aimed at creating and strengthening social services.

Even in those E&E countries where progress has been the strongest, many obstacles remain. Government officials and the general population often do not understand what social workers do and how their work differs from other professionals (*e.g.*, teachers or social pedagogues) .

In some countries, schools of social work are now producing cohorts of trained social workers, but these individuals are struggling to secure employment because those who provide social services may not understand the role of social workers or may not have the funds to pay them. Social workers in the region earn notoriously low wages and are often forced to seek higher paying employment in other fields.

BACKGROUND AND RATIONALE OF THE STUDY

Social work as a profession is relatively new to the E&E region, yet a well-trained social work workforce is a key component in the creation of an effective system of social services that can provide a continuum of community care to vulnerable individuals and groups. USAID Missions with an interest in social services programming (related to child welfare, disabilities, trafficking in persons (TIP) and domestic violence, social assistance social benefits and cash transfers, and related issues) , have identified an active and viable social work profession as critical in such programming.

As USAID Missions tackle programming related to a wide array of social services and social assistance programmes, there is a strong need for examples of best practices and creative thinking about how to facilitate recruitment and retention, training, and advocating for social workers in host countries.

The target countries for this study are: Albania, Armenia, Azerbaijan, Belarus, Bosnia, Bulgaria, Croatia, Georgia, Kazakhstan, Kosovo, Kyrgyzstan, Macedonia, Moldova, Romania, Russia, Serbia, Montenegro, Tajikistan, Turkmenistan, Ukraine, and Uzbekistan.

Purpose of Study: The purpose of this study is to collect, organize and report on information about model programmes, best practices, and lessons learned in addressing key issues related to social workers in the E&E Region.

Objectives: The study aims to:

- Describe the current status of social work practice and education in the region;

- Describe the gaps between the reality of social work practice and education (*i.e.*, what is desired, and strategies for improving outcomes for social work) ;
- Provide examples of model programmes and best practices from the region for key issues;
- Discuss lessons learned and challenges to implementation of best practices;
- Provide sources of information and tools to enhance programming related to key issues through websites, titles of documents and locations, professional guidelines, toolkits, and regional contacts.

Target Audience: The primary target audience for this activity is USAID Missions and USAID/Washington personnel who are involved in Investing in People (IIP) programming related to social services and vulnerable groups, or social assistance. Additional audiences include anyone interested in detailed information about the role of social workers in providing social services in the E&E region, such as the

U.S., Department of State, other U.S., government agencies, other donors, and NGOs that work on this issue in the region.

Data collection methodology: This desk-top study was conducted April-July, 2008 and utilized the following sources:

- *Literature review:* A review of the social work literature on the professionalization of social work, outcomes studies on models of practice, development and application of standards of practice, and studies on social work and social work practice specific to the Region was completed.
- *Documents review:* Copies of relevant policies, laws and strategic plans; descriptions of programmes and services available for different groups (public and private) ; USAID publications and websites; available data that included at-risk populations, workforce, financial resources, etc.; evaluation and research on programmes and services; curricula from education and training programmes and related documents were obtained through an internet search and personal contacts.
- *Field research:* Dr. Sergei Lukoshov, Technical Expert, EveryChild, worked directly in the field to provide information from selected countries of the Former Soviet Union. Dr. Rebecca Davis conducted field research in Romania during the Rutgers Study Abroad Programme. Interviews were held with university faculty, students, professionals, ministry and local level authorities, professional associations, and NGOs.
- *Individual and Group Interviews:* Interviews were completed using e-mail, telephone conferencing, and internet-based conversations and messaging through SKYPE. Included were representatives of UNICEF, advocacy groups, NGOs, university representatives such as Fulbright Scholars, Junior Faculty Development Programme (JFDP) Fellows, USAID Mission programme representatives, and association representatives. A complete list of those interviewed can be found in Appendix A and the questions discussed during interviews with USAID Mission representatives can be found in Appendix B.

SOCIAL WORK EDUCATION AND PRACTICE IN THE UNITED STATES

HISTORICAL DEVELOPMENT

The roots of the social work profession in North America and England are found in the Charitable Organization Societies (COSs) serving the poor and a desire by those leading that movement to provide consistent learning and education to “friendly visitors”. From its origins, social work recognized the value of real life/practical experience along with theoretical knowledge. So much so that when Mary Richmond first published *Social Diagnosis* in 1917, with a goal of establishing a professional base for casework through training, she was doubtful of university based education, which historically did not utilize field education as a model. In a paper published in 1897, when discussing the creation of the Charitable Organization Societies, Richmond spoke of the “right” of those doing the work to demand the right to further education and professional development, and a living wage. This article included a discussion on what should come first: professional standards of practice or a training school to establish them.

Essentially, in the United States the creation of formal social work education came from two paths, the COS’s and the Settlement Houses. Mary Richmond’s COS partnered with the New York School of Philanthropy to create technical training for case work and agency practice. Ultimately, this work came under the auspices of Columbia University in the 1940s. At the same time, Jane Addams’ Settlement House movement began working with the University of Chicago, having identified the need for an educational foundation based in social policy and social philosophy so that students would be prepared “for their role in constructing a better society”. The profession first emphasized attempting to determine the causes of individual problems through a focus on internal causes, and interventions were focused on rehabilitation of those internal problems. As time went on, the profession began to recognize the role societal causes played in individual problems, and began to focus its attention on social change.

SOCIAL WORK TODAY

Some might say that social work is a well developed field of practice in the US today. This perception is based on the profession meeting criteria used to measure the legitimacy of all professions: a systematic body of knowledge, a service mission, professional authority, self-regulation such as licensing or certification, and a regulative code of ethics. A quick review of the social work profession today indicates all those are in fact present. The higher education opportunities include two degrees, Bachelor in Social Work (BA) and Masters in Social Work (MSW) regulated by a national body, the Council on Social Work Education (CSWE) – an oversight body that sets standards for curriculum and field education. There is also a doctoral degree, which prepares social workers to carry out research in social work rather than relying on research in other disciplines. In

addition to the status gained by achieving one of these degrees, most social workers also enjoy the benefits of being identified as a social worker. In many ways, this professional identity was solidified with the emergence of social work licensure. And while it was originally developed as a consumer or public protection mechanism, it has also served to carve out a scope of practice for social workers particularly in direct or clinical practice. Both the education system established for social workers and the licensing system that has emerged support the premise that theory and practice go hand in hand. Every social work education programme accredited by the Council on Social Work Education (CSWE) includes both classroom study and field experience. Similarly, licensing requirements generally establish the need for continued classroom learning as well as peer or clinical supervision. The value placed on practical experience (through field education) is one of the defining attributes of social work in the US. However, like other professions, social work's status is determined to some degree by those they serve. The status of the profession is low because clients are typically held in low regard. Historically, this has impacted funding opportunities, work conditions, compensation, and prestige. As a result, recruitment and retention present unique challenges at any given time in the history of the profession. The research that underpins this report revealed the presence and/or emergence of similar elements as the profession of social work continues to develop in the former Soviet Bloc: specialization in educational programmes with a prescribed curriculum, recognition of particular degrees and training to assure certain knowledge and skill development; professional regulation through licensure or certification as a means of consumer protection; development and implementation of best practice standards in response to problem or crises identification. Also notable is the relationship of an identified crisis or social need and the development of social work practice and standards of care.

REGULATION OF PRACTICE: REGISTRATION, CERTIFICATION AND LICENSURE

In the U.S., The first social work regulatory bill was introduced in California in 1929, but failed to pass. In 1945, legislation was enacted in California for voluntary registration of social workers. In the meantime, Puerto Rico had enacted legislation in 1934, making it the first state to regulate social work practice. Since 1960, every state, Puerto Rico, the U. S. Virgin Islands and the District of Columbia have some form of social work regulation, either certification or licensure², which are used almost interchangeably. Voluntary registration is no longer used as a form of practice regulation in US jurisdictions. A major issue today is the lack of uniformity in the terminology and levels of licensure and certification across the US. The Association of Social Work Boards (ASWB) is the national association of social work boards from each state that regulates social work practice. ASWB develops and maintains the social work licensing examination used across the country and in several Canadian provinces. It is also a central resource for information on the legal regulation of social work. ASWB also

serves as a coordinating body for individual state boards to share information and work together on common issues of concern. ASWB has proposed a model social work practice act that has three levels of licensure: Licensed Baccalaureate Social Worker (LBSW) , Licensed Master Social Worker (LMSW) , and Licensed Clinical Social Worker (LCSW) .

SOCIAL WORK PROFESSIONAL ASSOCIATIONS

The National Association of Social Workers is the largest membership organization for professional social workers involved in all areas of social work practice. In the US prior to 1955, there were numerous professional associations existing at any given time, all with independent missions, focused on different specialty practice areas, and with varying numbers of members, offering varying levels of services to members. Then in the late 1940s, the association representing schools of social work initiated discussions with the membership organizations regarding possible unification. Over a period of several years, work was undertaken to develop one central organization, representing the needs of all professional social workers. The National Association of Social Workers (NASW) began operations on October 1, 1955. There is a chapter in every state as well as in a number of territories, each with a separate governing body, mission, and staff. They are all responsible for public education, professional development, advocacy, and public policy work. These functions are important to note as they are essential to the development of both policy and practice in the United States. It has been the collective voice of the professional associations and the educational institutions that helped raise the status and visibility of the profession.

The practice of social work is guided by the NASW Code of Ethics, which defines the facets of ethical social work practice at all levels. Professional social workers also receive guidance from their professional association through the Standards for Social Work Practice which have been developed for specific fields of practice (*i.e.*, child welfare, adolescence, substance use) as well as areas of social work knowledge (*i.e.*, cultural competence, technology, continuing education) .

Despite the intentions of the founders of NASW, social workers in the US have continued to form and join multiple organizations as they become concerned that NASW cannot or does not represent their individual needs. Some examples of the professional organizations that exist side by side with NASW are: The National Association of Black Social Workers, The Society for Clinical Social Workers, The Society for Transplant Social Workers, The National Network for Social Work Managers, The Association for the Advancement of Social Work with Groups, and perhaps the largest other entity – the Council on Social Work Education.

REFORMS OF HIGHER EDUCATION IN E&E: IMPACT ON SOCIAL WORK

The Bologna Process, initiated in 1999, is aimed at modernizing European Higher Education and creating a European Higher Education Area (EHEA) by 2010. The long

term objective is to facilitate speedy entrance of educated professionals into the job market. Other objectives of the programme are to enhance cross-border mobility of students and job seekers, and to increase the competitiveness of European higher education internationally. The Bologna Declaration calls for:

- Adoption of a system of easily readable and comparable degrees;
- Adoption of a degree structure based on two primary degrees (undergraduate and graduate) ; and
- Adoption of a system of credits such as European College Transfer System (ECTS) (for ease of transfer and accumulation of credits) .

Most of the social work programmes in universities in the formers Soviet Bloc countries, including the Former Soviet Union, have joined the Bologna Process. This means incorporating the required changes in programme structure and curriculum that aims to reform the nature of education – to make it more practical and applied. One major change is that the bachelor's degree in social work takes three years rather than four to complete, and the MA takes two years rather than one. By 2010, the plan is to have standard recognized European degrees at the bachelor's and master's or doctoral level in much the same way as exists in the United States. The implementation of this has proven to be difficult as there is resistance among some of the established universities in Western Europe as well as in the countries of the former Soviet Bloc. It is beyond the scope of this project to assess the impact of the Bologna Process. However, the benefits of a standardized curriculum and credit system may be outweighed by increased out-migration of social workers to countries where social work is more established in search of better working conditions and higher salaries.³

BEST PRACTICE MODEL APPLIED TO THE PROFESSION OF SOCIAL WORK

The current report builds on two previous USAID-funded studies on the evolution of community-based services in E&E countries that highlight good practice models that have been established as part of professional social work. This report explores the social work profession as a key dimension of social sector reforms and follows the same four-pillar framework and best practices matrix as the earlier Davis publications with an emphasis on the profession of social work as it is changing within the practice environment. The key principles of the framework and best practices model are consistent with international standards for social work education and social work practice.

BEST PRACTICES IN COMMUNITY-BASED SOCIAL WORK

The best practices framework consists of four pillars, deemed a comprehensive model of community-based social services for vulnerable groups. The framework incorporates the legal framework of the profession including laws related to practice within the

social welfare system, the structure of services and the work environment for social work, the human capacity dimension focused primarily on education and training, and outcomes and performance measured through research and evaluation of professional practice.

The specific elements for each pillar as they apply to social work include:

- *Policy and Legal Framework:* This pillar includes policies and laws that reflect internationally recognized standards for the social work profession, the legal mandate and policy mandates for social work, regulation of practice through legislated Code of Ethics and certification or licensure, and policies relating to standards of practice that reflect best practices in family and community care.
- *Structure of Services and the Practice Environment:* This pillar includes the programmes and services in which social workers practice, level of qualifications, job functions, salaries and status of social workers, relationships with clients, other professionals, and the public authorities. This pillar also includes development and implementation of standards of practice that are consistent with professional social work practice inclusive of community outreach and accessibility of services, citizen and user involvement through initiatives such as in-home services and volunteerism, and public education about social services.
- *Human Capacity:* This pillar focuses on the acquisition of knowledge, values and skills for social workers providing direct services and those managing and supervising services. This includes academic and professional education and training, curriculum development activities, field education, and conferences and workshops provided by a range of providers. In addition, this pillar includes role and activities of professional associations and quality improvement approaches to standard setting and professional accountability.
- *Performance Outcomes and Measures:* This refers to indicators used to measure outcomes for social work interventions, systems for monitoring social work inputs, cost-benefit analyses, development of evidence-based practices, research on the professionalization of social work such as salaries, standards, opinions and attitudes, client satisfaction, and client outcomes through research and evaluation.

The table below illustrates good practices in each of the four pillars. It describes how each pillar can function effectively to provide the best outcomes for society, from vulnerable populations to taxpayers.

OBSERVATIONS AND FINDINGS

This section is a synthesis of the information obtained through the social work literature, evaluation studies, project descriptions, and the ideas and perspectives presented by social work advocates, professionals, and educators that were interviewed. An observation that underscores all of the findings is the common goal among social workers and social work advocates to legitimize social work as a profession and clarify social

work's role in the service delivery system. Although, in a relatively short time, social work has been established as an occupation in the E&E region with specified job functions within the social welfare system, the degree to which it is viewed as a profession based in theory with a shared language, a code of ethics, and professional accountability varies greatly. From the first professional schools established in the Former Yugoslavia in the 1950's, to the more recently established social work jobs and schools established in Central Asia and the Caucasus, there is movement toward social work as a legitimate profession. But the progress of social work's transition from an occupation to a profession has been, and will continue to be, influenced by the social, political, and economic context of each country. This section is an attempt to present some of the approaches that are unique to individual countries as well as to identify trends relative to the development of the profession of social work within the environment of social sector reform. The authors have tried to capture, as much as possible, "lessons learned" through the eyes and experiences of those working "on the ground" in the region.

LEGAL AND POLICY FRAMEWORK

- *Social Work Legislation:* With the exception of Romania, which has a separate law on social work, the legal framework for the practice of social work is part of the social protection and social assistance legislation. Armenia's legislation on social assistance has a separate chapter defining a social work specialist as a person with an advanced degree. The law in Romania establishes four levels of licensure and also provides a possibility for independent practice. Both Armenia and Romania include a Code of Ethics as part of the legislation.
- *Regulation of Practice (Licensure):* Although Romania's social work licensing law went into effect in January 2005, implementation is described as slow and difficult. Georgia is slated to begin social work licensure in 2009. This study revealed that many countries are in the early discussion stages about the value of professional licensure for social workers, and indeed it is seen as a critical step for improving the status of the profession.
- *Laws on Public Protection:* Laws that establish clients' rights to make a complaint against a social worker, client's rights to confidentiality and involvement in decision-making are part of the social protection legislation. There is recognition that clients have this right although cases of malpractice are not known to those who were interviewed. Licensure laws, which philosophically are to "protect the public," are framed as providing "professional status to the social worker."

STRUCTURE OF SERVICES AND PRACTICE ENVIRONMENT

Definition of Social Work and Social Work Practice: Across the region, the meaning of "social worker," and related terms, as defined in the legislation and practice, is confusing. The Former Yugoslav Republics have a more consistent definition and understanding

of social work and social work practice, in part, due to the well-established social work schools and Centers for Social Work. For the most part, in other countries, there is confusion between social services, social assistance and social work. Social work as a profession and practice means persons that are working in social services providing basic services such as food, clothing and financial benefits (social assistance). In Ukraine, Russia and Armenia, the term “specialist of social work” is a degree that is somewhere between the BA and MA level degrees. It refers to someone that has higher education and credentials than “social workers.” In Romania, the “case manager” is at a specialist and supervisor, with an MA in Social Work at least 3-5 years, whereas in other countries the term usually refers to someone that is less experienced with lesser professional status (usually at the BA level).

- *Job Functions:* Most social workers are employed in public social services. Social work tasks primarily focus on completing administrative tasks related to eligibility for various social and financial benefits and providing concrete services such as housing, clothing, food, and meeting health-related needs such as medications. These tasks to require a lot of paperwork. Across all countries, there is concern that social workers are not performing the work that they are trained to do and they feel is needed such as family support and psycho-social counseling. There have been recommendations to separate the tasks related to eligibility for benefits (make that a clerical job) and family support and psychosocial services.
- *Salaries and Low Professional Status:* One of the most consistent and pervasive issues discussed was low salaries for social workers and difficult working conditions (large caseloads, excessive paper work, and limited resources for clients). Combined with low status when compared to other human services professionals (*i.e.*, doctors, nurses, teachers, psychologists), recruitment and retention of qualified social workers is affected across all countries. Social workers often leave traditional social service positions to take jobs in other sectors and ministries. Therefore, social service positions are also frequently filled by unqualified individuals, even if the position is for a qualified social worker. This is even more prevalent in rural towns and villages where the salary differentials may be even greater. Some caution against comparing social work salaries to those of health care professionals because of “under the table” payments given to doctors, nurses, etc., which makes the disparity even greater.
- *High Status as a “Helper:”* In spite of the negatives of salary and professional status, there is status and value in “being a helper” of those who have special needs. In Georgia, it was noted that there is a certain “prestige” to being a social worker as a new profession. A study on perceptions of social work in Croatia revealed that, although there was a negative attitude toward the profession of social work, there was a more positive attitude towards the work that they do.

- *Social Work as a Career:* The recognized trajectory for professional social workers is: social work as a job/occupation Æ social work as a profession Æ social work as a career. Experienced social workers expressed concern about their future with a career in social work. There are few opportunities to advance, or to make a career in social work, as there are few management and supervisory positions available and limited opportunities for diversified practice, and especially specialized services.
- *Role of Independent, Private Practice:* Seen as the hallmark of a legitimate profession and a career option, only Romania has a law that allows for private, independent practice although, in reality, there is not yet a market for private practice. Some emerging tension was observed related to the prestige associated with more psycho-social (clinical) work versus traditional case management within the public sector. This is an ethical and professional dilemma for an increasingly professional workforce as private practice is seen as serving those with middle and upper incomes. The authority to have an independent (clinical) practice is also viewed as another criterion of legitimate
- *Professional status:* But, the conflict remains concerning social work's mission to serve the poor and disenfranchised.
- *External Labor Migration:* For the countries in Central and Eastern Europe in which social work is more developed and the borders have opened for easier travel, trained and experienced social workers are leaving their countries to find social work jobs that have higher pay, better benefits, and more professional jobs.

Social workers are actively recruited by Western Europe, and especially the UK. Social workers find better working conditions and salaries and hope for a social work career that can provide financial and professional security. One of the aims of the higher education reforms of the Bologna Process is to make it easier for cross-border employment, at least between and among those countries of the EU.

- *Internal Labor Migration and Access to Services:* Internal migration, rural to urban, as experienced in Belarus, also contributes to the uneven distribution of a qualified workforce. Social protection strategies call for increased outreach and case-finding. This is especially a challenge in rural areas, not only due to transportation issues, availability of services and specialists, and overall access to services, but also due to increased isolation and workloads. Creative solutions need to be explored such as training paraprofessionals and integrating social workers into health and education services.
- *Risk Management and Safety of Social Workers:* An emerging issue noted within this framework relates to safety issues for social workers in particular regions. This has been highlighted in specialized fields of practice: human trafficking, domestic violence, and infectious diseases such as HIV/AIDS.

- *Transfer of Knowledge and Specialization:* Social work in most countries has grown out of child protection, directly in response to deinstitutionalization. Transferring knowledge and experience across fields of practice is beginning. In Romania, experience in community care for children was transferred to the fields of disability and aging. In Croatia, the experience in dealing with trauma recovery for war victims is now being applied to the field of domestic violence and child abuse. The emerging cross-discipline and cross-border models in work with victims of trafficking and domestic violence have direct application to child welfare, developmental disabilities and mental health and addictions, all growing fields of practice.
- *The Social Worker as Manager and Supervisor:* There is an overwhelming recognition about the significant contribution that social work supervision has on quality of services. There is also the reality that supervision, as practiced in the region, is primarily administrative in nature and does not address individual casework/clinical practice and professional coaching and mentoring. For those countries that have developed a Masters Degree in Social Work Supervision (*i.e.*, Romania, Bosnia and Herzegovina, and Croatia) , the challenge has been implementation of the supervisory job, with the title and function of a social work supervisor that reflects what is being taught. The model that has been developed in St. Petersburg through EveryChild is a promising model. In addition, managers and administrators do not receive formal training in strategic planning, programme development, and programme evaluation, all which are considered to be part of a supervisor and manager's professional job functions.

EDUCATION AND TRAINING

University Education: Social work education is alive and well across the Region. Although some social work education programmes are in early developmental stages, there are others that are well- established. The Former Yugoslav Republics have had established training programmes since the 1950s with Bulgaria, Romania, Russia and the Ukraine establishing programmes in the early 1990s. Other countries are just establishing both the BA and MA programmes, while some are just now establishing a Ph.D. programme, although primarily in related disciplines. The aspect of university education that causes the most concern is the heavy reliance on theory. With the emphasis on theory in universities, students expressed concern about learning how theory is integrated into practice. Field education in the majority of countries has not been established as formalized internships with trained field supervisors and clearly defined learning contracts. Rather the students are viewed as “passive learners.”

- *Teacher Training:* In selected countries of the region, there are faculty fellowship programmes through OSI and the U.S., Department of State that serve as a valuable resource for mutual exchange, curriculum development and teaching methodologies. Assisting faculty in accessing scholarships and grants that

currently exist and expanding the use of internet-based learning and exchanges need to be explored.

- *Social Work Literature:* There is a wealth of literature being published in English language journals and books by scholars and practitioners in the region. Publishing in English provides an avenue for communication and exchanges across borders and continents. The primary limitation is that the literature in the students' and practitioners' first language, reflective of the local reality, is more limited. Since there is some evidence that English language skills are more prevalent among urban city dwellers, this further increases the rural/urban divide in terms of access to social work practice literature.
- *Training and Continuing Education:* Opportunities for training and continuing education are limited for social workers due to resources and geographic constraints. Those opportunities that do exist tend to be project-specific, time-limited, and have limited expansion funding. These types of project-specific trainings do not help with career development/job mobility, whereas more generic training would be relevant across practice settings.
- *Professional Associations:* While the countries studied have established legal frameworks for the existence of professional associations and they exist in almost every country, they are not well-established as a "voice of the profession of social work." Most are involved in providing continuing education and conferences, with little emphasis on professional advocacy and standard setting for the profession. Also, as membership organizations, there is a general feeling that the organizations do not understand how to service the membership. This is a lost opportunity to utilize the professional network as a way to reduce personal and professional isolation.

OUTCOMES AND PERFORMANCE MEASURES

- *Social Work Practice Standards:* Although there is consensus in many countries about the need for standards, most of this work has been done through social protection legislation and the development of standards of care for services. There are some good models that have been developed but dissemination and adoption by public services entities has been difficult.
- *Outcomes in Deinstitutionalization:* Decreased deinstitutionalization results from the social worker gatekeepers being better trained and more professional and having a standardized case recording format and assessment forms. University and field partnerships need to be formed in order to
- Develop evidence-based research to scientifically test interventions and evaluate outcomes on specific populations.
- *Performance Measure (Indicators) and Client Outcomes:* The difficulty in developing standards is linked to the absence of performance measures (indicators) and client outcomes. In terms of evaluation and impact, the focus

has traditionally been on project outcomes as opposed to client outcomes and well-being. One example is the positive results seen in Georgia in reduction of child institutionalization in social services units that had a trained social worker. Yet the indicator was not directly related to child outcomes in terms of health, education, and quality of parenting. The problem of outcome tracking is exacerbated by the absence of functional information management systems available to public services and NGOs in the region.

RECOMMENDATIONS

Social work's service mission is at the heart and soul of the profession and this service mission presents a dilemma in social work's push for increased salaries and status. Changing the paradigm for legitimization of the profession from "promotion of social work" to "public protection" will provide a more consistent message to both the public and political decision-makers. If the overall goal of increased professionalization is improved quality of services and better outcomes for those being served, then competency-based education and evidence-based practices guide efforts to improve status, salaries and career opportunities.

Another change strategy is to work simultaneously on top-down and bottom-up approaches, a bottom-up approach in which changes are made through local, grassroots efforts such as public education campaigns that promote citizen involvement in advocating for investment in social work services and a top-down approach in which change is legislated through laws and/or policies. This list of recommendations are not presented in any priority order since the "entry point" for reforms are most successful when they are approached strategically within the political, social, and economic context of each individual country.

POLICY AND LEGAL FRAMEWORK

- *Develop and Promulgate a Clearly Articulated Scope of Social Work Practice:* Social services legislation and regulations should be reviewed and modified to assure that the social work scope of practice is clearly articulated. One approach is to develop a model social work law that can be shared across the region. Additional legal consultation, such as was done in Armenia by Ulbricht (2008), could address some of the outstanding questions related to the separation of job functions (*i.e.*, between social assistance and social support and counseling services) and the advantages and disadvantages of licensure for social workers. For the most part, countries are struggling to develop or refine legislation on social work and social services. Looking at the Romania social work law and other model legislation is recommended. One resource for model legislation and current legislation from
- U.S., states and Canadian provinces is the Association of Social Work Boards in North America. The time is right to bring in expertise in social work

legislation. This will further serve to legitimize the profession beyond its relationship to social protection.

- **Focus on the Development of Ethical Codes and Standards for Professional Practice and Mechanisms for Accountability:** As presented in the Ulbricht report on Armenia (2008), there is a strong push to develop licensure as a way to legitimize the profession and increase status. Licensure is a political process that requires consensus across various political entities. Standards and ethical codes requires consensus primarily within the profession. Therefore, from a developmental perspective, it would be easier to build consensus around a legally binding code of ethics and what the standards for practice should be. Before you can talk about accountability, there needs to be some consensus around criteria for what social workers are supposed to do. This will also allow the profession to begin to educate and train around competencies and ultimately lead to better outcomes for persons served. Once these have been established, the “profession of social work” would likely be more successful in building legal/political support for a social work regulatory law since they will already have established themselves as a “profession.”

STRUCTURE OF SERVICES AND THE PRACTICE ENVIRONMENT

- *Design and Implement a Social Work Labor Market Survey:* This survey would help move beyond anecdotal information and would provide objective data on educational qualifications, titles and job functions, salaries, perceived liabilities and assets of practicing social work, safety and security concerns, and also obtain more qualitative data on motivations and career aspirations. Additional research on alumni of social work schools would generate additional information on the range of factors that lead to exodus from the profession or staying in the profession that moves beyond the salary and status issue. More scientific investigations into attitudes and perceptions of the profession can provide valuable information for strategic and innovative thinking around increasing professional status.
- *Develop Interdisciplinary Programmes and Cross-Border Models:* More interdisciplinary practice models should be implemented to address practitioner and resource access challenges in remote, rural areas. Existing community buildings (*i.e.*, public libraries, health clinics, community centers, and schools) should be utilized to house and provide services in rural areas. Identifying practice options for using trained paraprofessionals needs to be explored. Targeted public awareness and education programmes related to social services should occur concurrent with the development of these resources.

HUMAN CAPACITY: EDUCATION AND TRAINING

- *Develop Social Work Competencies:* For those countries that will be governed by the Bologna Process, its professional competencies that guide social work

education should be adopted. These materials include a framework that social work education programmes should use, and provides guidance for competency-based learning and supervision.

- *Strengthen Social Work Supervision:* While education programmes to develop social work supervisors could and should be replicated, work is needed to create buy-in regarding the value of social work supervision from labor market, consumer protection, and professional development perspectives. The models used by the international NGOs should be considered for replication. Research should be considered as a means of validating the role of social work supervisors. Finally, curricula could be expanded or a separate track could be created to address administrative and management skills such as needs assessment, programme development, strategic planning, and programme evaluation (e.g., the NGO management tracks in some universities). These opportunities should be extended to continuing education programmes for more experienced social workers as well.
- *Create or Revise Curriculum to Link Social Work Practice with Field Education:* Strengthen field education and practice courses through curriculum development that links knowledge and skill competencies that are learned in the practice classes and applied in the field work. Providing additional training for field supervisors in development of learning objective that reflect the social work practice competencies and supervision of students will further improve the field education experience.

The field education experience may also be more strategically designed as a strategy for the recruitment of qualified and experienced social workers for a particular work setting.

- *Build Capacity of Social Work Associations to Service Members, Advocate, and Educate:* Clarity and consensus is needed around the role of professional associations, given the current climate of low status and salaries. Many existing associations are actually unions or service-providing NGOs with limited capacity to service the members, lobby the government, and educate the public. Helping social work associations and their members understand the real purpose of regulating social work practice through licensure as public protection, and finding ways to get the message out to the public and political leaders, would provide a purpose that could more easily be shared among all of the different constituencies. This would serve to strengthen efforts to help stakeholders and service recipients better understand the role of social workers in consumer protection and the preparation they receive for this work. A public image/education campaign or initiative should be developed with specific goals for specific regions. Work with media and social marketing professionals can provide some unique and creative ways to impact attitudes and behaviors related to social work.

PERFORMANCE MEASURES AND OUTCOMES

- *Develop Ph.D. Programmes in Social Work:* Building a cadre of social work researchers and developing social work research as a field of practice will further performance and outcome based approaches to service. This can have long-term benefits and lead to improved understanding of what kinds of interventions work better for what clients within the local reality. The scientific testing and reporting of applied theories and skills through research studies can also serve to build an evidence-based literature and further the development of the profession.
- *Develop and Apply Indicators and Mechanisms for Measuring Performance:* While there is an awareness of the value and the need to embrace evidence-based practice, work is needed to help government services agencies and NGOs build functional information management systems that will assist them in tracking data on services provided and outcomes. This would allow for the establishment of baseline data, from which the quality of services could be measured and informed decisions could be made regarding programme expansion, modification, or elimination. Concurrent with this work is the need to reach consensus on indicators for system performance and client outcomes. There are opportunities for collaboration between the E&E region and the United States, as these measures are universally needed.

4

CHAPTER

Global Standards for the Education and Training

INTERNATIONAL DEFINITION OF SOCIAL WORK

In July 2001, both the IASSW and the IFSW reached agreement on adopting the following international definition of social work:

- The social work profession promotes social change, problem solving in human relationships and the empowerment and liberation of people to enhance well-being. Utilising theories of human behaviour and social systems, social work intervenes at the points where people interact with their environments. Principles of human rights and social justice are fundamental to social work.

Both the definition and the commentaries that follow are set within the parameters of broad ethical principles that cannot be refuted on an ideological level. However, the fact that social work is operationalized differently both within nation states and regional boundaries, and across the world, with its control and status-quo maintaining functions being dominant in some contexts, cannot be disputed. Lorenz considered the ambiguities, tensions and contradictions of the social work profession, which have to be constantly negotiated and re-negotiated, rather than resolved, to constitute its success and challenge. It is, perhaps, these very tensions that lends to the richness of the local-global dialectic, and provides legitimacy for the development of global standards. According to Lorenz “It is its paradigmatic openness that gives this profession the chance to engage with very specific (and constantly changing) historical and Final Document Adopted at the

General Assemblies of IASSW and IFSW, Adelaide, Australia in 2004. political contexts while at the same time striving for a degree of universality, scientific reliability, professional autonomy and moral accountability”.

CORE PURPOSES OF THE SOCIAL WORK PROFESSION DEFINITION OF SOCIAL WORK

Social work in various parts of the world is targeted at interventions for social support, developmental, protective, preventive and/or therapeutic purposes. Drawing on available literature, the feedback from colleagues during consultations and the commentary on the international definition of social work the following core purposes of social work have been identified:

- Address and challenge barriers, inequalities and injustices that exist in society.
- Advocate for, and/or with people, changes in those policies and structural conditions that maintain people in marginalized, dispossessed and vulnerable positions, and those that infringe the collective social harmony and stability of various ethnic groups, insofar as such stability does not violate human rights.
- Advocate for, and/or with people, the formulation and targeted implementation of policies that are consistent with the ethical principles of the profession.
- Assist and educate people to obtain services and resources in their communities.
- Encourage people to engage in advocacy with regard to pertinent local, national, regional and/or international concerns.
- Engage in social and political action to impact social policy and economic development, and to effect change by critiquing and eliminating inequalities.
- Enhance stable, harmonious and mutually respectful societies that do not violate people's human rights.
- Facilitate the inclusion of marginalized, socially excluded, dispossessed, vulnerable and at-risk groups of people. Form short and longer-term working relationships with and mobilize individuals, families, groups, organisations and communities to enhance their well being and their problem-solving capacities.
- Formulate and implement policies and programmes that enhance people's well being promote development and human rights, and promote collective social harmony and social stability, insofar as such stability does not violate human rights.
- Plan, organize, administer and manage programmes and organizations dedicated to any of the purposes delineated above.
- Promote respect for traditions, cultures, ideologies, beliefs and religions amongst different ethnic groups and societies, insofar as these do not conflict with the fundamental human rights of people.
- Work towards the protection of people who are not in a position to do so themselves, for example children and youth in need of care and persons

experiencing mental illness or mental retardation within the parameters of accepted and ethically sound legislation.

STANDARDS REGARDING THE SCHOOL'S CORE PURPOSE OR MISSION STATEMENT

All schools should aspire toward the development of a core purpose statement or a mission statement which:

- Is clearly articulated so those major stakeholders who have an investment in such a core purpose or mission understand it.
- Reflects the values and the ethical principles of social work.
- Reflects aspiration towards equity with regard to the demographic profile of the institution's locality. The core purpose or mission statement should thus incorporate such issues as ethnic and gender representation on the faculty, as well as in recruitment and admission procedures for students.
- Respects the rights and interests of service users and their participation in all aspects of delivery of delivery of programmes.

STANDARDS REGARDING PROGRAMME OBJECTIVES AND OUTCOMES

In respect of programme objectives and expected outcomes, schools should endeavour to reach the following:

- A specification of its programme objectives and expected higher education outcomes.
- A reflection of the values and ethical principles of the profession in its programme design and implementation.
- Identification of the programme's instructional methods and how these cohere with achieving both the cognitive and affective development of social work students.
- An indication of how the programme reflects the core knowledge, processes, values and skills of the social work profession, as applied in context specific realities.
- An indication of how an initial level of proficiency with regard to self-reflective use of social work values, knowledge and skills is to be attained by social work students.
- An indication of how the programme coheres with nationally and/or regionally/internationally defined professional goals, and how the programme addresses local, national and/or regional/international developmental needs and priorities.
- As social work does not operate in a vacuum, the programme should reflect consideration of the impact of interacting cultural, economic, communication, social, political and psychological global features.
- Provision of an educational preparation that is relevant to beginning social work professional practice with individuals, families, groups and/or communities in any given context.

- Self-evaluation to assess the extent to which its programme objectives and expected outcomes are being achieved.
- External peer evaluation as far as is reasonable and financially viable. This may be in the form of external peer moderation of assignments and/or written examinations and dissertations, and external peer review and assessment of curricula.
- The conferring of a distinctive social work qualification at the certificate, diploma, first degree or post-graduate level as approved by national and/or regional qualification authorities, where such authorities exist.

STANDARDS WITH REGARD TO PROGRAMME CURRICULA INCLUDING FIELD EDUCATION

With regard to standards regarding programme curricula, schools should consistently aspire towards the following:

- The curricula and methods of instruction being consistent with the school's programme objectives, its expected outcomes and its mission statement.
- Provision for the inclusion and participation of field instructors in curriculum development.
- A partnership between the educational institution and the agency (where applicable) and service users in decision-making regarding field education and the evaluation of student's fieldwork performance.
- Making available, to fieldwork instructors or supervisors, a field instruction manual that details its fieldwork standards, procedures, assessment standards/criteria and expectations.
- Ensuring that adequate and appropriate resources, to meet the needs of the fieldwork component of the programme, are made available.
- Clear plans for the organization, implementation and evaluation of the theory and field education components of the programme.
- Involvement of service users in the planning and delivery of programmes.
- Recognition and development of indigenous or locally specific social work education and practice from the traditions and cultures of different ethnic groups and societies, insofar that such traditions and cultures do not violate human rights.
- Specific attention to the constant review and development of the curricula.
- Ensuring that the curricula helps social work students to develop skills of critical thinking and scholarly attitudes of reasoning, openness to new experiences and paradigms, and commitment to life-long learning.
- Field education should be sufficient in duration and complexity of tasks and learning opportunities to ensure that students are prepared for professional practice.
- Planned co-ordination and links between the school and the agency/field placement setting.
- Provision of orientation for fieldwork supervisors or instructors.

- Appointment of field supervisors or instructors who are qualified and experienced, as determined by the development status of the social work profession in any given country, and provision of orientation for fieldwork supervisors or instructors.

STANDARDS WITH REGARD TO CORE CURRICULA

In respect core curricula, schools should aspire toward the following:

- An identification of and selection for inclusion in the programme, curricula as determined by local, national and/or regional/international needs and priorities.
- Notwithstanding the provision of 4.1 there are certain core curricula that may be seen to be universally applicable. Thus the school should ensure that social work students, by the end of their first Social Work professional qualification, have exposure to the following core curricula which are organised into four conceptual components:

Domain of the Social Work Profession

- A critical understanding of how socio-structural inadequacies, discrimination, oppression, and social, political and economic injustices impact human functioning and development at all levels, including the global.
- Knowledge of human behaviour and development and of the social environment, with particular emphasis on the person-in-environment transaction, life-span development and the interaction among biological, psychological, sociostructural, economic, political, cultural and spiritual factors in shaping human development and behaviour.
- Knowledge of how traditions, culture, beliefs, religions and customs influence human functioning and development at all levels, including how these might constitute resources and/or obstacles to growth and development.
- A critical understanding of social work's origins and purposes.
- Understanding of country specific social work origins and development.
- Sufficient knowledge of related occupations and professions to facilitate interprofessional collaboration and teamwork.
- Knowledge of social welfare policies or lack thereof, services and laws at local, national and/or regional/international levels, and the roles of social work in policy planning, implementation, evaluation and in social change processes.
- A critical understanding of how social stability, harmony, mutual respect and collective solidarity impact human functioning and development at all levels, including the global, insofar as that stability, harmony and solidarity are not used to maintain a status quo with regard to infringement of human rights.

Domain of the Social Work Professional

- The development of the critically self-reflective practitioner, who is able to practice within the value perspective of the social work profession and shares responsibility with the employer for their well being and professional development, including the avoidance of “burn-out”.
- The recognition of the relationship between personal life experiences and personal value systems and social work practice.
- The appraisal of national, regional and/or international social work codes of ethics and their applicability to context specific realities.
- Preparation of social workers within a holistic framework, with skills to enable practice in a range of contexts with diverse ethnic, cultural, “racial” and gender groups, and other forms of diversities.
- The development of the social worker who is able to conceptualize social work wisdom derived from different cultures, traditions and customs in various ethnic groups, insofar that culture, tradition, custom and ethnicity are not used to violate human rights.

The development of the social worker who is able to deal with the complexities, subtleties, multi-dimensional, ethical, legal and dialogical aspects of power.

Methods of Social Work Practice

- Sufficient practice skills in, and knowledge of, assessment, relationship building and helping processes to achieve the identified goals of the programme for the purposes of social support, developmental, protective, preventive and/or therapeutic intervention – depending on the particular focus of the programme or professional practice orientation.
- The application of social work values, ethical principles, knowledge and skills to confront inequality, and social, political and economic injustices.
- Knowledge of social work research and skills in the use of research methods, including ethical use of relevant research paradigms, and critical appreciation of the use of research and different sources of knowledge about social work practice.
- The application of social work values, ethical principles, knowledge and skills to promote care, mutual respect and mutual responsibility amongst members of a society.
- Supervised fieldwork education, with due consideration to the provisions of Item 3 above.

Paradigm of the Social Work Profession

Of particular current salience to professional social work education, training and practice, are the following epistemological paradigms (which are not mutually exclusive) that should inform the core curricula:

- An acknowledgement and recognition of the dignity, worth and the uniqueness of all human beings.
- Recognition of the inter-connectedness that exists within and across all systems at micro, mezzo and macro levels.
- An emphasis on the importance of advocacy and changes in socio-structural, political and economic conditions that disempower, marginalize and exclude people.
- A focus on capacity-building and empowerment of individuals, families, groups, organisations and communities through a human-centred developmental approach.
- Knowledge about and respect for the rights of service users.
- Problem-solving and anticipatory socialisation through an understanding of the normative developmental life cycle, and expected life tasks and crises in relation to age related influences, with due consideration to socio-cultural expectations.
- The assumption, identification and recognition of strengths and potential of all human beings.
- An appreciation and respect for diversity in relation to “race”, culture, religion, ethnicity, linguistic origin, gender, sexual orientation and differential abilities.

STANDARDS WITH REGARD TO PROFESSIONAL STAFF

With regard to professional staff, schools should aspire towards:

- The provision of professional staff, adequate in number and range of expertise, who have appropriate qualifications as determined by the development status of the social work profession in any given country. As far as possible a Masters level qualification in social work, or a related discipline (in countries where social work is an emerging discipline) , should be required.
- The provision of opportunities for staff participation in the development of its core purpose or mission, in the formulation of the objectives and expected outcomes of the programme, and in any other initiative that the school might be involved in.
- Provision for the continuing professional development of its staff, particularly in areas of emerging knowledge.
- A clear statement, where possible, of its equity based policies or preferences, with regard to considerations of gender, ethnicity, “race” or any other form of diversity in its recruitment and appointment of staff.
- Sensitivity to languages relevant to the practice of social work in that context.
- In its allocation of teaching, fieldwork instruction, supervision and administrative workloads, making provision for research and publications.
- Making provision for professional staff, as far as is reasonable and possible to be involved in the formulation, analysis and the evaluation of the impact of social policies, and in community outreach initiatives.

STANDARDS WITH REGARD TO SOCIAL WORK STUDENTS

In respect of social work students, schools should endeavor to reach the following:

- Clear articulation of its admission criteria and procedures.
- Student recruitment, admission and retention policies that reflect the demographic profile of the locality that the institution is based in with active involvement of practitioners and service users in relevant processes. Due recognition should be given to minority groups that are under-represented and/or under-served. Relevant criminal convictions involving abuse of others or human rights violations must be taken into account given the primary responsibility of protecting and empowering service users.
- Provision for student advising that is directed toward student orientation, assessment of the student's aptitude and motivation for a career in social work, regular evaluation of the student's performance and guidance in the selection of courses/modules.
- Ensuring high quality of the educational programme whatever the mode of delivery, including distance, mixed-mode, decentralised and/or internet-based teaching. Mechanisms for locally-based instruction and supervision should be put in place, especially with regard to the fieldwork component of the programme.
- Explicit criteria for the evaluation of student's academic and fieldwork performance.
- Non-discrimination against any student on the basis of "race", colour, culture, ethnicity, linguistic origin, religion, political orientation, gender, sexual orientation, age, marital status, physical status and socio-economic status.
- Grievance and appeals procedures which are accessible, clearly explained to all students and operated without prejudice to the assessment of students.

STANDARDS WITH REGARD TO STRUCTURE, ADMINISTRATION, GOVERNANCE AND RESOURCES

With regard to structure, administration, governance and resources, the school and/or the educational institution should aspire towards the following:

- Social work programmes are implemented through a distinct unit known as a Faculty, School, Department, Centre or Division, which has a clear identity within the educational institution.
- The school has a designated Head or Director who has demonstrated administrative, scholarly and professional competence, preferably in the profession of social work.
- The Head or Director has primary responsibility for the co-ordination and professional leadership of the school, with sufficient time and resources to fulfil these responsibilities.
- The school's budgetary allocation is sufficient to achieve its core purpose or mission and the programme objectives.

- The budgetary allocation is stable enough to ensure programme planning and sustainability.
- There are adequate physical facilities, including classroom space, offices for professional and administrative staff and space for student, faculty and fieldliaison meetings, and the equipment necessary for the achievement of the school's core purpose or mission and the programme objectives.
- Library and, where possible, internet resources, necessary to achieve the programme objectives, are made available.
- The necessary clerical and administrative staff is made available for the achievement of the programme objectives.
- Where the school offers distance, mixed-mode, decentralized and/or internet based education there is provision of adequate infrastructure, including classroom space, computers, texts, audio-visual equipment, community resources for fieldwork education, and on-site instruction and supervision to facilitate the achievement of its core purpose or mission, programme objectives and expected outcomes.
- The school plays a key role with regard to the recruitment, appointment and promotion of staff.
- The school strives toward gender equity in its recruitment, appointment, promotion and tenure policies and practices.
- In its recruitment, appointment, promotion and tenure principles and procedures, the school reflects the diversities of the population that it interacts with and serves.
- The decision-making processes of the school reflect participatory principles and procedures.
- The school promotes the development of a cooperative, supportive and productive working environment to facilitate the achievement of programme objectives.
- The school develops and maintains linkages within the institution, with external organizations and with service users relevant to its core purpose or mission and its objectives.

STANDARDS WITH REGARD TO CULTURAL AND ETHNIC DIVERSITY AND GENDER INCLUSIVENESS

With regard to cultural and ethnic diversity schools should aspire towards the following:

- Making concerted and continuous efforts to ensure the enrichment of the educational experience by reflecting cultural and ethnic diversity, and gender analysis in its programme.
- Ensuring that the programme, either through mainstreaming into all courses/modules and/or through a separate course/module, has clearly articulated objectives in respect of cultural and ethnic diversity, and gender analysis.
- Indicating that issues regarding cultural and ethnic diversity, and gender analysis are represented in the fieldwork component of the programme.

- Ensuring that social work students are provided with opportunities to develop self-awareness regarding their personal and cultural values, beliefs, traditions and biases and how these might influence the ability to develop relationships with people, and to work with diverse population groups.
- Promoting sensitivity to, and increasing knowledge about, cultural and ethnic diversity, and gender analysis.
- Minimizing group stereotypes and prejudices and ensuring that racist behaviours, policies and structures are not reproduced through social work practice.
- Ensuring that social work students are able to form relationships with, and treat all persons with respect and dignity irrespective of such persons' cultural and ethnic beliefs and orientations.
- Ensuring that social work students are schooled in a basic human rights approach, as reflected in international instruments such as the Universal Declaration on Human Rights, the United Nations Convention on the Rights of the Child (1989) and the U.N. Vienna Declaration (1993). Ensuring that the programme makes provision for social work students to know themselves both as individuals as well as members of collective socio-cultural groups in terms of strengths and areas for further development.

STANDARDS WITH REGARD TO VALUES AND ETHICAL CODES OF CONDUCT OF THE SOCIAL WORK PROFESSION

In view of the recognition that social work values, ethics and principles are the core components of the profession, schools should consistently aspire towards:

- Focused and meticulous attention to this aspect of the programme in curricula design and implementation.
- Clearly articulated objectives with regard to social work values, principles and ethical conduct.
- Ensuring that every social work student involved in fieldwork education, and every professional staff member, is aware of the boundaries of professional practice and what might constitute unprofessional conduct in terms of the code of ethics. Where students violate the code of ethics, programme staff may take necessary and acceptable remedial and/or initial disciplinary measures, or counsel the student out of the programme.
- Registration of professional staff and social work students (insofar as social work students develop working relationships with people via fieldwork placements) with national and/or regional regulatory (whether statutory or non-statutory) bodies, with defined codes of ethics. Members of such bodies are generally bound to the provisions of those codes.
- Upholding, as far as is reasonable and possible, the principles of restorative rather than retributive justice in disciplining either social work students or professional staff who violate the code of ethics.

- Taking appropriate action in relation to those social work students and professional staff, who fail to comply with the code of ethics either through an established regulatory social work body, established procedures of the educational institution, and/or through legal mechanisms.
- Ensuring that regulatory social work bodies are broadly representative of the social work profession, including where applicable social workers from both the public and private sector, and of the community that it serves including the direct participation of service users.

PROCESS AND UNDERLYING APPROACH TO DEVELOPING GLOBAL STANDARDS

The Global Minimum Qualifying Standards Committee was formally established as a joint initiative of the International Association of Schools of Social Work (IASSW) and the International Federation of Social Workers (IFSW) at the joint IASSW/IFSW Conference in Montreal, Canada in July 2000. This discussion document was put together with the input of various Committee members, a review of relevant documents, e-mail consultations, and personal consultations with colleagues wherever possible. On the whole there was a favourable response to IASSW and IFSW developing a standards setting document that elucidates what social work represents on a global level. This document that identifies certain universals, may be used as guidelines to develop national standards with regard to social work education and training. Such a document should reflect some consensus around key issues, roles and purposes of social work. However, given the profession's historically fragmented strands; the contemporary debates around social work's intra-professional identity; its identity vis-à-vis other categories of personnel in the welfare sector such as development workers, child care workers, probation officers, community workers and youth workers (where such categories of personnel are differentiated from social work); and the enormous diversities across nations and regions, there was some scepticism about the possibility of identifying any such "universal". The suggestion was that such a document must be sufficiently flexible to be applicable to any context. Such flexibility should allow for interpretations of locally specific social work education and practice, and take into account each country's or region's socio-political, cultural, economic and historical contexts while adhering to international standards.

The main reasons for the development of global standards were to (stated in no particular order of priority) :

- Protect the "consumers", "clients" or "service users"¹⁷ of social work services;
- Take account of the impact of globalization on social work curricula and social work practice;
- Facilitate articulation across universities on a global level;
- Facilitate the movement of social workers from one country to another;

- Draw a distinction between social workers and non-social workers;
- Benchmark national standards against international standards;
- Facilitate partnerships and international student and staff exchange programmes;
- Enable IASSW and IFSW, in developing such guidelines, to play a facilitative role in helping those faculties, centres, departments or schools of social work¹⁸ that lack resources to meet such guidelines.
- Give practical expression to the aim of IASSW as some saw the formulation of international guidelines for social work education and training to be the core business of IASSW.

Clearly not all of the above expressed purposes are feasible *e.g.* it is not feasible via such an endeavour to draw a clear distinction between social workers and non-social workers, neither might we be able to realise the objective of protecting “clients” through the standards. Facilitating the movement of social workers from one country to another is a contentious issue in view of the direct recruitment of social workers from some countries to others *e.g.* from South Africa and the Caribbean to the United Kingdom to the disadvantage of South Africa and the Caribbean. However, from an ethical point of view the migration of those social workers that wish to practice in another country should be enabled and not blocked. Retention of social work skills within countries are dependent on such factors as service conditions, salaries and validation of the profession which need to be addressed on national levels.

A few participants expressed the view that the document should go further to include more practical guidelines. These practical guidelines should include: - a multi-tiered classification for the basic qualification *e.g.* with a range from the number of years of basic schooling + at least one year of full time social work education to a degree with 3 or 4 years of social work education (the minimum period of practical training should be specified in such a classification) ; the acknowledgement and recognition of prior learning experiences; and the identification of core competencies, knowledge and skills as applied to context specific realities. A very small minority went as far as asking for the global standards to prescribe texts and minimum number of hours that students need to spend on reading. This was clearly an impossible request to include at the global level as it would entail a complete denial of context specific realities. Indeed, it would perhaps be impossible to entertain such a request even at local or national levels, as it would contribute to the curtailment of academic freedom, restrict knowledge development and constrain the development of critical thinking. Other participants expressed concern that the proposed multi-tiered system may appear to be far too elitist, with perhaps social workers from the Two-Thirds World being more likely to be categorised into the lower ranks. Prescribing lengths of training or number of course credits are problematic, given the variations of the academic year across countries and regions, and the diversities in crediting courses in different contexts. Also, for example, a six to twelve month intensive social work programme, with careful selection of mature students with appropriate prior learning experiences and/or related qualifications, might prove to be

as valuable as a first degree social work programme with school leaving students. It is the quality of the educational programme that must not be compromised. From available information, it would appear that the academization of social work is becoming the norm, with many countries opting for either a 3 or 4 year Bachelors degree in Social Work, with a few countries like Chile being an exception with a 5 year Bachelors degree.

A minority view was that IFSW and IASSW begin with no document; that a grassroots approach be used in encouraging national bodies to formulate their own standards. These national standards, formulated for example via a five-year action plan, could then be processed into global standards. However, one does not have to adopt an either/or approach to the development of global standards. If we accept the premise that such standards do not represent a finite or static product, but a dynamic process through which we continue building a framework that we aspire towards then we accept that such an endeavour would involve a global-regional-national-local dialectical interaction. This must involve cross national and cross regional dialogue.

In developing global standards care needed to be taken that we did not further fragment and de-professionalize social work, as so clearly elucidated by Dominelli in her discussion on the impact of the competencies based approach to social work education and practice. This view was supported by Lorenz, who while not invalidating the need for quality control by having some benchmark criteria, warned that it might “trivialise social work skills even further”. To circumvent this possibility we made concerted efforts to transcend the kind of reductionist language, used within many national/regional contexts in their development of unit standards, designed to meet criteria for the competencies-based approach that fragments social work skills and roles into minute, constituent parts. We acknowledge that there might be merits to the competencies-based approach on national/regional levels. However, this is seen to be far too specific to be applied to the global level.

During consultations questions were raised regarding “minimum” by whose or what standards? Is it possible that “minimum standards” could decrease rather than enhance the profession’s standards? An alternative argument was that as “standards” represent an ideal, they could, in effect, come to be “maximum standards” that all Schools of Social Work in all countries and regions are put under pressure to attain. The experience of South Africa in the early 1990s is a case in point. The then Council for Social Work, which was actually a State apparatus designed to uphold the ideology of apartheid, proposed what it called “minimum standards”. However, the document actually reflected superior standards and proposed control mechanisms, which, if accepted, would have jeopardised the position and, perhaps, the very existence of Schools of Social Work at the historically disadvantaged black institutions, which were poorly resourced compared with the white universities. Fortunately there was sufficient solidarity among social work educators who rejected the document so it did not become part of the statutory requirement. These concerns provided further ground for omitting the “minimum” from

this document, and to move toward the use of: “global standards for the education and training of the social work profession”. This document does not purport to reflect minimum standards, but ideals that schools of social work should consistently aspire towards.

Some colleagues who engaged in the consultation process also expressed concern about the possibility of a western domination. Given the western hegemony in social work education and practice, and that “Western European countries and the USA perhaps have fairly settled views of what social work is and what it means to provide good social work education”, such fears are not merely speculative. We acknowledge that the claim to what constitutes good social work education in Western Europe and the USA may be based on ill-founded premises. Australia and Canada also seem to have made a great deal of progress in the development of national standards. In order to prevent such a Western domination the following were considered, and must continue to be considered in relation to the global standards:

- Ensuring representation from different regions of the world on the Committee in the formulation of the standards.
- Facilitating as much consultation and inclusion in the process as possible.
- Ensuring that the global standards take into account a country’s unique historical, political, cultural, social and economic contexts.
- Ensuring that the unique developmental needs of countries are considered in relation to the global standards.
- Ensuring that the profession’s developmental status and needs in any given country are considered.
- While encouraging schools to secure adequate resources to the extent possible, that we ensure that there is no assumption that schools with lesser resources provide poorer quality programmes.
- Facilitating open dialogue across national and regional boundaries.

Amongst those who participated during consultations, there was overwhelming concern that context specific realities, and the resources available to individual institutions to meet the global standards, are taken into consideration. In the development of global standards we should not create unintended consequences by disadvantaging some educational institutions. As much as global standards may be used to benchmark national norms and standards, as far as possible, national and regional experiences and practices (even where formal standards do not exist) were incorporated into the formulation of the global standards. Where national or regional standards do not exist, IASSW and IFSW should collaborate to facilitate the development of such standards. The circular, interactive and discursive processes in developing national and global standards can, in these ways, become and remain continuous and dynamic. The process-product dialectic, in the formulation of the global standards, has been vital. While we had necessary pre-determined timeframes, we tried, as far as possible, not to compromise consultation processes.

Two participants during consultations recommended a two-phased process; the first which would involve consultations to “get everyone on board” that might span a two to six year period. The second phase would consist of submissions by each region/ national body to IASSW to ensure compliance. The recommendations ranged from biannual submissions to submissions once in five years. The majority believed that beyond the formulation of a standards document, IASSW/IFSW could play no role, and that these bodies could not really effect any mechanisms to “ensure compliance”. Monitoring, conforming to the global standards and the possibility of downgrading or up grading of educational institutions were not seen as the tasks of IASSW/IFSW. The roles of IASSW and IFSW should be facilitative and supportive.

Payne pointed out that by virtue of membership with IASSW, educational institutions had to uphold at least the following minimum criteria:

- That social work education takes place after a school leaving certificate has been obtained; and that
- Social work education takes place at the tertiary level.

While these two criteria were accepted as valid for the purpose of this document, it must be remembered that various educational institutions do recognise prior learning experiences in the selection of students, where a school-leaving certificate has not been obtained. Where recognition of prior learning experiences is implemented (generally with policy or criteria determined at school, local or national levels) , this needs to be accepted and respected.

CONCLUSION

The development of global standards, by their very nature, generally tends to fall within the prescriptive, reductionist, logical-positivist paradigm. Efforts have been made to adopt an alternative and a more empowering, non-prescriptive language in this document. The main aim is to enhance social work education, training and practice on a global level, by facilitating dialogue within and across nations and regions. The document reflects global standards that schools of social work should consistently aspire towards, which (collectively and if met) would actually provide for quite sophisticated levels of social work education and training. This is, as it ought to be – the provision of the best possible education and training for social work students who, after qualifying, bear enormous responsibilities in their communities.

The extent to which schools of social work meet the global standards will depend on the developmental needs of any given country/region and the developmental status of the profession in any given context, as determined by unique historical, socio-political, economic and cultural contexts. These are given due consideration throughout the document. It is accepted that while some established schools might have surpassed the standards contained in this document, other schools might be in the process of beginning social work programmes. The document details ideals regarding what schools, such as latter, might aspire towards even if it takes the next twenty or more years to reach them.

In specifying the standards, there is no expectation that all schools in all parts of the world would measure up to them on an immediate level. Also whether or not schools measure up to them will not be determined at the international level. A school may engage in self-assessment to determine the extent to which its programme is consistent with the standards elucidated in this document. Quality assurance and accreditation criteria and procedures will have to be determined at national and/or regional levels. There is undoubtedly, on a global level, a move toward the creation of national and regional qualification frameworks.

In formulating the global standards, care was taken to ensure that we do not take on the dominant language of managerialism (this does not mean the negation of sound management) and marketization, which is seen to be inconsistent with the core values and purposes of social work. By locating the global standards against the international definition of social work and the core purposes of social work, the document ensures an approach to education and training that supports human rights, social justice, and an essential commitment to caring for, and the empowerment of individuals, groups, organisations and communities.

It also reflects a commitment to the personal and professional development of social work students, with particular emphases on the development of the critically self-reflective practitioner and the place of values and ethics in social work education and training. In the formulation of the global standards the challenge has been for them to be specific enough to have salience, yet broad enough to be relevant to any given context. While the standards have been formulated at the global level, the document allows for sufficient interpretation and application at the local levels. What would be a distinctive advantage is empirically based comparative international research involving the application and evaluation of the standards in different contexts. This would help in identifying gaps and limitations of the current document and for further revision and refinement.

Given the number of concerns that were raised in the preamble to this document and some additional issues that arose as the consultations proceeded caution must be exercised in how the document should and should not be used. There was some concern raised during the consultation process from a colleague in Canada that the development of global standards may be used for international trade purposes in relation to the Agreement on Trade in Service (GATS). IASSW and IFSW categorically assert that the standards are not to be used for such a purpose. Neither of these international bodies is bound to GATS regulations, and there was no external funding to the development of the global standards. The formulation of the standards represents the attempt to uphold the best possible standards for the social work profession on a global level, and to facilitate dialogue and debate within and across national and regional boundaries and is in no way linked to international trade in services. The intention is to enhance academic freedom and promote the development of locally specific theory and practice, rather than inhibit or constrain such development.

We concur with the view of Rossiter that there is a need to mobilize against such trade agreements as:

- Increase poverty,
- Contribute to environmental degradation,
- Reduce the power of labour,
- Contribute to national and international inequality,
- Constitute an un-elected, non-transparent series of agreements that weaken local power and government power to regulate economies towards human need rather than profit”.

Such agreements also reduce the power of individuals to control their own working and welfare environment. In the elucidation of standards on structure, administration, governance and resources neither IASSW nor IFSW endorses that view that those schools that lack material and infra-structural resources have poorer quality programmes. However, it is accepted that adequate human and material resources do facilitate easier achievement of programme purposes and objectives. Some colleagues in the Asia-Pacific, the African and the Nordic regions have indicated that they have used the draft document as leverage to lobby for more adequate resources from their institutions. That the document, even in its draft form and without adoption at the international level, was used for such a purpose is clearly an advantage. However, it must be borne in mind that such pressure from external sources may offend some educational institutions in other regions of the world. Thus, whether or not the document is used for the purpose of lobbying for adequate resources or how this is done should rest at the discretion of the individual school.

In formulating global standards for the education and training of the social work profession, neither the IASSW nor the IFSW will play any monitoring, control or accreditation function at the level of the school. Whether or not IASSW and IFSW would play any significant role at the national or regional level in the future will depend on how the document is interpreted in different contexts, how it is operationalised in different parts of the world, and on the requirements and expectations of its membership. One of the fears in producing a document such as this is that we recognise that a text, once written, is outside of the control of its authors. The roles of IASSW and IFSW are intended to be supportive and facilitative. There must be clear mechanisms of communication across national and/or regional social work educators’ associations and IASSW. Part of the developmental objective to the global standards, should be IASSW’s and IFSW’s commitment to developing guidelines about mechanisms to facilitate such communication. One of the objectives is that, through the assistance of the Census Commission, IASSW will develop a data bank containing the details and programmes of member schools and of national and/or regional standards and systems of quality assurance and accreditation. Such information may be shared on an international level on request and/or via the websites of IASSW and IFSW. It is hoped that such sharing would provide the impetus for schools of social work to aspire towards the global

standards for professional social work education and training elucidated in this document. The document is not intended to be a fixed timeless product; it is a dynamic entity subject to review and revision as and when the need arises. This can only be achieved with continued critical debate and dialogue within the profession on local, national, regional and global levels.

5

CHAPTER

Essence of Education

COLLEGE

A college is an educational institution or a constituent part of an educational institution. Usage varies in English-speaking nations. A college may be an degree-awarding tertiary educational institution, an institution within a federal university, an institution offering vocational, or a secondary school. In the United States and Ireland, “college” and “university” are loosely interchangeable, whereas in the United Kingdom, Canada, Australia and other Commonwealth countries, a “college” may be an institution between school and university level or a constituent school within a university.

ETYMOLOGY

In ancient Rome a collegium was a club or society, a group of persons living together, under a common set of rules. As well as an educational institution, the term can also refer to any group of colleagues, such as an electoral college, a college of arms or the College of Cardinals.

HOMEWORK

Homework, or homework assignment, refers to tasks assigned to students by their teachers to be completed mostly outside of class, and derives its name from the fact that most students do the majority of such work at home. Common homework assignments may include a quantity or period of reading to be performed, writing or typing to be completed, problems to be solved, a school project to be built or other skills to be practiced.

MAIN OBJECTIVES AND REASONS FOR HOMEWORK

The basic objectives of assigning homework to students are the same as schooling in general: To increase the knowledge and improve the abilities and skills of the students. However, opponents of homework cite homework as rote, or grind work, designed to take up children's time, without offering tangible benefit. Homework may be designed to reinforce what students have already learned, prepare them for upcoming lessons, extend what they know by having them apply it to new situations, or to integrate their abilities by applying many different skills to a single task. Homework also provides an opportunity for parents to participate in their children's education.

AMOUNT OF HOMEWORK REQUIRED

A review of over 60 research studies showed that, within limits, there is a positive correlation between the amount of homework done and student achievement. The research synthesis also showed that too much homework could be counterproductive. The research supports the "10-minute rule", the commonly accepted practice of assigning 10 minutes of homework per day per grade-level. For example, under this system, 1st graders would receive 10 minutes of homework per night, while 5th graders would get 50 minutes' worth, 9th graders 90 minutes of homework, etc. Many schools exceed these recommendations or do not count assigned reading in the time limit. In the United Kingdom, recommendations on homework quantities were outlined by the then Department for Education in 1998. These ranged from 10 minutes daily reading for 5-year-olds, to up to 2.5 hours per day for the pupils in Year 11 aged 15 or 16.

HOMEWORK RESOURCES

Internet Homework Resources

There are many homework-related resources available on the World Wide Web. There are web-sites dedicated to communicating about homework, for teachers to post assignments on-line for students, and to keep parents informed. Many schools host their own homework posting services on their web sites. There are non-profit organizations on-line that help students with their homework for free. There are also tutorials on most school subjects, especially math, which students can use if they don't understand their homework assignments.

Many libraries provide on-line resources which present subjects specifically for students who are looking for something to write about. And there are archives of ready-made homework assignments, including handouts, which teachers can use to provide homework to their students. Many other web sites are used for research, especially search engines, such as Google, and encyclopaedias. Apart from above given resources there are hundreds of web sites who are providing homework help at nominal rates. Such web sites claim to help students understand concepts. Some parents choose to monitor their students' usage of the internet, as some of the sites may be found deceptive

or inappropriate by academic institutions. Also, Internet resources offer students a wealth of opportunity for plagiarism.

Tutoring

With an enhanced emphasis on homework, parents and students are turning to customized solutions. Private institutions, such as Sylvan Learning Centers and Kaplan, help students through individually-tailored assignments. Other parents find help through their community where tutoring, study groups and other resources may be made available. Many libraries provide tutors for helping students with their homework, both in-person and on-line. If it is necessary to hire a tutor to assist with a child's homework, parents should also speak to the child's teacher about the amount and the appropriateness of the homework load.

PARENTAL HOMEWORK STRATEGIES

Students generally benefit when their parents become involved in the homework process. However, too much parental involvement can prevent the positive effects of homework. Setting a regular time to do homework and designating a specific place for doing homework helps keep the student well-focused on his or her studies. A flat surface, good lighting, school supplies and a dictionary are often essential. Teachers need to know what their students understand and can do independently, therefore they often advise parents not to do the children's homework assignments for them, nor correct their children's homework assignments and have them copy the corrections. Grades, and the teachers' other feedback, need to apply to the student's performance, not to the parents' performance, nor to student-parent co-performance.

On the other hand it is also fairly common for teachers to give assignments far beyond what students can do independently and for teachers to expect parents to go over homework and have the student make corrections before it is turned in. Practices vary. Independent learning is encouraged and improved by providing guidance rather than merely providing the answers to the child's homework-related questions. Having one's child read out loud allows the parent to provide corrections and help the student learn how to read better. When parents do "homework" of their own at the same time as their children, it sets a good example and helps to foster a good attitude towards learning. One key role for parents is to negotiate with teachers and schools should the homework burden be unmanageable or age-inappropriate for the students. This negotiation may take the form of speaking with the teacher individually, speaking to other school officials, or coordinating with other parents or with the PTA or school board to get the homework load for the entire class or school reduced.

TEACHING AND HOMEWORK EFFECTIVENESS

Student learning improves when homework serves a clear purpose and is matched to both the skills of each individual student and to the current topics being taught in class.

Feedback improves the effectiveness of homework, especially when given in a timely manner. Effective feedback improves student learning by correcting misunderstanding, validating process, and highlighting errors in thinking. Embedded comments provide much better feedback than a mere grade at the top of the paper. Homework must be concentrated to be effective: mastering takes days or weeks of practice. Fifty-percent mastery may be achieved after 4 practice sessions, but it takes 28 practice sessions to achieve approximately the eighty-percent mastery level.

Another way teachers can be more effective is by alerting parents to their students' homework, giving parents a chance to become familiar with the material and their child's progress.

This also encourages parents to become involved in the homework process. Messages tend to get lost in transit or even altered when using "pupil post" and therefore direct communication is much more effective and prevents frustration all around. Methods available for directly reporting homework assignments include the phone, e-mail, and centralized web-pages.

CRITICISM

There is research supporting the idea that homework is of little educational value, and that for young children it may have a negative effect on learning.

LEARNING

Learning is acquiring new knowledge, behaviours, skills, values, preferences or understanding, and may involve synthesizing different types of information. The ability to learn is possessed by humans, animals and some machines. Progress over time tends to follow learning curves. Human learning may occur as part of education, personal development, or training. It may be goal-oriented and may be aided by motivation. The study of how learning occurs is part of neuropsychology, educational psychology, learning theory, and pedagogy. Learning may occur as a result of habituation or classical conditioning, seen in many animal species, or as a result of more complex activities such as play, seen only in relatively intelligent animals. Learning may occur consciously or without conscious awareness.

There is evidence for human behavioural learning prenatally, in which habituation has been observed as early as 32 weeks into gestation, indicating that the central nervous system is sufficiently developed and primed for learning and memory to occur very early on in development. Play has been approached by several theorists as the first form of learning. Children play, experiment with the world, learn the rules, and learn to interact. Vygotsky agrees that play is pivotal for children's development, since they make meaning of their environment through play. While this article describes the different environments in which learning happens, there seems as yet to be no description of how a collection of neurons learns nor where, exactly, in this collection of neurons "memory" is located.

TYPES OF LEARNING

Simple Non-associative Learning

Habituation

In psychology, habituation is an example of non-associative learning in which there is a progressive diminution of behavioural response probability with repetition stimulus. An animal first responds to a stimulus, but if it is neither rewarding nor harmful the animal reduces subsequent responses. One example of this can be seen in small song birds—if a stuffed owl is put into the cage, the birds initially react to it as though it were a real predator. Soon the birds react less, showing habituation. If another stuffed owl is introduced, the birds react to it again as though it were a predator, demonstrating that it is only a very specific stimulus that is habituated to. Habituation has been shown in essentially every species of animal, including the large protozoan *Stentor coeruleus*.

Sensitization

Sensitization is an example of non-associative learning in which the progressive amplification of a response follows repeated administrations of a stimulus. An everyday example of this mechanism is the repeated tonic stimulation of peripheral nerves that will occur if a person rubs his arm continuously. After a while, this stimulation will create a warm sensation that will eventually turn painful. The pain is the result of the progressively amplified synaptic response of the peripheral nerves warning the person that the stimulation is harmful. Sensitization is thought to underlie both adaptive as well as maladaptive learning processes in the organism.

Associative Learning

Associative learning is the process by which an element is taught through association with a separate, pre-occurring element. It is also referred to as classical conditioning. Operant conditioning is the use of consequences to modify the occurrence and form of behaviour. Operant conditioning is distinguished from Pavlovian conditioning in that operant conditioning uses reinforcement/punishment to alter a behaviour. Behaviourism is a psychological movement that seeks to alter behaviour by arranging the environment to elicit successful changes and to arrange consequences to maintain or diminish a behaviour. Behaviourists study behaviours that can be measured and changed by the environment. However, they do not eliminate that there are thought processes that interact with those behaviours. Delayed discounting is the process of devaluing rewards based on the delay of time they are presented. This process is thought to be tied to impulsivity. Impulsivity is a core process for many behaviours. Making decisions is an important part of everyday functioning. How we make those decisions is based on what we perceive to be the most valuable or worthwhile actions. This is determined by what we find to be the most reinforcing stimuli. So when teaching an individual a response, you need to

find the most potent reinforcer for that person. This may be a larger reinforcer at a later time or a smaller immediate reinforcer.

Classical Conditioning

The typical paradigm for classical conditioning involves repeatedly pairing an unconditioned stimulus with another previously neutral stimulus. Following conditioning, the response occurs both to the unconditioned stimulus and to the other, unrelated stimulus. The response to the conditioned stimulus is termed a conditioned response.

The classic example is Pavlov and his dogs. Meat powder naturally will make a dog salivate when it is put into a dog's mouth; salivating is a reflexive response to the meat powder. Meat powder is the unconditioned stimulus and the salivation is the unconditioned response. Then Pavlov rang a bell before presenting the meat powder. The first time Pavlov rang the bell, the neutral stimulus, the dogs did not salivate, but once he put the meat powder in their mouths they began to salivate. After numerous pairings of the bell, and then food the dogs learned that the bell was a signal that the food was about to come and began to salivate just when the bell was rang. Once this occurs the bell becomes the conditioned stimulus and the salivation to the bell is the conditioned response.

Another influential person in the world of Classical Conditioning is John B. Watson. Watson's work was very influential and paved the way for B. F. Skinner's radical behaviourism. Watson's behaviourism stood in direct contrast to Freud. Watson's view was that Freud's introspective method was too subjective, and that we should limit the study of human development to directly observable behaviours. In 1913, Watson published the article "Psychology as the Behaviourist Views," in which he argued that laboratory studies should serve psychology best as a science. Watson's most famous, and controversial, experiment, "Little Albert," where he demonstrated how psychologists can account for the learning of emotion through classical conditioning principles.

Imprinting

Imprinting is the term used in psychology and ethology to describe any kind of phase-sensitive learning that is rapid and apparently independent of the consequences of behaviour. It was first used to describe situations in which an animal or person learns the characteristics of some stimulus, which is therefore said to be "imprinted" onto the subject.

Observational Learning

The learning process most characteristic of humans is imitation; one's personal repetition of an observed behaviour, such as a dance. Humans can copy three types of information simultaneously: the demonstrator's goals, actions and environmental outcomes. Through copying these types of information, infants will tune into their surrounding culture.

Play

Play generally describes behaviour which has no particular end in itself, but improves performance in similar situations in the future. This is seen in a wide variety of vertebrates besides humans, but is mostly limited to mammals and birds. Cats are known to play with a ball of string when young, which gives them experience with catching prey. Besides inanimate objects, animals may play with other members of their own species or other animals, such as orcas playing with seals they have caught.

Play involves a significant cost to animals, such as increased vulnerability to predators and the risk of injury and possibly infection. It also consumes energy, so there must be significant benefits associated with play for it to have evolved. Play is generally seen in younger animals, suggesting a link with learning. However, it may also have other benefits not associated directly with learning, for example improving physical fitness.

Enculturation

Enculturation is the process by which a person learns the requirements of their native culture by which he or she is surrounded, and acquires values and behaviours that are appropriate or necessary in that culture. The influences which as part of this process limit, direct or shape the individual, whether deliberately or not, include parents, other adults, and peers. If successful, enculturation results in competence in the language, values and rituals of the culture..

Multimedia Learning

Multimedia learning is where a person uses both auditory and visual stimuli to learn information. This type of learning relies on dual-coding theory.

E-learning and Augmented Learning

Electronic learning or e-learning is a general term used to refer to Internet-based networked computer-enhanced learning. A specific and always more diffused e-learning is mobile learning, which uses different mobile telecommunication equipment, such as cellular phones. When a learner interacts with the e-learning environment, it's called augmented learning. By adapting to the needs of individuals, the context-driven instruction can be dynamically tailored to the learner's natural environment. Augmented digital content may include text, images, video, audio. By personalizing instruction, augmented learning has been shown to improve learning performance for a lifetime.

Rote Learning

Rote learning is a technique which avoids understanding the inner complexities and inferences of the subject that is being learned and instead focuses on memorizing the material so that it can be recalled by the learner exactly the way it was read or heard. The major practice involved in rote learning techniques is learning by repetition, based

on the idea that one will be able to quickly recall the meaning of the material the more it is repeated. Rote learning is used in diverse areas, from mathematics to music to religion. Although it has been criticized by some schools of thought, rote learning is a necessity in many situations.

Informal Learning

Informal learning occurs through the experience of day-to-day situations. It is learning from life, during a meal at table with parents, play, exploring, etc.

Formal Learning

Formal learning is learning that takes place within a teacher-student relationship, such as in a school system.

Non-formal Learning

Nonformal learning is organized learning outside the formal learning system. For example: learning by coming together with people with similar interests and exchanging viewpoints, in clubs or in youth organizations, workshops.

Non-formal Learning and Combined Approaches

The educational system may use a combination of formal, informal, and non-formal learning methods. The UN and EU recognize these different forms of learning. In some schools students can get points that count in the formal-learning systems if they get work done in informal-learning circuits.

They may be given time to assist international youth workshops and training courses, on the condition they prepare, contribute, share and can proof this offered valuable new insight, helped to acquire new skills, a place to get experience in organizing, teaching, etc.

In order to learn a skill, such as solving a Rubik's cube quickly, several factors come into play at once:

- Directions help one learn the patterns of solving a Rubik's cube.
- Practicing the moves repeatedly and for extended time helps with "muscle memory" and therefore speed.
- Thinking critically about moves helps find shortcuts, which in turn helps to speed up future attempts.
- The Rubik's cube's six colours help anchor solving it within the head.
- Occasionally revisiting the cube helps prevent negative learning or loss of skill.

Tangential Learning

Tangential learning is the process by which some portion of people will self-educate if a topic is exposed to them in something that they already enjoy such as playing a musical instrument.

DOMAINS OF LEARNING

Benjamin Bloom has suggested three domains of learning:

- Cognitive – To recall, calculate, discuss, analyze, problem solve, etc.
- Psychomotor – To dance, swim, ski, dive, drive a car, ride a bike, etc.
- Affective – To like something or someone, love, appreciate, fear, hate, worship, etc.

These domains are not mutually exclusive. For example, in learning to play chess, the person will have to learn the rules of the game; but he also has to learn how to set up the chess pieces on the chessboard and also how to properly hold and move a chess piece. Furthermore, later in the game the person may even learn to love the game itself, value its applications in life, and appreciate its history.

LIBRARY

A library is a collection of sources, resources, and services, and the structure in which it is housed; it is organized for use and maintained by a public body, an institution, or a private individual. In the more traditional sense, a library is a collection of books. It can mean the collection, the building or room that houses such a collection, or both. The term “library” has itself acquired a secondary meaning: “a collection of useful material for common use.” This sense is used in fields such as computer science, mathematics, statistics, electronics and biology. It can also be used by publishers in naming series of related books, *e.g.* The Library of Anglo-Catholic Theology. Libraries often provide a place of silence for studying. Public and institutional collections and services may be intended for use by people who choose not to — or cannot afford to — purchase an extensive collection themselves, who need material no individual can reasonably be expected to have, or who require professional assistance with their research. In addition to providing materials, libraries also provide the services of librarians who are experts at finding and organizing information and at interpreting information needs.

Today’s libraries are repositories and access points for print, audio, and visual materials in numerous formats, including maps, prints, documents, microform, audio tapes, CDs, cassettes, videotapes, DVDs, video games, e-books, e-audiobooks and many other electronic resources. Libraries often provide public facilities to access to their electronic resources and the Internet. Thus, modern libraries are increasingly being redefined as places to get unrestricted access to information in many formats and from many sources. They are extending services beyond the physical walls of a building, by providing material accessible by electronic means, and by providing the assistance of librarians in navigating and analyzing tremendous amounts of information with a variety of digital tools.

TYPES OF LIBRARIES

By the entity that supports or perpetuates them:

- Academic libraries
- Corporate libraries

- Government libraries, such as national libraries
- Historical society libraries
- Private libraries
- Public libraries
- School libraries
- Special libraries.

By the type of documents or materials they hold:

- Data libraries
- Digital libraries
- Map libraries or collections
- Picture libraries
- Slide libraries
- Tool libraries.

By the subject matter of documents they hold:

- Architecture libraries
- Fine arts libraries
- Law libraries
- Medical libraries
- Aquatic science libraries
- Theological libraries.

By the users they serve:

- Military communities
- Users who are blind or visually/physically handicapped.

By traditional professional divisions:

- *Academic libraries:* These libraries are located on the campuses of colleges and universities and serve primarily the students and faculty of that and other academic institutions. Some academic libraries, especially those at public institutions, are accessible to members of the general public in whole or in part.
- *Public libraries or public lending libraries:* These libraries provide service to the general public and make at least some of their books available for borrowing, so that readers may use them at home over a period of days or weeks. Typically, libraries issue library cards to community members wishing to borrow books. Many public libraries also serve as community organizations that provide free services and events to the public, such as reading groups and toddler story time.
- *Research libraries:* These libraries are intended for supporting scholarly research, and therefore maintain permanent collections and attempt to provide access to all necessary material. Research libraries are most often academic libraries or national libraries, but many large special libraries have research libraries within their special field and a very few of the largest public libraries also serve as research libraries.

- *School libraries:* Most public and private primary and secondary schools have libraries designed to support the school's curriculum.
- *Special libraries:* All other libraries fall into this category. Many private businesses and public organizations, including hospitals, museums, research laboratories, law firms, and many government departments and agencies, maintain their own libraries for the use of their employees in doing specialized research related to their work. Special libraries may or may not be accessible to some identified part of the general public. Branches of a large academic or research libraries dealing with particular subjects are also usually called "special libraries": they are generally associated with one or more academic departments. Special libraries are distinguished from special collections, which are branches or parts of a library intended for rare books, manuscripts, and similar material.

Many institutions make a distinction between circulating libraries and collecting libraries. Many modern libraries are a mixture of both, as they contain a general collection for circulation, and a reference collection which is often more specialized, as well as restricted to the library premises.

PUBLIC LIBRARIES

The earliest example in England of a library to be endowed for the benefit of users who were not members of an institution such as a cathedral or college was the Francis Trigge Chained Library in Grantham, Lincolnshire, established in 1598. The library still exists and can justifiably claim to be the forerunner of later public library systems. The beginning of the modern, free, open access libraries really got its start in the U.K., in 1847. Parliament appointed a committee, led by William Ewart, on Public Libraries to consider the necessity of establishing libraries through the nation: In 1849 their report noted the poor condition of library service, it recommended the establishment of free public libraries all over the country, and it led to the Public Libraries Act in 1850, which allowed all cities with populations exceeding 10,000 to levy taxes for the support of public libraries. Another important act was the 1870 Public School Law, which increased literacy, thereby the demand for libraries, so by 1877, more than 75 cities had established free libraries, and by 1900 the number had reached 300.

This finally marks the start of the public library as we know it. And these acts led to similar laws in other countries, most notably the U.S., 1874 is a well known year in the history of librarianship in the United States. The American Library Association was formed, as well as The American Library Journal, Melvil Dewey published his decimal based system of classification, and the United States Bureau of Education published its report, "Public libraries in the United States of America; their history, condition, and management." During the post-Civil War years, there was a rise in the establishment of public libraries, a movement led chiefly by newly formed women's clubs. They contributed their own collections of books, conducted lengthy fund raising campaigns for buildings, and lobbied within their communities for financial support for libraries,

as well as with legislatures and the Carnegie Library Endowment founded in the 20th century. They led the establishment of 75-80 percent of the libraries in communities across the country.

The American Library Association continues to play a major role in libraries to this day, and Dewey's classification system, although under heavy criticism of late, still remains the prevailing method of classifying used in the United States. As the number of books in libraries increased, so did the need for compact storage and access with adequate lighting, giving birth to the stack system, which involved keeping a library's collection of books in a space separate from the reading room. This arrangement arose in the 19th century. Book stacks quickly evolved into a fairly standard form in which the cast iron and steel frameworks supporting the bookshelves also supported the floors, which often were built of translucent blocks to permit the passage of light. The introduction of electrical lighting had a huge impact on how the library operated.

The use of glass floors was largely discontinued, though floors were still often composed of metal grating to allow air to circulate in multi-story stacks. As more space was needed, a method of moving shelves on tracks was introduced to cut down on otherwise wasted aisle space. Library 2.0, a term coined in 2005, is the library's response to the challenge of Google and an attempt to meet the changing needs of users by using web 2.0 technology. Some of the aspects of Library 2.0 include, commenting, tagging, bookmarking, discussions, use of online social networks by libraries, plug-ins, and widgets.

Inspired by web 2.0, it is an attempt to make the library a more user-driven institution. Despite the importance of public libraries, they are routinely having their budgets cut by state legislature. Funding has dwindled so badly that some smaller public libraries have been forced to cut their hours and release employees. While most donations made to public libraries are from private benefactors, they still receive very little in the way of state funding.

LIBRARY USE

Patrons may not know how to fully use the library's resources. This can be due to some individuals' unease in approaching a staff member. Ways in which a library's content is displayed or accessed may have the most impact on use. An antiquated or clumsy search system, or staff unwilling or untrained to engage their patrons, will limit a library's usefulness. In United States public libraries, beginning in the 19th century, these problems drove the emergence of the library instruction movement, which advocated library user education. One of the early leaders was John Cotton Dana. The basic form of library instruction is generally known as information literacy. Libraries inform their users of what materials are available in their collections and how to access that information.

Before the computer age, this was accomplished by the card catalogue — a cabinet containing many drawers filled with index cards that identified books and other materials. In a large library, the card catalogue often filled a large room. The emergence of the

Internet, however, has led to the adoption of electronic catalogue databases which allow users to search the library's holdings from any location with Internet access. This style of catalogue maintenance is compatible with new types of libraries, such as digital libraries and distributed libraries, as well as older libraries that have been retrofitted. Electronic catalogue databases are criticized by some who believe that the old card catalogue system was both easier to navigate and allowed retention of information, by writing directly on the cards, that is lost in the electronic systems.

This argument is analogous to the debate over paper books and e-books. While libraries have been accused of precipitously throwing out valuable information in card catalogs, most modern ones have nonetheless made the move to electronic catalogue databases. Large libraries may be scattered within multiple buildings across a town, each having multiple floors, with multiple rooms housing the resources across a series of shelves.

Once a user has located a resource within the catalogue, they must then use navigational guidance to retrieve the resource physically; a process that may be assisted through signage, maps, GPS systems or RFID tagging. Finland has the highest number of registered book borrowers per capita in the world. Over half of Finland's population are registered borrowers. In the U.S., public library users have borrowed roughly 15 books per user per year from 1856 to 1978. From 1978 to 2004, book circulation per user declined approximately 50 per cent. The growth of audiovisuals circulation, estimated at 25 per cent of total circulation in 2004, accounts for about half of this decline.

Shift to Digital Libraries

In the past couple of years, more and more people are using the Internet to gather and retrieve data. The shift to digital libraries has greatly impacted the average person's use of physical libraries. Between 2002 and 2004, the average American academic library saw its overall number of transactions decline approximately 2.2 per cent. Libraries are trying to keep up with the digital world and the new generation of students that are used to having information just one click away. For example, The University of California Library System saw a 54 per cent decline in circulation between 1991 to 2001 of 8,377,000 books to 3,832,000. These facts might be a consequence of the increased availability of e-resources. In 1999-2000, 105 ARL university libraries spent almost \$100 million on electronic resources, which is an increase of nearly \$23 million from the previous year.

A 2003 report by the Open E-book Forum found that close to a million e-books had been sold in 2002, generating nearly \$8 million in revenue. Another example of the shift to digital libraries can be seen in Cushing Academy's decision to dispense with its library of printed books — more than 20,000 volumes in all — and switch over entirely to digital media resources. One claim to why there is a decrease in the usage of libraries stems from the observation of the research habits of undergraduate students enrolled in

colleges and universities. There have been claims that college undergraduates have become more used to retrieving information from the Internet than a traditional library. As each generation becomes more in tune with the Internet, their desire to retrieve information as quickly and easily as possible has increased.

There is no doubt that finding information by simply searching the Internet is much easier and faster than reading an entire book. In a survey conducted by NetLibrary, 93 per cent of undergraduate students claimed that finding information online makes more sense to them than going to the library. Also, 75 per cent of students surveyed claimed that they did not have enough time to go to the library and that they liked the convenience of the Internet. While the retrieving information from the Internet may be efficient and time saving than visiting a traditional library, research has shown that undergraduates are most likely searching only 0.03 per cent of the entire web. The information that they are finding might be easy to retrieve and more readily available, but may not be as in depth as information from other resources such as the books available at a physical library.

STUDENT

The word student is etymologically derived through Middle English from the Latin second-type conjugation verb *studĕre*, meaning “to direct one’s zeal at”; hence a student could be described as “one who directs zeal at a subject”. In its widest use, student is used for anyone who is learning.

MATURE STUDENTS

A mature, non-traditional, or adult student in tertiary education is normally classified as an student who is at least 21–23 years old at the start of their course and usually having been out of the education system for at least two years. Mature students can also include students who have been out of the education system for decades, or students with no secondary education. Mature students also make up graduate and postgraduate populations by demographic of age.

STUDENT PRANKS

University students have been associated with pranks and japes since the creation of universities in the Middle Ages. These can often involve petty crime, such as the theft of traffic cones and other public property, or hoaxes. It is also not uncommon for students from one school to steal or deface the mascot of a rival school. In fact, pranks play such a significant part in student culture that numerous books have been published that focus on the issue. Pranks may reflect current events, be a form of protest or revenge, or have no other purpose than for the enjoyment of the prank itself. A recent report has been released focusing on the misbehaviour of university students. The report, *Studentification: A Guide to Opportunities, Challenges and Practice*, by Universities UK, focuses on six British universities as case studies.

STUDY SKILLS

Study skills or study strategies are approaches applied to learning. They are generally critical to success in school, are considered essential for acquiring good grades, and are useful for learning throughout one's life. There are an array of study skills, which may tackle the process of organising and taking in new information, retaining information, or dealing with assessments. They include mnemonics, which aid the retention of lists of information, and effective notetaking. While often left up to the student and their support network, study skills are increasingly taught at High School and University level. A number of books and web sites are available, from works on specific techniques such as Tony Buzan's books on mind-mapping, to general guides to successful studying. More broadly, any skill which boosts a person's ability to study and pass exams can be termed a study skill, and this could include time management and motivational techniques. Study Skills are discrete techniques that can be learned, usually in a short time, and applied to all or most fields of study. They should therefore be distinguished from strategies that are specific to a particular field of study, *e.g.*, music or technology, and from abilities inherent in the student, such as aspects of intelligence or learning style.

TYPES OF STUDY SKILLS

Methods Based on Memorization such as Rehearsal and Rote Learning

One of the most basic approaches to learning any information is simply to repeat it by rote. Typically this will include reading over notes or a textbook, and re-writing notes.

Methods Based on Communication Skills, *e.g.*, Reading and Listening

The weakness with rote learning is that it implies a passive reading or listening style. Educators such as John Dewey have argued that students need to learn critical thinking - questioning and weighing up evidence as they learn. This can be done during lectures or when reading books. One method used to focus on key information when studying from books is the PQIRST method. This method prioritizes the information in a way that relates directly to how they will be asked to use that information in an exam. PQIRST is an acronym for Preview, Question, Read, Summary, Test.

- *Preview:* The student looks at the topic to be learned by glancing over the major headings or the points in the syllabus.
- *Question:* Then questions to be answered once the topic has been thoroughly studied are formulated.
- *Read:* Reference material related to the topic is read through, and the information that best relates to the questions is chosen.
- *Summary:* The student summarizes the topic, bringing his or her own ways of summarizing information into the process, including written notes, spider diagrams, flow diagrams, labeled diagrams, mnemonics, or even voice recordings.

- *Test*: Then the student answers the questions created in the question step as fully as possible, avoiding adding questions that might distract or change the subject.

Methods Based on Cues, e.g. Flashcard Training

Flash Cards are visual cues on cards. These have numerous uses in teaching and learning, but can be used for revision. Students often make their own flash cards, or more detailed index cards - cards designed for filing, often A5 size, on which short summaries are written. Being discrete and separate, they have the advantage of allowing students to re-order them, pick a selection to read over, or choose randomly to for self-testing.

Methods Based on Condensing Information, Summarising and the Use of Keywords

Summary methods vary depending on the topic, but most involve condensing the large amount of information from a course or book into shorter notes. Often these notes are then condensed further into key facts. Spider diagrams: Using spider diagrams or mind maps can be an effective way of linking concepts together. They can be useful for planning essays and essay responses in exams. These tools can give a visual summary of a topic that preserves its logical structure, with lines used to show how different parts link together.

Methods Based on Visual Imagery

Some learners are thought to have a visual learning style, and will benefit greatly from taking information from their studies which is often heavily verbal, and using visual techniques to help encode and retain it in memory. Some memory techniques make use of visual memory, for example the method of loci, a system of visualising key information in real physical locations, *e.g.*, around a room. Diagrams are often underrated tools. They can be used to bring all the information together and provide practice reorganizing what has been learned in order to produce something practical and useful. They can also aid the recall of information learned very quickly, particularly if the student made the diagram while studying the information. Pictures can then be transferred to flash cards that are very effective last minute revision tools rather than rereading any written material.

Methods Based on Acronyms and Mnemonics

A mnemonic is a method of organizing and memorizing information. Some use a simple phrase or fact as a trigger for a longer list of information. For example, the points of the compass can be recalled in the correct order with the phrase “Never Eat Shredded Wheat”. Starting with North, the first letter of each word relates to a compass point in clockwise order round a compass.

Methods Based on Exam Strategies

The Black-Red-Green method helps the student to ensure that every aspect of the question posed has been considered, both in exams and essays. The student underlines relevant parts of the question using three separate colours. BLAck denotes ‘BLAtant instructions’, *i.e.*, something that clearly must be done; a directive or obvious instruction. REd is a REference Point or REquired input of some kind, usually to do with definitions, terms, cited authors, theory, etc.. GREen denotes GREmlins, which are subtle signals one might easily miss, or a ‘GREEN Light’ that gives a hint on how to proceed, or where to place the emphasis in answers.

Methods Based on Time Management, Organization and Lifestyle Changes

Often, improvements to the effectiveness of study may be achieved through changes to things unrelated to the study material itself, such as time-management, boosting motivation and avoiding procrastination, and in improvements to sleep and diet. Time management in study sessions aims to ensure that activities that achieve the greatest benefit are given the greatest focus. A traffic lights system is a simple way of identifying the importance of information, highlighting or underlining information in colours:

- *Green*: Topics to be studied first; important and also simple
- *Amber*: Topics to be studied next; important but time-consuming
- *Red*: Lowest priority; complex and not vital.

This reminds students to start with the things which will provide the quickest benefit, while ‘red’ topics are only dealt with if time allows. The concept is similar to the ABC_analysis, commonly used by workers to help prioritise.

TEACHER

In education, a teacher is a person who provides schooling for others. A teacher who facilitates education for an individual student may also be described as a personal tutor. The role of teacher is often formal and ongoing, carried out by way of occupation or profession at a school or other place of formal education. In many countries, a person who wishes to become a teacher at state-funded schools must first obtain professional qualifications or credentials from a university or college. These professional qualifications may include the study of pedagogy, the science of teaching. Teachers will have to continue their education after they receive their degree from a college or university. Teachers may use a lesson plan to facilitate student learning, providing a course of study which helps cover a standardized test curriculum. A teacher’s role may vary between cultures. Teachers teach literacy and numeracy, or some of the other school subjects. Other teachers may provide help or certain education instruction in craftsmanship or vocational training, the Arts, religion or spirituality, civics, community roles, or life skills. In some countries, formal education can take place through home schooling. Informal learning may be assisted by a teacher occupying a transient or ongoing role, such as a parent or sibling or within a family, or by anyone with knowledge

or skills in the wider community setting. Religious and spiritual teachers, such as gurus, mullahs, rabbis pastors/youth pastors and lamas may teach religious texts such as the Quran, Torah or Bible.

PROFESSIONAL EDUCATORS

Teaching may be carried out informally, within the family which is called home schooling or the wider community. Formal teaching may be carried out by paid professionals. Such professionals enjoy a status in some societies on a par with physicians, lawyers, engineers, and accountants. A teacher's professional duties may extend beyond formal teaching. Outside of the classroom teachers may accompany students on field trips, supervise study halls, help with the organization of school functions, and serve as supervisors for extracurricular activities. In some education systems, teachers may have responsibility for student discipline.

Around the world teachers are often required to obtain specialized education, knowledge, codes of ethics and internal monitoring. There are a variety of bodies designed to instill, preserve and update the knowledge and professional standing of teachers. Around the world many governments operate teacher's colleges, which are generally established to serve and protect the public interest through certifying, governing and enforcing the standards of practice for the teaching profession. The functions of the teacher's colleges may include setting out clear standards of practice, providing for the ongoing education of teachers, investigating complaints involving members, conducting hearings into allegations of professional misconduct and taking appropriate disciplinary action and accrediting teacher education programmes. In many situations teachers in publicly funded schools must be members in good standing with the college, and private schools may also require their teachers to be college peoples. In other areas these roles may belong to the State Board of Education, the Superintendent of Public Instruction, the State Education Agency or other governmental bodies. In still other areas Teaching Unions may be responsible for some or all of these duties.

PEDAGOGY AND TEACHING

In education, teachers facilitate student learning, often in a school or academy or perhaps in another environment such as outdoors. A teacher who teaches on an individual basis may be described as a tutor. The objective is typically accomplished through either an informal or formal approach to learning, including a course of study and lesson plan that teaches skills, knowledge and/or thinking skills. Different ways to teach are often referred to as pedagogy. When deciding what teaching method to use teachers consider students' background knowledge, environment, and their learning goals as well as standardized curricula as determined by the relevant authority. Many times, teachers assist in learning outside of the classroom by accompanying students on field trips. The increasing use of technology, specifically the rise of the internet over the past decade, has begun to shape the way teachers approach their roles in the classroom. The objective

is typically a course of study, lesson plan, or a practical skill. A teacher may follow standardized curricula as determined by the relevant authority.

The teacher may interact with students of different ages, from infants to adults, students with different abilities and students with learning disabilities. Teaching using pedagogy also involve assessing the educational levels of the students on particular skills. Understanding the pedagogy of the students in a classroom involves using differentiated instruction as well as supervision to meet the needs of all students in the classroom. Pedagogy can be thought of in two manners. First, teaching itself can be taught in many different ways, hence, using a pedagogy of teaching styles. Second, the pedagogy of the learners comes into play when a teacher assesses the pedagogic diversity of his/her students and differentiates for the individual students accordingly. Perhaps the most significant difference between primary school and secondary school teaching is the relationship between teachers and children.

In primary schools each class has a teacher who stays with them for most of the week and will teach them the whole curriculum. In secondary schools they will be taught by different subject specialists each session during the week and may have 10 or more different teachers. The relationship between children and their teachers tends to be closer in the primary school where they act as form tutor, specialist teacher and surrogate parent during the course of the day. This is true throughout most of the United States as well. However, alternative approaches for primary education do exist. One of these, sometimes referred to as a “platoon” system, involves placing a group of students together in one class that moves from one specialist to another for every subject. The advantage here is that students learn from teachers who specialize in one subject and who tend to be more knowledgeable in that one area than a teacher who teaches many subjects. Students still derive a strong sense of security by staying with the same group of peers for all classes. Co-teaching has also become a new trend amongst educational institutions. Co-teaching is defined as two or more teachers working harmoniously to fulfill the needs of every student in the classroom. Co-teaching focuses the student on learning by providing a social networking support that allows them to reach their full cognitive potential. Co-teachers work in sync with one another to create a climate of learning.

RIGHTS TO ENFORCE SCHOOL DISCIPLINE

Throughout the history of education the most common form of school discipline was corporal punishment. While a child was in school, a teacher was expected to act as a substitute parent, with all the normal forms of parental discipline open to them. In past times, corporal punishment was one of the most common forms of school discipline throughout much of the world. Most Western countries, and some others, have now banned it, but it remains lawful in the United States following a US Supreme Court decision in 1977 which held that paddling did not violate the US Constitution. 30 US states have banned corporal punishment, the others have not. It is still used to a significant

degree in some public schools in Alabama, Arkansas, Georgia, Louisiana, Mississippi, Oklahoma, Tennessee and Texas.

Private schools in these and most other states may also use it. Corporal punishment in American schools is administered to the seat of the student's trousers or skirt with a specially made wooden paddle. This often used to take place in the classroom or hallway, but nowadays the punishment is usually given privately in the principal's office. Official corporal punishment, often by caning, remains commonplace in schools in some Asian, African and Caribbean countries. Currently detention is one of the most common punishments in schools in the United States, the UK, Ireland, Singapore and other countries. It requires the pupil to remain in school at a given time in the school day; or even to attend school on a non-school day, *e.g.*, "Saturday detention" held at some US schools. During detention, students normally have to sit in a classroom and do work, write lines or a punishment essay, or sit quietly.

A modern example of school discipline in North America and Western Europe relies upon the idea of an assertive teacher who is prepared to impose their will upon a class. Positive reinforcement is balanced with immediate and fair punishment for misbehaviour and firm, clear boundaries define what is appropriate and inappropriate behaviour. Teachers are expected to respect their students, and sarcasm and attempts to humiliate pupils are seen as falling outside of what constitutes reasonable discipline.

Whilst this is the consensus viewpoint amongst the majority of academics, some teachers and parents advocate a more assertive and confrontational style of discipline. Such individuals claim that many problems with modern schooling stem from the weakness in school discipline and if teachers exercised firm control over the classroom they would be able to teach more efficiently. This viewpoint is supported by the educational attainment of countries—in East Asia for instance—that combine strict discipline with high standards of education.

It's not clear, however that this stereotypical view reflects the reality of East Asian classrooms or that the educational goals in these countries are commensurable with those in Western countries. In Japan, for example, although average attainment on standardized tests may exceed those in Western countries, classroom discipline and behaviour is highly problematic. Although, officially, schools have extremely rigid codes of behaviour, in practice many teachers find the students unmanageable and do not enforce discipline at all. Where school class sizes are typically 40 to 50 students, maintaining order in the classroom can divert the teacher from instruction, leaving little opportunity for concentration and focus on what is being taught. In response, teachers may concentrate their attention on motivated students, ignoring attention-seeking and disruptive students. The result of this is that motivated students, facing demanding university entrance examinations, receive disproportionate resources, while the rest of the students are allowed, perhaps expected to, fail. Given the emphasis on attainment of university places, administrators and governors may regard this policy as appropriate.

OBLIGATION TO HONOUR STUDENTS RIGHTS

Sudbury model democratic schools claim that popularly based authority can maintain order more effectively than dictatorial authority for governments and schools alike. They also claim that in these schools the preservation of public order is easier and more efficient than anywhere else. Primarily because rules and regulations are made by the community as a whole, thence the school atmosphere is one of persuasion and negotiation, rather than confrontation since there is no one to confront. Sudbury model democratic schools' experience shows that a school that has good, clear laws, fairly and democratically passed by the entire school community, and a good judicial system for enforcing these laws, is a school in which community discipline prevails, and in which an increasingly sophisticated concept of law and order develops, against other schools today, where rules are arbitrary, authority is absolute, punishment is capricious, and due process of law is unknown.

TEACHER ENTHUSIASM

Since teachers can affect how students perceive the course materials, it has been found that teachers who showed enthusiasm towards the course materials and students can affect a positive learning experience towards the course materials. On teacher/course evaluations, it was found that teachers who have a positive disposition towards the course content tend to transfer their passion to receptive students. Teachers cannot teach by rote but have to find new invigoration for the course materials on a daily basis. Teachers have to keep in mind that they are teaching new minds every term or semester. Otherwise, teachers will fall into the trap of having done this material again and start feeling bored with the subject which in turn bore the students as well. Students who had enthusiastic teachers tend to rate them higher than teachers who didn't show much enthusiasm for the course materials.

Teachers that exhibit enthusiasm can lead to students who are more likely to be engaged, interested, energetic, and curious about learning the subject matter. Recent research has found a correlation between teacher enthusiasm and students' intrinsic motivation to learn and vitality in the classroom. Controlled, experimental studies exploring intrinsic motivation of college students has shown that non-verbal expressions of enthusiasm, such as demonstrative gesturing, dramatic movements which are varied, and emotional facial expressions, result in college students reporting higher levels of intrinsic motivation to learn. Students who experienced a very enthusiastic teacher were more likely to read lecture material outside of the classroom. There are various mechanisms by which teacher enthusiasm may facilitate higher levels of intrinsic motivation.

Teacher enthusiasm may contribute to a classroom atmosphere full of energy and enthusiasm which feed student interest and excitement in learning the subject matter. Enthusiastic teachers may also lead to students becoming more self-determined in their own learning process. The concept of mere exposure indicates that the teacher's

enthusiasm may contribute to the student's expectations about intrinsic motivation in the context of learning. Also, enthusiasm may act as a "motivational embellishment"; increasing a student's interest by the variety, novelty, and surprise of the enthusiastic teacher's presentation of the material.

Finally, the concept of emotional contagion, may also apply. Students may become more intrinsically motivated by catching onto the enthusiasm and energy of the teacher. Research shows that student motivation and attitudes towards school are closely linked to student-teacher relationships. Enthusiastic teachers are particularly good at creating beneficial relations with their students.

Their ability to create effective learning environments that foster student achievement depends on the kind of relationship they build with their students. Useful teacher-to-student interactions are crucial in linking academic success with personal achievement. Here, personal success is a student's internal goal of improving himself, whereas academic success includes the goals he receives from his superior. A teacher must guide his student in aligning his personal goals with his academic goals. Students who receive this positive influence show stronger self-confidence and greater personal and academic success than those without these teacher interactions. Students are likely to build stronger relations with teachers who are friendly and supportive and will show more interest in courses taught by these teachers.

Teachers that spend more time interacting and working directly with students are perceived as supportive and effective teachers. Effective teachers have been shown to invite student participation and decision making, allow humor into their classroom, and demonstrate a willingness to play. The way a teacher promotes the course they are teaching, the more the student will get out of the subject matter. The three most important aspects of teacher enthusiasm are enthusiasm about teaching, enthusiasm about the students, and enthusiasm about the subject matter.

A teacher must enjoy teaching. If they do not enjoy what they are doing, the students will be able to tell. They also must enjoy being around their students. A teacher who cares for their students is going to help that individual succeed in their life in the future.

The teacher also needs to be enthusiastic about the subject matter they are teaching. For example, a teacher talking about chemistry needs to enjoy the art of chemistry and show that to their students. A spark in the teacher may create a spark of excitement in the student as well. An enthusiastic teacher has the ability to be very influential in the young students life.

MISCONDUCT

Misconduct by teachers, especially sexual misconduct, has been getting increased scrutiny from the media and the courts. A study by the American Association of University Women reported that 0.6 per cent of students in the United States claim to have received unwanted sexual attention from an adult associated with education; be they a volunteer, bus driver, teacher, administrator or other adult; sometime during their educational

career. A study in England showed a 0.3 per cent prevalence of sexual abuse by any professional, a group that included priests, religious leaders, and case workers as well as teachers. It is important to note, however, that the British study referenced above is the only one of its kind and consisted of “a random... probability sample of 2,869 young people between the ages of 18 and 24 in a computer-assisted study” and that the questions referred to “sexual abuse with a professional,” not necessarily a teacher. It is therefore logical to conclude that information on the percentage of abuses by teachers in the United Kingdom is not explicitly available and therefore not necessarily reliable.

The AAUW study, however, posed questions about fourteen types of sexual harassment and various degrees of frequency and included only abuses by teachers. “The sample was drawn from a list of 80,000 schools to create a stratified two-stage sample design of 2,065 8th to 11th grade students” Its reliability was gauged at 95 per cent with a 4 per cent margin of error. In the United States especially, several high-profile cases such as Debra LaFave, Pamela Rogers, and Mary Kay Latourneau have caused increased scrutiny on teacher misconduct. Chris Keates, the general secretary of National Association of Schoolmasters Union of Women Teachers, said that teachers who have sex with pupils over the age of consent should not be placed on the sex offenders register and that prosecution for statutory rape “is a real anomaly in the law that we are concerned about.” This has led to outrage from child protection and parental rights groups.

UNIVERSITY

A university is an institution of higher education and research, which grants academic degrees in a variety of subjects. A university is a corporation that provides both undergraduate education and postgraduate education. The word university is derived from the Latin *universitas magistrorum et scholarium*, roughly meaning “community of teachers and scholars.” The definition of a university varies widely even within some countries. For example, there is no nationally standardized definition of the term in the United States although the term has traditionally been used to designate research institutions and was once reserved for research doctorate-granting institutions. Some states, such as Massachusetts, will only grant a school “university status” if it grants at least two doctoral degrees. In the United Kingdom, an institution can only use the term if it has been granted by the Privy Council, under the terms of the Further and Higher Education Act 1992. In India, a new tag deemed universities was created a few years ago, by the cabinet minister Arjun Singh during his tenure as the Minister for Human Resource Development. Through this provision many universities sprung up in India, which are commercial in nature and have been established just to exploit the demand of higher education.

6

CHAPTER

Aims of Education

AIMS AND OBJECTIVES OF EDUCATION

The importance of aims and objectives of education is recognised by all the educational, professional, political, non-political and religious associations, organisations and groups at various levels in their memoranda, letters and brochures. It is said that education without clear cut aims is like a rudderless ship. The following comparisons emphasise this point fully well. Every pilot has a route-chart and set timing of landing at predetermined destination. There is constitution or set of Principles and traditions through which a country is governed Similarly, there should be properly defined and declared principles, aims and objectives of education or the basis of which policies and programmes of education have to be formulated to achieve the set goals without wasting scarce energies and resources in chasing the wild goose.

It is generally felt that our educational system has not followed the desired aims as a result that it does not produce ideal citizens in the country. It has followed, rather a narrow aim of preparing individuals for livelihood, as mentioned in one of the documents received from an organisation. The main reason of failure of educational system is that it basically stands on pre-independence system. The main Objective of its products was how to take degree and to earn money and to be careerist without consideration of ethical values and national spirit. On the other hand, it has also been pointed out that it is unreasonable to criticise educational system alone because it is based on the other subsystems accepted by us. On the one hand we are developing and cultivating the British given economy, judicial system and system of administration and parliament,

and on the other we are decrying and Criticising the education system which merely fulfils the needs of the British systems that we are propagating. As pointed out in a memorandum of an association, “the main defect of the old. education policy is that it had completely ignored the Indian culture and the interest of the masses of India and have left them economically too backward and socially too fragmented to articulate their miseries....” The aims and objectives of education, suggested in the documents, include individual as well as social aims, with emphasis of social transformation aiming at reconstructing society to make it modernised, productive, partidipative, and value oriented nation committed to its constitutional obligations.

INDIVIDUAL DEVELOPMENT

Development of an individual - physically, mentally and spritually is well known aim of education. Objectives related to this aim of individual development have been expressed in various ways in the memoranda:

- Developing physical and mental faculties
- Acquiring the capacities of understanding, appreciation and expression through word and act, are the fundamental aims of education
- Aim of education should be to make children self- confident and self dependent, and to make them strong physically and mentally
- Education is meant to develop every child’s character, personality and culture and as much knowledge as the child can assimilate not merely memorize.

The best expression of complete development of an individual and the harmonious development of personality.

The policy should be directed to the aim of enlightenment of head and heart; illumination of consciousness for allround development of individual personality. Education should enable a human being to attain the greatest possible harmony, internal and external, spiritual and material, for the fullest possible development of human potentialities and capacities.

SOCIAL AND NATIONAL DEVELOPMENT

Social, aim of education in equally important because an individual lives in society and has his obligations towards his nation. There is a realisation that, “The present education system does riot yield required results mainly because it is divorced from the real social content and social goals”. It has, therefore, been suggested that education should be able, to discharge its natural functions and must correspond to its structure, goals and content in the interest of national development and social progress. It has also been suggested in this connection that students from young age should be made aware of the social responsibility cast on them. At the same time, there are certain constitutional commitments, which are intimately related to this aim.. We as the citizens of the republic, are constitutionally Committed to democracy, social justice, equality of opportunity, secularisum and above all to a welfare state. It has, therefore, been suggested that,

“Educational policy and educational programme should clearly reflect these commitments”. The objectives of developing a sense of national identity, unity and patriotism are advocated by many associations. It is pointed out that the national objectives of planning and programmes and development with special emphasis on popular participation and the national problems that we face in different fields should be taught at relevant stages. Individual and social aims of education are not contrary to one another. In fact they are complementary to one another. The following view strikes a balance between individual and social aims of education. The purpose of education should be the development of the fullest possible capacities and potentialities physical and spiritual of a ‘total man’. It should make a man capable of earning his livelihood reasonably well to enjoy a happy and secure life while making effective contributions to the society and national effort of making India strong, advanced and prosperous.

SOCIAL TRANSFORMATION

Education should not merely equip an individual to adjust with society to its customs and conventions, but it should enable him to bring desirable changes in the society. It has been, therefore, suggested that, “Every educational institution from secondary school to university college should be developed to become an agency of change....” However, it is essential that we should be quite clear about the purpose of change. It is, therefore, natural to ask the question, “Reform and change to achieve what”? What type of society we aim at and what type of citizens we wish to produce? The following ideas give an indication of the kind of changes education is expected to bring about.

Modernisation

Move with times and attain excellence in, science and technology. To quote an expression of this kind. Scientific and technological advances are, gaining momentum and conscious efforts are made to incorporate them into the development sectors. This calls for modernisation of education in order to make it in conformity with the modern times and to keep pace, with the advances in the world. Modernisation, however, is not interpreted and equated with westernisation. In fact, lot of emphasis is given to ‘Indianness’ while talking about modernisation. One of the suggestions explicitly points out that, “our education should integrate and unite the people of India, modernise society while preserving what is authentically Indian in our cultural and spiritual heritage”. The following suggestion beautifully reconciles the twin objectives of modern technical sophistication and the ancient spirituality. “New education policy of India should be built on the foundation of ancient spirituality and modern culture and technical sophistication. It should develop scientific temper and spirit of enquiry in the students”.

Productivity

Some documents have insisted on linking education with productivity and thus making individuals as productive citizens to build a productive society. One of the suggestions,

in a memorandum, for example, says. “It should bring about a social transformation, and enhance greater efficiency and productivity in all sectors: agri- cultural, industrial and service”.

It is in this context that Mahatma Gandhi’s system of basic education is still considered as a basically sound system and a suggestion has been made that with necessary modification elements of basic education may form part of education not only at the primary stage but at all stages in our national system of education.

These elements are:

- Productive activity in education.
- Correlation of the curriculum with productive activity and physical and social environment.
- Intimate contact between the school and the local community.

Community Participation

In a democracy education without community participation is barren. This aim of education is, therefore voiced by a number of groups and organisations. The change that is envisaged on this front is that of Integrating education with community in all respects. To quote a suggestion in this regard: The education system in all its branches and sectors should get itself involved in activities related to problems of local Community life and shall thus endeavour through the desirable community participation community involvement in the educational field to bring all education of its rightful place in community life.

ACQUISITION OF VALUES

Moral, cultural and spiritual values in education have been given immense importance in the Memoranda documents. One of the expressions emphatically point out that, “certain basic values as respect for others, responsibility, solidarity, creativity and integrity must be fostered in our children”. It is interesting that a number of specific values have been suggested in the documents.

The values which are considered important are mentioned below:

- Emphasis should be given in cultivating good qualities like cooperation, good will, forgiveness, tolerance, honesty, patience etc. in order to encourage universal brother-hood and to prepare students worthy citizens of the country.

Values of optimism and secularism, and service to the poor should be stressed on the young minds.

SUMMING UP

It is worth reproducing what a document mentions about the aims of education:

The aim of education is two-fold:

- Development of the individual in society and
- Consequent development of the society.

The aim of education in relation to individual may be spelt out as follows:

- To produce full human personality with courage, conviction, vitality, sensitivity and intelligence so that men and women may life in harmony with the universe;
- To bring out the fullest potential of child and prepare him. for life and its varied situations so that he becomes a cultured and responsible citizen dedicated in the service of community.

In relation to the society, the aim of education is to create:

- A sane and learning society where made of material production will be such that no section of the society remains unemployed. In the Indian context such a made of production will be necessarily based on a decentralised economy utilising all available manpower;
- A society. where the conditions of work and general environment will offer psychic satisfactions and effective motivations to its members.
- A society reconciling technological and scientific advancement with general well-being and securit of its members, enhancing joy of life and eliminating all forms of exploitation.

The broad objective of education should, therefore, be to look beyond the existing society and to develop men and. women amenable to the advent of a sane and healthier society of tomorrow.

While summing up, it may be pointed out that various dimensions individual and social development, social transformation, value- acquisition etc. have been well identified in the memoranda documents. The following words briefly summarise the various dimensions which are considered important indeed for marching into the 21st century: We are of the opinion that Indian education should aim at producing men and women of knowledge, character and cultural values and trained skills to achieve excellence in their career and life. Let us make it clear that we wish to prepare youth to march into the 21st century on the ideals of truth and non-voilence as shown to us by our great leaders.

AIMS OF EDUCATION IN PRESENT INDIAN CONTEXT WITH SPECIAL REFERENCE TO: (I) KOTHARI EDUCATION COMMISSION (1964-66) , (II) MUDALIAR COMMISSION (1952-53) AND (III) NPE (1986)

KOTHARI EDUCATION COMMISSION (1964-66)

The contribution of the Education Commission (1966) , popularly known as the Kothari Commission, to the issue of financing of education is very significant. From the monumental work of the commission that is regarded as a “turning point in India’s

educational life”, one can dig out a mine of relevant recommendations on financing of education in India.

There are:

- A few clear recommendations,
- A set of norms which may also be considered as valuable recommendations, and
- Several general intuitive normative observations which would suggest the need for a change in the approach of the policy-makers and planners.

Many of these recommendations and the premises on which they are based may be relevant still even after 40 years. Without claiming to be exhaustive, this short article revisits their premises, implementation and current relevance. It is also focused on issues relating to financing of education, though some aspects that are closely related to financing are also briefly referred to. Reviewing the report of the commission, J P Naik, who was the member-secretary and who had played an important role in drafting the voluminous report, classified the several recommendations into three categories:

- Recommendations that attracted wide attention,
- Recommendation that were opposed and rejected, and
- “other” recommendations.

Other recommendations include:

- Those that did not excite any major controversy and were accepted but implemented indifferently;
- Those that were simply ignored; and
- Other recommendations.

There is yet another category of recommendations, viz, recommendations agreed and approved, but not implemented. Some of the major recommendations on financing of education made by the commission belong to this last category; some belong to the first category that received wide attention, but were followed by little action; many to the second category that were opposed and rejected; and a few to the category which were either simply ignored and/or are of no significance. At the very outset it is important to note that the commission perhaps for the first time in India had emphasised the critical role of education in social and economic development. It was clearly recognised that “In a science-based world, education and research are critical to the entire developmental process of a country, its welfare, progress and security.” It is more emphatically noted that education “*determines* the level of prosperity, welfare and security of the people”. The commission seemed to have been influenced by the “human investment revolution in economic thought” created by Schultz, according to which investment in education leads to human capital formation which in turn contributes to economic growth. The whole approach to educational development in general and financing of education in particular was strongly influenced by its strong conviction on the role of education in development. Not only spatial disparities, the commission also realised the need to ensure equity between several socioeconomic classes in India.

Accordingly, it argued, “we should accord the highest priority to education and allocate the largest proportion of GNP possible to it”. It also warned, “In an age of science, there can be no greater risk than a policy of drift and niggardliness in education”. Second, it was very much concerned with the wide gap between India and the advanced countries, and the need to reduce it in education and through education in development between educationally and economically advanced countries and India on the one hand, and between several states and regions within India.

It resolved, “the gap between India and other rich countries needs to be reduced”. Third, an important aspect is that it was the first commission that was required to and did carefully look into the entire spectrum of education and adopted a comprehensive and holistic approach, rather than looking at education in a segmented and fragmented way at different levels of education. The whole approach of the commission assumes particular importance, as it worked in the backdrop of a politically fluid situation with the demise of a popular prime minister, a major war with a neighbouring country, agricultural drought and the accompanying severe economic problems including high rates of inflation and unemployment – all in a developing economy. Given this, the commission was both courageous in arguing for large investment in education and at the same time pragmatic in using austere and modest parameters and pleading for economy in use of resources, and to recommend regulating the expansion of higher education. Action on many of the recommendations was to be decided at a time of political and economic uncertainty, characterised by the defeat of the ruling Congress Party in many states, currency devaluation, powerful inflationary trends and the plan holiday. An important contribution of the commission is a detailed analysis of financing of education in India. The financial analysis attempted in the report was first of its kind in India. In fact, there were very few studies on economics and financing of education even in other countries at this time.

It made:

- A detailed expenditure analysis – total by levels and objects,
- A detailed source-wise analysis of funds,
- Unit cost analysis, and
- A detailed estimate of resources required for education for the next 20 years in constant prices.

Both the detailed framework provided and the insightful analysis made were of great significance and use for the researchers in economics and financing of education and for educational planners as well. The commission in fact, noted the absence of studies and the critical need for such studies, and recommended support to universities for research in these areas.

ALLOCATION TO EDUCATION: SIX PER CENT OF GNP

Of all, the most important recommendation made by the commission on financing of education refers to allocation of 6 per cent of national income to education. The

commission made a detailed analysis of the past trends in financing education in the post-independence period, estimated the financial requirements of the educational system in India up to 1985-86, and recommended that “if education is to develop adequately, ...the proportion of GNP allocated to education will rise ... to 6.0 per cent in 1985-86”. Of the several recommendations made by the commission, this 6 per cent of GNP is one that was accepted and resolved by the government of India in the National Policy on Education (NPE) 1968 “to increase the investment in education so as to reach a level of expenditure of 6 per cent of the national income as early as possible”.

Since the goal could not be reached, the government of India reiterated in 1986 its commitment to reach the target and stated in the NPE 1968: “It will be ensured that from the Eighth Five-Year Plan onwards it will uniformly exceed to 6 per cent of the national income”. Given the inadequate performance, the goal was to be reiterated again in the NPE 1992. The review committee on National Policy on Education made it clear that 6 per cent of national income should be devoted to education. The long underaccomplishment of the goal led the government to repeatedly reiterate the promise in subsequent years in every Five-Year Plan, in every policy statement, economic survey (s), reports of the ministry of education/human resource development, reports of several committees/commissions on education, and even in the Independence Day speeches of the PM from the ramparts of the Red Fort. Almost all political party manifestos and other agendas also endorsed this recommendation. All this shows some kind of a consensus among all in India towards fulfilling the recommendation of the commission.

However, the most often-cited recommendation is also subject to some controversies. Attempts were made to provide subverted definition and scope to the terms such as national income, educational expenditure, and 6 per cent and to misinterpret the letter and the spirit of the recommendation of the commission, the resolution of NPE 1968 and the NPE 1986, and finally to argue that India already spends about or more than 6 per cent of GNP on education, and we need not worry any more about this target. Significant attempts of this kind include, among others, a paper circulated by the Planning Commission, the national agenda of the BJP and alliance partners (1988), the draft Ninth Five-Year Plan of the Planning Commission and the *Economic Survey* of the ministry of finance. They tried to argue that the 6 per cent of national income, as recommended by the commission, consisted of not just government expenditure, but also all private expenditure including family expenditure on education and private sector expenditure, and even to show that as the goal is already over-achieved, it becomes redundant, and that it does not deserve attention any more.

As Tilak has shown, all these were attempts to misinterpret the facts, to quantitatively under-define the goals, to cover our dismal failures and to boast at our achievements. Tilak has further shown that these attempts have deliberately ignored the fact that the commission had referred mainly to public expenditure, and that the UNESCO and other international statistics that the commission used as a yardstick for comparison also refer to government expenditure only, and the recommendations made by the UNESCO,

UNDP, the Delors Commission, etc, in subsequent years refer to government expenditure alone. Anand Sarup, former education secretary, who was involved in the preparation of the critical review of education in India, titled 'Challenge of Education', and in the formulation of the NPE 1986 made the point clear. In a paper, circulated in a meeting at the Planning Commission, and later published elsewhere, Sarup stated, "Since it is public policy on education that is the crucial determinant of available educational places and opportunities in our country, *it is the Centre and State expenditure on education* that is used for policy planning and implementation. This includes both plan and non-plan outlays". Thus, it is clear that the attempts to redefine and reinterpret the commission's recommendation were to divert public attention from the very need to substantially increase the public allocations to education.

Finally, the controversy seems to have been buried recently with the common minimum programme of the UPA government pledging "to raise public spending in education to at least 6 per cent of the GDP". The second criticism of this recommendation is that this was not based on any sound basis and hence no sanctity needs to be attached to this recommendation. A careful look at the report shows that such a criticism is not tenable. The commission carefully reviewed in detail the trends in the expenditure on education in the past and based on certain reasonable assumptions regarding economic growth and population growth during the next 20-year period, it estimated the magnitude of the resources that should be available for educational development in India in the next 20 years.

It is a detailed analysis of estimating the requirements of the system. It noted, "the proportion of national income devoted to education in India is small in comparison with that in educationally advanced countries of the world". It compared the estimate of requirement with the corresponding figures of some specific countries, available in the UNESCO statistics: "Japan and the USA and the USSR are spending considerably more than 6 per cent of GNP on education"; and they spent no more than a small fraction of their GNP on education at the beginning of the century. The commission also felt that these countries might be spending about 10 per cent of GNP by 1986, and in fact more than 10 per cent, if comprehensive disarmament takes place. It further noted that "the absolute amount per capita spent by us on education is about one-hundred of that spent by a highly industrialised country like the USA". Methodological, including conceptual and definitional aspects of educational expenditure and the details of the analysis and the targets of the commission are unambiguously clear.

The rationale provided for its recommendation was also sound and it also gave enough time to the government for reaching the goal, providing a 20-year period. Some also found that the target of the 6 per cent of GNP was an ambitious one. But the commission felt that normally expenditure on education should grow at a rate of growth double to the rate of economic growth in the early stages of educational development. However, in a sense, the commission's recommendation does not have much sanctity on its own, as the estimate was made long ago and the requirement of the education system, based

on somewhat austere estimates of growth in enrolments, per student expenditure and other parameters. Nevertheless, it assumes importance mainly as the goal has remained unaccomplished so far.

The commission observed that taking into consideration changing circumstances, “the estimates will have to be continually revised”. The only valid criticism of the recommendation could be that the estimate was based on somewhat austere parameters, such as high pupil-teacher ratio at primary level, a smaller proportion of total expenditure on school education for construction of buildings and other items of capital expenditure, no provision of free uniforms, free stationery, free school meals, and health services in free and compulsory education, and so on. In case of higher education, the 1965-66 unit costs were used in estimating resource requirements and it recommended less expensive part-time and correspondence courses for about 30 per cent of the students. Essentially due to resource-scarcity, it recommended carefully planned expansion of higher education on the basis of manpower needs, which meant lower rate of expansion of higher education and even reduction in the total enrolment in higher education and cut in the number of places to be provided in fulltime education, and to economise the expenditure in terms of physical and financial investments without affecting standards. Therefore, any fresh estimate of the resources requirements may put the figure much above 6 per cent of GNP.

The main recommendation on allocating 6 per cent of GNP to education has been endorsed later by several other international organisations. The Delors Commission has clearly argued: “Increasing public spending on education, in place of expenditure under other budget heads, should be regarded as a necessity everywhere, and especially in developing countries, since it is a vital investment for the future. *As a rule of thumb, not less than 6 per cent of GNP should be devoted to education*”. UNESCO and UNDP also favoured it, as a desirable level for the developing countries. Despite the wide acceptance of the recommendation and despite making it a part of the NPE in 1968 and in 1986, which were approved by Parliament, the implementation has been very tardy.

The proportion of GNP spent on education was nearly trebled from 0.6 per cent in 1951-52 to 1.7 per cent in 1967-68. The slope of the line of increase was relatively reduced during the post-1968 period: the proportion was nearly doubled from 1.7 per cent in 1968-69 to 3.5 per cent in 1985-86; it was further flattened, as the corresponding proportion increased from 3.4 per cent in 1986-87 to 3.8 per cent in 2004-05. Though one does not expect a steep increase in this ratio for a long period, the fact is, the rate of increase has been reduced during the post-commission period, and was further reduced during the post-1986 period. As Shah commented in this regard, “the more unfortunate and disturbing long-term trend in this regard is the slackening of government effort to mobilise required resources during the period of high economic growth compared to that of low economic growth”. An analysis of such trends in India and in other countries led Tilak to conclude that the percentage of national income a nation allocates to education is not determined by the level of economic development, but by other factors,

the most important being political will. The proportion of national income allocated to education in India crossed 4 per cent in the early 1990s, but the level could not be maintained. Thus the major recommendation is one that belongs to the category of “Recommendations approved and received wide attention, but not implemented”. After all, no detailed financial plan of reaching the goal was ever thought of. Tilak has shown that if the past trends continue and no significant efforts are made in this direction, the situation might worsen and the goal remain elusive for a long period to come.

Interestingly, one may also note that the relative importance given to education in the Five-Year Plans has also declined during the post-commission period. While during the first three Five-Year Plans, on average about 7 per cent of the total Five-Year Plan expenditure was spent on education, the corresponding proportion declined to 5 per cent in the Fourth Five-Year Plan, the very First Plan after the report was submitted and the NPE 1968 was formulated. It further declined to 3.3 per cent in the Fifth Five-Year Plan and again down to 2.7 per cent in the Sixth Five-Year Plan. While there may be several factors, such as the war, drought and inflation that led to this trend, it is clear that the commission’s strong recommendation that “the efforts to increase allocations to education should be intensified” could not be given serious attention. It is only after the NPE 1986 was approved by Parliament, that this trend was reversed. The allocation in the Ninth, and probably in the Tenth Plan is still much below the allocation made in the very First Five-Year Plan!

Allocation to Different Levels of Education

The commission not only estimated the requirements of resources for the education sector as a whole, it also recommended “the best” pattern of intra-sectoral allocation of resources in education, i.e., allocation of resources between different levels of education. At the outset, it should be noted that the commission has acknowledged the interdependence of various levels of education and had adopted a balanced and holistic approach to educational development. By referring to the Japanese experience, the commission has indirectly advocated some kind of sequencing in the pattern of allocation of resources between different levels of education – first primary education, then secondary and then higher education.

Looking at the historical trends in India as they developed over the years, particularly during the post-independence period and given the then existing levels of development of education at various levels, and the path of development for the next 20 years, the commission felt that from 1965 to 1975, the relative emphasis should be on a larger expenditure at the school stage; during the decade of 1975 to 1985, emphasis will be on universal elementary education, vocationalisation of secondary stage, etc., “After 1985, there will be increased emphasis on the development of higher education and research”. It further added, “As societies become industrialised, the total expenditure on education begins to grow and an increasingly larger part of it comes to be devoted to higher education and research”.

But it appears that the pattern of intra-sectoral allocation followed was exactly the opposite: during the immediate postcommission period, the importance given to elementary education in the total Five-Year Plan expenditure on education either remained stable at around 30 per cent, or declined marginally, and the share of higher education increased from 15 per cent in the Third Plan to 25 per cent in the Fourth Plan. When the pressures on secondary and higher education were increasing in terms of increased demand, the allocation to higher education was drastically reduced from the Seventh Five-Year Plan onwards. Further, while the recommendation seems to be logical and hence appealing, it is nowadays being increasingly realised that the traditional sequencing of first primary education, then secondary education and then higher education may not work any more; higher education cannot wait until primary and secondary education becomes completely universal or well expanded. In a sense, the commission recognised this when it emphasised the importance of universal primary education along with laying special stress on improvement of quality in higher education and research, and recommended allocation of resources for various aspects of higher education and research, as described later.

In the long run, it favoured equal distribution of resources, as in 1965-66, approximately one-third to the first level of education, one-third to secondary and the remaining onethird to higher education. An important point worth noting is, that the commission recognised the importance of all levels of education, particularly primary education on the one side and the higher education and research on the other end, and did not pose one level of education against another in recommending allocation of resources. Clearly it did not favour increase in expenditure on primary education by cutting down expenditure on higher education and vice versa. It stated, "We realise the need for the development of higher education and the allocation of more resources to it. But it would not be proper to cut down for this purpose the expenditure on primary education".

It repeatedly stressed that "the provision of universal primary education is vital on grounds of social justice and to help the process of transformation of the national economy. Again, development of higher education and research is central to the entire developmental programme; and without an adequate provision for higher education there will be no adequate supply of competent teachers for primary and secondary education. What we want is a *balanced growth of education*". This is exactly what Surendranath Banerjee, stated in his presidential address to the Congress in Poona in 1895, "We are not in favour of higher education versus primary education.

We are in favour of all education, high and low. They act and react upon each other. They are part and parcel of a common and indissoluble system." Again, the government seemed to have ignored this wise dictum altogether and often juxtaposed one level against another in the allocation of resources in the Five-Year Plans, saying that the government can finance either elementary education, or higher education, but not both, and thereby adopted different and even contradictory, approaches to development of

different levels of education and their financing. Basically instead of having a holistic and integrated approach, the government has adopted a fragmented approach, looking at the different levels of education as if they compete with each other for resources.

Inter-Functional Allocation: Items of Priority

The commission recommended a significant raise in teachers' salaries, and also "non-teacher" items and their costs. The small proportion of "non-teacher" costs was regarded as the main reason "why our primary schools are so dull and drab". The commission recommended that these costs should be 20 per cent of the "teacher" costs at primary level; and one-third at the secondary level. It recommended that at least 2.5 per cent of the total expenditure on education should be devoted to buildings and other items of capital nature including major equipment in school education and 20-25 per cent in case of higher education. It also recommended providing at least 4 per cent of the total expenditure on school education for direction and inspection.

It placed great faith in teachers in educational reform and in the transformation of the society through them and emphasised the need to raise the economic status of the teachers. It felt that salary of the primary schoolteacher should be comparable to that of a public servant; it should be three-four times the per capita GNP. In case of teachers in higher education, it recommended a national salary structure. University teachers should receive the equivalent of what senior officers in government service get. The government acted upon this recommendation rather promptly for various reasons. Teacher's salaries have been upgraded, though not to the extent suggested by the commission. Important items of expenditure in education include teachers' salaries, recurring expenditure on "non-teacher" salary items, and capital expenditure for construction of buildings and purchase of major equipment. In case of school education, the commission considered the need to provide schools within reach of the children and evolved reasonable criteria like maximum distance a child can travel to reach a school. Though the commission recommended that an amount equivalent to 20 per cent of the teachers' salaries should be allocated to non-teacher costs, it did not seem to have paid sufficient attention to the provision of facilities within the schools, though it recognised that children drop out of schools as the schools are not able to attract them enough.

In fact, the commission felt poverty of the parents was as a major reason for children dropping out of schools. Therefore, it recommended continuation and setting up of new single-teacher schools, part-time primary education, etc, and did not feel the need to improve the school environment substantially. If at all this was recognised, it was left to district boards and municipalities to provide for infrastructure facilities; or they were simply taken as understood. As a result, many single-teacher schools and schools with poor infrastructure facilities continued to grow even during the post-commission period. Consequently, the deficit in expenditure on these items went on increasing, necessitating the countrywide launch of the massively expensive operation blackboard programme, as recommended in the NPE 1986. Among the several items, the commission focused

on two major items, one is scholarships for students and the other, improvement of quality in higher education and research.

Scholarships

One of the important items of expenditure that the commission paid serious attention to was scholarships for students – as a mechanism of searching and nurturing talent, and as a mechanism of equalisation of educational opportunities. The commission favoured expansion of scholarship programme – specifically expansion of the programme of national scholarships and expansion of the programmes of scholarships for the backward classes. It has recommended clear quantitative norms on the percentage of students to receive scholarships and the amounts as well. It felt that 2.5 per cent of the students at primary level should get scholarships.

Scholarships should be provided to at least 5 per cent of the students at secondary level, and 30 per cent of the students enrolled in vocational education. In higher education, the commission recommended that 25 per cent of the students in undergraduate courses in arts and commerce, 50 per cent of the students in undergraduate courses in science and professional courses and 50 per cent of the students enrolled in postgraduate courses should be provided with scholarships. In addition, the commission recommended increase in the number of national scholarships to about ten per cent of the students, and a wider coverage of university scholarships. Though the amounts may have to vary, given the change in the value of money, the proportion of students to receive scholarships, suggested by the commission seem to be still relevant. Presently only an insignificantly small fraction of students receive scholarships.

The purpose of the scholarship programme, as envisaged by the commission was to search and nurture talent and also to promote equity in the system. On both counts there is need to expand the present scholarship programme. In higher education, though the commission did not find any advantage of loan scholarships over that of outright scholarships, it recommended a liberal programme of loan scholarships to supplement the national and university scholarships. But it is important to note that “since an exclusive programme of loan scholarships is non-egalitarian”, it recommended loan scholarship programme to supplement a massively expanded programme of outright scholarships. A national loan scholarship scheme was launched in the subsequent years on the lines suggested. The commission was aware of the innumerable problems loan scholarships might create in recoveries and also the hardships they cause students. The problem of recovery was later found to be so severe that the programme was to be abandoned in the late 1980s and was to be replaced by a normal student loan programme in the early 1990s, operated by the commercial banks.

Higher Education and Research

The commission has laid special emphasis on higher education and research, as it strongly believed that it is higher education and research that will contribute to economic

development, and to bridge the gap between the rich countries and India. Specifically, it suggested larger allocations to the UGC for a few special programmes that will promote quality and excellence in higher education and research. For example, it recommended creation of new centres/schools in universities, and extra financial support to some specific activities.

The important ones that received its attention include:

- Creation of centres of advanced study and major universities,
- Creation of schools of education in a few selected universities,
- Promotion of postgraduate education and research,
- Provision of maintenance grant to state universities,
- Establishment of central testing organisation,
- Development of literature in modern Indian languages,
- Development of agricultural, engineering and medical education, and
- Promotion of educational research on all sectors of education.

The commission also recommended that UGC and state governments share the responsibility of providing development grants to universities. Some of these recommendations were followed up, though there might yet be a lot to do. The commission's recommendations that UGC provide maintenance grants to state universities, and that both UGC and state governments provide development grants to state universities did not receive any attention. UGC continues to provide only development grants to state universities and state governments tend to limit their grants to maintenance purposes.

Role of the State and Other Sources of Funds

The commission was emphatic that most of the responsibility for the support of education should be on government funds and not on the private sector. It rightly predicted and favoured, a big fall in the total revenue from fees. This was because it rightly stressed the need to provide free and compulsory education – free education up to grade X and provision of freestudentships in higher secondary and higher education, and expansion of scholarships at all levels of education. It also predicted that income from other sources, would not rise much and that local bodies would not be able to provide more than a very small percentage of the total expenditure, even after they have made the best effort to raise their contribution. As a result, “the funds of the central and state governments would have to bear about 90 per cent of the total educational expenditure”. Over the years we do notice that these predictions came true as far as the decline in the relative share of fees and other sources is concerned. The relative shares of fees, local bodies and “other” sources declined and the relative share of the government has increased in total expenditure on education. However, the latter has not reached the level of 90 per cent as recommended by the commission.

Though in case of school education the share of the government seems to be around or above 90 per cent, this is not the case in higher education. In recent years, it appears,

the trends are getting further disturbed, with the decline in the relative share of the government and a steep increase in the share of student fees, particularly in higher education. The commission went further and pleaded for a larger role for the union government in financing education. While the centre and states should meet 90 per cent of the total expenditure on education, the central government should assume primary responsibility and the states were to have the “residual responsibility to finance education”. Though elementary education was in the state list, the commission favoured larger role of the central government in funding elementary education. As Naik reiterated, drawing from the report, “it will not be possible for any state government to raise all the resources required for a programme of universal elementary education. It is, therefore, necessary to introduce a central grant earmarked for elementary education on the basis of equalisation.”

Though the commission argued for a larger responsibility of the central government in financing education, it was not in favour of changing the constitutional provisions regarding the role of centre and states in education. It clearly favoured continuation of education in the state list in the Constitution and felt that “there is plenty of scope, within the present constitutional arrangement, to evolve a workable centre-state partnership in education and this has not been exploited to the full”. The commission took a holistic view on every aspect of education, rather than looking at different levels of education in different compartments. It stated this clearly in the context of centre-state relations in education, “We are not in favour of fragmenting education and putting one part in the concurrent and the other in the state list; education should, under any circumstances, be treated as a whole”.

Contrary to what the commission suggested, education was made a concurrent subject with the 42nd amendment to the Constitution of India in 1976, though one fails to note any significant increase in the role of the centre in financing education in the following years. It was only since 1986 after the formulation of the NPE, the centre began assuming increasing role in financing of elementary education and contrary to the constitutional provisions there has been a diminution in the relative role of the centre in higher education in the areas of policy formulation, planning and financing.

Centrally Sponsored Sector

The larger responsibility of the central government should take, according to the commission, the form of expansion of the central and centrally sponsored sectors. The commission advocated that for any programme to be included in the central and in the centrally sponsored sectors, it should be of crucial importance and national in character. Programmes which need the adoption of a common policy in all parts of the country should preferably be included in the centrally sponsored sector. The commission also advocated dividing the total funds available in the centrally sponsored sector into two parts: about half of them being allocated to national programmes, and the other half should be made available to the states on some principle of equality. The states should

be free to use the later kind of funds, with the approval of the union government, for any scheme which is significant and urgent in their local situations. It made yet another important recommendation: central assistance for centrally sponsored schemes should be non-relapsable and should be available to the states on a five-year basis rather than on the basis of a plan period and for some important schemes in the centrally sponsored sector, the assistance may even be continued for a longer period, say 10 years.

Grants-in-aid to Local Bodies

In a supplementary note, the commission has suggested a detailed mechanism of grants-in-aid to local bodies – district level bodies and municipalities, in such a way that grants-in-aid to local authorities would stimulate local contributions to education. It recommended that “the assistance of the local communities should be fully harnessed for improving the physical facilities in schools”. The state government should provide for 100 per cent teacher grants, block grants per child to meet non-teacher costs, and separate grants for non-recurring expenditure. The commission nevertheless realised that the local bodies may not be able to generate more than a very small amount on their own, despite their best efforts.

The objective of the grants to the local bodies should be to ensure equality in expenditure per student and thereby to equalise educational opportunities. It also suggested that every school should continue to receive the maintenance grant on some egalitarian basis, so that all schools come up to a minimum level of performance, and those schools that do good work should receive additionally a special “jam” grant. Experience shows that the state grants to local bodies have not been able to stimulate generation of resources by the local bodies. The commission was right in predicting that local bodies would not be able to mobilise any significant amount of resources on their own, given the limited resource base, and competing needs of various sectors. Thirdly, there are wide disparities in educational development in general and educational expenditure per capita or per student between several districts and blocks.

Fees and Cost Recovery

The commission had a clear and progressive understanding of the role of fees in education and its implications. It stated, “It is undesirable to regard as a source of revenue. They are the most regressive form of taxation, fall more heavily on the poorer classes of society and act as an anti-egalitarian force”. It also dismissed a “progressive” or a discriminatory fee system based on economic levels of the students/their families as it “would not be administratively feasible and, ... their yield would be almost negligible”. Recognising the constitutional provisions, the commission reaffirmed the importance of providing free and compulsory education of a common school system.

There was no case of levy of any fees at primary stage, though the commission was confronted with arguments in favour of levying fees. After all, it was a constitutional commitment. The commission was not content simply to note that free education meant

only tuition fee-free education and provision of free textbooks, but felt constrained to recommend:

- Abolition of all kinds of fees in primary schools and
- Provision of incentives such as free stationery, uniforms, school meals, etc.

The then prevailing economic conditions seemed to have prevented the commission from making any such recommendation. These compulsions are clear. It however, recommended free education up to grade X and provision of free studentships in higher secondary and higher education. It also recommended all vocational education to be provided free. With respect to other levels of education also, the commission did not actually favour levy of fees; but it could not recommend against fee. In case of secondary education, though it agreed to the suggestion to levy of fees at higher secondary level as a pragmatic solution to the problem of resource scarcity, it did note very clearly that “the levy of fees in secondary schools prevents several children from the poorer classes of society, and particularly girls, from receiving education”. So it argued that in higher secondary and higher education, every attempt should be made to extend free education to cover all needy and deserving students. With respect to higher education, the commission found that the then existing levels of fee contributions were much higher in India than in the educationally advanced and richer counter such as US and UK.

The commission’s forward-looking progressive policies with respect to fees in higher education are worth noting, “We do not advocate the immediate general abolition of fees in higher education, although *this should be the ultimate goal of educational policy*....for the next 10 years, the main effort with regard to fees in higher secondary and university education should be to expand the provision of tuition-free education to cover all the needy and deserving students. To begin with, the proportion of free-studentships should be increased to at least 30 per cent of the total enrolment. We also commend, for general acceptance, policies which have been adopted in some areas to provide tuition-free higher education to underprivileged groups”. Nowadays fees are regarded as one of the most common measures of mobilising finances for education. In fact, student fees are being seen as a major potential source of funds. But confronted with the need to ensure that weaker sections of the society do not get neglected, often a programme consisting of levy of fees along with concessions and exemptions to needy students is proposed. This has been quite common. But the commission did not find merit in such arguments.

The commission observed, “such a system does not have much to commend itself and involves several administrative difficulties”. The commission also felt no need for any cost recovery mechanisms. In fact, in the long run education would become self-financing, not of course as being contemplated nowadays. The commission observed, “in the long run education to some extent is self-financing because the increased incomes generated by a relatively better educated labour force would provide resources for greater allocation to education...additional resource are generated through the process of economic growth”.

This is what Mishan also observed in a similar context, “higher education is an investment and will pay for itself; and will increase the earnings of the beneficiary students and the government will recover its costs through consequent higher tax receipts.” The commission has clearly recognised the significant economic contribution of education, when it observed, “the fact that education tends to augment the flow of national product, though with some time-lag, is of crucial importance”. Having noted that parents were required to incur “very heavy expenditure” on education, it suggested quite a few mechanisms of mitigating household costs on education, such as strengthening of the provision of free textbooks at primary stage, launching of a programme of book-banks at secondary stage and provision of book-grants in higher education.

The evidence on the practice of fees in education in the later period is in quite contrast to what the commission recommended. Fee, including tuition fee, besides many other types of fees, were continued to be charged in government, local body, government aided schools at primary level; fee levels in secondary education have been on the rise; and fees in higher education are going up. The commission also argued against over dependence on private sector in education development.

The commission felt that private sector has a limited and minor role in the national education system. It pleaded for control of private enterprise in education. Again, this suggestion is also not cared for very much by the government and in fact, this was opposed strongly. An unbridled growth of private education at all levels of education has been allowed, with all its ramifications.

Other Norms and Recommendations

A few other norms that were adopted by the commission, which have serious financial implications may be noted as follows: First the pupil-teacher ratios. Though for pragmatic reasons, the commission had adopted higher pupil-teacher ratios in estimating the resource requirements, it desired that to ensure reasonably good quality of education, the pupil-teacher ratio in primary education be 30 and 35 in higher primary schools. In lower secondary education it should be 25 students per every teacher and 20 in higher secondary education. In case of secondary/ vocational education the desirable ratio suggested was one teacher per every 11 students.

The commission suggested such norms for higher education as well – one teacher for every 15 students on average in undergraduate courses and eight students in postgraduate courses. Many of these norms are still relevant in contemporary educational planning. The present norms and the current actual pupil-teacher ratios are much higher than the norms suggested by the commission. Secondly, the commission realised the importance of vocational education and also to reduce pressures on higher education. It recommended a high degree of vocationalisation of education: 20 per cent of the total enrolments at secondary level, 50 per cent at higher secondary level and 30 per cent at higher education need to be in vocational streams.

Further, it argued that 60 per cent of the students in higher education should be enrolled in professional and sciences courses. Though the government often expressed in subsequent years its desire to vocationalise secondary education, the progress is not satisfactory. The reason is, fundamentally vocational education has been looked down upon and planned as second rate, cheap education for the poor. As a result, it suffered from both demand and supply side constraints. Vocational courses were introduced in colleges in the recent years. In all, the targets set by the commission still seem to be elusive to reach even in the near future after 40 years, as no serious attempts were made in this direction. Thirdly, the commission favoured promotion of excellence at all levels of education. Ten per cent of the schools at every level were to be provided with additional resources so that they function at optimal level of quality to become “pace-setting” institutions! This was recommended by the commission as “the highest priority programme”. Similarly, the commission favoured development of five or six “major” universities where “first class postgraduates work and research possible”.

This seemingly elitist idea was not favoured in case of university education, though the recommendation relating to school education was followed-up in the following decades. The scheme of Navodaya vidyalayas recommended in and set up after the NPE 1986, can be regarded as close to the proposal of the commission.

The proposal on universities was not received favourably by many, as it was felt that “better results would be obtained if minimum standards can be maintained in all institutions and special additional assistance, on the basis of proper criteria, given to institutions which show a high level of performance and promise” than from concentrating the efforts and resources on 10 per cent of the institutions. While the proposal of the commission *per se* was not accepted, some universities did receive support for their excellence in education and research, and centres of excellence and inter-university centres were created and supported with extra funds that offer first class postgraduate education and research, meeting to some extent the very objective that the commission had in mind.

There are several prescriptive observations made in the report which are of great significance in the present context. To cite a few, three such observations on financing of education may be noted: The commission rightly noted that “in the early stages of educational development the rate of growth of educational expenditure ought to be approximately twice the rate of growth of national income”. It is important to note that this is a desirable practice. For example, Schultz stated, “during the process of economic modernisation the rate of increase in human capital is higher than that of reproducible physical capital”.

If we follow the dictum and the educational expenditure increases at twice the rate of growth of the economy, reaching the goal of allocating 6 per cent of GDP to education will not be difficult at all; it can be reached very soon, as a committee of the government of India has shown. The commission also recommended analysis of education expenditure by converting the figures into the constant prices. Though official documents

of the government still do not present expenditure on education in constant prices, researchers have begun attempting analysis of education expenditures in real prices, using national income deflators or price indices. The commission stressed the need for efficient utilisation of resources. It repeatedly made the plea to eliminate underutilisation and wastage of resources in education and to introduce measures of economisation.

Concluding Observations

The report was considered a landmark in the history of Indian education. The commission made several important suggestions, which even after 40 years, are still relevant for development of education. They are relevant not just because their implementation is over due; they are relevant today for their intrinsic value, essentially because the commission had looked into the distant future, adopted a visionary approach, built its recommendations on strong empirical evidence and knowledge – national and international, with a strong conviction on the role of education in national development, and above all with a strong commitment to national development. Despite realising the need to be austere due to several reasons, as it worked under the overall atmosphere of austerity, the commission did not compromise on a few vital issues.

For instance, it strongly advocated a national system of education; and it pleaded for universal full-time education to all children of common school type, though development of alternative channels of education was also suggested, as full-time universal education was not immediately possible. It has argued for a free common school system of public education up to the end of grade X and the adoption of neighbourhood school concept at the elementary stage.

It also suggested that all private schools must be brought into the common school system. Third, it has recommended a large expansion of scholarship programme. It has also recommended a significant expansion of the centrally sponsored sector in education. It strongly recommended increase in the allocation of resources to education to the level of 6 per cent of GNP. Unfortunately, while the commission recommended a package of reforms, the government looked at the recommendations as piecemeal suggestions. As Naik observed, “It is thus a tragedy that the recommendations of the one commission which was directed to look *comprehensively* at education were considered mostly in a *piecemeal* fashion”.

While there can be several factors for the inaction of the government, the lack of strong political will seems to be the most important one. As a result, the Indian education system is still characterised with conspicuous failures – in eradicating illiteracy, in universalising elementary education, in vocationalisation of secondary education, in ensuring excellence and high standards in higher education, in allocating adequate resources for education, in improving the financial efficiency of the system, all these failures, along with of course some spectacular achievements in terms of student numbers, in building one of the largest reservoirs of scientific and technical manpower in the world, in “exporting” manpower, etc. The reason for widespread discontentment

with the education sector is obvious. The commission itself warned, “A report which is shelved or does not lead to action is worse than no report because it leads to frustration by arousing hopes that remain unfulfilled”.

MUDALIAR COMMISSION (1952-53)

The Secondary education commission was set up by a resolution dated 23rd September 1952. The commission was inaugurated on 6th October 1952. It submitted its report in June 1953.

Need for Appointment for Sec-1952-53

- No commission or committee appointed before dealt directly with secondary education problem. (The Indian education commission of 1882, University commission of 1902, the sadler commission of 1917 and the university education commission 1948-49 all dealt with certain aspect of secondary education.)
- The secondary education that time had unilinear and predominantly academic character. (There was need for the diversification of the courses to supplement the basic education at the primary level.)
- The secondary education prepares the teachers for the primary level and students for higher education level.

The SEC formulated four aims of education in India keeping in view the needs of democratic India as envisaged in the constitution.

- Development of democratic citizenship.
- Improvement of vocational efficiency.
- Development of personality.
- Development of the qualities for leadership.

Aims of Secondary Education Commission 1952-53

- *Development of democratic citizenship:* (Development of social, moral, intellectual qualities)
 - Clear thinking
 - a. A democrat to be effective must think clearly and be receptive to new ideas.
 - b. He should have the understanding and the intellectual integrity to shift truth from falsehood, facts from propaganda and to reject the dangerous appeal of fanaticism and prejudice.
 - c. He must develop a scientific attitude of mind to think objectively and base his conclusions on tested data.
- *Receptive to new ideas:* He should have an open mind receptive to new ideas. He should neither reject the old because it is old nor accept the new because it is new, but dispassionately examine both and courageously reject whatever arrests the forces of justice and progress.

- Clearness in speech and writing. This is an essential pre-requisite for successful giving is a democracy which is not base on force but on free discussion, persuasion and peaceful exchange of ideas.
- Education is the art of living us a community
 - Individual is a social being.
 - Development of the ability to live with others.
 - Inculcations of values of Co-operation.
 - Development of qualities necessary for living graciously, harmoniously and efficiently with ones fellowmen. Among the qualities which should be cultivated for this purpose are discipline, Co-operation, Social sensitiveness and tolerance.
- *Passion for social justice*: Development of sensitivity to social evils and exploitation which corrupts the grace of life.
- *Development of tolerance*: Essential to preserve the health and existence of democracy. For this it is essential to cultivate us the young generation an openness of mind and largeness of heart which would make them capable of entertaining and blending differences us ideas and behaviour into a harmonious pattern.
- *Development of true patriotism*: True patriotism involves three things,
 - A sincere appreciation of the social and cultural achievements of ones country.
 - A readiness to recognize its weaknesses frankly and to work for their eradication.
 - An earnest resolve to serve it to the best of one's ability, harmonizing and subordinating individual interest to national interest.
- *Improvement of vocation efficiency*:
 - Creating a new attitude that implies an appreciation of the dignity of all work however lowly.
 - Making students realize that self fulfillment and national prosperity are only possible through work in which every one must participate and a conviction that whenever a educated man take any piece of work in hand will try to complete it as efficiently and artistically as their powers permit.
 - Making attempts by all teachers to ensure that such an altitude on the part of students finds expression in every activity of the school.
 - Promotion of technical skills and efficiency at all stages of education so as to provide trained and efficient personnel to work out schemes of industrial and technological advancement.
 - Provision of diversified courses at secondary level such as agriculture, technical, commercial or other practical courses which will train their varied aptitudes.
- *Development of personality*:
 - Releasing the sources of creative energy with students so that they may be able to appreciate their cultural heritage.

- Cultivating rich interest which they can pursue in their leisure and contribute in later life to the development of this heritage.
- *Development of the qualities for leadership:*
 - Education must train our students for discharging their duties efficiently.
 - They must be trained in the art of leading and following others.
 - Our secondary education must train persons who will be able to assume the responsibility of leadership in the social, political, industrial or cultural fields in their own small groups of community or locality.
 - Inculcation of the following values: A deeper and cleaner understanding of social issues and greater technical efficiency.

7

CHAPTER

Educational Technology and Learning Theories

WHAT IS EDUCATIONAL TECHNOLOGY

The reason why the term ET is misconstrued is on account of the changing nature of ET's second component, *viz.*, technology. The basic tenet of ET, *viz.*, using all available resources in a systematic manner to find viable solutions to educational problems, does not change. However, as technologies change and newer ones are brought into service in education, the configurations, structures, and applications of ET will also change. This dynamic and ever-evolving nature of the discipline needs to be understood. Further, given the fact that educational problems are diverse, so are their solutions, ranging from providing resources in the classroom to distance education or using technologies to facilitate communication. These multiple facets of ET make a crucial difference to the way in which the term is interpreted. As the discipline continues to grow, we would like to give a short account of its evolution.

When the term was first coined it referred to “technology in education”, implying the use of a variety of audio-visual aids for teaching purposes. Implicitly relying on the then widely accepted sender–receiver construct, educational writers saw these aids primarily as transmitters of lesson content. As the concept of ET developed, the term “technology of education” came into vogue. This looked at education in a wider sense, and included various aspects such as entry behaviour of the learner, objectives, content analysis, evaluation, etc. By the mid 1970s, ET borrowed the terms “systems approach”

from management studies and “corrective feedback” from cybernetics. This widened the scope of ET as the teaching-learning process was examined in a holistic manner.

The arrival of digital convergent media encouraged interactivity and interconnectivity. This added a new dimension to ET. It gave an impetus to its further development as a discipline. While this field continues to evolve, we are faced with the problem of how to help learners to help themselves in learning in an effective and interactive manner.

As the Focus Group deliberated, we felt that when we talk of the role of ET, we should direct our thoughts to education, and not to any technology and the logistics of getting it set up. The emphasis has to be on a culture of learning rather than on technology per se. Our perspective is that, except in the case of teacher education where ET is taught as a subject, ET is a value addition to quality, relevance, appropriateness, and other such attributes, transforming education by making it dynamic and responsive to the passions that move the learners and arouse their curiosity and desire to learn.

The desire on the part of some people to leave the term ET behind is rooted in the awareness that historically the implementation of ET in India has been centered on machines and the educational software related to them. Such interpretations are easy to understand. More difficult to comprehend are approaches based on processes. Here we speak of the processes adopted by educationists to solve the difficulties that the prevalent system of education has in enhancing learning to meet the desired educational goals.

The universally accepted definition of ET involves processes, methods and techniques, products, resources and technologies organized into workable systems. The recognition of the need for a multilevel organisation of a classroom, for instance, along with the designing of an appropriate programme and its implementation, become as much an exercise in ET as the use of audio-visual aids or the information superhighway.

There is also a prevalent belief that modern technologies are better than older ones. Certainly, they offer many exciting possibilities, but both old and new serve different purposes of teaching learning, and are probably equally relevant in different given contexts.

The key phrases in ET are appropriate technology, that is, appropriate to the task in hand for meeting specific educational objectives, and the organization of all available resources into a workable system, which is checked again and again to ensure that it is appropriate and changing it where it is not working. Formative evaluation and summative evaluation are essential parts of ET. In applying the discipline of ET to the field of education, it is imperative that the media choice must relate to instructional design as well as to what is available and eminently usable.

The same is true of methods and techniques. For instance, the memorization of facts as a learning strategy still has a role to play. The ready availability of multiplication tables does speed up computation. It is the overgeneralization and unthinking application in inappropriate situations that has brought disrepute to memorization as a form of learning. Charts, graphs, textual materials, experimental kits, projected electronic aids, audio materials, computers, films, videos, the Internet, etc. can usefully serve the

purposes of education in their own special ways and together they can make learning an enriching experience. ET could be defined in simple terms as the efficient organisation of any learning system, adapting or adopting methods, processes, and products to serve identified educational goals.

This would involve:

- Systematic identification of the goals of education, taking into account nationwide needs, the system capabilities, and the learners' needs and potential.
- Recognition of the diversity of learners' needs, the contexts in which learning will take place, and the range of provisions needed for them.
- Recognition of not only the immediate needs of children but also their future needs in relation to the society for which we are preparing them.
- Designing, providing for, and enabling appropriate teaching-learning systems that could realise the identified goals.
- Developing a range of support systems and training, creating the enabling systemic conditions/materials, reaching these to the school system, and training teachers and students to use them.
- Research into existing and new techniques, strategies and technologies for solving problems of education, enabling judicious and appropriate application of technology.
- Appreciation of the role of ET as an agent of change in the classroom, influencing the teacher and the teaching-learning process, and its role in systemic issues like reach, equity, and quality.

HISTORICAL PERSPECTIVE

The basic elements of ET have always been present in any effective teaching-learning system, though it was not called ET. For example, the old gurukul system in India stressed individualized instruction and emphasised learning, which are also features of ET. The training programme in the gurukul was devised to suit the needs and abilities of the pupil. It was a one-to-one system, but it did not mean rote learning or following the guru blindly. A Sanskrit maxim states that the fervent wish of a guru should be that his disciple would better him. Moreover, the teaching programme was devised not only to suit the needs and potential of the pupil but also to suit the societal needs as expressed in identified learning goals.

One such goal, for example, in the days when writing was unknown was to maintain the oral tradition of the Vedas. Hence, it was important to preserve intact the accent and pronunciation of words. The method adopted was the memorisation of text, and the technique for doing so was using different pathas or sequences, which could be recited against each other to check any deviations.

The factory system of education with its centrally controlled curricula/ text books/ school hours/ holidays, etc. came with the British rule. Then, as Gandhiji ruefully noted, every alternative school system was abolished. This colonial legacy and control apparatus

continue to plague the educational system even today, almost six decades after India gained political independence. As a result, we are now in the sorry situation so well described in the Yash Pal Committee Report “Learning without Burden”.

EFFORTS TO MOBILIZE ET, LARGE AND SMALL

A number of groups doing innovative work in the field of ET in India and abroad made presentations of their work and experiences before the Focus Group.

INITIATIVES IN THE VOLUNTARY SECTOR

Several educators from Gandhiji onwards have sought to make education relevant and liberating by introducing alternative and experimental systems of learning. They have also tried to provide equity and quality in education by directing their efforts towards educating the marginalized child and providing the needed skills and knowledge in stimulating ways.

Gijubhai Badheka and Tarabai Modak worked in the sphere of early childhood education. The Tilonia programme in Rajasthan; the Hoshangabad project of Kishore Bharati/ Eklavya in Madhya Pradesh; Gram Mangal in western Maharashtra; the Bhandup project and the Avehi-Abacus project in Mumbai’s municipal school system—these are a few examples of such efforts undertaken and sustained at various times in different parts of the country.

However, even though the Tilonia and Hoshangabad models showed good results in the wider mode too, they were stopped because there was no political will to sustain them. The Bhandup project suffered neglect because the new municipal bureaucratic regime could not be bothered to continue it. The Bhandup programme continues to work well in those districts where the district education officers are supportive. It is sustained by teachers who have seen how children come alive and learn enthusiastically with multilevel teaching. This programme lets children move at their own pace, provides learning materials made at low or no cost, and encourages peer learning.

None of these projects are known as ET programmes of enhanced learning, but each of them displays most of the features of ET as described earlier, *viz.*, specific educational objectives, appropriate methods and materials, and emphasis on learning rather than on teaching. Indeed, the Centre for Educational Technology (CET) had funded the Tilonia project during its experimental stage as part of the Centre’s ET programme. Many teacher-training institutions also offer courses in different aspects of ET such as programmed instruction, individualized learning, microteaching, etc.

EFFORTS INITIATED BY THE GOVERNMENT

Mass media like radio and television have been used in a sporadic fashion for education for a long time. One of the earliest systematic and large-scale efforts in India to run an educational television channel was SITE (Satellite Instructional Television Experiment) in 1975–76, which was beamed to six states, and is well documented. Many innovations

were undertaken in SITE in both devising and deploying suitable hardware and making original software. This software was made by many agencies other than Doordarshan, which until then had a monopoly on video production and broadcasting in the country. In this connection, the work done by AIR in its Vigyan Vidhi programmes to disseminate scientific information to students and teachers, or state and AIR efforts in the project mode in Maharashtra and Rajasthan, have been prominent. However, the supportive structure that these programmes needed could not be maintained for long. The first television inputs in education did not have any worthwhile support systems. In 1970, the Ministry of Education took up a scheme of ET.

Both CET and DECU (Development and Educational Communications Unit) of the Space Application Centre (SAC), Ahmedabad conducted formative and summative research in respect of the programmes that they had carried out. CET launched a multimedia programme of in-service teacher training, which was highly successful.

There was excellent coordination between the state units and CET for the first few years. Every programme of CET was directed towards solving an educational problem or enhancing learning to achieve well-defined educational goals. The programmes were related to the education of marginalized communities. However, this vision was lost some time in the early 1980s. With new directives from the Ministry of Human Resource Development, the close coordination between the state units and the central unit virtually came to an end.

The launch of the Indian National Satellite (INSAT) in 1980, and its availability for educational purposes, led the Ministry of Education to take over the production of educational television programmes for transmission via Doordarshan. INSAT for Education was conceived as a tripartite project, and was supported by UNDP, UNESCO, and GOI.

Studios with adequate hardware for production were installed. CIET was charged with the task of undertaking educational television and radio production, conducting training and research, and performing as a central coordination agency for all production and utilization efforts. Along with SIETs, CIET aimed at the utilisation of mass communications in a major way to meet various educational objectives. These projects provided examples of the use of modern methods of media planning and application. The application of media in education in an Indian situation must take into account the availability of software and access to hardware. CIET conducted experiments in teleconferencing. The project seems to have reached sub optimal achievement levels as after the initial planning, support systems were found to be lacking; the educational system has failed to appreciate the usefulness of the media programmes in their educational plans.

Television and radio sets were supplied to schools over many years. AIR and Doordarshan were chosen as the carriers for the broadcasts. As production and broadcasts began, equipment and personnel were put in place. So far as CIET and the SIETs were concerned, the production of video and audio programmes became their main work.

Narrow field studies showed encouraging results, but the system failed to take root. Large-scale evaluations show gross underutilization. Studies have shown that a link between the broadcaster and the classroom teacher has failed to develop. The audio and video programmes do not indicate any definite patterns of suitability for supporting classroom transactions or supplementing them, either for particular age groups or for particular subjects.

The broadcasting organizations have their own priorities, allotting timings not always suited to the audiences concerned, often canceling these when other programming takes precedence. The institutions in government related to ET have been ailing for some time. Three committees have pointed out the problems that these institutions have been facing, but no effective action has been taken to revitalize them.

Under another scheme of the Ministry that was entirely equipment driven, between 1986 and 1990, the Ministry distributed 2,28,118 radio-cum-cassette players (RCCPs) and 31,129 colour television sets to schools at the cost of several crores of rupees. However, as a study conducted by Prof. M. Mukhopadhyay shows, this step did not yield the desired results, as it did not go beyond providing the equipment.

COMPUTERS IN EDUCATION

Indian experiments in taking computers to schools involved the participation of a large number of institutions for tasks such as the supply of hardware and software, the development of Computer Assisted Learning (CAL) packages, and the training of teachers.

It covered 42 Resource Centres and 2,582 schools. It made use of microcomputers provided by the BBC. The evaluation of the project by SAC revealed the need for greater interaction between resource centers and project schools, the need to reduce the time gap between the training of teachers, the installation of systems, and the initiation of activities in schools, the imparting of adequate hands-on experience to teachers and students, and the provision of computer literacy programmes in the timetable. The project had only a limited success, and has been described at best as a “spectator sport”.

A revised CLASS project during 1993–2004 saw the introduction of PC machines in keeping with broad global trends. Subsequently, the government initiated the CLASS 2000 programme with the aim of providing computer literacy in 10,000 schools, computer-assisted learning in 1,000 schools, and computer-based learning in 100 schools. These 100 schools were called smart schools, and were designed to be agents of change seeking to promote the extensive use of computers in the teaching-learning process. This, too, has not yielded the expected results. In the words of Prof. Utpal Mallik, “Ambiguity of purpose, tentative policies and faltering practices marked the major computing initiatives in India during the last two decades... Schools are using IT as an add-on, not as an integral part of a new pedagogy.” Though all these interventions did make some impact, where the schools and teachers went the extra mile to avail of the facilities provided using their own ingenuity, many of these schemes have been half-

hearted attempts even at the conceptual level. Computer literacy is not so much about knowing the technical jargon, but rather learning to use computers in a meaningful way, that is, meaningful to children. Two programmes illustrate this fact quite well. The first project—which the media has dubbed the Hole in the Wall—uses the method of Minimal Invasive Education (MIE). The government sponsored programmes lacked not only conceptual clarity but there were also no provisions for a number of other essential aspects, such as students and teachers having easy access to computers, problems of scalability, timetables, etc. None of these issues was discussed, nor were the relevant solutions worked out.

Without such clarity and preparation, and lacking the machinery to make mid-course corrections, these programmes failed to bring about the desired changes; some were given up half way. Given this void, many international corporations, and Indian companies as well, have entered the arena in recent years.

EMERGING LESSONS FROM PAST EXPERIENCES

The lessons emerging from the analysis of successes and failures are immense. Equipment-driven programmes do not work. In the macro mode, in which broadcasts are used, unless there is an interactive mechanism—such as the teacher-monitor figure in the in-service multimedia teacher-training programmes of SITE or intervention by teachers, equipped with programme notes or textual materials—the success of the intervention cannot be assured. Radio and Television broadcasts are effective when they cover something that the class teacher cannot show or otherwise do or demonstrate in a classroom. The broadcasts have to be planned to suit the instructional design of an ongoing educational programme that is designed to bring about change, with both the broadcaster and the teacher learning from each other. Both CIET and the SIETs have made good enrichment programmes that show what the class teacher would usually not be able to show in the classroom; to that end, they serve some purpose.

But any disseminative arrangement, however good it may be, provides only limited gains because it only offers partial solutions to problems of an educational nature. Synchronous broadcasts also have some inbuilt weaknesses. Often the planning of programmes is too thin. What could be the contribution of one programme a week in one subject and that, too, when its objectives and functions are no different from those of classroom teaching? With smaller interventions, things have usually worked fine during the peak periods, but often problems arise with respect to sustainability. Projects become person-centric. They become dependent on the central authority that runs the schools. *Good programmes once held up as exemplars are stifled because they run foul of the authorities, which have the power to close them down. When the colonial control mindset starts operating, innovations have no room to survive.*

An important lesson is that, whether one uses low technology or high technology, or micro or macro models, applying ET solutions to educational problems is a matter of a total fix. In other words, it is necessary to think of a total system. An important point to

remember is that to ensure the success of even good television and radio programmes, it is essential to meet certain necessary conditions. First, the programme must meet the needs of students; it must be such that teachers could use it as an aid; the broadcaster must schedule it during a convenient time slot. The school has to fit this broadcast into the timetable.

The production agency has to make appropriate programming available; post -telecast research has to support the production team and feed the data directly into the programme content. All these conditions will have to be met simultaneously, all the time, for such programming to achieve what its makers have planned. A piecemeal attempt will degenerate into tinkering, and the patches will begin to show much before the rust seeps out.

Some of these schemes, especially at the macro level, have been half-hearted attempts even at the conceptual level, and have failed in situations where the sharing of responsibilities by different agencies and the need to ensure that everything works in tandem are required. Constant evaluation and checking of systems is necessary because if mid-course corrections become necessary, they can be carried out in time.

India has now reached a stage where education must take recourse to the discipline of ET in organizing education because the challenges that the country faces today are far graver than ever before. Bringing about a sea change in education will be possible only if one applies ET in micro and macro models, using any technology—whether old or new— that is appropriate.

There is no other option if one's goals are not to leave any child behind and to provide learners with education that is high both in equity and quality. ET has the potential to make the educational system flexible not only in the teaching of subjects but also in the conduct of examinations, in research, in systemic reforms, and in teacher education. It will also provide a buffer against the homogenizing onslaught of the global market economy on our culture and values by making the peripheries much stronger than before and providing discerning media awareness. Let us consider the challenges that we face.

CHALLENGES IN EDUCATION

“Learning without Burden”, the report of the Yash Pal Committee of 1993, has extensively reported on the ills of the present education system. Briefly, it has shown how the education system has become highly centralized, examination driven, joyless, impersonal, and utterly irrelevant to the child's world.

The centralization deprives teachers of the freedom to organize teaching learning and meaningfully participate in the preparation of syllabi or textbooks. This in itself is bad enough, but now in addition to what is happening in India, it has become necessary to face the challenges of a rapidly changing world in the twenty-first century. As the world shrinks on account of developments in science and technology, these changes affect Indian society increasingly in many different ways. The world today is a global

village, and this represents unprecedented challenges for Indian Education. No society can live in isolation. This reality has a bearing on social processes in both the world as well as in India. This process has been going on for some time, but in the last ten years the pace at which the world is changing is becoming greatly accelerated.

KNOWLEDGE EXPLOSION

A decade ago, the knowledge base of humanity used to double every ten to twelve years; now it doubles every two to three years. Does this mean that there should be an even bigger load of material for children to learn? Or is there a way in which knowledge need not be served in different packages in schools? Is it possible to weave knowledge in a multi-disciplinary fashion into questions that are of relevance and interest to the learners? Can we provide sources for learning that are more open than the traditional ones?

TECHNOLOGICAL EXPLOSION, A DOUBLE-EDGED SWORD

A very important factor impelling change has been the technological explosion, particularly in the area of ICT (Information and Communication Technologies) . Such technologies are double-edged swords. They allow people to contact one another and exchange ideas very easily in order to create communities built around common interests and common causes. They also make it possible for global corporations to move billions of dollars around the world with the click of a button.

This gives them tremendous power over local and national economies, especially of Third World countries like India. Democratically elected national governments of Third World countries, even big ones like India, are no longer as sovereign as they were twenty years ago. With the profit motive reigning supreme, global corporations see human beings as consumers of their products and not as citizens with inalienable civic rights and duties. A byproduct of this phenomenon has been the loss of diversity in the biosphere, in cultural mores, and the ways in which we live. The world is becoming increasingly homogenized.

HOMOGENIZATION OF THE WORLD

The corporate world empire uses the tools of aggressive advertising and marketing campaigns and a centrally controlled media to turn around public opinion to support its agenda by means of what Noam Chomsky calls the “manufacturing of consent”. As mass production leading to profits means mass consumption, global corporations like to disinvest the world societies of diversities and pluralities, something that India has always cherished and deeply valued. The more homogenized the communities of the world become, the more effective their media and marketing reach can be. Multinational giants, therefore, pose a threat to diversities of both the environment and culture. The consumerism they help to promote a life style in which wasteful ways that the Earth cannot support become the needs created by advertising. The scenario of competition

leads to aggression and violence, which strip human beings of the essence of humanity, living in cooperation and harmony with others and with their surroundings.

This attack of the global corporate empire needs to be met by teaching young people democratic values and equipping them with a sense of discernment so that they can choose the right way to build a better world. Open sources, which are now increasingly becoming available, can help break corporate monopolies.

There is no end to the thirst of the corporate empire for power and control; it tries to use every means, fair and unfair to extend its reach in various ways. The latest thrust in this direction from the DPEP (District Primary Education Programme) sponsored by the World Bank has now become clear. Education has now fallen under the umbrella of the General Agreement on Trade in Services (GATS) . GATS and the World Trade Organisation (WTO) , education is a service, a marketable commodity. They have pushed the Indian government to accept this premise, and India is well on its way to participating in the WTO-controlled globalization of education. The only way to preserve our secular democratic values and way of life is to ensure that our people are enabled to become stronger so that they can fight back this menace that will surely destroy the only habitat of life that we know, Earth.

Already life has become worse on account of the crisis of potable water and environmental degradation. There are several ways in which people are fighting back all over the world, and also in India. Two presentations made before the Focus Group were electrifying. One was from Pastapur, where multiply disadvantaged women from 70 villages, organized in Sanghams, have linked food security with many social and educational concerns, and have used ET as a weapon to attain equity, preserve human dignity, and achieve development.. The second presentation was about the work being done by a group of engineers from IIT Madras to provide technology that is both cheap and suited to Indian conditions to enable people to access information and better their lot in life.

These new technologies used in the right way can empower ordinary people, and ET can thus become a tool in this struggle that is taking place all around us. The Internet and the Web provide alternative sources of information and connectivity across the world to people who share similar interests and concerns. This could help in creating several power centres in the peripheries that would correct the imbalance of central rule in every sphere, including education. It would then become possible to look at knowledge not as something that comes from a central source, but that emanates from all around us.

POPULATION EXPLOSION

Another important factor behind change is the population explosion. Globally, population has increased several folds, but the strange fact is that while the population in the developed world is declining, it is increasing at a phenomenal rate in the developing world. A special demographic feature in India is that the country has one of the world's

youngest populations. Furthermore, in the next decade, it is expected that over half of these young people will be below twenty years of age. This number itself will pose an unprecedented challenge, and we have nowhere to look for a solution, as no country in the history of the world has ever had to face this problem. Given our track record in bringing children to school, keeping them there, and attempting to provide them with a decent education, conventional solutions will not suffice. It is a major challenge to provide large-scale access to all parts of children, including the 10 per cent or so who are disabled, especially when education must also be equitable and of good quality.

SCARCITY OF RESOURCES

In the absence of assured access to alternative support materials—libraries, teaching aids, audio-visual material—textbooks have come to play a dominant role in the teaching-learning process. Textbooks combined with examinations, which test what has been memorized from textbooks, have exerted a stranglehold over the educational system in India; they have thwarted all attempts at curricular reform and have even undermined the goals of education. How can ET be harnessed to address these problems?

HOW CAN ET BE USED IN TODAY'S CONTEXT?

First, it is necessary to divest ourselves of the notion that ET means mass media or computers; no programme that is only equipment-driven works well. Our institutions have become the graveyards of a lot of useless equipment. We cannot afford to be as wasteful as we have been in the past, nor can we spend money on equipment without considering whether what we are buying is appropriate for the task at hand and whether the necessary support systems can be quickly set up.

We must also realise that knowledge springs from many sources, and that whatever is of importance in the learner's environment and suitable for his/her needs is what we must find and use in any teaching-learning system by employing effective instructional designs. Here considerable experimentation is necessary, and appropriate technologies for these designs will have to be worked out. The primary goal has to be an educational one. And to reach it, it might be necessary to tackle it by breaking it down into specific educational objectives. The same problem may exist in different localities and cultures.

The systems that ET specialists would have to think about would therefore have to be diverse. Efficient teaching-learning systems at every level, which use available resources and appropriate technologies and processes, and which are flexible enough to effect changes based on observations and evaluations, are the need of the hour.

Further, one should stop looking at knowledge as a packet to be delivered, and instead take up topics, at least at the earlier stages of the educational system, that are relevant to the child in his/her environment and let both teachers and children build a teaching-learning programme that is multidisciplinary. The saying of Jesus, "There are many mansions in my Father's house," is true in the case of ET systems as well. The vast numbers of children who need to be brought under educational systems pose a problem

of scalability. Here the new technologies and the mass media can help, but they must be woven into the system in such a manner that they give good results. Interactive rather than disseminative programmes are desirable.

This expertise needs to be built up. The Internet and the Web provide sources other than local ones. But it is necessary to inculcate media awareness in our children so that they do not replace the words of tradition by the mantras of advertisers. They must know that nothing is value-free.

The major responsibility for bringing about this change falls on the shoulders of teachers. The discipline of ET is an enabling discipline designed to make the teaching of any subject more efficient and effective to meet the goals for which the subject is being taught. ET is not a subject in any syllabus except in teacher-training institutions. Information with respect to the ET needs of the curriculum have been passed on to the Focus Group on Teacher Education. Networking of teacher-training institutions and universities that offer ET courses is necessary.

Building alternative systems of education in addition to schools is the need of the hour. Whatever alternative systems exist on the ground need to be made less bureaucratic in their operations, and they should also be reoriented to carry out their tasks more efficiently. Alternative models of education, distance and open-learning models, on-demand education, and similar flexible models of learning will have to be tried and established. Flexible systems, futuristic curricula, and a twenty-first-century career orientation have become necessary for young people today. Conventional definitions of livelihood options are too limited to cater to such a large number of people.

THE ET FOCUS GROUP'S PROPOSALS FOR ACTION

What are the things on which we could leverage? A vast number of institutions exist with ET components in one form or another—CIET, SIETs, state ET cells, SCERTs (State Councils of Educational Research and Training) , and more than 450 District Institutes of Education and Training (DIETs) .

Further, even in higher education, where it is recommended that teacher training in ET should form a part of their programme, there are media units. Under the University Grants Commission's (UGC's) Consortium of Educational Communication, we have a network of over 17 Educational Media Research Centre (EMRCs) and Audio Visual Research Centres (AVRCs) . More than 250 universities offer ET as an optional subject in B.Ed. and M.Ed. courses.

Technical Teachers Training Institutes (TTTIs) also have facilities for technical education. Several state open schools, the National Institute of Open Schooling (NIOS) , several state open universities, the national open university, Indira Gandhi National Open University (IGNOU) , and the distance education departments of conventional universities all have facilities to provide learning through alternative modes. However, all of them suffer from authoritative and exclusionary traditions. They must learn to collaborate, share, and revitalize themselves in order to meet the educational challenges

that the future will bring. The Focus Group proposes that serious thought should be given to making these institutions more effective and to gear them towards the need of providing equitable and highquality education with access to all.

RE-USING PROGRAMMES FOR INTERACTIVITY

Many organizations like CIET, SIETs, AVRCS, and EMRCs as well as many institutions like the Tata Institute of Social Sciences (TISS), for instance, have made some excellent programmes. The numbers game in which organizations like SIETs are caught up prevents them from reusing their old programmes or the programmes of other institutions. One of the most effective SIET has been Balchitravani in Maharashtra. It has a stock of nearly 3,000 programmes, many of which are of good quality. Starved of funds, it has been raising money by the sale of its programmes. It has made a profit of about ‘ 90 lakhs. Some of its programmes like “Get to know your trees” could easily be dubbed into different languages and used. The reorientation of such institutions, and bringing them under one network, is highly desirable. Many of the Balchitravani programmes need to be digitalized; otherwise they will be lost. The reorientation and strengthening of these institutions cannot happen if action is not taken on a war footing.

INFORMATION COLLECTION

There is as yet no aggregating function assigned to any institution either for programming or collecting data in respect of innovations, or an assessment of their worth. Many institutions—whether at the district level, or the state level, or the central level— could consider taking on this role. Programming for built-in interactivity is the need of the hour.

USING SATELLITES, DTH, AND OTHER TECHNOLOGIES

Another technology worth exploring is the use of satellites. India is perhaps the only country with a satellite completely dedicated to education, the EDUSAT. The present configuration of its structure is at variance with its declared aim of reaching backward areas and marginalized populations. Nevertheless, it does free educationists from the dictates of Prasar Bharati, which, on the one hand, declares itself as a social service, but, on the other hand, demands money from educational programmers as if they are advertisers of commercial products. EDUSAT has a limited capacity for interactivity, but it would enable a new type of programming that is better than disseminative programming. Developing interactive skills and programming could be a valuable programme for ET, for which a great deal of R&D would be necessary.

It is desirable to explore whether an educational channel that uses the DTH (direct to home) technology of Doordarshan might not be a better proposition since it costs only ‘ 3,000 to install a dish antenna. It provides all the DD channels. DTH technology will have transmission capabilities both for Doordarshan (video) and AIR (audio) channels. These channels can become available anywhere in India. DTH also provides

a number of free-to-air foreign channels similar to those offered by the BBC. This possibility should also be explored as a number of good documentaries on various subjects are available both in the public and private domains and could be shown on documentary or educational channels like the Discovery Channel.

Other resources that we have on the ground are a well-established publishing industry, desktop publishing, and know-how and capabilities in producing kits, teaching aids, etc. We have production capabilities for audio and video, multimedia, broadcast channels, Internet connectivity, trained manpower, and institutions with responsibilities to undertake the tasks.

If only we could start using these resources in efficient teaching-learning systems, we could achieve a great deal and show the way forward in providing large-scale access to all parts of children, even those out of school, and the disabled.

SPECIFIC PROPOSALS FOR THE SCHOOL SYSTEM

A paradigm shift has to be made regarding the notion of the teacher's role. She has to move from being a "teacher" to being a facilitator or guide. This means an emphasis on the learning culture rather than on the use of technology. One of the key concepts that should be kept in mind here is flexibility, as the appropriate use of facilities and the achievement of growth would be impossible without it. We need to pay special attention to the continuing education of in-service teachers. Here are the suggestions that arose during the discussions:

Revitalising and Reorienting Existing Resources

- At the risk of repeating ourselves, the Focus Group feels that it must emphasise that, while the macro attempts of the disseminative media have had limited effects, they have nevertheless led to the establishment of a large number of institutions and facilities, the founding of nationwide networks, and the emergence of trained professional and creative manpower in the area of ET. The challenge before us is to work out appropriate modes of re-engineering so that they can constitute a powerful and effective system.
- These institutions will have to play fresh roles. Apart from their role in producing audio and video materials, they will also have to be oriented towards action research, documentation, and assessment of innovative practices; undertake the in-service training of teachers; network with nationwide efforts in education and evaluation research; develop models for interactive classes, interactive multimedia, teleconferencing, and video conferencing; and lead the process whereby materials can be generated by teachers, parents, and children at every level.
- In recent years, ICT and the Internet have emerged as dependable media of interaction. Unlike the broadcast media, the Internet can facilitate the participation of the periphery in an eminently democratic discourse, which can

be empowering. And if properly deployed, quality concerns hitherto forced by economic and power considerations to be confined to the haves can now be within the reach of everyone. The need of the hour is, therefore, to recognise this potential, promote universal access, facilitate participatory forums, and develop communities and interest groups. Left to market forces alone, the reach is bound to remain limited. The Internet can be a sound investment for continuous on-demand teacher training and support, research and content repositories, value-added distance education, and online campuses aimed at increasing the access, equity, and quality of education.

- The model of education prevalent today presumes the existence of groups endowed with abilities, knowledge, and skills, which at times even subsume the right values, and which therefore acquire the mandate to educate. This separation of the centre and the periphery has led to the alienation and disempowerment of large communities of people. The fact that we continue to invest in adult education, that we continue to grapple with the problem of dropouts, that we continue to deal with issues relating to the provision of even minimum facilities can be traced largely to this chasm. Both for logical reasons and as a moral compulsion, it has become necessary to strengthen multiple, albeit shifting, centres. The challenge of population alluded to earlier can only be met if we overcome this centredominant thinking.
- The other implication of this idea is that knowledge is not centred at any powerful location, but is available everywhere. What constitutes education is an opportunity for every individual to develop his or her latent abilities and skills, to choose his or her teachers, and to benefit from select experiences. The problems of certification and standardisation, recognition of what constitutes legitimate knowledge, deciding who is a legitimate teacher, and determining what abilities, knowledge, and skills constitute legitimate education may all have to be looked at from drastically different perspectives.

In Systemic Reforms

- Ensure that technology is used in an equitable and democratic manner to enhance the selfworth and self-image of the poor and the disadvantaged.
- Counter the tendency to centralise; promote plurality and diversity.
- Ensure opportunities for autonomous content generation by diverse communities.
- Shift focus from fixed to flexible curricula with competencies and skills identified rather than specific factual content.
- Deploy ET to enhance open education, which implies openness in curriculum transactions.
- Work towards transforming all schools into ICT-rich environments.
- Create opportunities for administrators and educational leaders in the school system to become ET savvy and to be able to use ICTs competently.

In Refreshing the Skills of In-service Teachers

- Create a system of lifelong professional development and support, especially of educational leaders and managers such as headmasters and principals.
- Encourage ICT literacy for official and personal use to increase comfort and later enhance creativity in educational work.
- Support the development of and nurture teachers' self-help groups/ professional development groups on the ground as well as online.

In Pre-service Teacher Education

- Introduce teachers to flexible models of reaching curriculum goals.
- Introduce use of media and technology-enabled methods of learning, making it inherent and embedded in the teaching-learning process of teachers.
- Train teachers to evaluate and integrate available materials into the learning process.
- Enable trainee teachers to access sources of knowledge and to create knowledge.

The foremost challenge is to put in place a system of lifelong professional development and support. This has to replace the one-shot touch-and-go interaction, loaded with theory and almost no practice, into which the present teacher-preparation programmes have degenerated. Even while we set out to accomplish this goal, revamping the ET component of the course requires immediate attention. As long as ET is used in isolation from the other components related to teaching learning, it will fail to convince a teacher about the significance of her role in engineering the teaching-learning situation and the importance of making it a more meaningful experience for both herself and her pupils.

In School Education

Move from a predetermined set of outcomes and skill sets to one that enables students to develop explanatory reasoning and other higher-order skills:

- Enable students to access sources of knowledge, interpret them, and create knowledge rather than be passive users.
- Promote flexible models of curriculum transaction.
- Promote individual learning styles.
- Encourage use of flexible curriculum content, at least in primary education, and flexible models of evaluation.

Even within the confines of conventional schooling, helping children reach school and stay with it for a longer time will need to be addressed differently. Insights gained from various experiments aimed at reforming the school environment point towards the need for reform both in the system and within the classroom. ET will have a significant role to play here.

In Research

- Create a framework to identify the generic skills needed for the new initiatives to be undertaken in ET.

- Acquire knowledge about how learning takes place in ICT-rich learning environments, optimising learning paths for learners with different learning styles coming from a variety of social backgrounds, including gender differences. Examine possibilities of adopting mobile technologies for learning purposes.

POSSIBILITIES FOR THE FUTURE

Computers are programmable devices. This very fact makes it possible for users to make demands on these machines. This implies two things; first, that the computer ought to be capable of responding to intuitive demands, and second, that the user communicates in a language that the computer can interpret. Most software tools are designed with specific uses in mind. While this allows an ordinary user to concentrate on the task at hand, it is normally not flexible enough to respond to the different demands of the user.

Most computer education programmes degenerate into teaching students the art of punching the right buttons, which ends up making them glorified data-entry operators. The need is, therefore, to seek an interface with the computer in order to respond to the user's creative needs. 'Open Source' shows a way to achieve this goal, providing access to the source code of software.

The creative potential of the computer, and the liberating potential of the Internet, can only be unleashed when we actively make these kinds of demands of these technologies. The students of the future should be oriented to this possibility, allowing them to stand their ground amidst the technology-mediated onslaughts of the modern world. Integrating ICT into education will require that these aspects of the technology are catered to as a whole.

Hitherto we have used the media to pick up or beam out information depending on where we stood in the teaching-learning spectrum. Today, as learners, we can use the media to personalize our questions and to find the answers to our own questions as a matter of choice. As teachers, we can find out what intrigues our learners and we can orient our approaches accordingly. Institutions with the ET agenda s will now have to be on the look out to attract people and projects, functioning somewhat like lighthouses. They have to learn that their role is akin to that of an observatory or test-bed where new trends are sighted and experimented with. If they do not do this, they will either be frozen or become rusted like the machines in many such institutions, of no use to anyone except to those with vested interests.

The different Focus Groups involved in this exercise are deeply concerned about the ways in which education could be made more meaningful and joyful and how it should be designed in accordance with the learning pace and potential of children in a rapidly changing world. Education will continue to face the challenges of numbers, limited finances, knowledge explosion, corporate globalization, and the politics of power globally and locally, will continue to haunt education unless we can find a way out of this quagmire.

THE USE OF AUDIO-VISUAL AIDS IN EDUCATION

By audio-visual aids, we usually mean the most modern or the most recently used of these methods. This is a summary identification of very old methods and very modern instruments, and one should react against it. Visual aids are far older. They correspond to a profound tendency among the immense majority of men: to materialize their thoughts in the form of graphic or sonorous images or to give their thoughts a concrete frame of reference. Plato himself took care to set the scenery of his dialogues, and he used concrete words and concrete comparisons as foundations for his most abstract ideas.

In France, the *Très riches heures du duc de Berry* bring out the importance which ‘illustration’ can take in a work which would have otherwise sunk into oblivion. Xylographic images preceded the printing press by three-quarters of a century and the first illustrated book by nearly a century. The tremendous success of the ‘images of Epinal’ in books peddled from door to door in France was only a manifestation of popular taste in a society where illiterates continued to be in a majority and where images went with oral literature.

Films, radio and television, considered as educational instruments, have merely developed – at a rapid rate – alongside older means whose importance remains considerable. Their common denominator lies in their function as aids.

This is not a theoretical conclusion, for it is confirmed by the very attitude of the educator. The educator basically must contribute to the training of the individual with a view to his integration into a given society and teach new ideas, facts and techniques to a specific public. It is thus relatively easy to define the goals at which the educator aims. Achieving these goals is another task which brings him face to face every day with the basic problem of pedagogy – that of transmitting or communicating ideas or information. To solve this problem, the educator resorts to infinitely varied means, among them audiovisual aids. If our purpose, therefore, is to aid the educator, we must then offer him as complete an arsenal as possible of these means. But it is the educator and the educator alone who chooses the means which is best adapted to his subject, his audience and his circumstances. It is thus clear that audio-visual aids cannot be separated from educational materials in general.

This tendency towards the use of concrete examples has developed through a complex process. At first, graphic representation was probably only a way to enable man to capture fleeting thoughts and the sole way of transmitting thoughts, compared to oral transmission which was subject to rapid distortion.

The invention of writing, a perfect example of a visual aid at its origin, proceeded from the same necessity. It would be interesting to study, for example, in the light of Mayan writing – of the Codex Troano – how man progressed from the talking image to the letter. We can therefore conclude that ‘illustrations’ were looked upon at first, at least by the most educated persons, as a minor complement to thought. The entire history of publishing until the end of the eighteenth century confirms this. But, in the twentieth century, powerful means of reproduction, associated with radio, cinema and

television, have changed the aspect of the problem. Sound and visual ‘illustrations’ are no longer mere minor complements to thought but they directly influence the thoughts and the very conduct of millions of individuals. It was therefore inevitable that a desire should spring up to master such a powerful instrument, to discipline it for better or for worse. But this coveted mastery is still rather crude: it is often reduced to the creating of a few conditioned reflexes, satisfying the merchant but not the educator. Certain of these audio-visual aids are both means of education and media for information and propaganda, and it is not always easy to draw the line between what belongs to the educator and what is within the province of information or propaganda.

It is also probable that the child is more affected by the violence of street posters and by the shock techniques of radio and television at home than by the visual aids used in school. Should we conclude then that these means are harmful and should we condemn them? This negative attitude would be most unrealistic. The only possible conclusion is to accept the need for basic research in these fields. It should bear essentially upon psychology and upon the social sciences. We educators have already ventured forth, but timidly, onto this terrain. Systematic establishment of contacts with research workers and specialized institutes is the duty of all those who are responsible at the national level for audio-visual services.

It can be reasonably hoped that this basic research will lead to a better use of audio-visual aids and to more scientific pedagogy based upon them. It is not difficult to observe that their use is continuing and developing outside the pale of any research. As a result, a pragmatic pedagogy is taking shape and not necessarily in contradiction to the results of the most scholarly research. Establishing or stimulating closer collaboration between research workers and educators, stimulating the writing of theses or documents containing the fruits of the work of both, and publishing and distributing the results of this work should also be the common task of pedagogical and audio-visual services.

Until now, the problem of the use of audio-visual aids has been examined from an intellectual angle. It also includes important practical and technical aspects. To tell the truth, techniques cannot and should not be separated from pedagogy. We have seen that audio-visual aids cannot be separated from educational materials as a whole, this conclusion being thrust upon us by the attitude of the user when confronted by these materials. Now this same user – whether a teacher, a professor or an adult educator – does not act any differently when pedagogy and techniques are involved. He can never be purely a pedagogue or purely a technician. It is clear, therefore, that the pedagogy of audio-visual aids cannot be separated arbitrarily from audio-visual techniques. No one can hope to achieve good results unless he is a sensitive pedagogue and a skilled technician. The problem must be solved globally.

Unfortunately, this initiation into techniques is not always carried out in the institutions where future educators are trained. In underdeveloped countries, the lack of qualified personnel is the most frequent obstacle to such an initiation. But it is not the only one because similar shortcomings are often found in more favoured countries. Routine,

lack of initiative and administrative delays are the main factors responsible for educational sluggishness. There is no doubt that audio-visual aids produce their best results when they are used in connexion with active teaching methods. Here, the task of educators is to draw the attention of their governments to these methods and to the recommendations of previous seminars concerned with the introduction of an initiation into film and radio techniques into normal schools and similar institutions.

Finally, there are other questions which should be taken up in thorough and specific studies. They are related to the use of radio and television in the teaching of reading and writing and of languages.

A great deal has been said about the 'singular, specific and irreplaceable services' which can be rendered, for example, by 'teaching by radio'. A great many hopes were stirred as a result of statements repeated with such warm conviction that one could have believed them to be dictated by experience. First of all, we should note the ambiguity of the term 'teaching'. In the context of the statements to which we refer, this term covers both the teaching of subjects such as science and history as well as the teaching of reading and writing. One of the greatest problems which remains to be solved is the liquidation of illiteracy. Following hasty conclusions, a belief has grown that, thanks to radio, illiteracy can be liquidated quickly, easily and cheaply. But what do we know about it objectively? Until now, the number of experiments has been limited. Some of them were frankly admitted failures. Fortunately, we will soon be in possession of an exhaustive report of the results obtained during the best known of these experiments, the one at Colombia. One of our experts has made a global study of these results from which we think that we will be able to derive valuable lessons, if not definitive conclusions.

A few complimentary remarks should be made here. The first concerns the basic difference between teaching notions of history, geography, science, etc., by radio and the teaching of reading and writing. We say teaching by radio because the problem of teaching reading and writing by television is infinitely simpler and the results already acquired are sufficiently convincing. At any rate, it can never be said too often that the global use of audio-visual aids always gives the best results.

Opposing radio to television or both to films is a typical example of a false problem. In an educational campaign organized and carried out at a national level, all complementary means must be used if possible. It is also virtually certain that results are proportionate to the means employed in a geometrical, not an arithmetical, progression. In other words, overly strict economy does not pay. There lies a source of misunderstanding. Too often, it has been believed that making an expert and \$10,000 worth of equipment available to a government was enough to solve the problem of audio-visual aids in a given Member State. Audio-visual aids do not have this magic quality. They require serious study and, first of all, serious thought about the place which they should occupy in a budget. Pedagogical problems always end up by leading into budgetary problems – that is, in the long run, economic, social and political problems.

The educator must play his pedagogical role. The tool which is offered to him – and this is the case of audio-visual tools – can multiply the activity of the educator in large proportion. Therefore, the political authorities must be convinced of the necessity of a financial effort which is often considerably large. That is a point of view which often escapes the educator: he must also educate administrators and political authorities. Many mistakes begin here.

So far, we have talked about audio-visual ‘aids’ and ‘means’. Educators obviously consider them from this angle in the best of cases. But will this tremendous development of mass information media which we continue to allow itself to be domesticated?

Let us go to the heart of the matter: must we continue to consider these information media as blind forces whose unleashing – and, as far as the educator is concerned, the unleashing begins where his own control ends – would be an educational and cultural catastrophe? Or, to put things in a less impressive but equally embarrassing way, cannot audio-visual ‘means’ be allowed to play their role without the help or simply the intervention of the educator? Before issuing a condemnation without any possibility of appeal, perhaps it might be wise to remember that films, radio and television can only be arbitrarily separated from the social, economic and cultural context which gives them their means of existence.

No doubt, the study of these problems takes us a long way from modest film strips, flannelgraphs, and even traditional black-boards which are still a luxury for thousands of schools.

Let there be no misunderstanding. We know the importance of pedagogy in the use of audio-visual aids. We know that the training of good educators – in this case, good users of these didactic means – is a long and difficult matter. We know that we must think about the desperate problem of training teachers. But pedagogy itself is only a means whose end is education. And education, in the long run, is only a contribution – naturally, of capital importance – to the integration of the individual into a given society. It is in this perspective – from their production to their final use – that we must look at audio-visual aids and the various questions which they raise.

FILM, RADIO AND TELEVISION AS EDUCATIONAL FORCES IN MODERN CIVILIZATION

FILM, RADIO AND TELEVISION AS EDUCATIONAL INSTITUTIONS

Film, radio and television are usually discussed under the heading of the customary educational and teaching aids, *i.e.*, on the same level as abaci, wall maps, sand trays or sets of pictures, for the reason that films and tape recordings can be used in education in a way similar to the projection of slides. However, to lump them together with the traditional aids is to mistake completely their true importance and to ignore the social and psychological value of mass media in present-day civilization. A revision of thinking

about modern mass media is needed. Far from existing somewhere on the fringes of modern educational thought, they are right at the centre. Television, for example, is not just an alternative method of demonstration, but an independent and organically complete educational institution, like the school itself. It should be regarded as such and not simply as a vehicle for entertainment, political propaganda or commercial advertising. Educational subject-matter of the most varied kind falls within its range to a far greater extent than is generally realised. It is significant that most television organizations are themselves aware of this.

The large American broadcasting corporations, ABC, CBS and NBC, when questioned, described a large proportion of their programmes as educational. Indeed, the NBC view is that programmes should be adjusted to the possibilities of development and self-realization of individuals and should constantly serve the purposes of education.

A passage in the American Television Code in addition, stipulates that television networks, advertising agencies and, particularly significant, financial backers, must deliberately seek opportunities for the inclusion in broadcast programmes of factual material which will contribute to the enlightenment and education of the American public.

The situation in Germany is similar. An analysis of German afternoon programmes in the first half of 1960 shows that out of 203 programmes monitored, 104 were educational. Even among the 406 evening programmes viewed during the same period there were at least 155 with an educational slant. If television plays, and all theatrical and operatic performances are added, as they have some claim to be, our overall view of television almost suggests an educational bias, which may well be in conformity with the true nature of the medium.

The situation is possibly even more favourable with regard to sound broadcasting, although it is naturally completely different in the case of the film. The great majority of broadcasting stations, particularly in Germany, France and England, are public services of high standing. The cinema on the other hand is an industry guided by the profit motive; and while this does not necessarily restrict its practical educational influence, it does modify it in the sense that the educational responsibilities of the cinema are completely vague.

THE EQUINOXIAL NATURE OF THE MASS MEDIA

The assimilation of mass media by public education cannot be considered complete; in a deeper sense, it is only just beginning. Indeed, it is now being seriously overshadowed by large-scale public and expert criticism of these media, on medical and biological, psychological and sociological and aesthetic, cultural and educational grounds. Such criticism ranges from naive and ill-informed attacks to extremely subtle analyses like those of Giinther Anders and Adorno.

More particularly, the opposition of teachers to the impending threat of school television has recently been increasing. However, one good result of this has been to

produce specific affirmations and objections which it thus becomes possible to refute. What is particularly feared in connexion with school television is the encouragement of passive acceptance, educational impressionism, the growth of a dangerous type of conformity, the paralysis of creative imagination through the intensification of artificial stimuli, the replacement of first-hand by second-hand experience, loss of individuality, the mechanization of the pupil/teacher relationship, and the standardization of educational activity.

These objections are by no means to be lightly dismissed, for some of them are valid expressions of educational philosophy; nor can they easily be refuted by purely educational arguments.

The following are, however, useful criteria for establishing the value of visual documentation centres: reasoning based on cultural history and educational sociology, which goes beyond purely educational arguments and sets out to demonstrate the cultural legitimacy of the media; and a critical analysis of the very remarkable educational potential of these forms of communication, from which it might be possible to deduce methods of organizing them and using them for educational purposes.

Scientific and historical research provides ample evidence that the attitude of Europeans as to their surroundings, and their consciousness of the world have changed since the Renaissance, in the sense that the archaic, mythological outlook has gradually made way for the development of empirical and realistic ideas and concepts. The rapid expansion of the natural sciences is the most impressive intellectual reflection of this development.

This gradual development culminated in a consistent, if sensational way, though at a level beyond art, with the arrival of photography. This is the aesthetic correlation of a type of consciousness that has learned to think in terms of reality.

Photography projects a picture of the world as seen 'through the camera's eye' which has undoubtedly become representative of the twentieth century. Our consciousness has a strongly realistic bias and the greater the extent to which things and situations can be made to seem real or actually experienced, the more significant they will appear to us. It is this illusion of reality that film and television photography can create for us. Considered from the aspect of cultural history, all of us are at home in front of the screen and willingly become receptive to its influence.

The auditory stage of sound broadcasting is no exception to this rule, for the photographic or electronic camera lens and the radio microphone operate on the same aesthetic and psychological levels, though on different sensory levels. Our reality-bound consciousness regards them as identical instruments and, particularly in sound films and television, they have for a long time been fused in an excellent synthesis with an outstanding power of aesthetic and intellectual suggestion.

This evolution of consciousness and its aesthetic consequences may be regretted or may even be interpreted as a sign of regression. Nevertheless, along with the help of mass media, it is now enabling us to make the world our own, both physically and

spiritually, and on occasion to render it transparent. The French cinema theorist Henri Age1 correctly refers in this regard to a spiritual realism for which the film is something that can also dematerialize the world. In this connexion, if we consider that educational thought from Ratke and Comenius up to the American pragmatists of the Dewey school has followed a very similar road, even the most sensitive and conscientious educators should no longer find it too difficult to assimilate these media.

THE ARGUMENTS OF THE EDUCATIONAL SOCIOLOGISTS

The evolution of Western consciousness has been paralleled by a far-reaching social and cultural transformation of our lives, which in recent years has been proceeding at a staggering pace, owing to the all-pervading influence of technology. Sociologists are fond of describing this as a process of cultural change no longer conceived of as a normal periodic change in style but as a step through the gateway into a completely new cultural era.

Whether this is true or not, we are now being subjected to the pressure of a vast process of cultural disintegration. However, this is accompanied by a no less intensive building-up process, something which that is often forgotten. This latter process has already resulted in far-reaching changes in our work and consumption habits and in our social, ethical and cultural attitudes.

In the wake of these new living and world conditions, there are people who up until now have existed merely as pawns in a game of historical power and whose intellectual and political maturity has been gained at the rather dangerous moment when their own world is threatened with the collapse of its scale of values. Since becoming active on the historical scene, these masses have become shouldered with responsibilities and are conscious of educational needs which hitherto arose only in the training of an élite. This process, which Karl Mannheim has described as the 'fundamental democratization' of our society, can bear fruit only if the masses are provided with educational and teaching aids of a completely new type.

In this connexion, the mass media are already playing a tremendous part. For the masses, they really constitute signposts to a real and yet boundless world. They have become so because they make use of an audiovisual language that is understood internationally.

Mass media are, in fact, the catalysing agents in a vast process of cultural change, which they reflect and constantly further and modify. It is precisely this which makes them so outstandingly important for education, regardless of the direction in which their influence may radiate. They have contributed to the development of a revolutionary situation in education, which makes demands on the general educational perception of modern teachers-demands which cannot be ignored.

Cinema, broadcasting and television appeared in that order at almost exact quarter-century intervals, beginning at the close of the nineteenth century. At the same time, their educational and aesthetic potentialities became gradually enriched by increasing

technical progress. For example, as camera work and cutting-room techniques progressed, various standard types of film were created—narrative feature films, informational short subjects, educational films and documentaries. The film thus developed into a classical medium of concentrated narrative conveyed in visual and acoustic images.

The novelty of sound radio lay not so much in the fact that it created its own appropriate forms of presentation, as that it possessed practically unlimited possibilities for the diffusion of its message due to communication through electronic radiation. Its primary feature is reduced to a formula—its ubiquity, *i.e.*, the omnipresence of its broadcasts, whether in the form of original radio programmes or of concert relays. What matters is its reception at the greatest possible number of places simultaneously.

Television may be regarded as a kind of synthesis of these two stages of historical development. It has taken over both the ubiquity of electronic radiation and camera and cutting techniques and has therefore developed into a fully equipped universal stage for an era which in practice can draw upon the whole store of forms known to present and past civilizations. The television screen possesses an adaptability which is unlimited. This is why television is the richest and most versatile of the mass media when it comes to presenting a photographic and auditory picture and interpretation of the universe. This quality of differentiation naturally has educational consequences which are easy to recognize in connexion with the use of these media in teaching. Every audiovisual expert and every teacher discovers this as soon as he makes a serious study of either films or television.

THE EDUCATIONAL POTENTIAL OF THE MASS MEDIA

In spite of this differentiation in working with the mass media, the audio-visual expert and the teacher will time and again be referred back to an identical system of communication which is equally valid in all of its three variants—film, radio and television and which covers material as widely divergent as plays, sports reports, dance programmes, political reviews, interviews, quiz programmes and even religious services.

Adorno and Horkheimer, in their ‘*Dialektik der Aufklärung*’, show ironically that the complaints made by many art and culture historians about the disintegration of the power of stylistic invention in Europe are completely unfounded, as the stereotyped reduction of all the material communication into a form suitable for mechanical reproduction is something more rigorous than the demands of any genuine style. So far as our subject of discussion is concerned this means that things, faces, situations and events are so filtered for purposes of translation into audiovisual language of the mass media that they positively enforce a specific type of experience. The special effectiveness of these media is therefore to be sought in what has become a technical audio-visual language. One of the basic qualities of this language is its practically unlimited capacity for expressing and rendering virtually every aspect of the world. We all know that anything in this world that is designed to achieve significance and to arouse public

consciousness -whether in politics, art, beauty contests, sports records, ideas or even historical events-is compelled to employ this language. The universal appeal of these mass media has become so great that it can be accepted as an uncontested, basic feature of all mechanized communication.

In addition to this unlimited capacity, analysis of the phenomena involved brings to light other exceedingly important characteristics. For it is chiefly the combination of these that produce what may be called the educational potential of the mass media; and it is with this that we are really concerned. Reduced to four concepts, we have the following: the power of attraction of the audio-visual language, the almost unlimited extent to which it can be manipulated; its tendency towards the accumulation of stimuli which can be called in short the accumulation phenomenon and finally the topicality principle. These features are common to all mass media.

The four phenomena thus roughly defined are the essential structural principles of mass media. They are active at all levels, in programming, in production and presentation, in the action of the media on the public and, not least of all, in our own educational work. Indeed, they are the universal principles of modern communication and therefore deserve closer study and interpretation.

THE POWER OF ATTRACTION OF THE AUDIO-VISUAL LANGUAGE

The remarkable power of film fascination is probably the first thing we notice in considering mass media. This fascination is not merely a kind of diffuse moment of interest, but a very specific effect arising from the technological depiction of the world. In addition, we must remember not only the visual aspect but also the auditory components, which possess their own brand of fascination, principally of an emotional type.

This relationship of photography to our reality bound consciousness acts most powerfully through the fully-developed dynamism of film photography and television, in which it raises fascination to its ultimate level. In his research, Herbert Wölker proved empirically that the film multiplies the intensity of experience many times over, and thus facilitates identification with the matter shown. The influence of identification on the social process of learning has been made clear to us by Karl Heinrich's latest work on the effect produced by films.

The intensification of experience depends essentially on the visual and acoustic impact of photography. For that reason, in an age of increasing abstraction, increasing intellectualization and declining sensibility, as Gehlen calls it, film and television, through their power of making things concrete, have a compensatory function. By attracting audiences to the images and the stark graphic descriptions of commercial cinema and television, this function sometimes has a very negative effect; but it nevertheless retains an authentic and indispensable therapeutic quality.

Moreover, the relationship of photography to the world is not confined to this power of making things concrete. The subjects photographed have their own authenticity and

documentary value. Photography always relates to facts, to something that has actually happened or really existed. This enhances its fascination, but weakens its power of symbolism. According to Cassirer's theory of symbols photographs is never symbols used to express an intrinsically spiritual concept. They are signals pointing to some concrete fact. Therefore, the weak point of photography as a means of conferring reality and exercising fascination lies in its dubious relationship to abstraction.

Here we come to an argument applicable to all mass media and one which recurs in a variety of shapes. Photography makes possible a maximum of what English theorists call involvement, an entering into contact with the outside world; but it holds the mind captive in concrete situations. By drawing on real situations, it promotes what might be called emotional commitment, but at the same time weakens thought by making an assault on our senses. This sometimes appears in a crude form in popular science broadcasts, where lavish visual display, an abundance and complexity of pictorial matter, very easily gives the illusion of understanding, though in reality it merely builds up visual dummies in our consciousness. The intellect is frequently quite untouched by such visual exploits.

Educational responsibility and imagination must not capitulate in face of this dilemma. Indeed, they must use it consciously and even orient their educational activities on that basis, *e.g.*, by using films and broadcasts in teaching. Through their power of fascination, these are particularly suitable in the initial stages of a learning process, where the great thing is to give a powerful motivation and awaken interest from the start. The mass media are 'magnificent gateways', as the English publicist Hoggard once called them. However, they may prove to be a hindrance where abstractions and generalizations are concerned.

In such cases, they may almost act as barriers. On the other hand, visual barriers of this kind can be set up deliberately, to goad pupils into using their brains. Where an accurate observation of detail is the prerequisite for the correct training of perception, as in geology and geography, the use of films can be exceedingly profitable. It is also most valuable in the 'deepening stage', in the sense in which the term is used in Herbart's theory of phases. On other grounds, it is also valid for the teaching of politics and civics which, because of their abstract nature, are difficult to present in visual form. Here the mass media are absolutely essential.

The effect of the visual attraction of photography, as a means of realistic presentation and information, upon the film consumption of young people is best known to those who use films in teaching. A profit hungry industry exploits photography's power of attraction and conviction mainly to lend the unreal world of mediocre authors a semblance of reality and to draw the imagination of the audience into a dream world clothed in the garb of reality. We realise that there is at present no way of overcoming this evil except by preventive educational methods designed to immunize pupils against this form of mental poison. The ability to illustrate is transformed into the power to seduce. The basic meaning of the Latin word *fascinare* is 'to bewitch'. This conveys an

echo of the assault on our senses and, let us be honest, is not this power of enchantment present to some extent in every film and in every radio or television programme?

FACILITY OF MANIPULATION

The adaptability of mass media is a property which makes of them well-forged and efficient tools in the hands of experts. It gives them their wealth of expression and ensures their suitability for all purposes. However, its values are neutral and it may serve either truth or falsehood.

The concept itself is used here in a very broad sense, which makes it possible for us to penetrate to the reality underlying every phase of film and radio production. Facility of manipulation is a universal phenomenon in the mass media and is displayed in a great variety of ways.

Perhaps the clearest way of demonstrating it is by reference to the work of the film cutter, whose sole function is to cut and re-join single parts of film into intelligible sequences. The purpose in this case is to create sense. Cutting and mixing images in film and television are primitive forms of manipulation. They may be used to tell the truth, to lie, to camouflage, to excuse or anything else.

The early Russians were well aware of the power that lay in cutting and had good reasons for regarding it as the fundamental principle of film art. But manipulation begins as early as the shooting stage, in focusing, in lighting, in camera angles. No photography is possible without perspective. Even a snapshot is a form of interpretation.

Nowadays, the problem of manipulation also appears on an entirely different level in television, in the person of the interpreting speaker. Speech within the framework of a visual medium has in this way become a manipulatory power of the first rank. The oral description given in a visual broadcast establishes a very pronounced emotional accent. This is a very effective form of primitive manipulation.

It must however be granted that, on a higher level, television is nowadays campaigning for a new word picture relationship. This takes the form of an extremely interesting process where idea and observation are fused in a single simultaneous act. Words become concrete in picture sequences and images are intellectualized by the medium of the interpreting word.

This cultural phenomenon is of great importance and novelty and is rendered possible only through the medium of television. This experience gives an indication of educational resources which are still untapped but which will one day radically alter the educational style of television. On the highest level, again, the manipulation process is relentlessly carried through in radio programming and in the planning centres of the film industry. Here, decisions about spheres of influence are made, and whole populations are exposed to the magical influence of planned films. In this regard, everything depends on the personality of those who occupy these posts of command. At this point, the phenomenological method of examination ought to make way for a true sociological analysis of the balance of power in the 'brains trusts' who control the broadcasting

organizations and the film industry. In any event, a study of this power of manipulation at every stage of film and radio production might well provide a most valuable contribution to our knowledge of the inner educational structure of these systems. Manipulation is also important at a much humbler level, *e.g.*, in the preparation of our educational films. Here the problem is one which causes us great concern.

Should we take as our model the perfect educational film, which traditionally constitutes a system of knowledge so compact and as completely adaptable as almost to render superfluous both the teacher and any further work on the material? Or should we produce open fragments of film, where the subject matter is accompanied by questions, so that pupils are placed in a working situation which in terms of modern teaching theory is educationally ideal? Here we come up against the same ambivalence and qualitative difficulties as were encountered in considering the problem of fascination. In this case too, the search for truth or falsehood, the encouragement of sense or nonsense, the offer of an opiate or a challenge, are effected by the same method-the power of manipulation- and by that alone.

What educational maxim can we derive from this? Manipulation must always be uncovered. This must happen in two directions. In the first place, the manipulation of sense in the cinema must be discerned, *i.e.*, we must learn how to understand films. In the second place, it must be exposed. Whether we are concerned with exposing a dream structure or with seeing through the visual perfection and manipulation of the newsreels of the Third Reich and laying bare its ideology, we should be constantly engaged in an effort to uncover the serious consequences of manipulation.

THE CUMULATIVE EFFECT

The cumulative tendency, *i.e.*, the tendency to pile up stimuli, is something mass media share with the other constituents of modern civilization, which suffer in general from a plethora of facts, stimuli, material and opinions. Owing to their unlimited capacity, however, these media are predestined to a boundless accumulation of stimuli, precisely because the technique of providing them is so easily mastered.

In this regard, the thematic, visual and sound overloading of television broadcasts and films is typical of the production style of all the mass media. It is claimed that this ultimately induces a permanent blockage of the audience's capacity to take in what it sees and hears. It has been said that television broadcasts are nowadays only looked at, but no longer seen. The onlooker develops a layer of undigested secondary experiences, a hybrid form of consciousness, which believes that it knows everything but in actual fact merely contains masses of experience which can lead to absolutely nothing.

However, those who tend to frown upon the endless piling on of stimuli are overlooking the fact that it originates in a genuine spiritual need which is only awaiting satisfaction. This is the need for an expansion of consciousness and a widening of the horizon, without which it is becoming increasingly difficult to find our way and to act constructively in our complex world. In the recent development of Europe, our

consciousness has altered not only qualitatively but quantitatively. Educational theory seems at the moment to have no proposals for satisfying its demands. Even the schools are running aground at this point. If we consider seriously, even for a moment, what Henry Cassirer, head of the Unesco television part, put forward at the London Television Conference in 1957, namely that a revival of the universal mystic thought of the Renaissance, as represented by Leonardo, Michelangelo and Machiavelli, was required on sociological grounds, not merely on the level of genius but among the broad masses, we cannot casually dismiss television's possibilities in that direction. If there is any way out of the educational bottleneck that now exists all over the world, it must be by means of the mass media, which possess not only the technical capacity for the task, but also the methodical facilities for implementing it.

THE PROBLEM OF TOPICALITY

This problem is of the same magnitude as that posed by the power of fascination of the mass media. Topicality and the trend towards it have come to set the pattern for television. Live broadcasts are among the high points of television programmes. Arnold Hauser, in his *Sozialgeschichte der Kunst und Literatur* tries to account for the power of attraction exerted on the minds of spectators by the impression of spontaneity resulting from the coincidence of perception and event, and attempts to explain why the modern mind has this peculiar sensitivity towards actual events.

We may agree with Hauser that this is not so much a fashionable contemporary whim as a profound metamorphosis of our attitude to life, which experiences the world primarily as a process in which we can genuinely participate only through grasping significant moments—a truly Heraclitean view of the universe !

It seems that the television set has already become a concrete symbol of this process of simultaneous experience, for it provides a synchronized record of life as it flows by. Television enables millions of people to participate in all kinds of topical events, even if most of them are not 'Heraclitean' moments. This is not the place for an assessment of the value of this form of modern topicality. However, even if we regard it as no more than a potentially important anthropological phenomenon, we are faced with the introduction of completely new elements into our already complex world. Naturally, our educational thinking as a whole cannot remain unaffected. Try to imagine contemporary events becoming a legitimate, even an integral feature of modern general education. The schools would then be faced with a task for which their organization and present attitude towards teaching does not yet equip them.

Only the mass media, and television in particular, are already prepared to cultivate a relationship with the outside world that could conform to this topical pattern.

RIVAL EDUCATIONAL PATTERNS

Film, radio and television are cultural institutions which obviously cannot be considered as educationally neutral. On the contrary, they exert a lasting formative influence on

character, though this is often involuntary on the part of their organizers. Within the framework of their influence a completely novel educational situation has developed, because they use for their own purposes a technical audio-visual language that has created an entirely new educational atmosphere. Their boundless capacity for manipulation, their fascination for the senses, their tendency to intensify stimuli and their accent on topicality reveal the mass media as means of communication in which a hitherto unknown educational ideal has been given an institutional structure which seems clearly opposed to the traditional conception of the school.

We should be wise to accustom ourselves to the idea that we are now dealing with two rival forms of education, which not only follow different ideas of teaching but have become embodied in different social structures. On the one hand we have the state school system from kindergarten to university and on the other the large, more or less government-controlled broadcasting stations.

We need not hesitate to regard television stations as setting the pattern for the new type of teaching, as they not only embody a new educational trend but also display certain cultural, social and aesthetic developments in a representative combination. It is perhaps too soon to attempt an accurate definition of the differences between the two educational patterns, for the whole development is still in a state of flux. However, some of the differences are already beginning to emerge.

Television clearly supports an extensive concept of teaching, in which broad surveys, a wealth of facts, the piling up of stimuli, a receptive attitude towards information, topical value and the establishment of a pattern of learning with particular orientation are the decisive factors.

Schools, on the other hand, embody the ideal of an intensive, selective education, based on the independent activity of pupils, in which exemplary teaching conditions and methodical thoroughness are the valid objectives. They want to and must develop exemplary learning methods. Both types of education are based on the genuine needs of modern civilization. For a long time, our society has been subjected to spiritual and psychological stresses which find their concrete expression in this educational duality. As a result, these two types in conjunction form the first integrated educational system of modern times. The natural relationship between school and television might therefore be described as a sort of competitive co-operation, as in the long run schools will be unable to avoid making use of current educational material from outside.

However, the screen requires spectators who have been subjected to school educational methods and are equipped with faculties of comprehension without which the fluctuating world depicted in the daily flood of images cannot be intellectually revealed. This demands the cultivation of new educational qualities which until now have perhaps appeared suspect. Television requires promptness in orienting ourselves to a large mass of stimuli, alertness in our observance of rapid sequences of pictures, preparedness for the interpretation of quickly-changing superficial appearances and the ability to hold aloof from the assault on our senses. It cannot be denied that such

factors should be included in the spiritual and intellectual equipment of contemporary human beings even if only a few of us are prepared to admit that we already possess a training ground for these qualities in the form of the television screen.

It is therefore strange that, at any event in Germany, these two educational institutions still profess to ignore each other officially-though, paradoxically enough, they are already surreptitiously linked by the fact that schoolchildren participate in both. This temporarily illicit relationship need only be legalized at the level of educational policy to bring about a situation which could well be decisive for the future of education in our country.

School television broadcasts might possibly be one way of bringing about a temporary link, as is already the case in many countries. However, there are other possibilities, some of which are quite revolutionary. Developments in the United States are to some extent a model for the future everywhere. There, television stations have simply incorporated the school system, with its methods and curriculum content. The early-morning broadcast 'continental classroom' is an example of this, as is the Italian 'telescuola'. Moreover, some schools have begun to set up regional television systems of their own, with the help of which they are arranging a rich variety of educational programmes and are thus entering into competition with the television stations, partly through the adoption of radio methods. The head of an audio-visual centre should be responsible for an organ serving both types of education and both educational institutions. He represents the school and its classical teaching traditions and at the same time the novel requirements of the mass media. He leads an amphibian existence, which in this situation may be a valuable stimulus for a far-seeing government educational policy, as authorities cannot remain unaffected by these developments.

We must wait and see if they will know how to make best use of them. At any rate, the audio-visual centres are the appropriate clearing houses for these problems, the specialized agencies in a process which is only just beginning but which will become of national importance.

MULTIMEDIA AS AN EDUCATIONAL TOOL

For many of us, the lure of computers is a powerful one. However, many of us also refrain from using computers for fear of failure. We want to hone computer skills, but are scared to make the effort because we lack those very skills. Too many of us, especially in the field of learning, are caught in this modern tug-of-war.

Throughout the 1980s and 1990s, the concept of multimedia took on a new meaning, as the capabilities of satellites, computers, audio and video converged to create new media with enormous potential. Combined with the advances in hardware and software, these technologies were able to provide enhanced learning facility and with attention to the specific needs of individual users. A primary application of the interactive multimedia for instruction is in an instructional situation where the learner is given control so that he/she may review the material at his or her own pace and in keeping with his/her own individual interests, needs, and cognitive processes. The basic objective of interactive

multimedia material is not so much to replace the teacher as to change the teacher's role entirely. As such, multimedia must be extremely well designed and sophisticated enough to mimic the best teacher, by combining in its design the various elements of the cognitive processes and the best quality of the technology. With today's multimedia courseware, once a programme has been designed and built in with the appropriate responses, it should be flexible and permit change and alteration. We shall look at the usage, advantages and disadvantages of multimedia in education and training. Some of the prototype multimedia lessons are also given at the end as examples.

DEFINITIONS

Multimedia. is a term frequently heard and discussed among educational technologists today. Unless clearly defined, the term can alternately mean a judicious mix of various mass media such as print, audio and video. or it may mean the development of computer-based hardware and software packages produced on a mass scale and yet allow individualized use and learning. In essence, multimedia merges multiple levels of learning into an educational tool that allows for diversity in curricula presentation.

Multimedia is the exciting combination of computer hardware and software that allows you to integrate video, animation, audio, graphics, and test resources to develop effective presentations on an affordable desktop computer. (Fenrich, 1997) ..Multimedia is characterized by the presence of text, pictures, sound, animation and video; some or all of which are organized into some coherent programme. (Phillips, 1997) . Today's multimedia is a carefully woven combination of text, graphic art, sound, animation, and video elements. When you allow an end user, *i.e.* the viewer of a multimedia project, to control 'what' and 'when' and 'how' of the elements that are delivered and presented, it becomes interactive multimedia. As such multimedia can be defined as an integration of multiple media elements (audio, video, graphics, text, animation etc.) into one synergetic and symbiotic whole that results in more benefits for the end user than any one of the media element can provide individually.

Specific uses of multimedia include:

- Drill and practice to master basic skills
- The development of writing skill
- Problem solving
- Understanding abstract mathematics and science concepts
- Simulation in science and mathematics
- Manipulation of data
- Acquisition of computer skills for general
- Purposes, and for business and vocational training
- Access and communication to understand populations and students
- Access for teachers and students in remote locations
- Individualized and cooperative learning
- Management and administration of classroom activities.

THE NEED FOR MAKING MULTIMEDIA COURSEWARE

Why use multimedia at all? Of what use is multimedia in education? The answers to these questions could be sought through an understanding of the capabilities and limitations of the medium.

Besides being a powerful tool for making presentations, multimedia offers unique advantages in the field of education. For instance, text alone simply does not allow students to get a feel of any of Shakespeare's plays. In teaching biology, an instructor cannot make a killer whale come alive in a classroom. Multimedia enables us to provide a way by which learners can experience their subject in a vicarious manner. The key to providing this experience is having simultaneous graphic, video and audio, rather than in a sequential manner. The appeal of multimedia learning is best illustrated by the popularity of the video games currently available in the market. These are multimedia programmes combining text, audio, video, and animated graphics in an easy-to-use fashion.

Moreover, under conditions of chronic under-funding, multimedia can provide an enhanced or augmented learning experience at a low cost per unit. It is here that the power of multimedia can be unleashed to provide long-term benefit to all. Multimedia enables learning through exploration, discovery, and experience. Technology does not necessarily drive education. That role belongs to the learning needs of students. With multimedia, the process of learning can become more goal oriented, more participatory, flexible in time and space, unaffected by distances and tailored to individual learning styles, and increase collaboration between teachers and students. Multimedia enables learning to become fun and friendly, without fear of inadequacies or failure.

ADVANTAGES OF MULTIMEDIA

The pedagogical strength of multimedia is that it uses the natural information processing abilities that we already possess as humans. Our eyes and ears, in conjunction with our brain, form a formidable system for transforming meaningless sense data into information. The old saying that "a picture is worth a thousand words" often understates the case especially with regard to moving images, as our eyes are highly adapted by evolution to detecting and interpreting movement.

For example, a photograph of Ganges in Varanasi, apart from being aesthetically pleasing, can contain a wealth of information relating to the culture, religion, geography, geology, climate, history, and economics of the area. Similarly, a recording of a politician's speech can allow us to discern significant semantic features not obvious in a written transcript.

For the student, one advantage of multimedia courseware over the text-based variety is that the application looks better. If the courseware includes only a few images at least it gives relief from screens of text and stimulates the eye, even if the images have little pedagogical value. More often than not, the inclusion of nontextual media into courseware adds pedagogical value to the application.

For example, a piece of courseware describing a dig at an archeological site would be more valuable to the student, if it included images of the site, such as enhanced aerial images showing features like old field boundaries, or diagrams illustrating where the digging and scanning took place. In this respect, using the text only, even in a creative way, has obvious limitations as compared to the use of both text and pictures.

PRACTICAL DISADVANTAGES OF MULTIMEDIA

Multimedia requires high-end computer systems. Sound, images, animation, and especially video, constitute large amounts of data, which slow down, or may not even fit in a low-end computer. Unlike simple text files created in word processing, multimedia packages require good quality computers.

A major disadvantage of writing multimedia courseware is that it may not be accessible to a large part of its intended users if they do not have access to multimedia-capable machines. For this reason, courseware developers should think very carefully about the type of multimedia elements that need to be incorporated into applications and include only those that have significant value.

Multimedia has other weaknesses too. While proponents of this new technology are very enthusiastic about its potential, they often leave the financial and technical issues unattended. Development costs in multimedia are very high and the process of developing effective multimedia takes time. Time spent on developing the multimedia package requires money so that the true cost of an interactive programme mounts with each delay.

Further, if the prerequisites for using multimedia include access to computers with related software, the user must possess a minimum level of computer literacy in order to exploit the capabilities of this medium for learning. And finally, training of the educator who is unfamiliar with the production and design of multimedia courseware or packages can be equally complicating. The critical question, then, is: How do we overcome some of the identified barriers and begin the process of multimedia implementation alongside the instructor, textbook, and blackboard? It is the barriers rather than the technologies which we must address before multimedia, or for that matter, any media technology becomes as accepted as the printed text or guidebook.

USE OF MULTIMEDIA IN AN EDUCATIONAL SETTING

Let us look at some examples of what is called innovative use.. Let us say a student wants to write a paper on desert animals. Traditionally, the primary source for obtaining information would be the encyclopedia generally available in the library. With access to interactive multimedia, the student would collect various textual materials about the camel from sources on a CD-ROM.

In addition, the student may be able to copy a diagram or the skeleton and muscular structure of the camel and the ostrich to study what is common about the two creatures. With a multimedia approach, the student could also access Web sites on the Internet to

get more information. The student could then add film clips on these animals in their natural habitat (all may be from the same CD-ROM) and blend them into a report. Then by adding titles and credits, the student now has a new and original way of communicating his/her own individual perspective.

Besides student use, teachers should find multimedia of great use in delivering their lessons. For example, a history teacher could use a multimedia CD to create a lecture on the non-violence movement by using film clippings and audio tapes on Mahatma Gandhi or Martin Luther King, also by incorporating other audio visual information with text to make the subject come alive. All this material would be available on a videodisc.

Similarly, a university professor might use a multimedia CD to prepare or to update information or to teach so as to enliven and also add insight to his/her teaching, thereby improving the quality of the course. The uses of multimedia need not be seen as a tool for classrooms only. In an industry dealing with hazardous materials, workers need to be trained. It could be risky to provide hands on training. In this case, simulated learning can take the place of actual hands on training by using all the features of interactive multimedia. Training can thus take place individually at the learner's pace and on his/her own time.

Medical procedures, first-aid training and instruction of paramedics or even surgeons are made both simple and interesting through the use of multimedia. The doctor or paramedic can run through a complete procedure on videodisc and analyse all the possible outcomes and can evaluate the possibilities before treatment of the real life patient starts.

In all the instances, the user can and normally does work individually and in an interactive mode with the medium.

TEACHING AIDS

PRACTICAL EXAMMPLES: CONNECTING THEORY WITH APPLICATIONS

Students have expressed concern regarding the need for more industrial and practical examples to reinforce theory in the classroom.

The use of practical examples can help you connect engineering theory with practical applications for more effective teaching and learning. The introduction of practical examples does not imply an elimination of theory, but rather an enhancement of the theory taught in the classroom. It is important to simultaneously develop a theoretical and a practical base since neither is useful without the other.

The use of practical examples in the classroom is targeted at the following two main goals:

- Help illustrate and explain new material making the theoretical basis of the material more accessible to the students. Practical examples help students understand the new concepts being introduced.

- Teach students how to apply their knowledge of course material to new situations that are not directly covered in class. The goal here is to show the students not only that what they are learning has practical applications, but more importantly, how to apply their understanding of the basic principles to real engineering problems.

SCOPE

Practical examples can be included at all levels of the engineering curriculum. When determining examples to be used for instruction it is important to make the examples as clear and straight forward as possible. The key is to make the examples as simple as possible, and to make sure that they isolate the desired principle. Whenever possible, the examples should be designed so that the students' physical senses are brought into play. Examples that are likely to be enjoyed by the students include those that require them to use their sense of sight, feeling, hearing or smell.

The following guidelines should be remembered when implementing practical examples:

- Understand the example given and be able to explain it. If you cannot provide a clear explanation to the example, the example will confuse the students more than help them.
- Before giving a demonstration or take home assignment, carry out the assignment yourself. This will ensure that you know exactly what the students will "see". It will also help you to anticipate your students' questions. Giving an assignment or demonstration that doesn't work is frustrating to the students and is bad for your credibility.
- Choose examples that are relevant to the students. Examples that the students can observe first hand as opposed to those in a film or on TV are better. Try and find examples that the students can observe on campus or at home. Pull examples from current events like, for instance, explaining the cause for a design failure of a collapsed bridge recently in the news. Explain the basic principles behind a new or commonly used product like the fluid mechanics aspects of a Bernoulli disk drive in a computer.
- Allow ample time in class to discuss the example.
- Consider having the students prepare a written report to document what they have learned. Have them include a list of the basic principles involved.

CATEGORIES AND TYPES OF PRACTICAL EXAMPLES

Practical examples can be grouped into two broad categories:

- Those that help in the Explanation of Theory and New Concepts,
- Those that illustrate the Application of Basic Principles.

In addition, practical examples can also be broken down into different types based on the format in which they are used.

For example, one can design practical examples that are based on:

- Analogies,
- Observations,
- Demonstrations (experimental or mathematical) ,
- Sensing phenomena, and
- Observing secondary effects.

EXPLANATION OF PRACTICAL EXAMPLE TYPES

Analogy

The analogy is a very helpful tool for explaining new concepts. Here, the instructor links the new concept to an idea which the students can easily picture in their minds. An example of an analogy would be to explain the concept of the conservation of energy in terms of money in a bank. One can imagine the money in a checking account as being analogous to kinetic energy. Similarly, money in the savings and money market accounts can be thought of as being analogous to pressure and potential energies, respectively. Just as money can be transferred between the three different accounts, so can energy between the three different forms. The concept of frictional energy losses can now be easily related to the debiting of money from the accounts.

Observations

Observations that the student can make outside of class can help demonstrate basic principles being currently studied in class. The example can be carried out as a take home assignment where the students are required to go and observe a phenomena that they can readily see, feel, hear and smell, and later summarize their observations. The students bring their observations to class and the instructor leads a discussion of what the students observed and what those observations mean. This type of exercise not only helps with the understanding of a new concept or basic principle but teaches the students how to observe a phenomena before trying to analyse it.

Demonstrations (Experimental or Mathematical)

The demonstration example can be done either as an experimental exercise carried out in class with small experimental models, or as a mathematical exercise carried out on the “chalkboard” to explain a physical phenomena. This can be particularly instructive when the students are aware of the phenomena but are not able to explain the science behind it.

Experimental

An experimental demonstration requires physical equipment. While finding the right equipment may not always be possible, some examples require materials as simple as a paper clip or piece of paper. For instance, the factors affecting the aerodynamic drag

and lift forces on an object can be demonstrated with a simple piece of writing paper. Hold a flat sheet of paper parallel to the floor and drop it observing its rate of decent. Then take the same sheet of paper crumple it up, drop it and observe its rate of decent. In both cases you have the same material, the same mass, and the same gravitational force acting on the system. Therefore, these parameters can be eliminated from consideration. By further eliminating other parameters, the students can be lead to understand that the important parameter is the aerodynamic drag acting on the two different objects. Similarly, important governing parameters in other systems could be deduced. For instance, tests could be run with the same object shapes but with different projected areas. By observing how the time of fall depends on the various parameters, the students could arrive at the main governing parameters.

Mathematical

The purpose of a mathematical demonstration would be to explain, using the theory developed in class, the science behind some phenomena that the students have seen or heard of. This can be particularly enlightening if the phenomena is such that everyone knows about it, but few realise what really is happening. For instance, the term valve float in an Internal Combustion engine can be explained by modeling the valve as a train of solid links and springs, and then writing the equations of motion for the valve.

Sensing

Sensing examples are designed so that students can “feel” the science behind the phenomena. The goal here is to have the students carry out experiments that allow them to sense the different parameters that enter into the theory. An excellent example of this would be to study the relationship between speed and torque for a gear system using a ten-speed bicycle.

The students’ assignment would be to flip their ten speed bicycle upside down, switch through all the gear combinations while pedaling it by hand, and physically sense how the speed and torque for a particular gear setting are related. Clearly, the emphasis in this technique is not to teach or explain a new concept but to give a known concept more meaning by having the students sense it.

Secondary Effects

Secondary effects demonstrate the fact that sometimes the explanation of an engineering phenomenon is not obvious. The purpose here is to get the students to really consider all the possible explanations besides the most obvious one. A classic example of this would be the observation of the direction of movement of a helium balloon tied to the floor of a car when the car accelerates. Typically one would expect the balloon to move backwards when the car accelerates due to the inertia of the balloon. This would be the case if a steel ball were to be suspended from the ceiling of a car. In reality, the students will notice that the balloon moves forward as the car accelerates.

An investigation of the forces acting on the balloon can be done either as a homework assignment or as a class discussion. By doing so, the students should eventually come to realise that the balloon is pushed forward by the buoyancy force acting on it. As the car accelerates, the air in the back of the car is compressed slightly, resulting in a density gradient from the front to the rear of the car. The helium in the balloon is lighter than air and therefore experiences a buoyancy force in the horizontal direction.

SHOW AND TELL: REVERSING STUDENT ROLES

The “Show and Tell” technique is another form of the “Practical Examples” technique. However, in this technique the role of the student is reversed to that of a teacher, thereby changing their perspective of the problem. The basic premise of the “Show and Tell” technique is that if one can explain a concept to someone else then he/she truly understands the concept.

SCOPE

A typical “Show and Tell” project would require a student or a group of students to explain a given theory or phenomenon to the rest of the class and also demonstrate a physical example that helps visualize the phenomenon. Almost any example that you can convincingly demonstrate in a classroom would be appropriate for a “Show and Tell” project. However, it should be remembered that as with the case of the “Practical Examples” technique, the concept to be explained by the students should be relatively simple and straight forward. The purpose of this exercise is to challenge the students to come up with a creative solution to the problem at hand without overwhelming them. In order to avoid embarrassing situations and to ensure that the demonstrations are useful to the entire class, it is also important for you to know before hand what the students plan to present.

Example: Explain and Demonstrate the Magnus Effect

The magnus effect is a fluid dynamics phenomena observed when a projectile in flight is spinning. The spinning projectile moves in a direction perpendicular to both it’s main path and rotation axis.

Goal

The student or group would be given the assignment to both explain and demonstrate the magnus effect in class. The level of the expected explanation would depend on the level of the course and the philosophy of the teaching assistant. This particular example could be explained using potential flow theory (mathematically) or more intuitively using the ideal of streamlines and Bernoulli’s principle without rigorous proof.

For the demonstration the students use whatever equipment or apparatus that they can access. In the present case, one possible idea would be to use a cardboard tube from a paper towel or toilet roll and a piece of string. The student would wrap the roll with

the string, stand on a chair holding the free end of the string and let go of the roll, causing it to unwind from the string and move down. In addition to imparting a downward velocity, the unwinding of the roll also causes the roll to spin. When the string is completely unwound, the roll will tend to move perpendicular to the spin axis (horizontal to the floor) as it falls to the floor. The horizontal direction of movement of the roll will depend on the direction of rotation of the roll. The students could demonstrate this by starting with the string wound in clockwise or anti-clockwise directions. Other variations of this demonstration would be to bring in a ping pong ball and paddle and demonstrate the magnus effect by hitting the ball with different types of spin and watching the trajectory of the balls. The same idea could be demonstrated outdoors by pitching a baseball.

CASE STUDIES: BRINGING “REAL-LIFE” SCENARIOS INTO THE CLASSROOM

An engineering case study is an account of an actual engineering activity, event, or problem containing some of the background and complexities actually encountered by a practicing engineer. Since cases are accounts of “real-life” engineering activity, they help the students to better relate theory to the “real-world”. Cases often involve concepts from other disciplines like marketing and management, concepts that a practicing engineer needs to know anyway. In addition, the case method promotes discussion in class and feedback from the students.

Scope

Engineering case studies can be included at all levels of the engineering curriculum. The hardest part of using case studies is finding cases that fit with the class material. This however should not deter you from using the case method as there are over 200 cases in Wendt Library alone, not to mention many more texts on case studies. Professors, other TAs, or contacts in industry are other great ways to find cases. With a little work, it should not be too hard to find a good case for any class.

Strategies

Students are usually given written material regarding a case and asked to read it and answer a series of questions pertaining to various aspects of the case. The students can be required to work either individually or in groups.

The following are some tips to remember when using a case study:

- The case study questions may increase the amount of work the students have to do outside of class. Care must be taken to balance this extra workload against other homework assignments.
- When using case studies found in the library, do not stick to using the questions given with the case. Generate new questions that directly fit the topics covered in class.

- While the questions that are assigned form the basis of the discussion, be prepared with other questions in order to guide the discussion.
- The goal of the case should be kept in mind at all times. Keep the discussion from drifting away from this goal.
- Be thoroughly prepared for the discussion. Poor preparation will lead to frustration among the students.

Example: Design Change for a Walkway

The case study that follows was used in a design course taught in the College of Engineering. The case helped students discuss how changes in a design caused during manufacturing and construction can affect the safety of the overall design. The students considered the following problem: *The on-site engineers want to know if they can change part of a design to ease construction.*

GUIDED DESIGN PROJECTS: INTRODUCING PRACTICAL DESIGN EXPERIENCE IN CLASSROOMS

Guided designs projects aim to bring practical design experience into the classroom. Often conducted over a period of a semester, the projects give students an opportunity to work in a team environment, apply theory learned in the classroom, and learn about industrial design methodologies.

Scope

Guided design projects are appropriate for any level, but are often reserved for junior and senior levels. As with the case method, choosing a project is typically the hardest part. Using guided design projects usually requires a lot of preparation by the TA. One of the best ways to have students appreciate the industrial design methodology is to have them redesign existing systems or products.

Strategies

The following are some tips to remember when using a guided design project:

- Realise that the project is not as important as the thought processes that go into determining a design. It is not important that the students determine an optimum design. What is important, however, is that they experience the design process.
- Starting before the semester, determine the scope of the projects and the goals for the class. It is important that the scope of the project is reasonable, and care must be taken to ensure that the students are not overloaded.
- When possible, divide the design into parts. This spreads the work for the students and the grading duties over the semester. Design teams of two or three students are frequently used. This allows for in-depth projects, reduces the grading load of the TA, and promotes interaction among the students.

Example: Redesign of a Parking Garage Gate

The following guided design project was used in a course on the mechanics of materials. The goal of the project chosen for this class was to redesign a gate at a parking garage so that it deflected on impact from a car. The gate is required to deflect upon impact in order to avoid serious structural damage to the car. The project was divided into three parts so that the work would be spread over the semester. The first part involved brainstorming to determine the best methods to meet the requirements. The students worked together in teams generating ideas. Then the entire class spent part of a lecture discussing the various ideas. In the debate, the TA guided the class towards a consensus about a design that would satisfy all requirements and yet be easy to complete. The student assignment for the first part was to write up and hand in a summary of their two best designs, the advantages and disadvantages of each, and which design that they would choose and why.

The second part of the project involved determining the loads on the gate. This required the students to synthesize knowledge from several courses that they had taken earlier. Also, the TA had to devote extra time in class and during office hours to explain methods for determining loads on a structure. The final part of the project involved actual measurements and safety considerations.

Having determined the loads, the students proceeded to complete the final part of the project where they determined the actual sizes of the various components used in the design. As a last step, the students used failure theories discussed in class to ensure the safety of the design.

OPEN-ENDED LABS: MAKING STUDENTS THINK DEEPER

Open-ended laboratory classes can be broadly defined as classes where the students are encouraged to design their own experiments or devise their own experimental strategy, rather than required to follow a rigid set of experimental guidelines specified elsewhere as in a lab manual, for example.

Scope

Laboratory classes can be made open-ended to varying degrees depending on a number of factors including the autonomy that the TA has in changing the course structure, the facilities available, and the degree of difficulty that the TA perceives to be appropriate for the class.

One can think of three general areas where a laboratory class can be made more open-ended:

- The experiment setup itself where the students design an experimental setup to achieve certain goals,
- The experimental design where the students decide the scheme to be followed for data collection to achieve a prescribed goal, given a certain experimental setup,

- Data analysis and report writing, where the students decide how the data is to be analysed and reported. Additionally, if none of the options are feasible, the TA can consider making the laboratory briefing session open-ended.

Open-ended Experiment Setup

A simple but effective example of an open-ended lab setup comes from a mechanics of materials course. A few weeks into the semester, the TA divided the class into teams of two or three students. As a part of a course project, the teams were required to experimentally test the material properties of any household object that they were interested in, write a report, and present their results to the rest of the class.

The TA initially offered suggestions regarding the different experiments (like tensile testing, compression testing and bending tests) that could be run, different materials that could be tested (like golf balls, beer cans, eggs, plastic bags, scotch tape and fishing line) , while also encouraging the students to exercise their creativity in finding test materials or experiments.

The teams were allowed complete freedom in choosing the equipment, experimental parameters, and the data analysis methods. The TA mostly acted as a consultant, offering hints and suggestions. As expected, the project presented many challenges forcing the students to come up with creative means to solve the problem. For example, how can the thickness of a sheet of plastic be measured?

Student Response

At the end of the semester the students felt satisfied that they actually designed an experiment and performed “hands on” work. It was also satisfying to the TA that the students were able to apply theoretical concepts to practical applications, and develop a sense of appreciation for the difficulties involved in designing and executing an experimental project.

Time Constraints

The biggest problem with this method is that it is time-intensive. For safety reasons, the TA had to be in the lab at all times when students were testing samples.

Open-Ended Experiment Design

As an example of the open-ended experiment design lab, consider the following experimental setup available for performing heat transfer experiments in a junior level chemical engineering laboratory. A large cylindrical tank equipped with a heating coil, a stirrer with adjustable speed of rotation, baffles for efficient mixing, an external heat exchanger through which fluid in the tank can be circulated, a pump to circulate fluid through the external heat exchanger, a thermocouple at the bottom of the tank to measure the temperature, a source of low-pressure steam and equipment to measure the flow rates.

More equipment such as additional thermocouples and insulation are available in the lab stockroom. In the standard lab format, the class is divided into groups and assigned specific experiments to be performed by each group.

The goals of the experiments, the experimental procedure to be followed, the calculations to be performed, and the report format (including specific quantities or observations to be reported) are discussed in detail before the students perform the experiments.

In an attempt to make the lab more open-ended, one of the TAs decided to run the lab as follows:

- Students were asked to briefly study the experimental setup and arrive at an exhaustive list of experiments that could be performed. Clearly, this required the students to recollect all the heat transfer theory that they had studied and to involve themselves in collective brain-storming. The TA merely coordinated the brain-storming exercise, and offered hints and suggestions. For example, if additional thermocouples could be rigged up, the temperature distribution in the tank could be measured. The effect of evaporative losses could be determined if the tanks could be covered with insulating material.
- Next, the students were asked to briefly discuss the goals of each experiment and rank them according to their usefulness with respect to information content or time required to perform the experiment. For example, the students concluded that the experiment to study the effect of steam was more important than the experiment that would measure the heating effect of the pump. Therefore, they decided that they would perform the latter experiment only if time permitted.
- Subsequently, the students were asked to organize themselves into groups and select a subset of the experiments to work on. The groups were asked to briefly sketch the details of each particular experiment such as the flow rate or stirrer speed to be used, or making sure the right parameters are changed during the experiment. The TA acted as a consultant, reviewing the experimental plans proposed by each group and offering suggestions. After a briefing about general safety procedures the groups proceeded with the experiments.
- Finally, the students were required to turn in group reports in whatever format that they thought was appropriate. In addition, each group was required to briefly present their results to the class the following week.

Student Feedback

Student feedback regarding the open-ended lab was encouraging. Most of the students enjoyed the freedom they had in designing the experiments, and the opportunity to exercise their creativity. Some students were initially uncomfortable about the lack of direction and specific instructions, while a few expressed a preference for the standard lab protocol.

TA Response

From the viewpoint of the TA, the open-ended lab was a big success. The lab format forced the students to think deeply about the theoretical concepts that were to be applied, to generate ideas and evaluate them, and to design and successfully execute an experimental plan.

Open-Ended Data Analysis and Report Writing

Students in most lab classes are given detailed instructions regarding data analysis, and report formats such as specific quantities to be reported, and specific figures to be plotted. This traditional approach reduces the workload of the TAs. They have to spend less time explaining data analysis procedures. Grading lab reports with a standard set of protocols is easier. However, it does not allow the students to think deeply about the material or give them an opportunity to express their creativity.

In an effort to make the lab report writing process more open-ended, a COE TA adopted the following strategy:

- Hand out very brief information about expected report formats. Suggest possible calculations, tables, and figures to be presented.
- Have the students work on a rough draft or outline of the report and discuss their report outline with the TA during office hours.
- Suggest modifications and have the students turn in the final draft during lab the following week.

Student Response

As can be easily guessed, the initial student response was marked by frustration and pessimism, for the simple fact that they hadn't done anything like this before. The students' attempts at designing their own reports were half-hearted, and the students required a lot of hand-holding. However, as time progressed, they grew more confident and came to believe that they were capable of producing good quality work.

TA Response

The open-ended report writing format forced the students to think about different report writing schemes and evaluate them for their effectiveness in communicating the results. At the technical level, it forced the students to think deeply about the data analysis procedures and the interpretation of the results, in contrast to merely churning out numbers using a calculator.

Caveats

It should be noted that making the lab report writing process open-ended does not imply a "hands off" policy. Indeed, a TA stands to be perceived by the students as being uncooperative, or more damagingly, as trying to shirk his or her responsibilities, if such a policy were to be adopted. It is important for the TA to provide meaningful and

constructive feedback to the students, lest the students get discouraged and frustrated with the system.

Open-Ended Laboratory Briefing Sessions

Often it is not feasible for the TA to make a lab class open-ended. In this case, the goal of forcing the students to think deeply about the application of theoretical concepts and the design of the experimental plan can be achieved by making the laboratory lecture open-ended.

The thought process that goes into designing an experiment can be simulated in a lecture by adopting the following strategy:

- The TA introduces the goal of the experiment and organizes a brainstorming session by posing questions such as, “The goal of today’s experiment is to measure the viscosity of glycerol as a function of temperature. Can you suggest a few ways of doing this and explain the operating principle?” The TA then collates the ideas generated, has the students discuss the pros and cons. At this point, the TA directs the students’ attention on the technique prescribed in the syllabus by saying, “Now that we have discussed the different possible methods, let us focus on the method that we will be using today, the capillary viscometry technique”.
- In the next stage of the lecture, the TA generates a discussion of the details of the experimental apparatus and the experimental technique by posing questions such as: “Why is the capillary designed in such a fashion”, “How is the pressure drop problem addressed in this design”, “Why and how can we calibrate the viscometers”, “How can we maintain the viscometers at different temperatures”, “How should we process the data”, “What quantities do we plot in order to obtain the viscosity”.
- Finally, after a brief recap of the important aspects of the experiment the students are allowed to proceed with conducting the experiment.

Student Response

Most of the students agree that this technique “makes them think” and “helps them appreciate the science behind the experiments”. It also calls for a lot of class participation by the students. Occasionally, if the call for a discussion does not evoke sufficient responses, the TA may be required to call on students to answer questions or generate ideas. It should be remembered that some students dislike being called on to answer questions in class. Asking students to share ideas with a partner or in small groups first, generates more ideas and makes the students feel more confident about sharing their ideas.

Time Constraints

Organizing brain-storming sessions in lieu of the standard briefing sessions is bound to cut into the time available for conducting the experiments. However, the pay-off in

terms of the improved students' comprehension of the "science behind the experiments" makes the open-ended lecture technique well worth the investment in time.

THE FLOWCHART TECHNIQUE: ORGANIZING THE FLOW OF THOUGHT

The technique of flowcharting, as applied to a classroom scenario, is a tool for precisely and concisely representing the flow of information among various stages in the development of a theoretical concept or in the formulation or analysis of an engineering problem. Flowcharts are a tool to organize the flow of logic and thought in a classroom, much in the way that flowcharts help in presenting the flow of materials between various units of an industrial process.

Scope

Flowcharts are one of the most indispensable tools that practicing engineers use to organize large amounts of information. One cannot imagine even the smallest unit of a manufacturing concern being constructed or operated without the aid of flowcharts. Flowcharts can be equally useful and effective in conveying and presenting engineering information in a classroom or laboratory.

Flowcharts in Laboratory Instruction

Consider the following experiment in a junior level chemical engineering laboratory to determine the viscosity of a solution of glycerol in water using a capillary viscometer. In this method, a known volume of fluid is allowed to flow through a capillary of known dimensions, and the time taken for the fluid to flow a certain distance is related to the viscosity using a proportionality constant called the viscometer constant.

One of the main steps in this experiment is to determine the viscometer constant by running the experiment with a fluid of known viscosity such as sucrose. The sucrose solutions are prepared by dissolving crystals of sucrose in water and, therefore, have to be analysed for the sucrose concentration. An easy method for determining the concentration of sucrose is to measure its density and convert it to a concentration using published correlations. The density in turn is measured using a device called the pycnometer. The pycnometer is a carefully constructed volumetric flask which allows the weight of a constant volume of fluid to be measured and related to the density. The volume of the pycnometer itself needs to be calibrated using a fluid of known density such as distilled water.

The viscosity of sucrose can be interpolated from published tables once the density is known. This in turn allows the viscometer constant, and subsequently the viscosity of glycerol (the goal of the experiment) to be determined. Finally, as a check of consistency, the experimentally determined viscosity of glycerol is compared with published values. The published values of viscosity can be interpolated from tables if the concentration is known. The concentration of glycerol is determined along the same lines as that for sucrose. As an exercise, imagine that you are a student who has just

been briefed about these experiments in the standard lecture format. Now, try to recall the sequence of steps involved in the experiment. Which method is more efficient in communicating the flow of thought and logic in the experiment?!

Student Response

The students strongly agreed that the flow chart technique was more effective in communicating the overall goals of the experiment. In their words, they could now “see more clearly where they were headed”. The students felt more confident doing the experiments since they had a “road map” in front of them. Additionally, the students found that the flowcharts greatly helped them subsequently while analysing the data and writing the lab report.

Comments

It is possibly better to derive the flowchart on the chalk board during the lab briefing session and distribute previously prepared ones at the end of the class. This allows the students to “see where the TA is coming from” or understand why the flowchart looks the way it does. Better yet, the students can be required to create their own flowcharts during the briefing sessions.

Flowcharts in Lectures

Students often express concerns about not being able to understand where a lecture is headed or how the concepts taught in previous lectures relate to the concepts being presented in the current lecture. Flowcharts can be used effectively to address this concern. Flowcharts are very useful in emphasizing the flow of thought in a presentation, or in relating previously covered topics to the present material. One COE TA has made flowcharts an integral part of every lecture. The lecture starts off with the TA partitioning the chalkboard into two parts; one for presenting the lecture and the other for developing a flowchart as the lecture progresses. At the start of the lecture the flowchart consists of a single block where the goals of the lecture are clearly spelled out. As each concept is presented and the lecture proceeds towards the goals, the flowchart is updated by adding more blocks to indicate the significant concepts and arrows connecting the blocks to indicate the flow of thought. Since the flowchart always stays in a corner of the chalkboard, the students are constantly reminded of the goals of the lecture and the concepts that have been covered in the lecture. The flowchart, which is essentially a summary of the lecture, also makes it easy for the TA to recap the lecture at the end of the class. For the same reason, flowcharts come in handy for students to review lecture material before exams.

Student Use of Flowcharts to Tackle Engineering Problems

Flowcharts can be especially helpful in solving open-ended problems. Students can prepare flowcharts alone or in groups to show processes and help solve open-ended

problems. A new module to help students create flowcharts will be available for 1995-96 from the Technical Communication Division, Department of Engineering Professional Development. The module was developed with support from IBM.

OPEN-ENDED QUIZZES: MOVING STUDENTS AWAY FROM MEMORIZATION

The open-ended quiz is intended to stimulate students' creativity and to help students to think deeply about the material covered in lectures. In contrast, straightforward "Given this, calculate that" or "Plug and chug" type of quizzes merely encourage students to memorize equations and formulae.

Scope

As in the case of the other open-ended teaching methods, the scope of the open-ended quiz method is only limited by the creativity of the instructor and the constraints imposed by the system.

Examples

An interesting example of the open-ended quiz method comes from Dr. Felder, professor of chemical engineering at the North Carolina State University. For one of the mid-semester quizzes in a Graduate level course in chemical reactor design, Professor Felder gave a five-week take-home exam that asked students to make up and solve a final examination for the course! The students were told that if they produced a straightforward "Given this and that, calculate that", they would receive a minimum passing grade, and that to receive more credit, they would have to demonstrate a deeper understanding of the material, the ability to apply techniques from other disciplines, and the ability to evaluate the value of a design, product, or system.

The example, while probably one of the most open-ended exercises that can be thought of, is also time consuming. The goal of making students think deeply about the material can be achieved even in a 15 minute quiz. For example, students can be asked to solve problems that do not specify all the information that is needed to arrive at the answer. The students are then forced to think about what other information may be needed, and how they might go about obtaining the needed information.

Some other variants of open-ended questions are:

- Make up a problem which has more information than what is needed to solve the problem. The students must now think about what pieces of information are critical to solving the problem at hand.
- Make up a problem that has the students come up with a list of different ways to accomplish a specific task. For example, come up with different ways to measure the flow rate of water in a pipe. The discussion of the feasibility of various methods will help students acquire the ability to critically evaluate different solutions.

- Have the students assume that they are teaching the class and come up with creative ways to teach that class. In addition to helping the students think creatively, this also provides useful feedback about the direction that the students would like the TA to take.
- Have the students come up with creative quiz questions.
- Have students come up with different analogies to teach difficult technical concepts.
- Assign or make up a problem that requires the students to use their engineering judgment.

BRAINSTORMING: ENCOURAGING CREATIVITY

The brainstorming technique is widely used in industry and academia to encourage participants to generate ideas in an unhindered manner. In an academic context, brainstorming encourages students to participate actively in idea-generation exercises and experience benefits of a multi-dimensional approach to analysing problems or solutions.

Scope

The brainstorming technique is applicable to all levels of the engineering curriculum and to all teaching scenarios - labs, lectures or discussion parts. It is especially useful in design courses since it calls for a multiple-answer-multiple-dimension methodology rather than the usual single-answer approach to problems. The brainstorming technique can be implemented in a number of different ways as follows:

- *Structured*: The whole class is given a topic to discuss and each student is called upon to contribute an idea. The advantage of this method is that all students participate and the more vocal students tend not to dominate the discussions. The disadvantage is that the discussions usually do not flow freely as in an unstructured session, and can make some students feel pressured and uncomfortable.
- *Unstructured*: Students are allowed to contribute ideas as and when they think of them. This approach allows for a freer flow of ideas and a more relaxed environment. The drawbacks to this approach are that it can lead to the students not responding at all or to a few students dominating the discussions.
- *Group*: This is a structured approach except that the class is broken into small groups and each group presents its ideas after an allotted amount of time. The advantage of this method is that the students are likely to be more at ease and willing to express their ideas. This also promotes synergy and communication among the students. One obvious drawback to this method is that it more time-intensive than the other two methods.

Guidelines

The following guidelines should be followed with any of the methods:

- Make sure that everyone agrees on the question or issue being brainstormed. Write it down on a chalkboard.

- Never criticize students' ideas or allow students to criticize each other.
- Do not allow students to reject ideas initially. Ideas should not be weeded out until the brainstorming is completed. This keeps the solution path from becoming prematurely narrowed.
- Write every idea down. Use a flip chart, blackboard, overheads, post-it notes, or other visual methods.
- Use the words of the speaker when recording; do not interpret.

Examples of Brainstorming Applications

Analysis of Lab Equipment

In the ME 370 “Energy Systems Laboratory”, Senior level ME lab course, students run experiments using equipment that have complete computerized data acquisition systems. While data acquisition systems allow the students to collect a large amount of precise data, they also reduce the creativity and thought that is required of the students to run the experiments. To better involve the students in the experiments and to help them think more deeply about the experimental setup, the TA initiated an impromptu brainstorming session on the instrumentation for an Ingersol Rand double-acting reciprocating air compressor.

Specifically, the TA asked the students to consider the following questions:

- What are the important performance parameters needed to characterize the performance of the compressor?
- What measurements are necessary to determine or measure the important performance parameters?
- What instrumentation could be used to take the necessary measurements?

The students initially considered the first question and suggested several ideas. The TA then initiated a discussion of the generated ideas and guided the class towards a consensus about the correct set of parameters. The class then moved on to consider the second and third questions using the same brainstorming format. After discussing the final question the class proceeded to review the actual equipment instrumentation and compare it with the instrumentation scheme that they had arrived at.

The entire brainstorming exercise required only 45 minutes. It served as a review for the students, and proved to be a very interesting learning experience for both the students and the instructor. The students came away with an appreciation for the amount of detail and planning that goes into designing experiments.

Optimal Design of a Journal Bearing

The critical step in optimal design is in determining the proper objective function that is to be minimized or maximized. The objective function is an equation in the independent variables and design parameters of the problem. The objective function is minimized or maximized by varying the independent parameters within the given constraints while

keeping the design parameters constant. Professor Seireg organized a brainstorming session on the optimal design of a journal bearing. The independent variables, the design parameters and, the constraints imposed on the design are summarized in the table below:

Independent Variables (To be optimized)	Length of the bearing - L Radius of the bearing - R Radial clearance between journal and bearing - C
Design Parameters (To be kept constant)	Load to be supported - F Speed (or speed range) bearing must operate at - N
Constraints on Design (Physical or other limitations)	Minimum oil film thickness allowable - h_{min} Maximum oil temperature allowable - T_{max} Bearing stability criteria

The goal of the brainstorming session was to determine the proper objective function $U = f(L, R, C; F, N)$ which can be minimized or maximized by varying the three independent variables L, R and C.

A summary of the class responses are given below:

- Minimize the oil flow rate (Q) required for lubrication
- Minimize the maximum temperature rise of the bearing (DT)
- Minimize the overall volume of the bearing (V)
- Minimize frictional losses encountered while running the bearing. The frictional losses are equivalent to the energy generated in the oil ($= Q.DT$).

All of the solutions were on the right track, but were not complete. At first glance, it appears that the minimization of the friction losses is the best strategy since the function of a bearing is to minimize frictional losses. However, Professor Seireg showed that the best objective function was one that combined two of the four factors suggested by the class. The objective function $U = Q + K.DT$, where K is an empirical constant, accounts for the costs associated with pumping the oil and cooling hot oil back to the original temperature. It can be seen from the description of the problem that the task of determining the optimal design of a journal bearing is not trivial. There are a number of variables and parameters that enter the calculations, and it not very obvious what the best objective function is. However, the process of brainstorming allowed the students to discuss various solutions and understand the pros and cons of each approach.

QUESTION-AND-ANSWER METHOD: ENCOURAGING STUDENT PARTICIPATION

The goal of the question-and-answer method is to draw students into active participation in teaching and learning processes. The technique also encourages students to move beyond memorization to higher levels of learning that require clarification, expansion,

generalization, and inference. Often students are conditioned to simply sit in class, take notes and then study and learn the information on their own. With the question-and-answer methodology, the students can learn in real-time, as they are being taught, which helps them understand and integrate the material better.

Scope

The question-and-answer methodology is useful in any teaching environment. The approach used in applying this methodology is of course influenced by the course being taught and the teaching style of the instructor. This methodology can be used to structure an entire class, where the class consists entirely of a question-and-answer session. This approach is especially useful for discussion parts and exam reviews. Effective and efficient use of this method requires that you have a thorough understanding of the material being discussed. It also requires that you devote time and thought into devising a set of appropriate questions for use during a presentation. It is a good strategy for you to anticipate student responses to your questions so that you can stay “a step ahead” of the students.

Goals

Thoughtful questioning attempts to achieve one or more of the following goals:

- Stimulate analytical thought
- Diagnose student difficulties
- Determine progress towards specific goals
- Motivate students
- Clarify and expand concepts
- Encourage new appreciation and attitudes
- Give specific direction to thinking
- Relate cause to effect
- Encourage student self-evaluation
- Encourage the application of concepts
- To arouse interest and curiosity
- To focus attention on an issue
- Promote thought and understanding of ideas
- Manage or remind students of a procedure.

Guidelines

A well-planned list of questions is only the first step towards achieving the goals. Merely posing questions is not enough to motivate students to move to higher levels of learning.

The following guidelines may be useful in conducting successful question-and-answer sessions:

- Probe deeper after a student answers a question.
- Structure the follow-up questions ahead of time. It is a good idea to write down such questions on note cards or the margins of the lecture notes.

- Use “who”, “what”, “when”, and “where” questions to check information possessed by students. For higher thought levels, use “why” and “how” questions.
- Push students’ responses to “why” and “how” questions to higher levels of thought by asking for more explanation.
- When using questions with individuals, state the question, pause, then call on a student to answer. This leads all students to listen to the question. The pause provides time to think — respect that period of silence.
- Summarize complicated or ambiguous answers to questions.
- Do not embarrass a student by repeatedly asking questions the student is unable to answer.
- Be reasonably lavish in the use of “good”, or other words of praise to students who give correct answers. Avoid making any negative comments after an incorrect answer — this is sure to ensure low response on future questioning.
- Allow students the opportunity to formulate questions in response to answers from you or other students.
- Never use questions as a form of punishment. Do not call on a student just because they forgot their homework, never volunteers, disrupts the class or provides a careless response.

SOFTWARE: INCREASING TEACHING EFFICIENCY

Software tools are widely used today in the classroom and are proving to be very effective teaching aids. One of the benefits of software tools is that they offer a means for visualization of abstract concepts and ideas. If used properly software tools can improve teaching efficiency in many instances. As a teaching tool, computer software can provide a new way to link abstract concepts with tangible visualizations. Specialized computer software offers students the option of self-learning and can also be instrumental in motivating them to learn abstract engineering concepts.

Scope

A variety of software options for classroom use are currently available. These range from software for improving communication with students to specialized software for teaching an entire course. The Computer Aided Engineering (CAE) Center in the College of Engineering and the Department of Information Technology (DoIT) at UWMadison have access to numerous teaching-related software. Any interested TA should contact these units for further information. The CAE also maintains an on-line help system which can be accessed from CAE computers (by typing “caehelp”) , from “gopher”, or from “Wisc-Info”. Meanwhile the TA may consider using the following three types of software in the classroom:

E-mail

Electronic mail offers a way for increased two-way communication between the TA and the students. Mailing lists for distribution of e-mail to an entire class are now

available from the CAE. Effective use of e-mail requires that the students and the TA be diligent about checking their mail and responding to questions. One pitfall with e-mail is the tendency to use it as a substitute for office hours and direct contact. Although this approach may seem to be efficient on the surface, it tends to distance the TA from the students and discourages interaction.

Presentation

Currently available word processors like Word, WordPerfect and AMI Pro and presentation software like Powerpoint, Persuasion and Harvard Graphics offer very powerful presentation capabilities not available only a few years ago. This means that the TA can potentially present more information in a clearer manner and in lesser time. The combination of new presentation software and equipment like optical scanners (available at CAE) gives the TA the ability to select and condense material from a wide variety of sources and present it in a convenient and professional manner. The following caveat is in order when using presentation software:

- Avoid shifting the emphasis of the presentation from the content to the appearance. It must be remembered that in engineering *function should precede form*.

Modeling and Analysis

Software programmes for modeling and analysis of large amounts of data or complex systems are now widely available. One such programme used extensively for lecture and laboratory courses in the Mechanical Engineering department is Engineering Equation Solver (EES) developed by Professor S. Klein. The advantage to EES and other mathematical analysis software such as Mathcad and Matlab is that the instructor can assign homework problems that are more in-depth in nature. These programmes allow students to model systems and see the effect of varying system parameters, thereby, aiding them in developing a deeper understanding of the theory. The advanced computing capabilities of these programmes also allow the TA to demonstrate complicated phenomena in the classroom.

The following are some tips to keep in mind while using such software:

- Completely solve problems before assigning them for homework. Ensure that the problem is reasonable and can be completed by the students in a reasonable amount of time. It is easy to design problems that seem simple on the surface but are actually quite complicated.
- Avoid assigning problems that fit the software. The goal is to solve problems that are relevant to the course and seek software help to do so.

EXAMPLE OF SOFTWARE USE IN A LABORATORY CLASS

The following example is from a junior level ECE laboratory course in electromechanical power conversion. The course is intended to teach the basic principles of electromechanical power conversion and power electronics through the use of studentrun

experiments. The students are given the lab handout detailing the procedures and goals of the experiment. The students are required to setup the experiment and acquire data, process the data, and analyse the results. The experiments involve the use of a number of electronic measuring instruments such as LCR meters (for measuring inductance) , oscilloscopes, electric power supplies, function generators, and other related instruments.

However, a majority of the students taking the course are from Mechanical Engineering who are typically unfamiliar with the detailed functions and use of these instruments. Quite naturally, the students have been unsure about proceeding with the experiments because of this reason. Recently a user-friendly software module was developed by the TAs under the supervision of Professor R. D. Lorenz in the ME department.

The module guides the students through the process of setting up the experiments and collecting the data. Using a graphical approach, the module guides the students in a step-by-step fashion through all phases of the experiment including connecting the equipment, assembling circuits, setting parameters, and collecting the data. The module also gives the students the option of learning more about the concepts behind the experiments as they proceed with collecting information. In addition, the software eliminates the drudgery of recording and manipulating the data since the data collected is now available on the computer in a ready-to-use format.

CONCLUSION: TEACHING IMPROVEMENT: MONITORING YOUR PROGRESS

Just as this Handbook will continue to evolve, so will your teaching style continue to evolve as you strive to match the learning styles of your students. We hope that this Handbook has sparked your creativity and provided you a basic resource from which to share teaching strategies that have worked for others. Developing a variety of teaching strategies will help you and your students build a learning community.

To develop this variety of strategies that will work for you and your students, we suggest that you monitor your progress. The following framework is one that you could use to document strategies you use and assess how they work. The framework follows the familiar “plan, do, check, act cycle” which is a continuous improvement process used in business, industry, government, and higher education.

Plan:

- *Concept:* The concept that you plan to teach
- *Strategy:* The teaching strategy that you plan to use
- *Date:* The day you plan to use the strategy
- *Materials Needed:* The teaching materials that you will need
- *Time Needed:* Plan your teaching activity so that you can accomplish all your goals
- *Feedback:* Decide on a strategy to obtain student feedback. Consider fast feedback, written reports and observing students’ reactions.

Do: Execute your plan:

- *Check:* Review student evaluations
- *Act:* Decide on what you would do next time. Stick with the strategy? Change?

8

CHAPTER

Curriculum

In education, a curriculum is broadly defined as the totality of student experiences that occur in the educational process. The term often refers specifically to a planned sequence of instruction, or to a view of the student's experiences in terms of the educator's or school's instructional goals. In a 2003 study Reys, Reys, Lapan, Holliday and Wasman refer to curriculum as a set of learning goals articulated across grades that outline the intended mathematics content and process goals at particular points in time throughout the K–12 school programme. Curriculum may incorporate the planned interaction of pupils with instructional content, materials, resources, and processes for evaluating the attainment of educational objectives. Curriculum is split into several categories, the explicit, the implicit (including the hidden) , the excluded and the extra-curricular.

Curricula may be tightly standardized, or may include a high level of instructor or learner autonomy. Many countries have national curricula in primary and secondary education, such as the United Kingdom's National Curriculum. UNESCO's International Bureau of Education has the primary mission of studying curricula and their implementation worldwide.

DEFINITIONS AND INTERPRETATIONS

There is no generally agreed upon definition of curriculum. Some influential definitions combine various elements to describe curriculum as follows:

- Kerr defines curriculum as, “All the learning which is planned and guided by the school, whether it is carried on in groups or individually, inside or outside of school.”

- Braslavsky states that curriculum is an agreement among communities, educational professionals, and the State on what learners should take on during specific periods of their lives. Furthermore, the curriculum defines “why, what, when, where, how, and with whom to learn.”
- Outlines the skills, performances, attitudes, and values pupils are expected to learn from schooling. It includes statements of desired pupil outcomes, descriptions of materials, and the planned sequence that will be used to help pupils attain the outcomes.
- The total learning experience provided by a school. It includes the content of courses (the syllabus) , the methods employed (strategies) , and other aspects, like norms and values, which relate to the way the school is organized.
- The aggregate of courses of study given in a learning environment. The courses are arranged in a sequence to make learning a subject easier. In schools, a curriculum spans several grades.
- Curriculum can refer to the entire programme provided by a classroom, school, district, state, or country. A classroom is assigned sections of the curriculum as defined by the school.
- Through the readings of Smith, Dewey, and Kelly, four curriculums could be defined as:
- *Explicit curriculum*: Subjects that will be taught, the identified “mission” of the school, and the knowledge and skills that the school expects successful students to acquire.
- *Implicit curriculum*: Lessons that arise from the culture of the school and the behaviors, attitudes, and expectations that characterize that culture, the unintended curriculum.
- *Hidden curriculum*: Things which students learn, ‘because of the way in which the work of the school is planned and organized but which are not in themselves overtly included in the planning or even in the consciousness of those responsible for the school arrangements (Kelly, 2009) . The term itself is attributed to Philip W. Jackson and is not always meant to be a negative. Hidden curriculum, if its potential is realized, could benefit students and learners in all educational systems. Also, it does not just include the physical environment of the school, but the relationships formed or not formed between students and other students or even students and teachers (Jackson, 1986) .
- *Excluded curriculum*: Topics or perspectives that are specifically excluded from the curriculum.
- *Extracurricular*: May include school-sponsored programmes, which are intended to supplement the academic aspect of the school experience, or community-based programmes and activities. Examples of school-sponsored extracurricular programmes include sports, academic clubs, and performing arts. Community-based programmes and activities may take place at a school (after hours) but

are not linked directly to the school. Community-based programmes frequently expand on the curriculum that was introduced in the classroom. For instance, students may be introduced to environmental conservation in the classroom. This knowledge is further developed through a community-based programme. Participants then act on what they know with a conservation project. Community-based extracurricular activities may include “environmental clubs, 4-H, boy/girl scouts, and religious groups” (Hancock, Dyk, & Jones, 2012).

Curriculum can be ordered into a procedure:

- *Step 1:* Diagnosis of needs.
- *Step 2:* Formulation of objectives.
- *Step 3:* Selection of content.
- *Step 4:* Organization of content.
- *Step 5:* Selection of learning experiences.
- *Step 6:* Organization of learning experiences.
- *Step 7:* Determination of what to evaluate and of the ways and means of doing it.

Under some definitions, curriculum is prescriptive, and is based on a more general syllabus which merely specifies what topics must be understood and to what level to achieve a particular grade or standard. The word Syllabus originates from Greek. The Greek meaning of the word basically means a “concise statement or table of the heads of a discourse, the contents of a treatise, the subjects of series of lectures.

‘Curriculum’ has numerous definitions, which can be slightly confusing. In its broadest sense a curriculum may refer to all courses offered at a school, explicit. The intended curriculum, which the students learn through the culture of the school, implicit. The curriculum that is specifically excluded, like racism. Plus, the extra curricular activities like sports, and clubs. This is particularly true of schools at the university level, where the diversity of a curriculum might be an attractive point to a potential student.

A curriculum may also refer to a defined and prescribed course of studies, which students must fulfill in order to pass a certain level of education. For example, an elementary school might discuss how its curriculum, or its entire sum of lessons and teachings, is designed to improve national testing scores or help students learn the basics. An individual teacher might also refer to his or her curriculum, meaning all the subjects that will be taught during a school year.

On the other hand, a high school might refer to a curriculum as the courses required in order to receive one’s diploma. They might also refer to curriculum in exactly the same way as the elementary school, and use curriculum to mean both individual courses needed to pass, and the overall offering of courses, which help prepare a student for life after high school.

Curriculum can be envisaged from different perspectives. What societies envisage as important teaching and learning constitutes the “intended” curriculum. Since it is usually presented in official documents, it may be also called the “written” and/or

“official” curriculum. However, at classroom level this intended curriculum may be altered through a range of complex classroom interactions, and what is actually delivered can be considered the “implemented” curriculum. What learners really learn (*i.e.* what can be assessed and can be demonstrated as learning outcomes/learner competencies) constitutes the “achieved” or “learned” curriculum. In addition, curriculum theory points to a “hidden” curriculum (*i.e.* the unintended development of personal values and beliefs of learners, teachers and communities; unexpected impact of a curriculum; unforeseen aspects of a learning process) . Those who develop the intended curriculum should have all these different dimensions of the curriculum in view. While the “written” curriculum does not exhaust the meaning of curriculum, it is important because it represents the vision of the society. The “written” curriculum is usually expressed in comprehensive and user-friendly documents, such as curriculum frameworks; subject curricula/syllabuses, and in relevant and helpful learning materials, such as textbooks; teacher guides; assessment guides.

In some cases, people see the curriculum entirely in terms of the subjects that are taught, and as set out within the set of textbooks, and forget the wider goals of competencies and personal development. This is why a curriculum framework is important. It sets the subjects within this wider context, and shows how learning experiences within the subjects need to contribute to the attainment of the wider goals.

There are many common misconceptions of what curriculum is and one of the most common is that curriculum only entails a syllabus. Smith (1996,2000) says that, “A syllabus will not generally indicate the relative importance of its topics or the order in which they are to be studied. Where people still equate curriculum with a syllabus they are likely to limit their planning to a consideration of the content or the body of knowledge that they wish to transmit”. Regardless of the definition of curriculum, one thing is certain. The quality of any educational experience will always depend to a large extent on the individual teacher responsible for it (Kelly, 2009) .

Curriculum is almost always defined with relation to schooling. According to some, it is the major division between formal and informal education. However, under some circumstances it may also be applied to informal education or free-choice learning settings. For instance, a science museum may have a “curriculum” of what topics or exhibits it wishes to cover. Many after-school programmes in the US have tried to apply the concept; this typically has more success when not rigidly clinging to the definition of curriculum as a product or as a body of knowledge to be transferred. Rather, informal education and free-choice learning settings are more suited to the model of curriculum as practice or praxis.

Historical Conception

In the early years of the 20th century, the traditional concepts held of the “curriculum is that it is a body of subjects or subject matter prepared by the teachers for the students to learn.” It was synonymous to the “course of study” and “syllabus”.

In *The Curriculum*, the first textbook published on the subject, in 1918, John Franklin Bobbitt said that curriculum, as an idea, has its roots in the Latin word for *race-course*, explaining the curriculum as the course of deeds and experiences through which children become the adults they should be, *for success in adult society*. Furthermore, the curriculum encompasses the entire scope of formative deed and experience occurring in and out of school, and not only experiences occurring in school; experiences that are unplanned and undirected, and experiences intentionally directed for the purposeful formation of adult members of society. (cf. image at right.)

To Bobbitt, the curriculum is a social engineering arena. Per his cultural presumptions and social definitions, his curricular formulation has two notable features: (i) that scientific experts would best be qualified to and justified in designing curricula based upon their expert knowledge of what qualities are desirable in adult members of society, and which experiences would generate said qualities; and (ii) curriculum defined as the deeds-experiences the student *ought to have* to become the adult he or she *ought to become*.

Hence, he defined the curriculum as an ideal, rather than as the concrete reality of the deeds and experiences that form people to who and what they are.

Contemporary views of curriculum reject these features of Bobbitt's postulates, but retain the basis of curriculum as the course of experience (s) that forms human beings into persons. Personal formation via curricula is studied at the personal level and at the group level, *i.e.* cultures and societies (*e.g.* professional formation, academic discipline via historical experience) . The formation of a group is reciprocal, with the formation of its individual participants.

Although it formally appeared in Bobbitt's definition, curriculum as a course of formative experience also pervades John Dewey's work (who disagreed with Bobbitt on important matters). Although Bobbitt's and Dewey's idealistic understanding of "curriculum" is different from current, restricted uses of the word, curriculum writers and researchers generally share it as common, substantive understanding of curriculum. Development does not mean just getting something out of the mind. It is a development of experience and into experience that is really wanted.

Robert M. Hutchins, president of the University of Chicago, regarded curriculum as "permanent studies" where the rules of grammar, rhetoric and logic and mathematics for basic education are emphasized. Basic education should emphasize 3 Rs and college education should be grounded on liberal education. On the other hand, Arthur Bestor as an essentialist, believes that the mission of the school should be intellectual training, hence curriculum should focus on the fundamental intellectual disciplines of grammar, literature and writing. It should also include mathematics, science, history and foreign language.

This definition leads us to the view of Joseph Schwab that discipline is the sole source of curriculum. Thus in our education system, curriculum is divided into chunks of knowledge we call subject areas in basic education such as English, Mathematics,

Science, Social Studies and others. In college, discipline may include humanities, sciences, languages and many more. Curriculum should consist entirely of knowledge which comes from various disciplines. To learn the lesson is more interesting than to take a scolding, be held up to general ridicule, stay after school, receive degrading low marks, or fail to be promoted.

Thus, curriculum can be viewed as a field of study. It is made up of its foundations (philosophical, historical, psychological, and social foundations); domains of knowledge as well as its research theories and principles. Curriculum is taken as scholarly and theoretical. It is concerned with broad historical, philosophical and social issues and academics. Under a starting definition offered by John Kerr and taken up by Vic Kelly in his standard work on the curriculum, curriculum is “all the learning which is planned and guided by the school, whether it is carried on in groups or individually, inside or outside the school.

Four ways of approaching curriculum theory and practice:

- Curriculum as a body of knowledge to be transmitted.
- Curriculum as an attempt to achieve certain ends in students – products.
- Curriculum as a process.
- Curriculum as praxis.

In recent years the field of education, and therefore curriculum, has expanded outside the walls of the classroom and into other settings such as museums. Within these settings curriculum is an even broader topic, including various teachers such as other visitors, inanimate objects such as audio tour devices, and even the learners themselves. As with the traditional idea of curriculum, curriculum in a free choice learning environment can consist of the explicit stated curriculum and the hidden curriculum, both of which contribute to the learner’s experience and lessons from the experience. These elements are further compounded by the setting, cultural influences, and the state of mind of the learner. Museums and other similar settings are most commonly leveraged within traditional classroom settings as enhancements to the curriculum when educators develop curriculum that encompasses visits to museums, zoos, and aquarium.

PROGRESSIVIST VIEWS

On the other hand, to a progressivist, a listing of school subjects, syllabi, course of study, and list of courses of specific discipline do not make a curriculum. These can only be called curriculum if the written materials are actualized by the learner. Broadly speaking, curriculum is defined as the total learning experiences of the individual. This definition is anchored on John Dewey’s definition of experience and education. He believed that reflective thinking is a means that unifies curricular elements. Thought is not derived from action but tested by application.

Caswell and Campbell viewed curriculum as “all experiences children have under the guidance of teachers.” This definition is shared by Smith, Stanley and Shores when they defined “curriculum as a sequence of potential experiences set up in schools for

the purpose of disciplining children and youth in group ways of thinking and acting.” Curriculum as a process is when a teacher enters a particular schooling and situation with: an ability to think critically, in-action; an understanding of their role and the expectations others have of them; and a proposal for action which sets out essential principles and features of the educational encounter. Guided by these, they encourage conversations between, and with, people in the situation out of which may come a course of thinking and action. Plus, the teacher continually evaluates the process and what they can see of outcomes. Marsh and Willis on the other hand view curriculum as all the “experiences in the classroom which are planned and enacted by teacher, and also learned by the students.

Any definition of curriculum, if it is to be practically effective and productive, must offer much more than a statement about knowledge-content or merely the subjects which schooling is to teach or transmit or deliver. Some would argue of the course that the values implicit in the arrangements made by schools for their pupils are quite clearly in the consciousness of teachers and planners, again especially when the planners are politicians, and are equally clearly accepted by them as part of what pupils should learn in school, even though they are not overtly recognized by the pupils themselves. In other words, those who design curricula deliberately plan the schools’ ‘expressive culture’. If this is the case, then, the curriculum is ‘hidden’ only to or from the pupils, and the values to be learnt clearly from a part of what is planned for pupils. They must, therefore, be accepted as fully a part of the curriculum, and most especially as an important focus for the kind of study of curriculum with which we are concerned here, not least because important questions must be asked concerning the legitimacy of such practices. Currently, a spiral curriculum is promoted as allowing students to revisit a subject matter’s content at the different levels of development of the subject matter being studied. The constructivist approach proposes that children learn best via proactive engagement with the educational environment, *i.e.* learning through discovery.

PRIMARY AND SECONDARY EDUCATION

A curriculum may be partly or entirely determined by an external, authoritative body (*e.g.*, the National Curriculum for England in English schools). Crucial to the curriculum is the definition of the course objectives that usually are expressed as learning outcomes and normally include the program’s assessment strategy. These outcomes and assessments are grouped as units (or modules), and, therefore, the curriculum comprises a collection of such units, each, in turn, comprising a specialised, specific part of the curriculum. So, a typical curriculum includes communications, numeracy, information technology, and social skills units, with specific, specialized teaching of each.

Core curricula are often instituted, at the primary and secondary levels, by school boards, Departments of Education, or other administrative agencies charged with overseeing education. A core curriculum is a curriculum, or course of study, which is deemed central and usually made mandatory for all students of a school or school

system. However, even when core requirements exist, they do not necessarily involve a requirement for students to engage in one particular class or activity. For example, a school might mandate a music appreciation class, but students may opt out if they take a performing musical class, such as orchestra, band, chorus, etc.

Australia

In Australia, the Australian Curriculum took effect nationwide in 2014, after a curriculum development process that began in 2010. Previously, each state's Education Department had traditionally established curricula. The Australian Curriculum consists of one curriculum covering eight subject areas through year 10, and another covering fifteen subjects for the senior secondary years.

Canada

In Canada each province and territory has the authority to create its own curriculum. However, the Northwest Territories and Nunavut both choose to use the Alberta Curriculum for select parts of their curriculum. The territories also use Alberta's standardized tests in some subjects.

South Africa

In South Africa the Caps curriculum is used in Public schools. Private schools use IEB, Cambridge, etc.

South Korea

The National Curriculum of Korea covers kindergarten, primary, and secondary education, as well as special education. The version currently in place is the 7th National Curriculum, which has been revised in 2007 and 2009. The curriculum provides a framework for a common set of subjects through 9th grade, and elective subjects in grades 10 through 12.

Japan

The Japanese educational system is based off traditional values from their heritage with curriculum ideas borrowed from England, Germany, France and the United States. The Japanese curriculum is world-famous. Their math and science standards are among the most demanding in the developed countries. Students in Japan are expected to know more about another country's history, economics, and geography than their own country. Japanese students cannot skip grades and are not held back. They are expected to master the curriculum at every level. Due to their meritocratic nature, all students are funded equitably and follow exactly the same curriculum with the same expectations. Students that are ahead in class are expected to help those that are not. Beyond the academics, students are expected to clean the classrooms and the hallways to teach respect and responsibility.

The Netherlands

The Dutch system is based on directives coming from the Ministry of Education, Culture and Science (OCW) . Primary and secondary education use key objectives to create curricula. For primary education the total number of objectives has been reduced from 122 (back in 1993) to 58 in 2006. Starting in 2009/2010 all key objectives are obligatory for primary education. The key objectives are oriented towards subject areas such as language, mathematics/arithmetic, orientation towards self and the world, art, and physical education. All of the objectives have accompanying concrete activities. Also final exams are determined by the OCW and required. Parts of those exams are taken in a national setting, created by a committee: Centrale examencommissie vaststelling opgaven. Furthermore, OCW will determine the number of hours to be spent per subject. Apart from these directives every school can determine its own curriculum.

Nigeria

In 2005, the Nigerian government adopted a national Basic Education Curriculum for grades 1 through 9. The policy was an outgrowth of the Universal Basic Education programme announced in 1999, to provide free, compulsory, continuous public education for these years. In 2014, the government implemented a revised version of the national curriculum, reducing the number of subjects covered from 20 to 10.

Sweden

In Sweden since 2011, the primary school curriculum is Lgr 11, while secondary schools use Lgy 11.

Scotland

In Scotland, the Curriculum for Excellence (CfE) was introduced in August 2010 in all schools. The national qualifications were introduced in 2013 by the Scottish Qualifications Authority (SQA) . The national qualifications include: Life Skills Coursework (SFL) , National 3 (NAT3) , National 4 (NAT4) , National 5 (NAT5) , Higher, Advanced Higher.

United Kingdom

The National Curriculum was introduced into England, Wales and Northern Ireland as a nationwide curriculum for primary and secondary state schools following the Education Reform Act 1988. Notwithstanding its name, it does not apply to independent schools, which may set their own curricula, but it ensures that state schools of all local education authorities have a common curriculum. Academies, while publicly funded, have a significant degree of autonomy in deviating from the National Curriculum.

The purpose of the National Curriculum was to standardise the content taught across schools to enable assessment, which in turn enabled the compilation of league tables detailing the assessment statistics for each school. These league tables, together with

the provision to parents of some degree of choice in assignment of the school for their child (also legislated in the same act) were intended to encourage a ‘free market’ by allowing parents to choose schools based on their measured ability to teach the National Curriculum.

United States

In the U.S., each state, with the individual school districts, establishes the curricula taught. Each state, however, builds its curriculum with great participation of national academic subject groups selected by the United States Department of Education, *e.g.* National Council of Teachers of Mathematics (NCTM) nctm.org for mathematical instruction. The Common Core State Standards Initiative promulgates a core curriculum for states to adopt and optionally expand upon. This coordination is intended to make it possible to use more of the same textbooks across states, and to move toward a more uniform minimum level of educational attainment.

Higher Education

Many educational institutions are currently trying to balance two opposing forces. On the one hand, some believe students should have a common knowledge foundation, often in the form of a core curriculum; on the other hand, others want students to be able to pursue their own educational interests, often through early specialty in a major, however, other times through the free choice of courses. This tension has received a large amount of coverage due to Harvard University’s reorganization of its core requirements.

Many labor economics studies report that employment and earnings vary by college major and this appears to be *caused* by differences in the labor market value of the skills taught in different majors. Majors also have different labor market value even after students complete graduate degrees such as law degrees or business degrees.

An essential feature of curriculum design, seen in every college catalog and at every other level of schooling, is the identification of prerequisites for each course. These prerequisites can be satisfied by taking particular courses, and in some cases by examination, or by other means, such as work experience. In general, more advanced courses in any subject require some foundation in basic courses, but some coursework requires study in other departments, as in the sequence of math classes required for a physics major, or the language requirements for students preparing in literature, music, or scientific research. A more detailed curriculum design must deal with prerequisites within a course for each topic taken up. This in turn leads to the problems of course organization and scheduling once the dependencies between topics are known.

Russia

Core curriculum has typically been highly emphasized in Soviet and Russian universities and technical institutes.

United States

Core Curriculum

At the undergraduate level, individual college and university administrations and faculties sometimes mandate core curricula, especially in the liberal arts. But because of increasing specialization and depth in the student's major field of study, a typical core curriculum in higher education mandates a far smaller proportion of a student's course work than a high school or elementary school core curriculum prescribes.

Amongst the best known and most expansive core curricula programmes at leading American colleges and universities are that of Columbia University, as well as the University of Chicago's. Both can take up to two years to complete without advanced standing, and are designed to foster critical skills in a broad range of academic disciplines, including: the social sciences, humanities, physical and biological sciences, mathematics, writing and foreign languages.

In 1999, the University of Chicago announced plans to reduce and modify the content of its core curriculum, including lowering the number of required courses from 21 to 15 and offering a wider range of content. When *The New York Times*, *The Economist*, and other major news outlets picked up this story, the University became the focal point of a national debate on education.

The National Association of Scholars released a statement saying, "It is truly depressing to observe a steady abandonment of the University of Chicago's once imposing undergraduate core curriculum, which for so long stood as the benchmark of content and rigor among American academic institutions." Simultaneously, however, a set of university administrators, notably then-President Hugo Sonnenschein, argued that reducing the core curriculum had become both a financial and educational imperative, as the university was struggling to attract a commensurate volume of applicants to its undergraduate division compared to peer schools as a result of what was perceived by the pro-change camp as a reaction by "the average eighteen-year-old" to the expanse of the collegiate core.

As core curricula began to diminish over the course of the twentieth century at many American schools, some smaller institutions became famous for embracing a core curriculum that covers nearly the student's entire undergraduate education, often utilizing classic texts of the western canon to teach all subjects including science. Four Great Books colleges in the United States follow this approach: St. John's, Shimer, Thomas Aquinas, Gutenberg College and Thomas More.

Distribution Requirements

Some colleges opt for the middle ground of the continuum between specified and unspecified curricula by using a system of distribution requirements. In such a system, students are required to take courses in particular fields of learning, but are free to choose specific courses within those fields.

Open Curriculum

Other institutions have largely done away with core requirements in their entirety. Brown University offers the “New Curriculum,” implemented after a student-led reform movement in 1969, which allows students to take courses without concern for any requirements except those in their chosen concentrations (majors) , plus two writing courses. In this vein it is certainly possible for students to graduate without taking college-level science or math courses, or to take only science or math courses. Amherst College requires that students take one of a list of first-year seminars, but has no required classes or distribution requirements. Similarly, Grinnell College requires students to take a First-Year Tutorial in their first semester, and has no other class or distribution requirements. Others include Evergreen State College, Hamilton College, and Smith College.

Wesleyan University is another school that has not and does not require any set distribution of courses. However, Wesleyan does make clear “General Education Expectations” such that if a student does not meet these expectations, he/she would not be eligible for academic honors upon graduation.

GENDER INEQUALITY IN CURRICULA

Gender inequality in curricula shows how men and women are not treated equally in several types of curricula. More precisely, gender inequality is visible in the curriculum of both schools and Teacher Education Institutes (TEIs) . Physical education (PE) is an example where gender equality issues are highlighted because of preconceived stereotyping of boys and girls. The general belief is that boys are better at physical activities than girls, and that girls are better at ‘home’ activities such as sewing and cooking. This is the case in many cultures around the world and is not specific to one culture only.

9

CHAPTER

Peer Education

Peer education is an approach to health promotion, in which community members are supported to promote health-enhancing change among their peers. Peer education is the teaching or sharing of health information, values and behaviour in educating others who may share similar social backgrounds or life experiences. Rather than health professionals educating members of the public, the idea behind peer education is that ordinary lay people are in the best position to encourage healthy behaviour to each other.

AREAS OF APPLICATION

Peer education has become very popular in the broad field of HIV prevention. It is a mainstay of HIV prevention in many developing countries, among groups including young people, sex workers, men who have unprotected sex with men, or people who use intravenous drugs. Peer education is also associated with efforts to prevent tobacco, alcohol and other drug use among young people. Peer educators can be effective role models for young adolescents by promoting healthy behaviour, helping to create and reinforce social norms that support safer behaviors, and also serve as an accessible and approachable health education resource both inside and outside the classroom. Peer education is useful in promoting healthy eating, food safety and physical activity amongst marginalized populations. Peer education is also favorably used in medical education.

THE PROCESS

A peer education programme is usually initiated by health or community professionals, who recruit members of the *target* community to serve as peer educators. The recruited

peer educators are trained in relevant health information and communication skills. Armed with these skills, the peer educators then engage their peers in conversations about the issue of concern, seeking to promote health-enhancing knowledge and skills. The intention is that familiar people, giving locally-relevant and meaningful suggestions, in appropriate local language and taking account of the local context, will be most likely to be able to promote health-enhancing behaviour change.

There is a great variety in the support provided to peer educators. Sometimes they are unpaid volunteers, sometimes they are given a small honorarium, sometimes they receive a reasonable salary. The peer educators may be supported by regular meetings and training, or expected to continue their work without formal supports.

Theories

A variety of theories are offered regarding the question of how peer education is supposed to achieve positive results. Positive results of a Peer Educator are achieved through educating individuals to enhance their understanding about diseases, alcohol and drug abuse etc. and effectively communicate what they've learned to other at-risk peers in their community. The goal is to educate and reach as many people to limit the spread of disease and to learn how to overcome drug and alcohol abuse etc.

Kelly's Popular Opinion Leader Theory

The popular opinion leader theory suggests a parallel between peer education and the marketing of commercial products. Peer educators are seen as *opinion leaders*—respected and admired by other members of the community. These opinion leaders espouse a certain lifestyle (such as safer sex, or not smoking, etc.) —and their peers wish to emulate them.

Critical Consciousness

Campbell argues that what peer education ought to do is to promote the kind of critical consciousness theorised by Paulo Freire. This means that peers use the peer education process to critically discuss their circumstances, especially the social factors impacting upon their health. Becoming critically aware of these forces is the first step to tackling them. So, for instance, if local norms regarding sexuality and gender put people's health at risk, this approach argues that peers should critically discuss those norms, so that they can then collectively seek to establish new more health-enhancing norms.

Social Learning Theory

Based on the work of Bandura and colleagues, social learning theory claims that modelling is an important component of the learning process. In the most basic sense, people observe behaviour taking place and then go on to adopt similar behaviour. Participants require the opportunity to practice modelled behaviour and positive reinforcement if it is to be adopted successfully.

Differential Association Theory

Based on the work of Sutherland and Cressy, differential association theory has been applied to the study of crime. Rather than the result of biological or psychological disorders, crime is a learned behaviour. This learning happens in social situations by associating with those who can teach the necessary skills and techniques needed. Through this theory it can be understood that peers can be very influential for both positive and negative behaviours. Young people can learn both good and bad habits from each other. In differential association theory the mere association with others provides a learning opportunity. If social learning theory is essentially psychological, differential association theory is essentially sociological.

Role Theory

Sarbin argues that peer educators will adapt to the role expectations of a tutor and behave appropriately. Furthermore, through adopting a role, individuals develop a deeper understanding and commitment to it. The potential is that Peer educators can develop a stronger commitment and a greater appreciation of the relevance of the health topic. Role theory is also based on the premise that communication can be blocked by differences in culture between the teacher and learner. Peer educators who have a similar set of experiences and culture are therefore likely to be more effective in promoting learning.

Communication of Innovations Theory

Developed by Rogers and Shoemaker, the communication of innovations theory explains how innovations come to be adopted by communities and what factors influence the rate of adoption. These factors include the characteristics of those who adopt the innovation, the nature of the social system, the characteristics of the innovation and the characteristics of change agents. Rogers and Shoemaker argue that all innovations follow a similar pattern of adoption, with one group of people—the *innovators*—taking it up immediately. Then there are *early adopters*, the *early majority*, the *late majority* and finally the *laggards*, including some who never adopt the innovation. In this theory key people influence the opinion leaders within a community. Change agents can be viewed as health professionals while opinion leaders correlate with peer educators. Rogers and Shoemaker argue that effective communication occurs when the source and receiver are *homophilous*, that is, are similar in certain attributes. These include beliefs, values, education and social status. This would suggest that peers communicate better than those who are unequal or different.

Support

Peer educators are seen as credible sources of information. This has been shown to be particularly effective amongst the youth population. Peers and peer education are an important influence and approach in changing health behaviours.

One of the beliefs of peer education is that it is cost effective. Peer education has been identified as a more economical way to deliver health training.

A team of peer educators can extend health promotion outreach and be more accessible than paid health professionals. Peer educators help to bridge many of the gaps in service that occur through fear and suspicion of official health care providers, and to facilitate effective communication with community members and professional provider. Engaging youth peer educators helps professionals to extend their outreach of programmes and services to ensure their efforts are impactful.

Peer education is empowering from both the standpoint of the peer educator and the individual receiving service. Peer education has been operative in encouraging knowledge, attitudes and intention to change behaviour in AIDS prevention. Furthermore, nondirective peer support has been identified as the best way to motivate individuals in the preparation, action, or maintenance stages of readiness to change. Researchers have acknowledged that trained peer tutors were more effective than the untrained peers in influencing positive health outcomes. Peer education offers the educators the opportunity to benefit from taking on meaningful roles. Peer educators can act as enthusiastic advocates for the programme and have a sense of purpose in their community outreach efforts.

Peer education is sustainable. This has been found to be an important issue for community-based health promotion interventions to make a difference over time. A grassroots initiative involving volunteers means that the health issue is reaching the target audience continuously with less of a threat of financial cutbacks impacting on their work. Research findings support the use of volunteer peer educators as a feasible and effective healthcare delivery strategy and as having promising indicators of sustainability over time. Sustainability through the engagement of peer educators can strengthen the social environment so that it is supportive of healthy behaviors.

Debates

Despite its popularity, the evidence about peer education is mixed, and there is no consensus on whether it works or how it works. Researchers have questioned the validity of the assumption that peer education influences behaviour.

One important line of inquiry suggests that peer education may work in some contexts but not in others. A study comparing peer education among sex workers in India and South Africa found that the more successful Indian group benefited from a supportive social and political context, and a more effective community development ethos, rather than the biomedical focus of the South African intervention.

A key issue concerns what a peer is and who defines this. In some instances age is a central factor but in other contexts, commonalities such as status may be more relevant. Caution has been noted regarding selection of peer educators. Some argue that there can be a stigma held against peer educators who have faced adversities in their own lives, particularly by mainstream health service organizations and professionals.

Alternatively, peers educators would need to have high status within their social group to be effective. Researchers have argued that peer educators sometimes receive inadequate training, which limits their ability to educate their peers effectively and further state that peer selection and training is very important.

An important analysis on the development of many peer education projects is that it is led by adult constructions of adolescence and adolescent health behaviour. A central question should therefore be whose agenda is being served by using peer education projects which manipulate and exploit the social worlds of young people?

10

CHAPTER

Educational Values

VALUES: MEANING, DEFINITION, CONCEPTS OF VALUES

MEANING

In ethics, value denotes the degree of importance of some thing or action, with the aim of determining what actions are best to do or what way is best to live (normative ethics) , or to describe the significance of different actions. It may be described as treating actions themselves as abstract objects, putting value to them. It deals with right conduct and living a good life, in the sense that a highly, or at least relatively highly, valuable action may be regarded as ethically “good” (adjective sense) , and an action of low in value, or somewhat relatively low in value, may be regarded as “bad”. What makes an action valuable may in turn depend on the ethic values of the objects it increases, decreases or alters. An object with “ethic value” may be termed an “ethic or philosophic good” (noun sense) .

Values can be defined as broad preferences concerning appropriate courses of action or outcomes. As such, values reflect a person’s sense of right and wrong or what “ought” to be. “Equal rights for all”, “Excellence deserves admiration”, and “People should be treated with respect and dignity” are representative of values.

Values tend to influence attitudes and behaviour. Types of values include ethical/moral values, doctrinal/ideological (religious, political) values, social values, and aesthetic values. It is debated whether some values that are not clearly physiologically

determined, such as altruism, are intrinsic, and whether some, such as acquisitiveness, should be classified as vices or virtues.

DEFINITION

- Important and lasting beliefs or ideals shared by the members of a culture about what is good or bad and desirable or undesirable. Values have major influence on a person's behaviour and attitude and serve as broad guidelines in all situations. Some common business values are fairness, innovation and community involvement.
- The monetary worth of something in areas such as accounting, economics, marketing or mathematics.

CONCEPTS OF VALUES

Ethical value may be regarded as a study under ethics, which, in turn, may be grouped as philosophy. Similar to that ethics may be regarded as a subfield of philosophy, *ethical value* may be regarded as a subgroup of the more broad (and vague) philosophic value. Ethical value denotes something's degree of importance, with the aim of determining what action or life is best to do, or at least attempt to describe the value of different actions. It may be described as treating actions themselves as abstract objects, putting value to them. It deals with right conduct and good life, in the sense that a highly, or at least relatively highly, valuable action or may be regarded as *good*, and an action of low, or at least relatively low, value may be regarded as *bad*. The study of ethical value is also included in value theory. In addition, values have been studied in various disciplines: anthropology, behavioral economics, business ethics, corporate governance, moral philosophy, political sciences, social psychology, sociology and theology.

Similar Concepts

Ethical value is sometimes used synonymously with goodness. However, goodness has many other meanings as well, and may be regarded as more ambiguous.

Personal versus Cultural Perspectives

Personal values exist in relation to cultural values, either in agreement with or divergence from prevailing norms. A culture is a social system that shares a set of common values, in which such values permit social expectations and collective understandings of the good, beautiful and constructive. Without normative personal values, there would be no cultural reference against which to measure the virtue of individual values and so cultural identity would disintegrate.

Personal Values

Personal values provide an internal reference for what is good, beneficial, important, useful, beautiful, desirable and constructive. Values generate behaviour and influence

the choices made by an individual. Values may help common human problems for survival by comparative rankings of value, the results of which provide answers to questions of why people do what they do and in what order they choose to do them. Moral, religious, and personal values, when held rigidly, may also give rise to conflicts that result from a clash between differing world views.

Over time the public expression of personal values that groups of people find important in their day-to-day lives, lay the foundations of law, custom and tradition. Recent research has thereby stressed the *implicit nature of value communication*.

Cultural Values

Individual cultures emphasize values which their members broadly share. One can often identify the values of a society by noting which people receive honor or respect. In the United States of America, for example, professional athletes at the top levels in some sports receive more honor (measured in terms of monetary payment) than university professors. Surveys show that voters in the United States would not willingly elect an atheist as president, suggesting belief in a God as a generally shared value.

Values clarification differs from cognitive moral education:

- Value clarification consists of “helping people clarify what their lives are for and what is worth working for. It encourages students to define their own values and to understand others’ values.”
- Cognitive moral education builds on the belief that students should learn to value things like democracy and justice as their moral reasoning develops.

Values relate to the norms of a culture, but they are more global and abstract than norms. Norms provide rules for behaviour in specific situations, while values identify what should be judged as good or evil. While norms are standards, patterns, rules and guides of expected behaviour, values are abstract concepts of what is important and worthwhile. Flying the national flag on a holiday is a norm, but it reflects the value of patriotism. Wearing dark clothing and appearing solemn are normative behaviors to manifest respect at a funeral. Different cultures reflect values differently and to different levels of emphasis. “Over the last three decades, traditional-age college students have shown an increased interest in personal well-being and a decreased interest in the welfare of others.” Values seemed to have changed, affecting the beliefs, and attitudes of the students.

Members take part in a culture even if each member’s personal values do not entirely agree with some of the normative values sanctioned in that culture. This reflects an individual’s ability to synthesize and extract aspects valuable to them from the multiple subcultures they belong to.

If a group member expresses a value that seriously conflicts with the group’s norms, the group’s authority may carry out various ways of encouraging conformity or stigmatizing the non-conforming behaviour of that member. For example, imprisonment can result from conflict with social norms that the state has established as law.

Furthermore, institutions in the global economy can genuinely respect values which are of three kinds based on a “triangle of coherence”. In the first instance, a value may come to expression within the World Trade Organization (WTO) , as well as (in the second instance) within the United Nations – particularly in the Educational, Scientific and Cultural Organization (UNESCO) – providing a framework for global legitimacy through accountability. In the third instance, the expertise of member-driven international organizations and civil society depends on the incorporation of flexibility in the rules, to preserve the expression of identity in a globalized world.

Nonetheless, in warlike economic competition, differing views may contradict each other, particularly in the field of culture. Thus audiences in Europe may regard a movie as an artistic creation and grant it benefits from special treatment, while audiences in the United States may see it as mere entertainment, whatever its artistic merits. EU policies based on the notion of “cultural exception” can become juxtaposed with the policy of “cultural specificity” on the liberal Anglo-Saxon side. Indeed, international law traditionally treats films as property and the content of television programmes as a service. Consequently, cultural interventionist policies can find themselves opposed to the Anglo-Saxon liberal position, causing failures in international negotiations.

TRANSMISSION

Values are generally received through cultural means, especially transmission from parents to children. Parents in different cultures have different values. For example, parents in a hunter–gatherer society or surviving through subsistence agriculture value practical survival skills from a young age. Many such cultures begin teaching babies to use sharp tools, including knives, before their first birthdays. Italian parents value social and emotional abilities and having an even temperament. Spanish parents want their children to be sociable. Swedish parents value security and happiness. Dutch parents value independence, long attention spans, and predictable schedules. American parents are unusual for strongly valuing intellectual ability, especially in a narrow “book learning” sense. The Kipsigis people of Kenya value children who are not only smart, but who employ that intelligence in a responsible and helpful way, which they call *ng’om*. Luos of Kenya value education and pride which they call “nyadhi”. The reception of values can be regarded as a part of socialization.

Properties and Forms

Relative or Absolute

Relative values differ between people, and on a larger scale, between people of different cultures. On the other hand, there are theories of the existence of *absolute values*, which can also be termed *noumenal values* (and not to be confused with mathematical absolute value) . An absolute value can be described as philosophically absolute and independent of individual and cultural views, as well as independent of whether it is known or

apprehended or not. Ludwig Wittgenstein was pessimistic towards the idea that an elucidation would ever happen regarding the absolute values of actions or objects; “*we can speak as much as we want about “life” and “its meaning,” and believe that what we say is important. But these are no more than expressions and can never be facts, resulting from a tendency of the mind and not the heart or the will*”.

Intrinsic or Extrinsic

Philosophic value may be split into *instrumental value* and *intrinsic values*. An instrumental value is worth having as a means towards getting something else that is good (e.g., a radio is instrumentally good in order to hear music) . An intrinsically valuable thing is worth for itself, not as a means to something else. It is giving value intrinsic and extrinsic properties. An *ethic good* with *instrumental value* may be termed an ethic mean, and an *ethic good* with *intrinsic value* may be termed an end-in-itself. An object may be both a mean and end-in-itself.

Summation

Intrinsic and instrumental goods are not mutually exclusive categories. Some objects are both good in themselves, and also good for getting other objects that are good. “Understanding science” may be such a good, being both worthwhile in and of itself, and as a means of achieving other goods. In these cases, the sum of instrumental (specifically the all instrumental value) and intrinsic value of an object may be used when putting that object in value systems, which is a set of consistent values and measures.

Intensity

The *intensity* of philosophic value is the degree it is generated or carried out, and may be regarded as the prevalence of the good, the object having the value. It should not be confused with the amount of value per object, although the latter may vary too, e.g. because of instrumental value conditionality. For example, taking a fictional life-stance of accepting waffle-eating as being the end-in-itself, the intensity may be the speed that waffles are eaten, and is zero when no waffles are eaten, e.g. if no waffles are present. Still, each waffle that had been present would still have value, no matter if it was being eaten or not, independent on intensity.

Instrumental value conditionality in this case could be exemplified by every waffle not present, making them less valued by being far away rather than easily accessible. In many life stances it is the product of value and intensity that is ultimately desirable, i.e. not only to generate value, but to generate it in large degree. Maximizing lifestances have the highest possible intensity as an imperative.

Homology in Physics

When comparing to the homologous measure in physics, then intensity in physics may not be the best example, but may better be described as its area. In this sense,

power in physics may be compared to the amount of value per object, and physical intensity the product of value per object and ethic intensity. If there is no physical area, then no energy is generated, regardless of physical power. In the same way, if there is no ethic intensity, then no total value is generated, regardless of value per object.

Positive and Negative Value

There may be a distinction between positive and negative philosophic or ethic value. While positive ethic value generally correlates with something that is pursued or maximized, negative ethic value correlates with something that is avoided or minimized. Negative value may be both intrinsic negative value and/or instrumental negative value.

Protected Value

A *protected value* (also sacred value) is one that an individual is unwilling to trade off no matter what the benefits of doing so may be. For example, some people may be unwilling to kill another person, even if it means saving many others individuals. Protected values tend to be “intrinsically good”, and most people can in fact imagine a scenario when trading off their most precious values would be necessary. If such trade-offs happen between two competing protected values such as killing a person and defending your family they are called *tragic trade-offs*. Protected values have been found to be play a role in protracted conflicts (*e.g.*, the Israeli-Palestinian conflict) because they can hinder businesslike (“utilitarian”) negotiations. A series of experimental studies directed by Scott Atran and Ángel Gómez among combatants on the ISIS frontline in Iraq and with ordinary citizens in Western Europe suggest that commitment to sacred values motivate the most “devoted actors” to make the costliest sacrifices, including willingness to fight and die, as well as a readiness to forsake close kin and comrades for those values if necessary. From the perspective of utilitarianism, protected values are biases when they prevent utility from being maximized across individuals. According to Jonathan Baron and Mark Spranca, protected values arise from norms as described in theories of deontological ethics (the latter often being referred to in context with Immanuel Kant). The protectedness implies that people are concerned with their participation in transactions rather than just the consequences of it.

Value System

A *value system* is a set of consistent values used for the purpose of ethical or ideological integrity.

Consistency

As a member of a society, group or community, an individual can hold both a personal value system and a communal value system at the same time. In this case, the two value systems (one personal and one communal) are externally consistent provided they bear no contradictions or situational exceptions between them.

A value system in its own right is internally consistent when

- its values *do not contradict* each other and
- Its exceptions are or could be
 - *Abstract* enough to be used in all situations and
 - *Consistently* applied.

Conversely, a value system by itself is internally inconsistent if:

- Its values *contradict* each other and
- Its exceptions are
 - Highly *situational* and
 - *Inconsistently* applied.

Value Exceptions

Abstract exceptions serve to reinforce the ranking of values. Their definitions are generalized enough to be relevant to any and all situations. Situational exceptions, on the other hand, are ad hoc and pertain only to specific situations. The presence of a type of exception determines one of two more kinds of value systems:

- An idealized value system is a listing of values that lacks exceptions. It is, therefore, absolute and can be codified as a strict set of proscriptions on behaviour. Those who hold to their idealized value system and claim no exceptions (other than the default) are called *absolutists*.
- A realized value system contains exceptions to resolve contradictions between values in practical circumstances. This type is what people tend to use in daily life.

The difference between these two types of systems can be seen when people state that they hold one value system yet in practice deviate from it, thus holding a different value system. For example, a religion lists an absolute set of values while the practice of that religion may include exceptions. Implicit exceptions bring about a third type of value system called a formal value system. Whether idealized or realized, this type contains an implicit exception associated with each value: “as long as no higher-priority value is violated”. For instance, a person might feel that lying is wrong. Since preserving a life is probably more highly valued than adhering to the principle that lying is wrong, lying to save someone’s life is acceptable. Perhaps too simplistic in practice, such a hierarchical structure may warrant explicit exceptions.

Conflict

Although sharing a set of common values, like hockey is better than baseball or ice cream is better than fruit, two different parties might not rank those values equally. Also, two parties might disagree as to certain actions are right or wrong, both in theory and in practice, and find themselves in an ideological or physical conflict. Ethnomonics, the discipline of rigorously examining and comparing value systems, enables us to understand politics and motivations more fully in order to resolve conflicts. An example

conflict would be a value system based on individualism pitted against a value system based on collectivism. A rational value system organized to resolve the conflict between two such value systems might take the form below. Note that added exceptions can become recursive and often convoluted.

- Individuals may act freely unless their actions harm others or interfere with others' freedom or with functions of society that individuals need, provided those functions do not themselves interfere with these proscribed individual rights and were agreed to by a majority of the individuals.
- A society (or more specifically the system of order that enables the workings of a society) exists for the purpose of benefiting the lives of the individuals who are members of that society. The functions of a society in providing such benefits would be those agreed to by the majority of individuals in the society.
- A society may require contributions from its members in order for them to benefit from the services provided by the society. The failure of individuals to make such required contributions could be considered a reason to deny those benefits to them, although a society could elect to consider hardship situations in determining how much should be contributed.
- A society may restrict behaviour of individuals who are members of the society only for the purpose of performing its designated functions agreed to by the majority of individuals in the society, only insofar as they violate the aforementioned values. This means that a society may abrogate the rights of any of its members who fails to uphold the aforementioned values.

Economic and Philosophic Value

Philosophical value is distinguished from economic value, since it is independent on some other desired condition or commodity. The economic value of an object may rise when the exchangeable desired condition or commodity, *e.g.* money, become high in supply, and vice versa when supply of money becomes low. Nevertheless, economic value may be regarded as a result of philosophical value. In the subjective theory of value, the personal philosophic value a person puts in possessing something is reflected in what economic value this person puts on it. The limit where a person considers to purchase something may be regarded as the point where the *personal philosophic value* of possessing something exceeds the personal philosophic value of what is given up in exchange for it, *e.g.* money. In this light, everything can be said to have a "personal economic value" in contrast to its "societal economic value."

VALUE EDUCATION: NEED, IMPORTANCE AND OBJECTIVES

Value education is rooted in Indian philosophy and culture and ingrained in every tradition of Indian culture. Educational institutions play a significant role in the promotion of

value. The Vedas and Upanishads form the source of inspiration for value education. In the Vedic period, In Ashram education, the Guru insists his sishya to follow certain values throughout his life.

Socialist, Secular, Democratic, Justice, Liberty Equality, Fraternity, Dignity of the individuals and integrity of the nation are the ideal conditions in the Constitution. Our values in life must draw their inspiration from these ideals. University education commission 1948-49 mentioned the various aspects of morality as: loyalty, courage, discipline, self-sacrifice and spirituality.

The Secondary Education Commission 1952-53 laid special emphasis on the following values in the formation of character of the students:

- Efficiency
- Good Temper
- Cooperation
- Integrity
- Discipline.

NEED FOR VALUE EDUCATION

- Moral awareness should be endorsed to orient the progress in science & technology towards the welfare of mankind.
- Common values should be re-discovered to unite human beings with the general decline of traditional values.
- Teachers pass values to the students both consciously and unconsciously through their conduct in and out of class rooms. Therefore the need for a consciously planned value education programme is obvious to establish a formal learning.
- The students might face more complicated decision making situations about issues involving values. They should be helped in developing the ability to make proper choices in such situations through value education.
- Increase in Juvenile delinquency is a crisis to youth who under goes the process of personal growth. In such situation value education assumes a special significance.
- Value Education awakens curiosity, development of proper interests, attitudes, values and capacity to think and judge about oneself.
- Value Education helps in Promoting Social and Natural Integration.

IMPORTANCE AND OBJECTIVES OF VALUE EDUCATION

Value education should aim at the development of values of the following type.

- Scientific temper of mind.
- Large heartedness.
- Co-operation.
- Tolerance
- Respect for the culture of other groups.

MORAL VALUES: NEED AND THEORIES OF VALUES

Moral values are set of principles guiding us to evaluate that what is right or wrong. It builds the personal character, governs behaviour and choices of a person as well as helps his/her to justify decisions, goals, and actions all through the life. Moral values are something that teaches us good virtues including honesty, truthfulness, kindness, integrity, helpfulness, compassion, respectfulness, love, respect for others, hard-work, co-operation, forgiveness, etc.

Moral values should be taught to everyone especially the younger generations from childhood by the parents at home and teachers in the schools. Future of any student depends on the moral values imparted to him during the student's life. Thus, lessons regarding moral values should be well implemented and taught to the students in school and colleges. Teaching those moral values in the schools should not be neglected in any condition as it is not good for the individual, society or country. How a person without knowing moral values can be able to distinguish the differences between right and wrong.

NEED AND VALUE THEORY

Value theory is a range of approaches to understanding how, why, and to what degree persons value things; whether the object or subject of valuing is a person, idea, object, or anything else.

This investigation began in ancient philosophy, where it is called axiology or ethics. Early philosophical investigations sought to understand good and evil and the concept of "the good". Today, much of value theory aspires to the scientifically empirical, recording what people do value and attempting to understand why they value it in the context of psychology, sociology, and economics.

At the general level, there is a difference between moral and natural goods. Moral goods are those that have to do with the conduct of persons, usually leading to praise or blame. Natural goods, on the other hand, have to do with objects, not persons. For example, the statement "Mary is a good person" represents a very different sense of the word 'good' than the statement "That was some good food".

Ethics is mainly focused on moral goods rather than natural goods, while economics has a concern in what is economically good for the society but not an individual person and is also interested in natural goods. However, both moral and natural goods are equally relevant to goodness and value theory, which is more general in scope.

Sociology

In sociology, value theory is concerned with personal values which are popularly held by a community, and how those values might change under particular conditions. Different groups of people may hold or prioritize different kinds of values influencing

social behaviour. Major Western theorists who stress the importance of values as an analytical independent variable include Max Weber, Émile Durkheim, Talcott Parsons, and Jürgen Habermas. Classical examples of sociological traditions which deny or downplay the question of values are institutionalism, historical materialism (including Marxism) , behaviorism, pragmatic-oriented theories, postmodern philosophy and various objectivist-oriented theories. Methods of study range from questionnaire surveys to participant observation.

Economics

Economic analysis emphasizes goods sought in a market and tends to use the consumer's choices as evidence (revealed preference) that various products are of economic value. In this view, religious or political struggle over what "goods" are available in the marketplace is inevitable, and consensus on some core questions about body and society and ecosystems affected by the transaction, are outside the market's goods so long as they are unowned.

However, some natural goods seem to also be moral goods. For example, those things that are owned by a person may be said to be natural goods, but over which a particular individual (s) may have moral claims. So it is necessary to make another distinction: between moral and non-moral goods. A non-moral good is something that is desirable for someone or other; despite the name to the contrary, it may include moral goods. A moral good is anything which an actor is considered to be morally obligated to strive toward.

When discussing non-moral goods, one may make a useful distinction between inherently serviced and material goods in the marketplace (or its exchange value) , versus perceived intrinsic and experiential goods to the buyer. A strict service economy model takes pains to distinguish between the goods and service guarantees *to the market*, and that of the service and experience *to the consumer*.

Sometimes, moral and natural goods can conflict. The value of natural "goods" is challenged by such issues as addiction. The issue of addiction also brings up the distinction between economic and moral goods, where an economic good is whatever stimulates economic growth. For instance, some claim that cigarettes are a "good" in the economic sense, as their production can employ tobacco growers and doctors who treat lung cancer. Many people would agree that cigarette smoking is not *morally* "good", nor *naturally* "good," but still recognize that it is economically good, which means, it has exchange value, even though it may have a negative public good or even be bad for a person's body (not the same as "bad for the person" necessarily – consider the issue of suicide.)

In Ecological Economics value theory is separated into two types: Donor-type value and receiver-type value. Ecological economists tend to believe that 'real wealth' needs a donor-determined value as a measure of what things were needed to make an item or generate a service. (H.T. Odum 1996) . An example of receiver-type value is 'market

value', or 'willingness to pay', the principal method of accounting used in neo-classical economics. In contrast both, Marx's Labour Theory of Value and the 'Emergy' concept are conceived as donor-type value. Emergy theorists believe that this conception of value has relevance to all of philosophy, economics, sociology and psychology as well as Environmental Science.

Ethics and Axiology

Intuitively, theories of value must be important to ethics. A number of useful distinctions have been made by philosophers in the treatment of value.

Intrinsic and Instrumental Value

Many people find it useful to distinguish *instrumental value* and *intrinsic values*, first discussed by Plato in the "Republic". An instrumental value is worth having as a means towards getting something else that is good (*e.g.*, a radio is instrumentally good in order to hear music). An intrinsically valuable thing is worth having for itself, not as a means to something else. It is giving value intrinsic and extrinsic properties.

Intrinsic and instrumental goods are not mutually exclusive categories. Some things are both good in themselves, and also good for getting other things that are good. "Understanding science" may be such a good, being both worthwhile in and of itself, and as a means of achieving other goods.

A prominent argument in environmental ethics, made by writers like Aldo Leopold and Holmes Rolston III, is that wild nature and healthy ecosystems have intrinsic value, prior to and apart from their instrumental value as resources for humans, and should therefore be preserved. This line of argument has been articulated further in recent years by Canadian philosopher John McMurtry within the *Encyclopedia of Life Support Systems* (<http://www.eolss.net>) published by UNESCO.

Pragmatism and Contributory Goodness

John Dewey (1859-1952), in his book *Theory of Valuation*, sees goodness as the outcome of ethic valuation, a continuous balancing of "ends in view". An end in view is said to be an objective potentially adopted, which may be refined or rejected based on its consistency with other objectives or as a means to objectives already held.

Dewey's empiricist approach evinces absolute intrinsic value denial; *i.e.* not accepting intrinsic value as an inherent or enduring property of things. Instead, Dewey sees the appearance of intrinsic value as an illusory product of our continuous valuative activity as purposive beings. In addition to denying categorically that there is anything like intrinsic value, Dewey held the same position with regard to moral values - for Dewey, moral values are also based on a learning process, and are never intrinsic or absolute.

Another contribution of pragmatism to value theory is the idea of contributory goods with a contributory conditionality. These have the same qualities as the good thing, but need some emergent property of a whole state-of-affairs in order to be good. For example,

salt is food on its own, but is far better as part of a prepared meal. In other words, such goods are only “good” when certain conditions are met. This is in contrast to other goods, which may be considered “good” in a wider variety of situations.

Kant: Hypothetical and Categorical Goods

The thinking of Immanuel Kant (1724-1804) greatly influenced moral philosophy. He thought of moral value as a unique and universally identifiable property, as an absolute value rather than a relative value. He showed that many practical goods are good only in states-of-affairs described by a sentence containing an “if” clause, *e.g.*, in the sentence, “Sunshine is only good if you do not live in the desert.” Further, the “if” clause often described the category in which the judgment was made (art, science, etc.) . Kant described these as “hypothetical goods”, and tried to find a “categorical” good that would operate across all categories of judgment without depending on an “if-then” clause.

An influential result of Kant’s search was the idea of a *good will* being the only intrinsic good. Moreover, Kant saw a good will as acting in accordance with a moral command, the “Categorical Imperative”: “Act according to those maxims that you could will to be universal law.” but should not be confused with the Ethic of Reciprocity or Golden Rule, *e.g.* Mt. 7:12. Whereas the golden rule states that “One should treat others as one would like others to treat oneself,” Kant asks us to analyze whether an act can be performed simultaneously by everyone without exception. For example, murder cannot be performed simultaneously by everyone, one set of people would have to live and the other die. That disparity is an exception. The act cannot be performed without exception, therefore it fails the categorical imperative. Contrast this with the golden rule which is subjective to the individual. Following the logic of the golden rule, if I wanted someone to kill me, then it would be acceptable for me to kill others, because I would be doing to others what I would want done to me. This is very important to keep in mind, because Kant’s categorical imperative avoids this flaw. From this, and a few other axioms, Kant developed a moral system that would apply to any “praiseworthy person”. Kantian philosophers believe that any general definition of goodness must define goods that are categorical in the sense that Kant intended.

CLASSIFICATION OF VALUES: BASIC VALUES OF RELIGION, CLASSIFICATION OF VALUES

BASIC VALUES OF RELIGION

The value of religion depends, of course, on what you mean by “religion”. If religion is merely euphemization of escapism or nihilism, as it so often manifests itself, then it probably has a net negative value—“probably” only because I can imagine some poor unfortunate souls that are constituted in ways that are painfully incompatible with the

world as presently or possibly configured. Too many of us use religion or are used by religion to stop caring about the world and each other, except to the extent it and we happen to be “good” already.

Too many of us wait on God (or an analog) to solve our problems. Too many of us demand that others wait for God to solve our problems. Too many of us posit heaven (or an analog) far away in time or space — or even as negation of time and space. Is it any surprise, then, that our world in practice reflects this escapism and nihilism?

If religion is merely an instigator of conflict over convoluted propositions about supernatural entities, as it also so often manifests itself, then again it probably has a net negative value — here, “probably” only because I can imagine some that are constituted in ways that are stunningly incomprehensible to the rest of us, and whose expressions, if they form propositions at all, must inescapably baffle us while nonetheless being the greatest possible fulfillment of their will to relate with us. Some celebrate faith as irrationality, deriding science and reason, supposing their sacrifice to be required by God. Some indulge in superstition, carelessly embracing their own interpretation of experience without concern for reconciling others’ interpretations and experiences. Some mindlessly repeat and otherwise consent to the positions of authorities under duress of social pressures.

If, on the other hand, religion is any practice that provokes a communal strenuous mood, as I define it consequent to observing and considering others’ observations of its actual function, there’s a lot more to be said about the value of religion. While we can engage in practices that provoke us to a communal strenuous mood toward escapism and nihilism, we can also engage in practices that provoke us to a communal strenuous mood toward life in all its presence, embodiment, relations, and ecology. As religion can provoke us to terrible atrocities, so it can provoke us to wonderful charities. Religion, from this perspective, is a tool, a social tool, and the most powerful of social tools. And like all tools, it can be used for good or evil, and surely has been.

So we have a choice, individually, whether to engage the tool or not. Will we expose ourselves to the risks and opportunities, persuading each other, and joining in religious community? And yet, whether we recognize it or not, there’s a risk and opportunity both ways. What kinds of communities and consequent experiences can arise only within the context of a shared strenuous mood? Do we value those? Can we pursue those opportunities in ways that mitigate associated risks? Can we, with wisdom and inspiration, design our religions for the better? Alternatively, can we mitigate the risks of irreligion? What could we lose by choosing against the religious impulse and drive? When the most vital moments of life come calling, perhaps when the very existence of our community is on the line, or when nothing less than our greatest aspirations suffice, will we have what it takes to come together?

It’s interesting to note how many of my apparently technophilic peers are actually quite technophobic when it comes to social technologies. Many love the powerful gadgets of the introverts, while becoming squeemish at even a hint of the powerful processes of

the extroverts. Many others engage the form of religion, ritualizing their gadget love beyond mere marketing, and yet cannot or will not see their behaviour for what it is. Even now in this post, I'm engaging some of these social technophobes by employing secular descriptors for the religious phenomenon, as if they were somehow the most accurate descriptors, as if they could present the sublime in a general way rather than at best merely proxy for accounts that have evolved across many generations within the vocabularies of those who actually experience the sublime in ways that move them together. The problem with secularism is not that it's evil, as some religious persons claim, but rather just that it's weak in itself. How could anything so bland be evil? How could anything devoid of the strenuous mood be a worthy foe for a God? Secularism certainly is useful in its own ways, and most particularly as a context in which to regulate competing religions and encourage them to non-destructive expressions. But of course it can do so only by aspiring to the disinterestedness of a referee at a sporting event, or by being as dry as a moderator at a debate. And in both cases the real value of the dry disinterestedness arises entirely from its relation to and service of the wet interestedness of the contenders.

How about Humanism? Is that an alternative to religion that rises above the weakness of mere secularism? Yes and no. Yes, it can be an alternative to religion, as many non-strenuous practices can be alternatives to strenuous practices. No, it doesn't rise above the weakness of secularism unless it is a religious Humanism, which of course is quite possible — and in some forms even quite recommended, so far as I'm concerned. Recall that this assessment is not hypothetical, here, in context of my definition of "religion", in context of defining "religion" in terms of its observable historical and present function across the various phenomena we've commonly labeled as "religion". Only if we require a definition that narrows religion to presecular, supernatural, or irrational behaviors can we claim some forms of Humanism to be strenuous non-religious practices.

CLASSIFICATION AND VALUES

The concept of value has different definitions. Because value that is a general and abstract concept is often used to express different meanings. Value derived from "valere" a Latin word means "adopted, admired, overrated and upheld thing" or "a thing which is important and valid for human". The value can be defined as "a fact believed, desired and used as a scale" or "as a belief shows individual and social preferences".

Many definitions of value have been made by researchers and different value classifications have been made. This shows there is not any classification accepted by everyone. Lickona expressed the values as moral and non-moral. Additionally, values were classified as instrumental and terminal by Rokeach, as theoretical, utilitarian, aesthetic, social, individualistic and traditional by Spranger and as power, achievement, hedonism, stimulation, self-direction, universalism, benevolence, tradition, conformity, security and super-grouping by Schwartz. Values education is a value development process. An important question in values education is that which values should be taught.

There is no universal answer of this question. The more accepted answer is that views of students, teachers, parents and the other partners should be taken to determine the values taught in schools. The family has an important role in values education. Important task of the schools is to teach values, to discipline students and to contribute to moral and character development. Both the school and parents are required to put more effort than before. Because factors which affect the child's values system are diversified.

Purpose

There are two aims of this study. The first aim is to access a new and comprehensive value classification. The second is to analyse the views of teachers and parents related to this classification.

Method

The general survey model has been employed in the study. General survey model designs to be administered on an entire population, a group selected from the whole population, sample or sampling in order to make a general judgment about the population consisting of many items. The population of this study composed of school teachers working in primary schools and parents of their students in Eskişehir. The present study has adopted a stratification sample, known as a purposeful sampling. The sample consists of 575 teachers and 1726 parents selected from the primary schools in Eskişehir. The Living Values Table has been used to collect data. The data obtained have been analysed on the computer with "SPSS for Windows, ver: 18.0" programme. Signification of the analyses we have made has been tested at the level of .05. Hierarchical cluster analysis has been used to access the value classification.

ENVIRONMENTAL EDUCATION

DEFINITION

The word environment is derived from the french word 'environ' meaning surroundings. Hence, everything surrounding us is called "Environment". Every organism is surrounded by materials and forces that constitute its environment. It is the environment from where every organism must derive its requirement. The environment creates favourable conditions for the existence and development of living organisms. The survival of any organism requires a steady supply of materials and removal of waste products from its environment. The degradation of the environment has become a serious problem for the existence of human beings. Pollution of soil, water and air causes harm to living organisms as well as loss to valuable natural resources. Environmental studies involves educating the people for preserving the quality of environment.

The scope of environmental studies include:

- Developing an awareness and sensitivity to the total environment and its related problems

- Motivating people for active participation in environmental protection and improvement
- Developing skills for active identification and development of solutions to environmental problems
- Imbibe and inculcate the necessity for conservation of natural resources
- Evaluation of environmental programmes in terms of social, economic, ecological and aesthetic factors.

SCOPE

Scope of **environmental studies** is so wide that it has got relation with every science and scientific aspects in general and biology in particular.

The scope of environmental studies in numerous fields is given below:

- Conservation and management of natural resources like forest, water, air, mineral, etc.
- Conservation of biodiversities like genetic diversity, species diversity, ecosystem diversity, landscape diversity, etc.
- Control of environmental pollutions like air pollution, water pollution, soil pollution, solid waste pollution, noise pollution, electronic waste pollution, e-pollution, etc.
- Replacement of development (like green revolution, urbanization,) economic growth, industrialization, etc.) With sustainable development
- Control of human population.
- Ecology

Main Objectives of Environmental Studies are given below:

- Acquire an awareness of the environment as a whole and its related problems.
- Gain a variety of experiences and acquire a basic understanding and knowledge about the environment and its allied problems.
- Acquire an attitude of concern for the environment.
- Acquire the skills for identifying and solving environmental problems.
- Participate in improvement and protection of environment.
- Develop the ability to evaluate measures for the improvement and protection of environment.

To sum up, the objectives of environmental studies are to develop a world in which persons are aware of and concerned about environment and the problems associated with it, and committed to work individually as well as collectively towards solutions of current problems and prevention of future problems.

NEED AND IMPORTANCE OF ENVIRONMENTAL STUDIES

Need

Increasing population, Urbanization and poverty have generated pressure on the natural resources and lead to a degradation of the environment. To prevent the environment

from further degradation, the supreme court has ordered and initiated environmental protection awareness through government and non-government agencies to take part in protecting our environment. Environmental pollution cannot be prevented by laws alone. Public participation is equally important with regard to environmental protection.

Environmental Education (EE) is a process of learning by giving an overall perspective of knowledge and awareness of the environment. It sensitizes the society about environmental issues and challenges interested individuals to develop skills and expertise thereby providing appropriate solutions. Climate change, loss of biodiversity, declining fisheries, ozone layer depletion, illegal trade of endangered species, destruction of habitats, land degradation, depleting ground water supplies, introduction of alien species, environmental pollution, solid waste disposal, storm water and sewage disposal pose a serious threat to ecosystems in forest, rural, urban and marine ecosystems.

Both formal and informal education on the environment will give the interested individual the knowledge, values, skills and tools needed to face the environmental challenges on a local and global level.

Importance of Environmental Studies

In the industrialized era that we live today, every component that we intake - be it, air, water or food are contaminated by industrial activities. There is no zero pollution. To minimize this problem, knowledge of environmental studies is essential.

An in-dept study of environmental studies will help us in the following ways:

- We will begin to appreciate and adopt the idea of “Development without Destruction of the Environment”
- Knowledge about “Various Types of Environments & Different Environmental Hazards”
- Playing an effective role in protecting the environment by “Demanding Changes in Law and Enforcement Systems”.
- Having a “Positive Impact” on “Quality of Life”.
- Creating a “Concern and Respect for the Environment”.

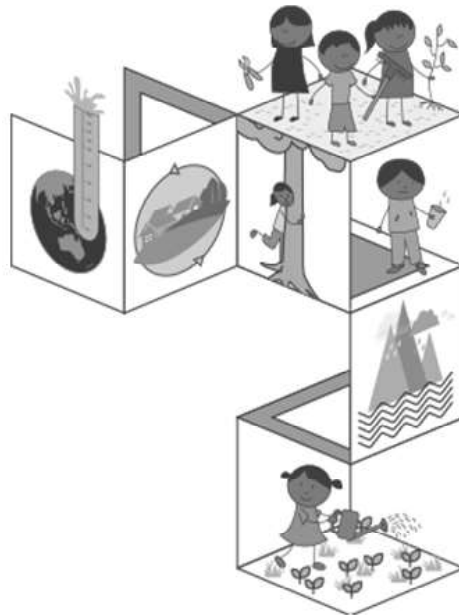
CONCEPT OF ENVIRONMENTAL EDUCATION

The term environment refers to our surrounding which includes physical, biological and socio-cultural aspects. The nonliving things like air, water, land, etc. Come under physical aspects and plants and trees come under biological aspects and the man-made aspect like culture, religion, customs, etc. come under socio cultural aspects. There is a harmonious relationship among the components of the environment. The status of soil, water and other elements determines the biodiversity because different animals and plants have special adaptability cyclic relationship. Environmental condition determines our health also. Environment is the source of all our needs. It provides all the things for our livelihood. However, the activities of human beings are not environment friendly. Environment Education is the educational process that teaches everything about the

environment. It teaches us the way to keep our environment clean and preserve the environmental elements. According to UNESCO in 1975, environment education can be defined as, Education that concerns about the total environment and its associated problems and commitment to work individually and collectively toward the solution of current problems and prevention of new one.

Importance of Environment Education

Environment helps to fulfill all our demands. Human activities have resulted into natural calamities, environmental degradation, environmental pollution and many other problems. Environment education plays a significant role to control these problems and maintain the balance of the environment.



The importance of education environment are:

- *Develops the concept of wise use of natural resources:* Many people do not have the concept that their activities interfere the status of the environment. They use the natural resources randomly. They do not even know the hazards of their malpractices. They do not try to replace the lack by the means of afforestation and other regeneration programme mes. Environment education teaches people about wise use of natural resources.
- *Encourage for the creation of healthy environment:* Healthy environment refers to the environment which is free from the dust, germ and other elements that cause many hazardous diseases. Environment education provides knowledge about various things such as personal hygiene, environmental sanitation and other topic related to healthy environment which prompts us to create a healthy environment.

- *Cultivates awareness about environmental:* Environmental degradation has been a great threat to the existence of all flora, fauna and human beings. Environment education gives knowledge about environmental pollution, create awareness.
- *Helps to provide integrated knowledge:* Environment education incorporates the elements that are related with health and population education. It helps to control the rapidly growing population, to keep the surrounding clean and to give emphasis on personal hygiene.

HISTORICAL BACKGROUND OF ENVIRONMENTAL EDUCATION

The roots of environmental education can be traced back as early as the 18th century when Jean-Jacques Rousseau stressed the importance of an education that focuses on the environment in *Emile: or, On Education*. Several decades later, Louis Agassiz, a Swiss-born naturalist, echoed Rousseau's philosophy as he encouraged students to "Study nature, not books." These two influential scholars helped lay the foundation for a concrete environmental education programme, known as nature study, which took place in the late 19th century and early 20th century.

The nature study movement used fables and moral lessons to help students develop an appreciation of nature and embrace the natural world. Anna Botsford Comstock, the head of the Department of Nature Study at Cornell University, was a prominent figure in the nature study movement and wrote the Handbook for Nature Study in 1911, which used nature to educate children on cultural values. Comstock and the other leaders of the movement, such as Liberty Hyde Bailey, helped Nature Study garner tremendous amounts of support from community leaders, teachers, and scientists and change the science curriculum for children across the United States.

A new type of environmental education, Conservation Education, emerged as a result of the Great Depression and Dust Bowl during the 1920s and 1930s. Conservation Education dealt with the natural world in a drastically different way from Nature Study because it focused on rigorous scientific training rather than natural history. Conservation Education was a major scientific management and planning tool that helped solve social, economic, and environmental problems during this time period.

The modern environmental education movement, which gained significant momentum in the late 1960s and early 1970s, stems from Nature Study and Conservation Education. During this time period, many events – such as Civil Rights, the Vietnam War, and the Cold War – placed Americans at odds with one another and the U.S., government. However, as more people began to fear the fallout from radiation, the chemical pesticides mentioned in Rachel Carson's *Silent Spring*, and the significant amounts of air pollution and waste, the public's concern for their health and the health of their natural environment led to a unifying phenomenon known as environmentalism. Environmental education was born of the realization that solving complex local and global problems cannot be accomplished by politicians and experts alone, but requires "the support and active participation of an informed public in their various roles as

consumers, voters, employers, and business and community leaders” One of the first articles about environmental education as a new movement appeared in the *Phi Delta Kappan* in 1969, authored by James A. Swan. A definition of “Environmental Education” first appeared in *The Journal of Environmental Education* in 1969, authored by William B. Stapp. Stapp later went on to become the first Director of Environmental Education for UNESCO, and then the Global Rivers International Network.

Ultimately, the first Earth Day on April 22, 1970 – a national teach-in about environmental problems – paved the way for the modern environmental education movement. Later that same year, President Nixon passed the National Environmental Education Act, which was intended to incorporate environmental education into K-12 schools. Then, in 1971, the National Association for Environmental Education (now known as the North American Association for Environmental Education) was created to improve environmental literacy by providing resources to teachers and promoting environmental education programmes.

Internationally, environmental education gained recognition when the UN Conference on the Human Environment held in Stockholm, Sweden, in 1972, declared environmental education must be used as a tool to address global environmental problems. The United Nations Education Scientific and Cultural Organization (UNESCO) and United Nations Environment Programme (UNEP) created three major declarations that have guided the course of environmental education.

Stockholm Declaration

June 5–16, 1972 - The Declaration of the United Nations Conference on the Human Environment. The document was made up of 7 proclamations and 26 principles “to inspire and guide the peoples of the world in the preservation and enhancement of the human environment.”

Belgrade Charter

October 13–22, 1975 - The Belgrade Charter was the outcome of the International Workshop on Environmental Education held in Belgrade, Yugoslavia (now Serbia) . The Belgrade Charter was built upon the Stockholm Declaration and adds goals, objectives, and guiding principles of environmental education programmes. It defines an audience for environmental education, which includes the general public.

Tbilisi Declaration

October 14–26, 1977 - The Tbilisi Declaration “noted the unanimous accord in the important role of environmental education in the preservation and improvement of the world’s environment, as well as in the sound and balanced development of the world’s communities.” The Tbilisi Declaration updated and clarified The Stockholm Declaration and The Belgrade Charter by including new goals, objectives, characteristics, and guiding principles of environmental education.

Later that decade, in 1977, the Intergovernmental Conference on Environmental Education in Tbilisi, Georgia emphasized the role of Environmental Education in preserving and improving the global environment and sought to provide the framework and guidelines for environmental education. The Conference laid out the role, objectives, and characteristics of environmental education, and provided several goals and principles for environmental education.

CELEBRATION OF VARIOUS DAYS IN RELATION WITH ENVIRONMENT

- World Wetlands Day: 02-02: February 2
- International Polar Bear Day: 02-27: February 27
- World Wildlife Day: 03-03: March 3
- Solar Appreciation Day: second Friday of March
- International Day of Action for Rivers: 03-14: March 14
- World Consumer Rights Day: 03-15: March 15
- World Sparrow Day: 03-20: March 20
- International Day of Forests: 03-21: March 21
- World Planting Day: 03-21: March 21
- World Wood Day: 03-21: March 21
- World Water Day: 03-22: March 22
- Earth Day: 04-22: April 22
- World Fish Migration Day: April 21, 2018
- Day of Remembrance for all Victims of Chemical Warfare: 04-29: April 29
- Green Up Day: 05: first Saturday of May in Vermont
- Greenery Day: 05-04: May 4 in Japan (previously April 29)
- World Migratory Bird Day: second Saturday in May
- International Migratory Bird Day: second Saturday in May (in the U.S. and Canada) and second Saturday in October (in Mexico, Central and South America, and the Caribbean)
- International Recycling Day: May 17th, 2017
- Endangered Species Day: 05-19: May 19
- International Day for Biological Diversity (World Biodiversity Day) : 05-22: May 22
- World Turtle Day: 05-23: May 23
- European day of parks (Europarc) : 05-24: May 24
- Bike-to-Work Day: 05: third Friday in May
- World Environment Day: 06-05: June 5
- World Oceans Day: 06-08: June 8
- Coral Triangle Day: 06-09: June 9
- Global Wind Day: 06-15: June 15
- World Sea Turtle Day: 06-16: June 16
- World Day to Combat Desertification and Drought: 06-17: June 17

- World Giraffe Day: 06-21: June 21
- World Population Day: 07-11: July 11
- International Tiger Day: 07-29: July 29
- World Lion Day: 08-10: August 10
- World Elephant Day: 08-12: August 12
- World Orangutan Day: 08-19: August 19
- National Honey Bee Day: 08-22: August 22
- Amazon Day: 09-05: September 5
- World Cleanup Day 2018: 09-08: September 8, 2018
- International Day for the Preservation of the Ozone Layer: 09-16: September 16
- Clean Up the World: 09-17 to 09-19: weekend - September 16-18
- World Water Monitoring Day: 09-18: September 18
- Zero Emissions Day: 09-21: September 21
- Car Free Day: 09-22: September 22
- World Rhino Day: 09-22: September 22
- Ecological Debt Day (Earth Overshoot Day) 09-23: September 23 in 2008, but receding
- World Environmental Health Day: 09-26: September 26 since 2011 (IFEH)
- World Rivers Day: 09: last Sunday in September
- World Habitat Day: 10: first Monday in October
- World Animal Day: 10-04: October 4
- International Day for Natural Disaster Reduction: 10: second Wednesday in October, October 13.
- Sustainability Day: 10: fourth Wednesday of October
- International Snow Leopard Day: October 23
- International Day of Climate Action: 10-24: October 24
- World Vegan Day: 11-01: November 1
- International Day for Preventing the Exploitation of the Environment in War and Armed Conflict: 11-06: November 6
- America Recycles Day: 11-15: November 15
- World Soil Day: 12-05: December 5
- International Mountain Day: 12-11: December 11
- Ozone Action Day: at certain times during the summer months
- eDay: variable date

PLASTIC RECYCLING & PROHIBITION OF PLASTIC BAG/COVER

Plastic Recycling

Plastic recycling is the process of recovering scrap or waste plastic and reprocessing the material into useful products. Since the vast majority of plastic is non-biodegradable,

recycling is a part of global efforts to reduce plastic in the waste stream, especially the approximately eight million metric tonnes of waste plastic that enter the Earth's ocean every year. This helps to reduce the high rates of plastic pollution.

Plastic recycling includes taking any type of plastic, sorting it into different polymers and then chipping it and then melting it down into pellets. After this stage, it can then be used to make items of any sort such as plastic chairs and tables. Soft Plastics are also recycled such as polyethylene film and bags. This closed-loop operation has taken place since the 1970s and has made the production of some plastic products amongst the most efficient operations today.

Compared with lucrative recycling of metal, and similar to the low value of glass, plastic polymers recycling is often more challenging because of low density and low value. There are also numerous technical hurdles to overcome when recycling plastic.

A macro molecule interacts with its environment along its entire length, so total energy involved in mixing it is largely due to the product side stoichiometry. Heating alone is not enough to dissolve such a large molecule, so plastics must often be of nearly identical composition to mix efficiently.

When different types of plastics are melted together, they tend to phase-separate, like oil and water, and set in these layers. The phase boundaries cause structural weakness in the resulting material, meaning that polymer blends are useful in only limited applications. The two most widely manufactured plastics, polypropylene and polyethylene behave this way, which limits their utility for recycling. Recently, the use of block copolymers as “molecular stitches” or “macromolecular welding flux” has been proposed to overcome the difficulties associated with phase separation during recycling. Another barrier to recycling is the widespread use of dyes, fillers, and other additives in plastics. The polymer is generally too viscous to economically remove fillers, and would be damaged by many of the processes that could cheaply remove the added dyes. Additives are less widely used in beverage containers and plastic bags, allowing them to be recycled more often. Yet another barrier to removing large quantities of plastic from the waste stream and landfills is that many common but small plastic items lack the universal triangle recycling symbol and accompanying number. An example is the billions of plastic utensils commonly distributed at fast-food restaurants or sold for use at picnics.

The percentage of plastic that can be fully recycled, rather than downcycled or go to waste, can be increased when manufacturers of packaged goods minimize mixing of packaging materials and eliminate contaminants. The Association of Plastics Recyclers have issued a “Design Guide for Recyclability”. The use of biodegradable plastics is increasing.

Processes

Before recycling, most plastics are sorted according to their resin type. In the past, plastic reclaimers used the resin identification code (RIC), a method of categorization

of polymer types, which was developed by the Society of the Plastics Industry in 1988. Polyethylene terephthalate, commonly referred to as PET, for instance, has a resin code of 1. Most plastic reclaimers do not rely on the RIC now; they use automatic sort systems to identify the resin, ranging from manual sorting and picking of plastic materials to mechanized automation processes that involve shredding, sieving, separation by rates of density *i.e.* air, liquid, or magnetic, and complex spectrophotometric distribution technologies *e.g.* UV/VIS, NIR, Laser, etc. Some plastic products are also separated by colour before they are recycled. The plastic recyclables are then shredded. These shredded fragments then undergo processes to eliminate impurities like paper labels. This material is melted and often extruded into the form of pellets which are then used to manufacture other products.

Thermal Depolymerization

Another process involves the conversion of assorted polymers into petroleum by a much less precise thermal depolymerization process. Such a process is able to accept almost any polymer or mix of polymers, including thermoset materials such as vulcanized rubber tire separation of wastes and the biopolymers in feathers and other agricultural waste. Like natural petroleum, the chemicals produced can be made into fuels as well as polymers. A pilot plant of this type exists in Carthage, Missouri, United States, using turkey waste as input material. Gasification is a similar process, but is not technically recycling, since polymers are not likely to become the result.

Waste Plastic Pyrolysis to Fuel Oil

Plastic pyrolysis can convert petroleum-based waste streams such as plastics into quality fuels and carbons. Given below is the list of suitable plastic raw materials for pyrolysis:

- Mixed plastic (HDPE, LDPE, PE, PP, Nylon, Teflon, PS, ABS, FRP etc.)
- Mixed-waste plastic from waste paper mill
- Multi-layered plastic.

Heat Compression

Yet another process that is gaining ground with startup companies (especially in Australia, United States and Japan) is heat compression, which takes all unsorted, cleaned plastic in all forms, from soft plastic bags to hard industrial waste, and mixes the load in tumblers (large rotating drums resembling giant clothes dryers) . The most obvious benefit to this method is that all plastic is recyclable, not just matching forms. However, criticism rises from the energy costs of rotating the drums, and heating the post-melt pipes.

Distributed Recycling

For some waste plastics, technical devices called recyclebots enable a form of distributed recycling. Preliminary life-cycle analysis (LCA) indicates that such distributed recycling of HDPE to make filament of 3-D printers in rural regions is energetically favorable to

either using virgin resin or conventional recycling processes because of reductions in transportation energy

Other Processes

A process has also been developed in which many kinds of plastic can be used as a carbon source in the recycling of scrap steel. There is also a possibility of mixed recycling of different plastics, which does not require their separation. It is called compatibilization and requires use of special chemical bridging agents compatibilizers. It can help to keep the quality of recycled material and to skip often expensive and inefficient preliminary scanning of waste plastics streams and their separation/purification.

Applications

PET

Post-consumer polyethylene terephthalate (PET or PETE) containers are sorted into different colour fractions, and baled for onward sale. PET recyclers further sort the baled bottles and they are washed and flaked (or flaked and then washed) . Non-PET fractions such as caps and labels are removed during this process. The clean flake is dried. Further treatment can take place *e.g.* melt filtering and pelletising or various treatments to produce food-contact-approved recycled PET (RPET) .

RPET has been widely used to produce polyester fibres. This sorted post-consumer PET waste is crushed, chopped into flakes, pressed into bales, and offered for sale.

One use for this recycled PET is to create fabrics to be used in the clothing industry. The fabrics are created by spinning the PET flakes into thread and yarn. This is done just as easily as creating polyester from brand new PET. The recycled PET thread or yarn can be used either alone or together with other fibers to create a very wide variety of fabrics. Traditionally these fabrics are used to create strong, durable, rough, products, such as jackets, coat, shoes, bags, hats, and accessories since they are usually too rough for direct skin contact and can cause irritation. However, these types of fabrics have become more popular as a result of the public's growing awareness of environmental issues. Numerous fabric and clothing manufacturers have capitalized on this trend.

Other major outlets for RPET are new containers (food-contact or non-food-contact) produced either by (injection stretch blow) moulding into bottles and jars or by thermoforming APET sheet to produce clam shells, blister packs and collation trays. These applications used 46% of all RPET produced in Europe in 2010. Other applications, such as strapping tape, injection-moulded engineering components and building materials, account for 13% of the 2010 RPET production.

In the United States the recycling rate for PET packaging was 31.2% in 2013, according to a report from The National Association for PET Container Resources (NAPCOR) and The Association of Postconsumer Plastic Recyclers (APR) . A total of 1,798 million pounds was collected and 475 million pounds of recycled PET used out of a total of 5,764 million pounds of PET bottles.

HDPE

Plastic #2, high-density polyethylene (HDPE) is a commonly recycled plastic. It is typically downcycled into plastic lumber, tables, roadside curbs, benches, truck cargo liners, trash receptacles, stationery (*e.g.* rulers) and other durable plastic products and is usually in demand.

PS

Most polystyrene products are not recycled due to the lack of incentive to invest in the compactors and logistical systems required. As a result, manufacturers cannot obtain sufficient scrap. Expanded polystyrene (EPS) scrap can easily be added to products such as EPS insulation sheets and other EPS materials for construction applications. When it is not used to make more EPS, foam scrap can be turned into clothes hangers, park benches, flower pots, toys, rulers, stapler bodies, seedling containers, picture frames, and architectural molding from recycled PS.

Recycled EPS is also used in many metal casting operations. Rastra is made from EPS that is combined with cement to be used as an insulating amendment in the making of concrete foundations and walls. Since 1993, American manufacturers have produced insulating concrete forms made with approximately 80% recycled EPS.

Other Plastics

The white plastic polystyrene foam peanuts used as packing material are often accepted by shipping stores for reuse.

Successful trials in Israel have shown that plastic films recovered from mixed municipal waste streams can be recycled into useful household products such as buckets.

Similarly, agricultural plastics such as mulch film, drip tape and silage bags are being diverted from the waste stream and successfully recycled into much larger products for industrial applications such as plastic composite railroad ties. Historically, these agricultural plastics have primarily been either landfilled or burned on-site in the fields of individual farms.

CNN reports that Dr. S. Madhu of the Kerala Highway Research Institute, India, has formulated a road surface that includes recycled plastic: aggregate, bitumen (asphalt) with plastic that has been shredded and melted at a temperature below 220 degrees C (428 °F) to avoid pollution. This road surface is claimed to be very durable and monsoon rain resistant. The plastic is sorted by hand, which is economical in India. The test road used 60 kg of plastic for an approximately 500-metre-long, 8-metre-wide, two-lane road.

The process chops thin-film road-waste into a light fluff of tiny flakes that hot-mix plants can uniformly introduce into viscous bitumen with a customized dosing machine. Tests at both Bangalore and the Indian Road Research Centre indicate that roads built using this 'KK process' will have longer useful lives and better resistance to cold, heat, cracking, and rutting, by a factor of three.

Recycling Rates

The quantity of post-consumer plastics recycled has increased every year since at least 1990, but rates lag far behind those of other items, such as newspaper (about 80%) and corrugated fiberboard (about 70%) . Overall, U.S., post-consumer plastic waste for 2008 was estimated at 33.6 million tons; 2.2 million tons (6.5%) were recycled and 2.6 million tons (7.7%) were burned for energy; 28.9 million tons, or 85.5%, were discarded in landfills.

Economic and Energy Potential

In 2008, the price of PET dropped from \$370/ton in the US to \$20 in November. PET prices had returned to their long-term averages by May 2009. Recycling one ton of plastic can save 5,774 kWh of energy, 98,000,000 btu of energy, 1,000–2,000 gallons of gasoline, 685 gallons of oil, 30 cubic yards of landfill space, 48,000 gallons of water.

Consumer Education

In the UK, the amount of post-consumer plastic being recycled is relatively low, due in part to a lack of recycling facilities. The Plastics 2020 Challenge was founded in 2009 by the plastics industry with the aim of engaging the British public in a nationwide debate about the use, reuse and disposal of plastics, and hosts a series of online debates on its website framed around the waste hierarchy.

There is a facility in Worksop capable of recycling 60–80 thousand metric tonnes a year. In Northern Ireland, the rate of recycling is 37.4%. However, emerging technologies are helping to increase the recycling rates of items previously landfilled *e.g.* mixed hard plastics.

Plastic Identification Code

Seven groups of plastic polymers, each with specific properties, are used worldwide for packaging applications. Each group of plastic polymer can be identified by its plastic identification code (PIC) , usually a number or a letter abbreviation. For instance, low-density polyethylene can be identified by the number “4” or the letters “LDPE”. The PIC appears inside a three-chasing-arrow recycling symbol. The symbol is used to indicate whether the plastic can be recycled into new products.

The PIC was introduced by the Society of the Plastics Industry, Inc., to provide a uniform system for the identification of various polymer types and to help recycling companies separate various plastics for reprocessing. Manufacturers of plastic products are required to use PIC labels in some countries/regions and can voluntarily mark their products with the PIC where there are no requirements. Consumers can identify the plastic types based on the codes usually found at the base or at the side of the plastic products, including food/chemical packaging and containers. The PIC is usually not present on packaging films, since it is not practical to collect and recycle most of this type of waste.

United States

Low national plastic recycling rates have been due to the complexity of sorting and processing, unfavorable economics, and consumer confusion about which plastics can actually be recycled. Part of the confusion has been due to the use of the resin identification code which is not on all plastic parts but just a subset that includes the recycling symbol as part of its design. The resin identification code is stamped or printed on the bottom of containers and surrounded by a triangle of arrows. The intent of these symbols was to make it easier to identify the type of plastics used to make a particular container and to indicate that the plastic is potentially recyclable. The question that remains is which types of plastics can be recycled by local recycling centers. In many communities, not all types of plastics are accepted for sidewalk recycling collection programmes due to the high processing costs and complexity of the equipment required to recycle certain materials. There is also sometimes a seemingly low demand for the recycled product depending on a recycling center's proximity to entities seeking recycled materials. Another major barrier is that the cost to recycle certain materials and the corresponding market price for those materials sometimes does not present any opportunity for profit. The best example of this is polystyrene (commonly called styrofoam), although some communities, like Brookline, Massachusetts, are moving toward banning the distribution of polystyrene containers by local food and coffee businesses.

Prohibition of Plastic Bag/Cover

The ACT banned plastic bags on 1 November 2011. Single-use plastic shopping bags litter our environment, harm wildlife and require valuable resources to manufacture. The Northern Territory, South Australia and Tasmania have also banned single-use plastic shopping bags. More states and territories will likely follow.

The ban has reduced plastic bag use, with shoppers encouraged to bring their own bags. Overall the ban has reduced the volume of plastic bag waste going to landfill by around one-third. The ban applies to all retailers in the ACT for single-use, lightweight polyethylene polymer plastic bags that are less than 35 microns in thickness (these are the thin plastic bags with handles that were typically supplied at supermarkets check-outs). The ban does not apply to other bags such as barrier bags for fruit and vegetables.

The ACT's plastic bag ban was reviewed in 2012 and 2014, including through community surveys. The 2014 survey reported that:

- more than 70% of people surveyed did not want the ban overturned
- 65% of Canberra grocery shoppers supported the ban for environmental reasons and agreed it had a positive effect on the environment.

ROLE OF SCHOOL IN ENVIRONMENTAL CONSERVATION AND SUSTAINABLE DEVELOPMENT

As we close in on Earth Day 2012, it seems fitting to reflect on the school's dual role in environmental protection. Like all entities, schools have an environmental footprint.

Those in the school generate trash. They use energy for heating, lighting, photocopying and so on. Schools are cleaned using chemicals that have environmental impacts. The list continues. But schools also have a very unique role in environmental protection: They help students become conscientious stewards of (and advocates for) the environment, protecting it for themselves and generations to come. As we look to schools in the effort to preserve our environment, we should consider both roles to maximize their impact. Of course, simply by minimizing their own environmental footprint, schools can do a great deal to help the environment. As Redmond (WA) environmental science teacher Mike Town has pointed out, “in Redmond and in most of the small cities in the United States — suburban cities, rural cities — the single biggest greenhouse point source is the local high school.” In addition to raising thermostats in the summer and turning out lights when classrooms and auditoriums are not in use, one relatively easy step that schools and districts can take to lessen their environmental impact is “cleaning green.” In a recent edition of *The State Education Standard*, the National Association of State Boards of Education (NASBE) offered “A Roadmap to Implementing Green Cleaning in Districts and Schools,” suggesting a few simple steps to help schools (and districts) start a green cleaning programme:

- Develop a comprehensive programme focusing on low-hanging fruit or a significant challenge. It should include prevention (in terms of both dirt entering the building and exposure of students and staff to chemicals) , conservation (of energy, water and chemicals) , education (of the community and staff) and evaluation (of both cost and quality of cleaning and products)
- Switch to green cleaning products as identified by standards set by third-party organizations like Green Seal, EcoLogo and the EPA’s Design for the Environment. Such a switch is often cost-neutral
- Introduce green equipment and supplies such auto scrubbers that reduce water consumption and chemical use and high-quality entryway mats that reduce dirt contaminations. While the upfront costs of such materials may be higher than traditional options, they can save money as well as the environment in the long-term
- Adopt green cleaning procedures adjusting the frequency, technique or time of a cleaning. Such changes can have environmental and cost impacts as large as switching products and equipment
- Share the responsibility to ensure that teachers, administrators, students, outside contractors and others join the custodial staff in efforts to maintain a green cleaning programme.

Inspire

In addition to acting themselves, schools can also educate and inspire students on environmental issues, equipping them to make informed choices about the impact of their (and others) actions.

Some might argue that this task does not fall under the purview of the school. But, as Town has said, there are a number of other reasons to teach environmental science. It is a great hook for getting students interested in science and math. And there is an occupational component — green jobs, some claim, are the way of the future. Schools need to ensure students are prepared to fill them. The Partnership for 21st century skills considers environmental literacy, a key requirement for success in the global community.

Sometimes opportunities for environmental education occur by chance. For example, at Maryland's Crellin Elementary School, science camp students noticed orange water seeping from a playground area into a stream. Investigation revealed that the seepage was acid mine drainage from water flowing under coal and mine waste that had been spilled back when Crellin was a coal mining town. The school and community joined forces, working together to reclaim the land and develop an environmental education lab for use by the school.

But environmental issues should also be deliberately addressed, either in science, math, social studies or other courses or in dedicated environmental science courses, like Town's. His students have designed green buildings and developed new, greener transportation plans for their community. He also engaged his students in creating the Cool School Challenge. They measured the carbon footprint for everything that happened in their school (from transportation to electricity to heating and waste). With partners, they developed a replicable model for reducing the carbon footprint of a school (available for free on the web) and got buy-in from their school community to implement it.

As is evidenced by the Cool School Challenge (and energy conservation programmes in St. Tammany Parish, Louisiana, York County, Virginia, and Scarsdale, New York, to name just a few examples), the school's two roles in environmental protection are not mutually exclusive. Most schools and districts that have successfully decreased their environmental footprint have done so in a comprehensive manner that includes both administrator actions and educational experiences for students. As we look forward to Earth Day, hopefully more schools and districts will follow their lead.

POLLUTION FREE ECO-SYSTEM

Pollution is the introduction of contaminants into the natural environment that cause adverse change. Pollution can take the form of chemical substances or energy, such as noise, heat or light. Pollutants, the components of pollution, can be either foreign substances/energies or naturally occurring contaminants. Pollution is often classed as point source or nonpoint source pollution.

Air pollution has always accompanied civilizations. Pollution started from prehistoric times when man created the first fires. According to a 1983 article in the journal *Science*, "soot" found on ceilings of prehistoric caves provides ample evidence of the high levels of pollution that was associated with inadequate ventilation of open fires." Metal forging appears to be a key turning point in the creation of significant air pollution levels outside the home. Core samples of glaciers in Greenland indicate increases in pollution associated

with Greek, Roman and Chinese metal production, but at that time the pollution was comparatively small and could be handled by nature. The burning of coal and wood, and the presence of many horses in concentrated areas made the cities the cesspools of pollution. The Industrial Revolution brought an infusion of untreated chemicals and wastes into local streams that served as the water supply. King Edward I of England banned the burning of sea-coal by proclamation in London in 1272, after its smoke became a problem. But the fuel was so common in England that this earliest of names for it was acquired because it could be carted away from some shores by the wheelbarrow.

It was the industrial revolution that gave birth to environmental pollution as we know it today. London also recorded one of the earlier extreme cases of water quality problems with the Great Stink on the Thames of 1858, which led to construction of the London sewerage system soon afterward. Pollution issues escalated as population growth far exceeded view ability of neighborhoods to handle their waste problem. Reformers began to demand sewer systems, and clean water.

In 1870, the sanitary conditions in Berlin were among the worst in Europe. August Bebel recalled conditions before a modern sewer system was built in the late 1870s:

- “Waste-water from the houses collected in the gutters running alongside the curbs and emitted a truly fearsome smell. There were no public toilets in the streets or squares. Visitors, especially women, often became desperate when nature called. In the public buildings the sanitary facilities were unbelievably primitive....As a metropolis, Berlin did not emerge from a state of barbarism into civilization until after 1870.”

The primitive conditions were intolerable for a world national capital, and the Imperial German government brought in its scientists, engineers and urban planners to not only solve the deficiencies but to forge Berlin as the world’s model city. A British expert in 1906 concluded that Berlin represented “the most complete application of science, order and method of public life,” adding “it is a marvel of civic administration, the most modern and most perfectly organized city that there is.”

The emergence of great factories and consumption of immense quantities of coal gave rise to unprecedented air pollution and the large volume of industrial chemical discharges added to the growing load of untreated human waste. Chicago and Cincinnati were the first two American cities to enact laws ensuring cleaner air in 1881. Pollution became a major issue in the United States in the early twentieth century, as progressive reformers took issue with air pollution caused by coal burning, water pollution caused by bad sanitation, and street pollution caused by the 3 million horses who worked in American cities in 1900, generating large quantities of urine and manure. As historian Martin Melosi notes, The generation that first saw automobiles replacing the horses saw cars as “miracles of cleanliness.” By the 1940s, however, automobile-caused smog was a major issue in Los Angeles. Other cities followed around the country until early in the 20th century, when the short lived Office of Air Pollution was created under the Department of the Interior. Extreme smog events were experienced by the cities of Los

Angeles and Donora, Pennsylvania in the late 1940s, serving as another public reminder. Air pollution would continue to be a problem in England, especially later during the industrial revolution, and extending into the recent past with the Great Smog of 1952.

Awareness of atmospheric pollution spread widely after World War II, with fears triggered by reports of radioactive fallout from atomic warfare and testing. Then a non-nuclear event, The Great Smog of 1952 in London, killed at least 4000 people. This prompted some of the first major modern environmental legislation, The Clean Air Act of 1956.

Pollution began to draw major public attention in the United States between the mid-1950s and early 1970s, when Congress passed the Noise Control Act, the Clean Air Act, the Clean Water Act and the National Environmental Policy Act.

Severe incidents of pollution helped increase consciousness. PCB dumping in the Hudson River resulted in a ban by the EPA on consumption of its fish in 1974. Long-term dioxin contamination at Love Canal starting in 1947 became a national news story in 1978 and led to the Superfund legislation of 1980. The pollution of industrial land gave rise to the name brownfield, a term now common in city planning.

The development of nuclear science introduced radioactive contamination, which can remain lethally radioactive for hundreds of thousands of years. Lake Karachay, named by the Worldwatch Institute as the “most polluted spot” on earth, served as a disposal site for the Soviet Union throughout the 1950s and 1960s. Chelyabinsk, Russia, is considered the “Most polluted place on the planet”.

Nuclear weapons continued to be tested in the Cold War, especially in the earlier stages of their development. The toll on the worst-affected populations and the growth since then in understanding about the critical threat to human health posed by radioactivity has also been a prohibitive complication associated with nuclear power. Though extreme care is practiced in that industry, the potential for disaster suggested by incidents such as those at Three Mile Island and Chernobyl pose a lingering specter of public mistrust. Worldwide publicity has been intense on those disasters. Widespread support for test ban treaties has ended almost all nuclear testing in the atmosphere.

International catastrophes such as the wreck of the Amoco Cadiz oil tanker off the coast of Brittany in 1978 and the Bhopal disaster in 1984 have demonstrated the universality of such events and the scale on which efforts to address them needed to engage. The borderless nature of atmosphere and oceans inevitably resulted in the implication of pollution on a planetary level with the issue of global warming. Most recently the term persistent organic pollutant (POP) has come to describe a group of chemicals such as PBDEs and PFCs among others. Though their effects remain somewhat less well understood owing to a lack of experimental data, they have been detected in various ecological habitats far removed from industrial activity such as the Arctic, demonstrating diffusion and bioaccumulation after only a relatively brief period of widespread use. A much more recently discovered problem is the Great Pacific Garbage Patch, a huge concentration of plastics, chemical sludge and other debris which has

been collected into a large area of the Pacific Ocean by the North Pacific Gyre. This is a less well known pollution problem than the others described above, but nonetheless has multiple and serious consequences such as increasing wildlife mortality, the spread of invasive species and human ingestion of toxic chemicals. Organizations such as 5 Gyres have researched the pollution and, along with artists like Marina DeBris, are working toward publicizing the issue.

Pollution introduced by light at night is becoming a global problem, more severe in urban centres, but nonetheless contaminating also large territories, far away from towns.

Growing evidence of local and global pollution and an increasingly informed public over time have given rise to environmentalism and the environmental movement, which generally seek to limit human impact on the environment.

Forms of Pollution

The major forms of pollution are listed below along with the particular contaminant relevant to each of them:

- Air pollution: the release of chemicals and particulates into the atmosphere. Common gaseous pollutants include carbon monoxide, sulfur dioxide, chlorofluorocarbons (CFCs) and nitrogen oxides produced by industry and motor vehicles. Photochemical ozone and smog are created as nitrogen oxides and hydrocarbons react to sunlight. Particulate matter, or fine dust is characterized by their micrometre size PM_{10} to $PM_{2.5}$.
- Light pollution: includes light trespass, over-illumination and astronomical interference.
- Littering: the criminal throwing of inappropriate man-made objects, unremoved, onto public and private properties.
- Noise pollution: which encompasses roadway noise, aircraft noise, industrial noise as well as high-intensity sonar.
- Soil contamination occurs when chemicals are released by spill or underground leakage. Among the most significant soil contaminants are hydrocarbons, heavy metals, MTBE, herbicides, pesticides and chlorinated hydrocarbons.
- Radioactive contamination, resulting from 20th century activities in atomic physics, such as nuclear power generation and nuclear weapons research, manufacture and deployment.
- Thermal pollution, is a temperature change in natural water bodies caused by human influence, such as use of water as coolant in a power plant.
- Visual pollution, which can refer to the presence of overhead power lines, motorway billboards, scarred landforms (as from strip mining) , open storage of trash, municipal solid waste or space debris.
- Water pollution, by the discharge of wastewater from commercial and industrial waste (intentionally or through spills) into surface waters; discharges of untreated domestic sewage, and chemical contaminants, such as chlorine, from treated

sewage; release of waste and contaminants into surface runoff flowing to surface waters (including urban runoff and agricultural runoff, which may contain chemical fertilizers and pesticides) ; waste disposal and leaching into groundwater; eutrophication and littering.

- Plastic pollution: involves the accumulation of plastic products in the environment that adversely affects wildlife, wildlife habitat, or humans.

Pollutants

A pollutant is a waste material that pollutes air, water or soil. Three factors determine the severity of a pollutant: its chemical nature, the concentration and the persistence.

Cost of Pollution

Pollution has cost. Manufacturing activities that cause air pollution impose health and clean-up costs on the whole society, whereas the neighbors of an individual who chooses to fire-proof his home may benefit from a reduced risk of a fire spreading to their own houses. If external costs exist, such as pollution, the producer may choose to produce more of the product than would be produced if the producer were required to pay all associated environmental costs. Because responsibility or consequence for self-directed action lies partly outside the self, an element of externalization is involved. If there are external benefits, such as in public safety, less of the good may be produced than would be the case if the producer were to receive payment for the external benefits to others.

Sources and Causes

Air pollution comes from both natural and human-made (anthropogenic) sources. However, globally human-made pollutants from combustion, construction, mining, agriculture and warfare are increasingly significant in the air pollution equation.

Motor vehicle emissions are one of the leading causes of air pollution. China, United States, Russia, India Mexico, and Japan are the world leaders in air pollution emissions. Principal stationary pollution sources include chemical plants, coal-fired power plants, oil refineries, petrochemical plants, nuclear waste disposal activity, incinerators, large livestock farms (dairy cows, pigs, poultry, etc.) , PVC factories, metals production factories, plastics factories, and other heavy industry. Agricultural air pollution comes from contemporary practices which include clear felling and burning of natural vegetation as well as spraying of pesticides and herbicides

About 400 million metric tons of hazardous wastes are generated each year. The United States alone produces about 250 million metric tons. Americans constitute less than 5% of the world's population, but produce roughly 25% of the world's CO₂, and generate approximately 30% of world's waste. In 2007, China has overtaken the United States as the world's biggest producer of CO₂, while still far behind based on per capita pollution - ranked 78th among the world's nations. In February 2007, a report by the Intergovernmental Panel on Climate Change (IPCC) , representing the work of 2,500

scientists, economists, and policymakers from more than 120 countries, said that humans have been the primary cause of global warming since 1950. Humans have ways to cut greenhouse gas emissions and avoid the consequences of global warming, a major climate report concluded. But to change the climate, the transition from fossil fuels like coal and oil needs to occur within decades, according to the final report this year from the UN's Intergovernmental Panel on Climate Change (IPCC) .

Some of the more common soil contaminants are chlorinated hydrocarbons (CFH) , heavy metals (such as chromium, cadmium—found in rechargeable batteries, and lead—found in lead paint, aviation fuel and still in some countries, gasoline) , MTBE, zinc, arsenic and benzene. In 2001 a series of press reports culminating in a book called *Fateful Harvest* unveiled a widespread practice of recycling industrial byproducts into fertilizer, resulting in the contamination of the soil with various metals. Ordinary municipal landfills are the source of many chemical substances entering the soil environment (and often groundwater) , emanating from the wide variety of refuse accepted, especially substances illegally discarded there, or from pre-1970 landfills that may have been subject to little control in the U.S., or EU. There have also been some unusual releases of polychlorinated dibenzodioxins, commonly called *dioxins* for simplicity, such as TCDD.

Pollution can also be the consequence of a natural disaster. For example, hurricanes often involve water contamination from sewage, and petrochemical spills from ruptured boats or automobiles. Larger scale and environmental damage is not uncommon when coastal oil rigs or refineries are involved. Some sources of pollution, such as nuclear power plants or oil tankers, can produce widespread and potentially hazardous releases when accidents occur. In the case of noise pollution the dominant source class is the motor vehicle, producing about ninety percent of all unwanted noise worldwide.

Effects

Human Health

Adverse air quality can kill many organisms including humans. Ozone pollution can cause respiratory disease, cardiovascular disease, throat inflammation, chest pain, and congestion. Water pollution causes approximately 14,000 deaths per day, mostly due to contamination of drinking water by untreated sewage in developing countries. An estimated 500 million Indians have no access to a proper toilet, Over ten million people in India fell ill with waterborne illnesses in 2013, and 1,535 people died, most of them children. Nearly 500 million Chinese lack access to safe drinking water. A 2010 analysis estimated that 1.2 million people died prematurely each year in China because of air pollution. The WHO estimated in 2007 that air pollution causes half a million deaths per year in India. Studies have estimated that the number of people killed annually in the United States could be over 50,000. Oil spills can cause skin irritations and rashes. Noise pollution induces hearing loss, high blood pressure, stress, and sleep disturbance.

Mercury has been linked to developmental deficits in children and neurologic symptoms. Older people are majorly exposed to diseases induced by air pollution. Those with heart or lung disorders are at additional risk. Children and infants are also at serious risk. Lead and other heavy metals have been shown to cause neurological problems. Chemical and radioactive substances can cause cancer and as well as birth defects.

Environment

Pollution has been found to be present widely in the environment. There are a number of effects of this:

- Biomagnification describes situations where toxins (such as heavy metals) may pass through trophic levels, becoming exponentially more concentrated in the process.
- Carbon dioxide emissions cause ocean acidification, the ongoing decrease in the pH of the Earth's oceans as CO₂ becomes dissolved.
- The emission of greenhouse gases leads to global warming which affects ecosystems in many ways.
- Invasive species can out compete native species and reduce biodiversity. Invasive plants can contribute debris and biomolecules (allelopathy) that can alter soil and chemical compositions of an environment, often reducing native species competitiveness.
- Nitrogen oxides are removed from the air by rain and fertilise land which can change the species composition of ecosystems.
- Smog and haze can reduce the amount of sunlight received by plants to carry out photosynthesis and leads to the production of tropospheric ozone which damages plants.
- Soil can become infertile and unsuitable for plants. This will affect other organisms in the food web.
- Sulfur dioxide and nitrogen oxides can cause acid rain which lowers the pH value of soil.

Environmental Health Information

The Toxicology and Environmental Health Information Programme (TEHIP) at the United States National Library of Medicine (NLM) maintains a comprehensive toxicology and environmental health web site that includes access to resources produced by TEHIP and by other government agencies and organizations. This web site includes links to databases, bibliographies, tutorials, and other scientific and consumer-oriented resources. TEHIP also is responsible for the Toxicology Data Network (TOXNET) an integrated system of toxicology and environmental health databases that are available free of charge on the web.

TOXMAP is a Geographic Information System (GIS) that is part of TOXNET. TOXMAP uses maps of the United States to help users visually explore data from the

United States Environmental Protection Agency's (EPA) Toxics Release Inventory and Superfund Basic Research Programmes.

Worker Productivity

A number of studies show that pollution has an adverse effect on the productivity of both indoor and outdoor workers.

Regulation and Monitoring

To protect the environment from the adverse effects of pollution, many nations worldwide have enacted legislation to regulate various types of pollution as well as to mitigate the adverse effects of pollution.

Pollution Control

Pollution control is a term used in environmental management. It means the control of emissions and effluents into air, water or soil. Without pollution control, the waste products from overconsumption, heating, agriculture, mining, manufacturing, transportation and other human activities, whether they accumulate or disperse, will degrade the environment. In the hierarchy of controls, pollution prevention and waste minimization are more desirable than pollution control. In the field of land development, low impact development is a similar technique for the prevention of urban runoff.

Practices:

- Recycling
- Reusing
- Waste minimisation
- Mitigating
- Preventing
- Compost

Pollution control devices:

- Air pollution control
 - Thermal oxidizer
- Dust collection systems
 - Baghouses
 - Cyclones
 - Electrostatic precipitators
- Scrubbers
 - Baffle spray scrubber
 - Cyclonic spray scrubber
 - Ejector venturi scrubber
 - Mechanically aided scrubber
 - Spray tower
 - Wet scrubber

- Sewage treatment
 - Sedimentation (Primary treatment)
 - Activated sludge biotreaters (Secondary treatment; also used for industrial wastewater)
 - Aerated lagoons
 - Constructed wetlands (also used for urban runoff)
- Industrial wastewater treatment
 - API oil-water separators
 - Biofilters
 - Dissolved air flotation (DAF)
 - Powdered activated carbon treatment
 - Ultrafiltration
- Vapor recovery systems
- Phytoremediation.

Perspectives

The earliest precursor of pollution generated by life forms would have been a natural function of their existence. The attendant consequences on viability and population levels fell within the sphere of natural selection. These would have included the demise of a population locally or ultimately, species extinction. Processes that were untenable would have resulted in a new balance brought about by changes and adaptations. At the extremes, for any form of life, consideration of pollution is superseded by that of survival.

For humankind, the factor of technology is a distinguishing and critical consideration, both as an enabler and an additional source of byproducts. Short of survival, human concerns include the range from quality of life to health hazards. Since science holds experimental demonstration to be definitive, modern treatment of toxicity or environmental harm involves defining a level at which an effect is observable. Common examples of fields where practical measurement is crucial include automobile emissions control, industrial exposure (*e.g.* Occupational Safety and Health Administration (OSHA) PELs), toxicology (*e.g.* LD₅₀), and medicine (*e.g.* medication and radiation doses).

“The solution to pollution is dilution”, is a dictum which summarizes a traditional approach to pollution management whereby sufficiently diluted pollution is not harmful. It is well-suited to some other modern, locally scoped applications such as laboratory safety procedure and hazardous material release emergency management. But it assumes that the dilutant is in virtually unlimited supply for the application or that resulting dilutions are acceptable in all cases.

Such simple treatment for environmental pollution on a wider scale might have had greater merit in earlier centuries when physical survival was often the highest imperative, human population and densities were lower, technologies were simpler and their byproducts more benign. But these are often no longer the case. Furthermore, advances have enabled measurement of concentrations not possible before. The use of statistical

methods in evaluating outcomes has given currency to the principle of probable harm in cases where assessment is warranted but resorting to deterministic models is impractical or infeasible. In addition, consideration of the environment beyond direct impact on human beings has gained prominence.

Yet in the absence of a superseding principle, this older approach predominates practices throughout the world. It is the basis by which to gauge concentrations of effluent for legal release, exceeding which penalties are assessed or restrictions applied. One such superseding principle is contained in modern hazardous waste laws in developed countries, as the process of diluting hazardous waste to make it non-hazardous is usually a regulated treatment process. Migration from pollution dilution to elimination in many cases can be confronted by challenging economical and technological barriers.

Greenhouse Gases and Global Warming

Carbon dioxide, while vital for photosynthesis, is sometimes referred to as pollution, because raised levels of the gas in the atmosphere are affecting the Earth's climate. Disruption of the environment can also highlight the connection between areas of pollution that would normally be classified separately, such as those of water and air. Recent studies have investigated the potential for long-term rising levels of atmospheric carbon dioxide to cause slight but critical increases in the acidity of ocean waters, and the possible effects of this on marine ecosystems.

Most Polluting Industries

The Pure Earth, an international non-for-profit organization dedicated to eliminating life-threatening pollution in the developing world, issues an annual list of some of the world's most polluting industries.

- Lead-Acid Battery Recycling
- Industrial Mining and Ore Processing
- Lead Smelting
- Tannery Operations
- Artisanal Small-Scale Gold Mining
- Industrial/Municipal Dumpsites
- Industrial Estates
- Chemical Manufacturing
- Product Manufacturing
- Dye Industry

World's Worst Polluted Places

The Pure Earth issues an annual list of some of the world's worst polluted places.

- Agbogbloshie, Ghana
- Chernobyl*, Ukraine
- Citarum River, Indonesia

- Dzershinsk*, Russia
- Hazaribagh, Bangladesh
- Kabwe*, Zambia
- Kalimantan, Indonesia
- Matanza Riachuelo, Argentina
- Niger River Delta, Nigeria
- Norilsk*, Russia.

11

CHAPTER

Research in Educational and Behavioral Statistic

INTRODUCTION

Currently, there is considerable research into the teaching and learning of statistics at all levels of education. In higher education, statistics education is a fast growing and important area. In the next part, we introduce our perspective on activity theory pertaining to statistics education. I summarise some of the literature on statistics education relevant to my investigation and outline some issues arising from the research. Following that I introduce some aspects of my research design and approach.

While full details of my methodology and procedures, I give some background to the two studies making up the project in this part—who the participants were and the setting for each of the studies. I explain the fundamentals of the Gothenburg phenomenographic method and my adaptation of it for Study Two. I outline the ideas underlying my development of the Approaches to Learning Statistics Questionnaire. I also explain how my study fits into an emerging form of research concerned with investigating individuals engaged in socially meaningful practices. I introduce my theoretical model, drawing on Leont’ev’s

framework. I firstly look at other important theories informing mathematics education and how activity theory fits with them. I describe some key elements of activity theory and include some major influences on its development in the former Soviet Union. I explain the structure of activity and its analysis. I frame my project, including the

principal research question, in terms of an activity theory approach. Finally I summarise the key ideas of this approach.

PERSPECTIVE ON ACTIVITY THEORY

The theory of activity, based on the work of Vygotsky was expounded mainly by A. N. Leont'ev. It was an extremely influential theory in the former Soviet Union, and was developed and extended by important psychologists there, including Luria, Gal'perin, P. Ia. Zinchenko, and, after them, Davydov, V. P. Zinchenko, as well as many others. It emphasises the role of external, practical activity in cognition.

According to Leont'ev it is an important proposition that:

- Internal psychological activities originate from practical activity, historically accumulated as the result of the education of man based on work in society, and that in separate individuals of every new generation they are formed in the course of ontogenetic development.

That is, higher mental abilities develop in the individual from activities that are rooted in the ongoing practical and communal life by which societies organise and reorganise themselves.

The philosophical foundations for this approach are the theories of Marx, Engels and Lenin. Their concern was with the economic foundations from which development arises, both individual development and societal progress. Following their theories, Leont'ev explained that people actively develop knowledge on the basis of life experiences, which have an economic, social and political context, and are mediated by cultural tools. The idea of mediation arises from Engel's proposition that the work done by humans is mediated by the tools they develop. Vygotsky extended the idea of physical tools as mediators to include mediation of cognitive processes by psychological tools—speech and semiotic systems.

An eminent Western analyst of Leont'ev's activity theory, Wertsch, describes the activity framework as being concerned with how abilities are developed:

- To carry out socially formulated, goal-directed actions with the help of mediating devices.

While the Russian word *deyatel'nost'* is commonly translated into English as “activity”, the ideas incorporated in the construct are, to me, better embodied in the word “engagement”. This word conveys notions of purpose and affect and includes intellectual, as well as physical, processes. All of these are bound up with the social world surrounding the individual. Activity, in ordinary English usage, usually refers to physical activity—behaviour.

Indeed, in early Russian publications, activity referred to physical labour, which was mediated by tools. Later, the notion of activity was developed to include mental actions, such as remembering and reflection, and included Vygotsky's idea of mental tools as mediators, rather than only material tools of work. Activity, in the sense of engagement, both produces thought and is a product of the individual's awareness—her

reflection on her environment. According to Leont'ev activity, with its corresponding goals, means and conditions, dialectically forms and reforms individuals and their social worlds. That is, activity both orients individuals in the world in which they live and changes that world in cycles of mutual transformations.

Activity is described by Leont'ev as a functional unit of life:

- A system with its own structure, its own internal transformations, and its own development.

Throughout this thesis, I will try to explicate Leont'ev's ideas about the configuration of an activity system as a functional and dynamic unit. In this way I try to shed light on how students' learning activities develop and how they both shape and are organised by the setting surrounding them. The activity of learning is a process in which people grapple with new information—to make it meaningful, to solve problems and to adapt to new conditions. Rather than stressing the nature of the information received by the learner, the emphasis of my investigation is on the individual acting in her social and cultural world—what the learner does, why she takes those actions and how her actions relate to the learning arena.

Varela, Thompson and Rosch describe cognitive capacities as:

- Paths that exist only as they are laid down in walking.

Their metaphor beautifully illustrates the inseparability of personal meaning and setting; the co-emergence of thought and action. These notions are fundamental to my interpretation, application and extension of activity theory. Activity theory emerged from a culture and era very different to my own. The insights offered by this theory, in part due to these very differences, highlight the problematic nature and complexity of human learning.

STATISTICS EDUCATION AT UNIVERSITY

"Statistics" said Jane, a mathematics student:

- Really fits my logic and it feels right. Maybe the men have finally invented a good mathematical study for a change.

The comment highlights the tension between knowledge as culturally endorsed and the individual's personal appraisal of it, a tension not always as agreeably resolved as in Jane's case. Statistical literacy and appreciation, particularly an understanding of data gathering, presentation and interpretation, are important components of undergraduate and graduate education in many fields. They are also essential for modern living in technologically developed countries. Statistics is often a compulsory unit of university courses, such as Psychology, Economics, Business or a Health Science, because it is an important tool for analysing the "uncertainties and complexities of life and society". Despite this importance, many students in statistics courses are reluctant to study it.

Indeed, Cotts maintains that it is:

- Almost unarguable that the introductory statistics course is the most widely feared course on most university campuses. This conviction is echoed in several studies.

RESEARCH IN STATISTICS EDUCATION

There has been considerable research into statistics education in the last thirty years. This interest in the teaching and learning of statistics is not only a part of the general growth spurt in mathematics education but is also a field of research in its own right. Four international conferences, the International Conferences on the Teaching of Statistics have been devoted to grappling with issues relating to statistics education. Much topical literature concerns the teaching and learning of statistics at all levels of education. For example, in a special edition on teaching statistics of the *Journal of Educational and Behavioural Statistics*, Becker reviews 530 articles and dissertations which are documented on electronic databases such as the Educational Resources Information Centre. Almost all these articles (97%) were published after 1970 and about one third have appeared since 1990.

A bibliography on the available literature published between 1987 and 1994, concerning the teaching of probability and statistics, is provided by Sahai, Khurshid & Misra. This shows that the body of work being done in this area is extensive and growing. Truran & Truran review Australasian research into the learning and teaching of stochastics, for the period 1992-1995, which reflects the current importance of probability, statistics and combinatorics to education in my region of the world. Electronically based journals such as the *Journal of Statistics Education* and Web discussion groups facilitate the extensive exchange of ideas and the development and distribution of research among participants in statistics education.

The quantity and scope of the published literature, as well as the ongoing discussion and research, is a strong indication of current interest in statistics education. The literature and discourse concerns many topics. One important topic concerns teaching strategies. For example, Lan writes about self monitoring and Smith discusses the use of writing assignments in teaching statistics.

The role of computers, multimedia materials and other technological aids in teaching statistics is an area of increasing prominence. Individual differences, such as gender differences in learning statistics, have also been investigated. For instance, Clark proposes that female students prefer statistical questions which have a people-orientation. Preliminary results by Forbes however, suggest that, in statistical examinations in New Zealand, the form and type of examination questions, for example, whether questions involve essay writing or require using Calculus, are at least as important as their contextual embedding in determining gender preferences and performance.

As with any other field in education, what constitutes statistical knowledge and how to enhance the quality of students' learning of it, are key problems in statistics education. A major theme that recurs in the literature on higher education in statistics pertains to teaching and assessing statistics. There is general consensus in the community of statistics educators that reforms are urgently needed in the teaching of introductory statistics courses at university. Some university educators describe teaching activities and/or methods of assessment intended to address the problems - to make statistical concepts

meaningful and useful - to relate statistics to the real world. For example, Anderson & Loynes and Pfannkuch discuss what abilities and skills are needed by practising statisticians and how to teach these. Garfield Giraud and Keeler & Steinhorst describe new ways of teaching statistics using co-operative learning activities. Garfield and Hubbard present frameworks for developing assessment instruments and procedures which are appropriate for measuring students' understanding and applications of concepts in probability and statistics.

Many university educators report on the success of their statistics courses by describing enhanced performance and/or increased student satisfaction. However, as some researchers point out, little is known about how such courses would transfer to other settings. Becker concurs with this view. In her review of the literature and other resources on teaching statistics, she focuses mainly on statistics education at the university level. She reports that this literature is largely anecdotal, with less than 30% of the reviewed articles describing the results of empirical studies. Hence, extensive information on instructional strategies and resources are available to tertiary educators in statistics, either in print or through electronic media. However, research empirically evaluating the teaching and learning of statistics is less copious. Further, and in stark contrast to the literature on mathematics education, there is a dearth of studies on the teaching and learning of statistics which are framed in terms of theories of education. The literature on teaching and learning statistics at the tertiary level tends to concentrate on the "knowledge craft" of the teachers.

A further and major area of research in statistics education concerns students' understanding of particular concepts in statistics and probability theory. This research shows that many students have difficulties with and misconceptions about statistical ideas. Konold reviewed research showing that the intuitions of adults are at odds with accepted probability and statistical theory. Fischbein and Schnarch investigated the stability of students' misconceptions in probability across different age levels from grade 5 to college students. Surprisingly, they found three different outcomes: some misconceptions grew stronger with age, some grew weaker and one remained stable.

Vallecillos & Batanero presented a study on the persistence of conceptual errors in university students' understanding of levels of significance in tests of hypotheses. There are also parts on students' understanding and misunderstanding of concepts relating to probability and statistics in books on the teaching of statistics such as Green; Holmes; Hawkins, Jolliffe & Glickman. These attest to the importance of conceptual understanding as a goal of statistics instruction.

In summary, research on student learning in statistics shows that students have difficulties with statistical concepts and that they are often anxious about and have poor attitudes to learning statistics. The body of research on teaching and learning statistics is large, vibrant and growing. However, much of it still lacks the systematic methodology and theoretical foundations that are often of the most lasting value in any field of education. Further, a concern with describing teaching strategies or students' attitudes

or their misconceptions, while indicating an awareness of the problems facing statistics education, does not offer a model for understanding the impasse—a first step towards alleviating it. My approach is to explore the issues underlying this impasse within a theoretical framework which relates students, their actions and the context.

ISSUES FOR EXPLORATION IN RESEARCH

The acknowledgment, in the education literature, of the prevalence of students' misinterpretations of key statistical concepts leads me to question how statistical thinking is embedded in academic settings. That is, how do students experience statistical reasoning in the academic setting and how does this reasoning relate to that of working statisticians in practical situations? According to one statistician the components of statistical thinking are: understanding the dynamics of the real world problem, moving towards a statistical model and using statistical tools.

If students in a university statistics course are not provided with experiences enabling them to understand the relationship between the statistical model and the “real life” situation, they may not be adequately equipped to use the statistical tools with which they are presented and so may be unable to appreciate the statistical reasoning process. For example a lack of understanding about variation could lead students either to trust implicitly their own intuition about data, without attempting to use statistical reasoning to evaluate the findings more critically, or to distrust universally any statistical statement—without the means to assess its reliability. It is important for me in this investigation to understand how students' concepts of statistical knowledge relate to the contexts in which they apply this knowledge.

The ways that students experience learning statistics are also likely to contribute to the difficulties that many students have with the subject and to their objections to studying it. How students understand statistics is related to how they interpret it in the context in which it is presented. This is an aspect of students' learning of statistics which has received scant attention in the research literature and, in my experience, is rarely accorded importance in the teaching of statistics courses in higher education. Skills and facts are easily identified in the design and assessment of statistics courses at university. However, students' conceptions of the subject matter as a whole, approaches to learning it and their perceptions and beliefs are rarely evaluated in any formal sense. These aspects of students' learning of Statistics are major topics for exploration in my project as it is a prime postulate of my research that these are critical dimensions of students' emerging statistical knowledge.

In learning Statistics, a student's thinking and problem solving is accompanied by affective elements: feelings, beliefs, desires and attitudes which will affect how long she will persist with a task or a problem, how easily distracted she is likely to be, how long she will remember facts and skills and other important factors which can facilitate or hinder her cognition. Affect is therefore of importance and interest in my investigation which seeks to understand learning Statistics from the activity theory perspective. In

this framework, affective elements are not extraneous or secondary to intellectual processes but indivisible from them. Vygotsky describes this inseparability of intellectual processes and affective elements as follows:

- Thought itself is engendered by motivation, *i.e.* by our desires and needs, our interests and emotions. Behind every thought there is an affective-volitional tendency, which holds the answer to the last ‘why’ in the analysis of thinking. A true and full understanding of another’s thought is possible only when we understand its affective-volitional basis.

Hence, the affective components of students’ cognitive processes give those processes their “attitudinal colour”. Statistics is included in the curricula of many disciplines at university in order that students may use it and appreciate its relevance to their chosen fields. In Western countries, statistical thinking is bound up with technology, commercial concerns and other matters of cultural importance. The social interests and preoccupations of a culture are not, however, automatically assimilated by individuals, but are monitored.

Students regulate their thinking and actions according to their evaluations of a learning task. What is culturally accepted may be personally abhorrent. There may even be a conflict between studying statistics and a student’s perception of her own self-realisation or values. For example, in the following survey response one student expressed the idea that real psychologists don’t do mathematics.

She wrote:

- I don’t even see the point. In psych why must maths infiltrate itself??? Studies have shown that those who have high maths abilities have low or poor communication & perception skills—shouldn’t psychologists be exceptionally perceptive & able to communicate well? It seems that if there aren’t silly numbers to justify things then they aren’t plausible in our computer/maths/science promotive society.

Leont’ev makes an important distinction between meaning and personal sense, which relates to conflicts such as the one expressed by this student. To Leont’ev, meanings are connected with the reality of the outside objective world, the life of society. Leont’ev calls them the “crystallization of social experience”. Personal sense, on the other hand, is connected with the reality of the person’s own life and motives. That is, personal sense involves the incorporation of socially constituted meanings into the psychology of the individual.

Personal sense, according to Leont’ev “does not have its own ‘supraindividual’, ‘nonpsychological’ existence”. Sometimes, there is a mismatch between societal and personal meaning. According to Leont’ev, it is affect that signals to the individual the fit between the two.

Lerman argues that the valuing of decontextualised, intellectual processes, divorced from personal and social elements, is expressive of oppressive discourse. It is this privileging of abstract thought, such as academic mathematics, that is disempowering

for some students. Sierpinska & Lerman refer to the vested interest mathematicians have in maintaining the status of mathematics in society. This idea was passionately expressed by a participant in my pilot study for Study Two.

She wrote:

- Maths is an exercise in agony, because the people who teach it make one feel as though maths belongs in a higher plane of evolution. Even though the number system is for everyone, and the concepts are there for everyone, the feeling that you do not deserve to know anything runs rampant. Maths, in short, is a lofty pain and a real headache to study.

In summary, the issues that I am concerned with in this project relate to the ways that students orient themselves to and interpret the task of learning Statistics and to how personal dimensions and individual actions are linked to the wider social, institutional and cultural arenas surrounding them.

RESEARCH DESIGN AND APPROACH

The activity theory perspective suggests that a focus on teaching, unrelated to students' perceptions about their own learning and its context, is unlikely to be successful in changing outcomes. My approach focuses on the experience of learning Statistics at university from the point of view of the student. Consistent with my theoretical framework based on the ideas of Leont'ev and Vygotsky, I explore the network of relationships between learners, the subject matter and context.

My investigation took place at a large and traditional metropolitan university, the oldest in Sydney. The participants were second year Psychology students, all of whom were required to study Statistics. Statistics is introduced in first year Psychology at The University of Sydney, but it is as a compulsory component of the second year course that it plays a major role, contributing one quarter of the final assessment mark. The instruments I used included participant observation, interviews and surveys. My project consists of two investigations. Study One is entitled "Exploring Mature Students' Learning of Statistics". The context for this study was the Mathematics Learning Centre where I work. The participants in this study were of great interest to me as an educator concerned with ameliorating students' difficulties with Statistics. Study Two is called "Understanding University Students Learning Statistics as a Service Course to Psychology". In this study, I widened the context of my inquiry to include the general Statistics class. For my investigations, I drew on a triad of research tools—naturalistic inquiry, procedures based on the phenomenographic method and quantitative methods for describing data and exploring patterns and relationships. Among the instruments I developed was the Approaches to Learning Statistics Questionnaire.

PSYCHOLOGY STUDENTS AT THE MATHEMATICS LEARNING CENTRE

In Study One I utilise a naturalistic perspective to draw on the rich accounts of five students who were studying Psychology after many years away from the educational

setting. My aim is to tell the story of these students in a way that is generalisable to others—that is, I try to uncover recognisable insights. I was in no way involved with the formal assessment of these students and worked very intensively and closely with them. Hence I was able to develop a close relationship with the participants of Study One and had more access to their perceptions and feelings than is usually the case for university educators in statistics. My analysis is based mainly on interview data, complemented by written expressions from various sources. The five participants were selected “purposively” for their range of experiences and attributes. As is often the case with qualitative research, the data for this study consisted of descriptions, direct quotations and excerpts, obtained by close psychological contact with those being studied.

The study is an example of action research in that my actions as a researcher were inseparable from those designed to assist the students in their learning of Statistics. As I have outlined in Gordon this means that the methodology for the study developed through the “action research spiral”. They describe this spiral as recurring cycles of:

- Planning, acting, observing and reflecting, with each of these activities being systematically and self-critically implemented and interrelated.

VIEW OF PSYCHOLOGY STUDENTS LEARNING STATISTICS

Study Two was grounded in Study One and broadened the perspective on students learning Statistics. The participants of Study Two were Psychology II Students who took part in a survey during their Statistics lecture, near the end of the first semester. The survey was completed during 20 minutes of the lecture time and was the major source of data for this study. Other data for this study included assessment results and vignettes from interviews with students selected from those who agreed to contribute further to my research.

Background of Phenomenography as a Research Tool

One major focus of Study Two was to elicit students’ conceptions of the subject matter, in the context in which they were studying it. To do this I adopted a phenomenographic approach. The term phenomenography was first used by Ference Marton to describe programmes of research which had the common aim of describing peoples’ conceptions.

The results of phenomenographic analysis are categories of description of the qualitatively different ways students conceive of the phenomenon being investigated. This focus on describing conceptions is not the sole aim of the investigations in which phenomenography is used as a research tool. Rather it is learning and teaching that are being investigated based on the focusing of conceptions. Conceptions are the lens through which the phenomenographic researcher views learning.

Marton describes phenomenography as a method:

- For investigating the qualitatively different ways in which people experience, conceptualise, perceive and understand various aspects of, and phenomena in, the world around them.

Phenomenography was developed within the framework of education research at the University of Gothenburg, Sweden. It is concerned with the relations between people and the objects of their perceptions or content of their thoughts. Marton calls phenomenography a “research specialization”, that is, a combination of a research orientation and a research approach to describing and comparing conceptions and understandings.

The research orientation emphasises the importance of description and the use of categories as forms of expressing the conceptions. The research approach is characterised by the open explorative form of data collection and the interpretative analysis of data. Saljo, a forerunner in the development of the phenomenographic approach, comments that in more recent studies, the term “way of experiencing” is used in preference to the term “conception”.

This emphasises that:

- The prime interest of phenomenographic research is in finding and delimiting the variation in ways of experiencing reality.

Hence, in deriving categories of description for students’ conceptions of Statistics, I am attempting to find relationships between the students and the subject matter which express their ways of experiencing the subject, Statistics, as they were learning it. Svensson argues that underpinning the phenomenographic specialisation is an assumption that knowledge is fundamentally a “question of meaning in a social and cultural context”. Such an approach views phenomena systemically and avoids separating person and context.

As Saljo in an editorial of Learning and Instruction, concludes:

- Human experiences are inescapably cultural in nature, and learning and growth take place within cultural boundaries.

This view of the intertwining of culture, mind and action is consistent with a Vygotskian assumption that there is no duality between self and context; between thinking and acting. The phenomenographic view of knowledge is that it is relational, created through human thinking and activity which relates to the external world—a relation between thought and social and cultural life. Hence my choice of phenomenography as one of my research tools is consistent with the theoretical perspective with which I am framing this investigation.

The relational view of learning furnished by phenomenography has formed the basis of much of the research into student learning in higher education. Phenomenography has been fruitful in exploring students’ conceptions and understanding in a number of topics, including physics; literature interpretation; essay writing and computer programming. In summary, this research indicates that learners’ experiences should be considered as involving the ways students relate to the learning environment, their goals and intentions, as well as their learning strategies. My colleagues and I have found phenomenographic methods useful to explore the qualitatively different ways in which students experience learning mathematics. Two studies were conducted.

In the first study we investigated the conceptions of mathematics and approaches to learning it of students entering university. In the second investigation we explored these students' conceptions of a fundamental mathematical concept, namely that of a function. Traditionally, phenomenographic studies are built around interviews. However, in these investigations, my colleagues and I extended previous research by constructing categories from the written responses of a large number of students. This developed phenomenography as a research tool.

In Study Two, I built on our previous phenomenographic research by looking at the variation in the way second year Psychology students conceived of the nature of the subject matter, Statistics. The phenomenographic perspective provided my basis for identifying categories of description of students' conceptions. These revealed qualitatively different ways of experiencing Statistics. In much research, the categories of description or variables are imposed on the data by an expert researcher.

In Study Two, in keeping with phenomenographic practice, the categories of description were constituted in relation to the data—from interpretations of the survey responses. This does not imply that I, as researcher, viewed the data free of ideas based on prior experience or theoretical orientation.

Indeed:

- The act of categorisation is by its very nature subjective.

Further, my research is deliberately set in the framework of activity theory. However, instead of characterising students' responses to the three open-ended questions in the survey by means of an established taxonomy, the categories emerged in the context of the study. By keeping the questions as general as possible, I tried not to shape students' responses. Rather I tried to explore the students' points of view. This enhances the ethnographic validity of this research, that is, the sense of the research in terms of the lives and awareness of the participants in it.

The phenomenographic method analyses meanings constituted within a particular context and content domain. My categories were derived in a particular setting. However, they and their structure as an organised set, may prove useful in contexts other than the one from which they emerged.

In accord with the phenomenographic approach:

- The categories of description are supposed to be replicable, but the way in which they were found is not.

Conceptions may be expressed in different forms of action but they are most accessible through language. Variation in conceptions, expressed by people, immediately brings in the cultural and social context in which these conceptions are reported. The students' reports on their ways of conceiving Statistics, although limited to the students' awareness and abilities in written verbal expression, provided me with a rich source of information for characterising these experiences. This experiential perspective is referred to as a "second order" perspective by Marton. He contrasts this with phenomenology, based on Husserl's imperative to return to the essence of a phenomenon, through immediate

experience and free from conceptual thought—a first order enterprise. Phenomenography is a perspective on how things appear to people—how phenomena are experienced, rather than what they are “in reality” or whether or not such a reality can be described.

Phenomenography was derived from empirical and pragmatic research in education. Its commitment to philosophical foundations is therefore not always clear. The procedures defining its use are neither prescriptive nor indisputable. Indeed, Hasselgren & Beach argue that “there is no genuine consensus method of phenomenography”.

To them:

- Phenomenography is research which is simply concerned with how things are understood, the experiences of the process of formation of understandings at individual levels, and their distributions in specific collectivities.

While my way of applying this research tool is individual, I have tried to maintain the essence of phenomenographic research as a way of understanding the multiple realities of students.

That is, my appropriation of Marton’s term signifies this approach. I view phenomenography as a research tool which bridges the gap between empirical, analytic methods, which are recognised by the research community, and students’ subjective awareness of their own diverse conceptions and experiences.

Introduction to Analysis of Students’ Approaches to Learning Statistics

I analysed students’ approaches to learning Statistics by means of a questionnaire, the Approaches to Learning Statistics Questionnaire. My questionnaire was modified from the Approaches to learning Mathematics Questionnaire which, in turn, was derived by my colleagues and myself from the Study Process Questionnaire. Biggs refers to three approaches to learning: deep, surface and achieving. He differentiated each of these into two aspects—the motive, or intention, and the strategy. Hence, in the Study Process Questionnaire the items are divided into those referring to motive or intention and those referring to strategy.

Each motive-strategy combination defines a distinct approach to learning—deep, surface or achieving. Early work by Biggs, on samples of students from universities and Colleges of Advanced Education, indicated that the achieving approach was equally related to the surface and deep approaches in tertiary students.

The surface and deep approaches, however, are referred to by Biggs as:

- Independent ways in which students may become involved in learning.

In this project I was concerned with deep and surface approaches to learning. Achieving approaches were of less theoretical interest to me and were not investigated. Considerable research has been done on the Study Process Questionnaire. For example, a literature review on approaches to studying indicates that mature students seem more likely than younger students to adopt deep approaches to learning, while surface approaches decline with age. However, completion rates and possible bias in these studies, for example, in the postal returns of surveys in Biggs’ investigations, renders

this finding problematic. Richardson also reviewed the literature on cultural specificity of approaches to studying in higher education. Investigations using the Study Process Questionnaire in countries as diverse as Australia, Nepal, Britain and Nigeria yielded a similar two factor structure for this questionnaire—one indicating a deep approach and the other a surface approach. The research also suggests that orientations towards understanding the meaning of learning materials, that is, deep approaches to learning, reflect a consistent agreement across different cultures with regard to the goals of higher education.

On the other hand, this research indicates that patterns of responses on the subscales measuring surface approaches to learning, are more distinctive to differing cultural contexts. Concern has been expressed about the factor structure underlying Biggs' differentiation of the Study Process Questionnaire into motive and strategy components.

In my analysis of the Approaches to Learning Statistics Questionnaire I did not find evidence for a breakdown into intention and strategy factors. I therefore analysed the items of the ALSQ in terms of two subscales: one measuring a general surface approach and the other a deep approach to learning Statistics. Biggs characterises a surface approach and a deep approach to an academic task by the following.

A surface approach is distinguished by:

- A view of the task as imposed, a demand to be met;
- A fragmented view of the task—parts are unrelated to each other and the task is unrelated to other tasks;
- Worries about the time taken by the task;
- A lack of recognition of any personal meaning in the task;
- Reliance on memorisation and ways of reproducing the details.

A deep approach reflects:

- Interest and enjoyment in the task;
- A search for underlying meaning;
- Efforts to make the learning personally meaningful—relating task to personal experience or to the real world;
- Holistic approach—relating parts of the task to each other and to other knowledge;
- Theorising about the task.

The surface approach and the deep approach, as defined by Biggs, correspond closely with the surface and deep levels of processing distinguished by Marton and Saljo. Later, Marton and Saljo referred to these as surface and deep approaches to learning. In contrast to Biggs, however, Marton and Saljo see surface and deep approaches as the opposite ends of a single continuum, rather than as independent ways of learning. Marton and Saljo showed that students would adopt one of two methods of processing information according to their intentions.

If their aim was merely to memorise and reproduce material, or, as Biggs puts it:

- To display the symptoms of having learned,

They would adopt a surface level of processing; if they wanted to maximise their understanding of the underlying meaning, they would adopt a deep strategy. Hence Marton and Saljo inferred a relationship between strategy and intention. What the student intends to gain from the learning determines the strategy used. In recent research, Marton, Watkins & Tang make the point that while a surface approach to learning is often associated with rote learning, this is not what characterises it.

- Rather, a surface approach is characterised by a focus on the learning material or task in itself and not, as would be the case for the deep approach, on the meaning or purpose underlying it.

Their ideas accord with Leont'ev's proposition that what distinguishes actions are the goals driving them.

INVESTIGATING INDIVIDUALS IN SOCIALLY SIGNIFICANT PRACTICES

My project is an example of what Chaiklin describes as “investigating individuals in socially significant practices”.

These are studies in a “yet-to-be-embodied” research tradition that:

- Try and develop an account of the actions of individuals participating in a societally significant practice while it is occurring, by an analysis that locates the practice in a social, societal and/or historical perspective.

Chaiklin proposes that many different theoretical perspectives and methods could contribute to such research. He summarises such projects as having five common characteristics, which I assert are fulfilled by my project.

- Firstly, these studies take concrete, meaningful, societal practices as the direct object of study. That is, the setting for the study is an actual example of human practice and would be so whether or not it was the object of study. In my research I have tried to get as close as possible to the participants and interpret situations which were as typical as possible for students learning Statistics.
- Secondly, the practice takes place in a recognisable societal institution, in this case an institution of higher learning.
- Thirdly, such studies have a definite theoretical interest. Insights into the practice of learning Statistics are important elements of my research but so are insights into and extensions of the activity theory framework in which I have interpreted this practice.
- Fourthly, there is the idea that knowledge and actions are social as well as individual. I draw on Vygotsky's and Leont'ev's idea of “co-knowing” to illustrate this notion—statistical knowledge is socially created and re-created through the consensus of practitioners. This collective development of knowledge is facilitated through tools, for example, by the easy and quick exchange of ideas through electronic media. In my university, Statistics is continually being transformed by each lecturer and committee involved in organising the curriculum and setting the examinations. Students come to know Statistics

through interacting with peers and teachers, as well as through their own reflections and actions. Socially organised constraints and resources surround individual actions.

- *Finally, the particular practice selected for such studies has:*
 - Significant consequences for the people participating in these practices.

Learning Statistics can change the lives of those participating in this practice. In some cases, the outcomes could be instrumental in determining students' academic and career paths. For example, students' examination results in Statistics could determine whether they would attain a degree or be accepted into Psychology Honours. Learning Statistics can also powerfully affect students' personal development as was the case for some of the participants of Study One.

In this project I have tried to piece together practical and theoretical perspectives to contribute to the building of a research framework—to an understanding of the relations between theory and practice in education.

THEORETICAL MODEL

OVERVIEW

In order to present a model of how learning Statistics is derived from statistical activity, it is important for me to explain both the context of the learning I am investigating and the theory by which I interpret the findings. The scene for my two investigations. The research tools I use, such as naturalistic inquiry, phenomenography and statistical analyses, provide the means for investigating local occurrences. I use these tools to explore relationships and events as they occurred for particular individuals in a setting which was bounded spatially and in time.

It is by means of the theoretical framework, that is, by my application of activity theory, that I try to provide insights that transcend the particular context. To start off, I locate frameworks which draw on Vygotsky's theory in terms of some other major theories of education, particularly those which are currently important to research in mathematics education. An outline of activity theory, as expounded by Leont'ev. I introduce my approach to understanding students learning Statistics from an activity theory perspective.

COMPARISON WITH OTHER THEORETICAL APPROACHES TO LEARNING

Western researchers are increasingly finding that a Vygotskian perspective provides a helpful lens with which to view the complexities of mathematical learning. This approach takes into account the fundamental role of social interactions and historical or cultural influences on learning. It aims to understand how language and symbols mediate meaning. Vygotsky's work on cognitive development contributed to the evolution of cognitive psychology. In this framework, not only are the actions of learners important but also their goals and how the actions are situated in a sociohistorical context. For

Vygotsky and the activity theorists who followed him, the context for all activity is important, and collective external activity precedes individual internal activity.

The emphasis on the melding of the individual with society is particularly relevant to the analysis of adult learning, as adults are cognitively fully developed in the biological sense but generate new knowledge through participation in activities which are socially and culturally rooted. Hence, while Vygotsky's interest was mainly in the development of the child his explanation of learning through interpersonal interactions, and Leont'ev's interpretations and ideas, are useful for extending research on adult development.

The Vygotskian framework provided a stimulus for theories of situated cognition in which a setting is defined as the relationship between an actor and the arena in which she or he acts. Cognitive skills take shape in the course of individual participation in socially organised practices. Research into situated learning recognises the cultural dimensions of learning and practices, for example, the cultural aspects of literacy; numeracy; apprenticeship and interactive expertise or working intelligence.

Issues addressed by researchers working in this paradigm often concern the effects of cultural artefacts—the devices and technologies, such as computers or internet systems, through which we “manipulate reality”. Varela et al propose that knowledge is “amplified” by technology.

To them, knowledge is seen as:

- Tangibly and inextricably linked to a technology that transforms the social practices which make that very knowledge possible.

These issues are highly pertinent to a study of statistical learning, as statistics is an artefact of Western culture. It is linked to our technology and it both moulds our thinking—the way we understand our world—and is used to produce changes in our environment. Theories of situated cognition resonate with activity theory approaches in their recognition of cognition as constituted in the social, vocational and cultural life of people. Consonant with Leont'ev's theory research in this area acknowledges the role of tools in mediating cognition. However, in theories of situated learning, consideration is given to the importance of informal learning, an area little addressed by Vygotsky or Leont'ev.

One of the strengths of the conceptual frameworks provided by Vygotsky and Leont'ev is the integration of affective and motivational dimensions into explanations of cognition. There have been many researchers in diverse areas of education, who seek to show that motivational goals and intentions give meaning to learning—they are what drives the process. For example, Ausubel, Novak & Hanesian characterise all learning as meaningful learning.

Ames & Ames argue that goals:

- Provide the mechanism for filtering perceptions and other cognitive processes.

Bandura explains the link between intentions and behaviour by means of the construct of self efficacy:

- The conviction that one can successfully execute behaviour.

Studies by Volet; Volet & Chalmers and Volet & Lawrence show the importance of goals as mediators of university students' learning. Education research increasingly recognises that the attitudes, beliefs, motivation and intentions expressed by learners are important, not only for understanding the actions of those learners, but also for the insights provided into cognitive theory. However, much of the literature in the area of affect and motivation lacks a unifying theory. In Leont'ev's theory, the role of motivation and of goals in cognitive processes is systematised by means of his three levels of analysis of activity.

In mathematics education, research into affect and related areas, such as beliefs and perceptions, is extensive. The breadth and depth of these studies reflect an awareness of the importance of charting the changing trends and many dimensions of affect in learning. For example, Mcleod reviewed the research since 1970 in the *Journal for Research in Mathematics Education* on affective issues in mathematics learning, showing that such studies are central to the goals of mathematics education. These studies explore many different aspects of the affective domain. One major area of interest to mathematics educators is that of students' attitudes to learning mathematics, particularly gender effects. Fennema summarises work using attitude scales, such as the Fennema-Sherman Scale, for measuring gender effects in affective issues and achievement in mathematics.

Her research showed major differences between males and females on a number of variables, such as confidence in learning mathematics, or perceived usefulness of mathematics. Recent work on the Fennema-Sherman Scale by Forgasz, Leder, and Gardner suggests that changes in response patterns challenge the reliance that has been placed on this important instrument. They recommend revising it, particularly the scale referring to mathematics as a male domain. This indicates that important transformations are taking place in students' attitudes to mathematics. An area of research related to attitude studies concerns students' perceptions and beliefs about mathematics. Schoenfeld pointed out that students' performances on mathematical problems were often undermined by their beliefs about mathematics, such as the belief that a mathematical problem should always yield a solution quickly.

There have also been recent studies on factors affecting students' motivation to learn mathematics showing that classroom environment, goal setting and type of task are critical elements. Research in statistics education, too, confirms the importance of students' attitudes, beliefs, interests, expectations and motivation to their learning.

Affective issues are related to the ways in which students evaluate and regulate their learning—key elements in defining their activities. In this regard, a helpful distinction was made by Semenov, a Soviet psychologist, who applied activity theory to the study of thought processes. He distinguished the “intellectual plane of thought”, which pertains to the development of the content of a problem, from the “personal plane of thought”, which refers to the individual's evaluation of his or her efforts—reflections on the meaning and success of the ongoing mental activity. While Semenov was referring to problem solving, his distinction extends usefully to students' appraisals of studying a statistics

course. In my project I will differentiate between students' so called "objective" or practical evaluations of Statistics—evaluations in terms of extrinsic and culturally framed factors such as jobs, or higher study, and their personal assessment of it in terms of their liking for or interest in it and feelings about learning it.

A framework for the role of values in learning mathematics was developed by Southwell. This extends McLeod's research on the relationship between beliefs, attitudes and emotions in mathematics education. Southwell sees values as closely related to beliefs but "more complicated and encompassing". She denotes values by a triangle consisting of the three elements of valuing: cognition, affect and volition.

She does not distinguish between values and valuing, which renders her framework problematic to me. However, the three aspects of valuing she specifies, tie in with Semenov's notion that the plane of thought has intellectual and personal aspects. Southwell's notion that volition is an aspect of valuing also accords with Leont'ev's insistence that activity is purposeful. Leont'ev proposes that need always stands behind thought or action. However, unlike Southwell, Leont'ev explains how purpose, needs and goals are socioculturally framed. That is, he explicates these relations in terms of individual's activities, rather than simply stating that such relations exist.

The focus on learning as "embedded in social situations, practices and cultures" is one of the key elements differentiating activity theory from constructivism, a major theory in mathematics education. While activity theorists assert that all learning is socially constituted, constructivism emphasises that all learning is the construction of meaning by the individual. In both cases action is required on the part of the learner. Constructivist ideas are drawn initially from Piaget's work, but constructivism has developed in various forms, such as radical constructivism or social constructivism.

Constructivist thinking has been instrumental in challenging the way educators regard learning—and therefore teaching. Piaget's work led to the idea that a mathematics classroom could be a place where children could direct their own learning, pose their own problems and discover their own mathematics. The impact of these ideas on research into mathematics education has been extensive, so much so that Ellerton asks whether we have become too comfortable with constructivism. What do we really mean by the constructivist teacher?

Relation between Constructivism and Vygotskian Approaches

Vygotskian perspectives are often contrasted with theories of learning based on constructivism. A Piagetian approach considers learning as the result of individual constructions of the environment, namely assimilation and accommodation. Assimilation is the process whereby the individual integrates new perceptions or situations into her or his existing individual schemes, and accommodation is the individual's effort to adjust schemes to the environment. The essence of this approach is the focus on the individual and on stages of maturation as prerequisites for the development of mental facilities.

Vygotsky recognised the genius of Piaget's work and was influenced by it, though disagreeing with Piaget's conception of:

- The role of egocentrism in the developmental relationship of language and thought.

According to Vygotsky:

- The developmental uniformities established by Piaget apply to the given milieu, under the conditions of Piaget's study. They are not laws of nature but are historically and socially determined.

Thus Vygotsky criticised Piaget for his failure:

- To take into account the importance of the social situation and milieu.

Piaget, in turn, was aware of Vygotsky's theories. Indeed, Piaget modified some of his theories in the light of Vygotsky's criticisms. However, Bodrova & Leong point out that this happened after Vygotsky's death. For this reason, the works of Vygotsky's students, more than of Vygotsky himself, have some common ground with Piaget's ideas.

Bodrova & Leong express the opinion that this:

has caused many psychologists to erroneously consider

- The Vygotskian framework as part of Piaget's constructivist tradition.

Piaget, like Vygotsky, explained cognitive development, such as mathematical development, as a result of children interacting with their environments. However, while Piaget viewed learning as subordinate to development, Vygotsky insisted that learning, by which he meant formal or school learning, directly influences development.

According to Vygotsky:

- Instruction is one of the principal sources of a schoolchild's concepts and is also a powerful force in directing their evolution; it determines the fate of his total mental development.

To Vygotsky, thinking itself, together with the cognitive structure required for thought and indivisible from the function of that thinking, is generated by internalisation of social relations. Meaning making takes on different forms in the interpretation by constructivists and Vygotskian theorists. In the former, it is a personal contribution of the learner who is involved in the education process.

In a Vygotskian approach, meaning making is bound up with cultural values, such that:

- The qualities of thinking are actually generated by the organizational features of the social interaction.

Vygotsky explained that higher mental processes develop through interaction with others.

He expressed this as follows:

- From the very first days of the child's development his activities acquire a meaning of their own in a system of social behaviour and, being directed toward a definite purpose, are refracted through the prism of the child's environment.

The path from object to child and from child to object passes through another person. This complex human structure is the product of a developmental process deeply rooted in the links between individual and social history.

Knowledge to a constructivist is an individual construction; to the activity theorist it is a collective representation. This is not to intimate that constructivists deny the importance of the societal context, nor that there is no place for the individual in the Vygotskian viewpoint. Rather it is a shift in emphasis.

Saxe suggests of the Piagetian perspective:

- Social life is related to cognitive development as an external process, and the way sociocultural life may be deeply interwoven with the character of intellectual functioning is unanalyzed.

In contrast, Bauersfeld calls activity theory a prototype of collectivist perspectives. A simplified overview of this perspective, provided by Bauersfeld is that learning is enculturation into pre-existing societal structures. This suggests a ready made world and knowledge, a notion for which activity theory is criticised. This enculturation is not a passive process. The individual develops through effective participation in activities. In contemporary research the term “appropriate” is used rather than “internalise” to emphasise the active role of the learner.

Radical constructivism, developed by Von Glasersfeld, has been criticised for studying:

- Human mental functioning as if it exists in a cultural, institutional and historical vacuum.

To its critics, radical constructivism presents a view of the human being as a closed system: “self-organising, self regulating, self-contained”. Communication is problematic in this approach with meanings “taken as shared” rather than shared. There have been attempts to integrate social aspects in accounts of the construction of mathematical knowledge. These add on to radical constructivist perspectives by acknowledging the “important but secondary” place of social interactions in knowledge construction. Lerman argues that adding the “social” to constructivism leads to incoherence. It fails to account for how people understand each other, or to explain the social dimension in a personal world.

Lerman proposes that the Vygotskian approach presents a different world view to the mentalism that underlies constructivist thinking, one in which:

- It is necessary to recognize the shift from a view of the autonomous cognizing subject constructing her or his subjectivity and knowing to one of the construction of human consciousness in and through communication.

Hence, the difference in thinking between Piaget’s and Vygotsky’s followers does not simply lie in the latter taking greater account of social interactions, as social constructivists do, anyway. Rather it is in acknowledging the primary sense in which subjectivities and positionings are constituted and reconstituted by the social events and cultural history surrounding the individual. To activity theorists, socially and

culturally linked goals and needs are seen as integral to cognition, rather than as the “interference” of subordinate issues.

Ernest proposes that any form of constructivism retains portions of the radical constructivism metaphor of an:

- Evolving and adapting, but isolated organism, a cognitive alien in an hostile environment.

Hence the individual’s cognition takes place in a private domain of experience. Transference from the public or social realm to this personal domain cannot be explained. This problem is not, however, restricted to constructivism. Davydov an important student of Leont’ev, postulates that the structural difference between individual and collective activity is also an unsolved problem of activity theory.

The two poles of thought represented by constructivism and Vygotskian approaches are major current theories which have in common a rejection of the transmission view of teaching and learning. They both focus on the actions of the individual in learning. Their points of difference have led to a multiplicity of models. Current approaches include: synthesising the two approaches superseding models and emerging connectionist theories of cognition or enactivism. One example of a middle position between individualism and collectivism is an “interactionist” perspective.

It takes the position that the culture of the classroom is constituted in social interactions among teachers and students. Another strategy has been to adopt complementary perspectives. Bartolini Bussi argues that rigid adherence to the principles of constructivism or to a Vygotskian perspective does not recognise the richness and complexity of the ideas of the founders. Moreover she feels that theoretical coherence should not take precedence over real life problems. Indeed Bartolini Bussi moves that such a pragmatic view is “deeply Vygotskian”, as Vygotsky himself was concerned with pressing social and cultural issues, rather than theorising. Lerman presents a critique of this view, arguing that adding bits of one theory to another, or slipping from one to the other, ignoring the contradictions, or plastering over theoretical holes in each, do not do justice to the insights and coherence of either.

There is considerable debate between protagonists of Vygotskian approaches and those of other current theories. I believe that the usefulness of activity theory lies in its systemic approach. Activity theory posits a view of learning in which personal experiences, goals, subjective perceptions and sociohistorical factors are interwoven. Rather than focusing on separate facets of learning and context, activity theory implies a commitment to investigating the learning process as a “dynamic system of meaning” in which intellect and affect unite and through which society and the individual interact and evolve. This view of learning includes notions of growth and diversity.

SOME INFLUENCES ON THE THEORY OF ACTIVITY

In developing a Marxist psychology, Leont’ev drew on the work of many philosophers, such as Marx himself and Engels, and of psychologists, particularly Rubinshtein

and Vygotsky. Leont'ev's orientation to Marxism is a critical element in his theories about human consciousness and cognition. He explained the revolution brought about by Marx in the theory of cognition in terms of the role Marx ascribed to human practice in cognition. That is, Leont'ev stressed that to Marx cognition and activity were inseparable.

Leont'ev wrote:

In reality the philosophic discovery of Marx consists not in identifying practice with cognition but in recognising that cognition does not exist outside the life process that in its very nature is a material, practical process. Leont'ev concurred with Vygotsky who was convinced of the Marxian concept that the "human essence is constituted by social relations". That is, awareness, thought and other higher psychological functions, arise and develop in the interactions among people and their ties to the world in which they live. In Marx's philosophy, the object of knowledge is to transform the world, not just to understand it. In Soviet education, these ideas were translated into attempts "to construct a new socialist man". Both Vygotsky and Leont'ev built their psychology on a conception of Man developed by Engels and Marx.

This conception is expressed by Mellin-Olsen as follows:

- The conception is one of man as an acting person, at one time being both determined by history and determining it, being both created by society and creating it.

It was in this context of historical and dialectical materialism that Leont'ev considered activity as an object for psychology. Leont'ev's formulation of activity owes much to his many debates with Rubinshtein. Rubinshtein, who lived from 1889 to 1960 explored the 'mind body' problem stressing that activity is not only external behaviour but is inseparably linked with consciousness.

He expressed this in the formula:

- External causes act through internal conditions.

Leont'ev disagreed with Rubinshtein in that the latter considered that practical activity was the subject of study for psychology only to the extent that such activity included internal mental processes, such as perception or thinking. Leont'ev characterised this as a one sided view, in which mental activity is seen as directing physical activity, or external activity is viewed as dependent on psychological images. In Leont'ev's view, the "circle" of mental processes opens up "to meet" the external world of objects through external practical processes or activity. For example, bouncing a ball leads to a perception of its resilience.

To Leont'ev the function of activity is "transforming this reality into the form of subjectivity". That is, activity enters psychology not as one of its elements, nor as an aspect of internal processes, but through its function—the function of linking the individual's external and internal worlds. At the early stages of its development, activity must have an external form. Activity develops through reflection and regulation. Through processes of interacting with others and the environment, mental images and thoughts

take place. According to Leont'ev these “deflect, change and enrich this activity”. Vygotsky's ideas and images permeate activity theory as a melody runs through a complex orchestrated score, with the harmonies, counterpoint, new melodies and even discordant notes, being provided by his students, mainly Leont'ev and Luria. It is therefore important for me to interpret Vygotsky's notions as they apply to activity theory in general and to my application of it in particular.

Vygotsky did not himself develop the concept of activity as a theoretical construct. However, he conducted empirical investigations based on the assumption of activity, emphasising the role of speech and semiotics in causing

Fundamental changes to the nature of the activity. According to Davydov and Radzikhovskii:

- The true methodological significance of Vygotsky's work consists of the assertion that activity is the explanatory principle in psychological theory.

This remark suggests that Vygotsky adopted Marx' approach to formulating a new psychology. In the last years of his short life, however, Vygotsky moved away from strict adherence to Marxist principles—at least this was the view of the Soviet authorities—resulting in his works being banned for many years in the Soviet Union. Some of Vygotsky's colleagues and students, including Leont'ev, went on to interpret and propagate Vygotsky's sociohistorical theory of psychology within a Marxist framework, linking individual consciousness and personality with social and practical activities.

OUTLINE OF LEONT'EV'S ACTIVITY THEORY

Wertsch, commenting on Leont'ev's exposition of the tenets of activity theory summarises three major original contributions of Leont'ev to the works of Vygotsky and other theorists. The first of these is the construct of activity itself. Secondly, Leont'ev presents levels of analysis of human activity, arguing that activity can be viewed from the perspective of the milieu or context in which it takes place, from the goal directed actions which make up the activity and from the operations which depend on the conditions under which actions take place. Thirdly, Leont'ev replaces sign systems, which are central to Vygotsky's theories, with activity as the mediator between humans and their worlds. Thus an analysis of activity explains the dynamics of how humans relate to their physical and social environments.

Activity

Activity, as a construct, occupies a major explanatory role in the psychology that dominated the Soviet Union in the sixties and seventies.

The very process of living is described by Leont'ev as:

- The system of activities that succeed one another.

Activity refers to the functional unit of human behaviour that relates the individual to his or her social and cultural world. Leont'ev and his fellow activity theorists explain that humans understand their world and develop knowledge about it by acting

purposefully in it. In turn this activity changes the world. Leont'ev rejects "positivist" notions of activity which stress activity as purely adaptation to the external world. His main criticism of this perspective is that if psychology is limited to the concept of socialisation of individuals, the structure and transformations which link humans with their society will remain a mystery. To understand these links he posits that we must investigate activity—its structure, its specific dynamics and its various forms, both overt and cognitive.

The Structure of Activity

Leont'ev identified three levels of analysis of activity.

Further expansions were provided by Zinchenko and Gordon.

- The first level is concerned with the global aspect of activity. This defines activity as a unit of life mediated by reflection; a frame outlining the context in which the activity takes place, for example, play, formal education or work. This level identifies the socioculturally defined milieu in which the actions occur. At this level analysis is therefore concerned with the motivation underlying and engendering the actions, the "energising function" of the activity.
- The second level relates to the actions which make up and realise the activity: how the task was carried out. To Leont'ev, actions are always directed toward a goal. He did not acknowledge that we sometimes do things without a conscious reason. He postulated that there may be many intermediate or partial goals to be satisfied on the way to achieving the final aim. These intermediate goals are determined consciously and with regard to the social relations in which they are set. Leont'ev gives the example of individuals beating the bushes to scare out an animal which will be caught by other, strategically placed, members of the community. The actions of beating the bushes do not in themselves have direct relevance to satisfying hunger, the motive of the activity. It is the connection of these actions to those of others and to the main purpose of the exercise which explain the actions.
- As outlined by Leont'ev actions have an intentional aspect and an operational aspect and these are carried out to realise a conscious motive. The same motive can give rise to different goals and accordingly can produce different actions. Conversely the same actions can realise different activities. For example, a statistical procedure such as a "t - test" may be carried out by a student to get an answer, with little reflection about understanding the procedure while different goals may lead to the same actions being carried out but with the focus on the underlying rationale of the procedure.
- The means by which actions are carried out are labelled operations by Leont'ev. Operations are therefore components of actions. While actions are determined by the goals which they fulfil, operations depend on the conditions, especially the tools, which delineate the exact mechanisms for carrying out the action.

Tools can be physical, or “extra-cerebral” for example, computers or calculators for carrying out statistical procedures. They can also be cognitive, for example an algorithm, followed mechanically in order to get a result. Leont’ev posits that it is the fate of operations to become automatic by their mechanisation. Operations, however, are inseparable from the actions and in turn the goals which they serve. They are under conscious control as part of the actions and, through these actions, the activity in which the individual is engaged. The issue here is that different conditions lead to different compositions of actions. Solving a linear algebraic equation, for example, may be performed on an operational plane if the student has practised the process many times but not if the student is a novice in algebra. This level of analysis is particularly pertinent to the analysis of learning Statistics where the aim of many students is to become familiar with, and so automate, processes and algorithms.

The implication of viewing activity from different levels is that these provide different vantage points for investigation. At the global level, Leont’ev distinguishes activities on the basis of motive and the object toward which they are oriented. This provides one vantage point from which to understand students’ learning. Activities are comprised of actions which are determined by the goals impelling them. Hence the conscious goals or needs driving the actions provides another perspective for analysis. Actions, in turn, consist of operations where the notion of operation has to do with the routine automatic aspects of carrying out a task. This level of analysis, therefore, relates to the means or resources for carrying out the actions.

Dynamics

To Leont’ev, the problem of understanding the dynamics of activity, is one of discovering the relationships that connect the individual’s behaviour with physiological functions, “the work of the brain”. By this, Leont’ev is not proposing a quick course in neuro-physiology. Rather he suggests an indirect approach—investigating the functional development of the mechanisms in the brain as a product of activity. This analysis must include an understanding of both phylogenetic development, that is, development which is sociohistorical or evolutionary, and ontogenetic development—the maturing of the organism. Leont’ev posits that higher order mental functions take place as a result of “mastering tools and operations”. These mental systems are instated by or generated by activity.

This means that mental images are not formed by the brain; they are the function of the brain. They are generated in the transition from the “extracerebral to the intracerebral sphere”. That is, there is no landing pad in the brain awaiting the arrival of images. It is by means of our actions, including mental actions such as reflection, that we transfer images from the external world to the cognitive sphere. Activity mediates between the environments surrounding humans and our internal domains. Leont’ev stresses that activity is characterised, not by its units which are meaningless if studied in isolation,

but by the systemic connections between the units and their transformations. That is, activity is not simply a sum of actions. Rather, the connections between the parts and the goal formation determine activity. The levels of activity are mobile or dynamic; actions can become activities in their own right. For example, driving a car is an activity if the person is learning to drive in order to pass a driving test.

It is purposeful, effortful action. The same actions—such as steering, braking, accelerating, may, however, be carried out as part of another activity, for example driving to work. Here the setting is work; the partial goal of the driving actions is to arrive at work—actions such as steering or braking, are the means of achieving this aim. Further, if the driver is experienced, the actions taken in driving may be automatic. They are now operations. However, in a difficult situation, such as a driver would experience in heavy traffic and wet conditions, conscious decisions will again regulate the actions, rather than a mechanical set of operations being performed. The example illustrates that in the course of attaining an overall goal, activity may be split up into separate, successive actions which are consciously carried out with the help of operations which, in turn, may have been formed under different circumstances. The opposite happens when the individual is no longer conscious of intermediate results—the overt actions and mental reflection merge together or are consolidated in carrying out an activity.

According to Leont'ev, investigating both these aspects:

- The breaking down of the activity and the integration of its actions and operations, can only be done by studying the links. These may be all internal, as in cognitive activities, but, more often, internal activity is implemented by external actions.

The word “activity”, as used throughout this thesis, refers to this framework. That is, a student’s statistical activity encompasses her actions within a specific context, her goals, and the resources available to her, whether technological or her own expertise.

THE ACTIVITY OF LEARNING STATISTICS

While activity may be viewed as an abstract construct, Leont'ev emphasises that in reality we always deal with specific activities, within a finite time, space and setting. His three levels of analysis provide three vantage points on a fundamental question for any investigation from the perspective of activity theory: the question of what an individual or group is doing in a particular setting.

In order to understand what a particular group of students is doing in learning Statistics, responses can be formulated in terms of each of these three levels. The first level concerns the cultural or historical milieu of the students’ actions. This is an account of students’ learning from the perspective of the global level of activity. To respond to the question at this level it is necessary to understand students’ actions in the institutional context—while studying a university course. A university has its own well defined social practices. Students’ interpretations of these practices are bound up with their perceptions of what is expected of them and will be fundamental to their actions.

According to Wertsch:

- An activity setting is grounded in a set of assumptions about appropriate roles, goals and means used by the participants in that setting.

This setting guides the selection of actions and choice of tools. It also determines the function of the activity. Wertsch explains that settings are not determined by the physical context but are created by the participants in the activity. Further, assumptions about the setting are often implicit, rather than consciously identified. Participants may not identify what organises their performance. An individual's understanding of the setting emerges as a "byproduct" of interacting with others in it.

To say that a student is engaged in the activity of learning a university subject, Statistics, simply tells us that the student is working in a particular socioculturally defined setting.

To understand activity at Leont'ev's second level of analysis one must look at the actions which, in his view, are defined by goals or partial goals. In my view, goals are not automatic nor fixed in advance. They are tested by action and remain fluid throughout the process of selection and testing. This notion of goals as in flux - as being socially tested hypotheses - was expressed explicitly by one student whom I interviewed during her first year at university.

I've never really studied before this at all, and it's only been since last year that I actually applied myself to anything in academic terms. And so for me everything's just - what's the word... an experiment. An experiment to see what works and what doesn't. Finally, a response to the question about a student's learning of Statistics can be formulated in terms of the tools and operations that the student has at her command.

This is the third level of the activity of learning Statistics that must be investigated. The availability of tools as mediating devices will partly be determined by the setting - for example what calculators are deemed appropriate for students to use. Tools and operations also depend on the student's personal history - her experiences, including the student's repertoire of skills, such as her ability to use a calculator or mathematical symbols. Affective elements are important as well. If the student's memory of using a compass at school is linked to negative emotions, this tool will not be readily utilised. The tools or operations used by a student affect how successfully the student is able to carry out her intentions.

FRAMING THE RESEARCH QUESTION

Guided by activity theory and within the methodological framework outlined, I explore the qualitatively different ways students relate to their learning of a statistics course at university. Statistics, with a capital "S" refers specifically to the statistics component of the Psychology II course. Hence this usage immediately specifies the setting of the activity as university learning of a particular topic at a defined time and place.

The main research question which I investigate is:

- What are students' orientations to learning Statistics?

I am interpreting the word “orientations” as signifying students’ positioning of themselves with respect to the learning task. This question is important because, from the theoretical perspective I have outlined, students’ orientations to learning Statistics are integral to the ways students engage with the learning task and hence relate to the quality of their emerging knowledge. To investigate the research question I explore three interdependent aspects. Firstly, I investigate different facets of students’ orientations to learning Statistics, including affective elements, students’ conceptions of Statistics and their approaches to learning the subject.

Secondly, I look at the connections between students’ orientations to learning Statistics and the outcomes of learning it, mediated by their activities. Thirdly, investigating this question intrinsically includes exploring the relationships between the students’ perceptions and actions and the contexts surrounding their learning. For example, if a student’s conception of Statistics is “algorithms”, this is not a characteristic of the student or of the subject Statistics or of the educational institution. It is a relationship between the student’s way of experiencing Statistics and the contexts surrounding these experiences. From the activity perspective, how a student orients herself in the learning arena her actions and her evaluations of the learning task, develop together and coherently and, with their accompanying outcomes, are organised within and contribute to the wider sociocultural setting.

A student’s orientation to learning Statistics could be regarded as a particular mind set individually generated in response to her experience of a certain context. My view of orientation is in contrast to this view of it as an individual and internal construction. Consistent with an activity theory framework, I regard a student’s orientation to learning Statistics as inseparable from her actions to learn Statistics and as part of a wider and dynamic societal “system of activities”. Statistical thinking cannot occur in isolation, in the head. In learning Statistics, an individual is participating in a cultural practice even when she appears to be acting in isolation, such as doing examples or reading a text book. Further, the statistics that the student is studying has itself developed historically through interactions which are culturally founded. The methodology for statistical inference, for example, has been cultivated and improved by successive generations and continues to undergo major transitions as a result of current technology.

My interpretation of the activity of learning Statistics, drawing on Leont’ev’s sense of the construct, refers to the actions a student takes to grapple with it—to engage, purposefully, with the problem of learning it. A student’s engagement with the task of learning Statistics is expressed by both cognitive actions and practical actions. These actions are directed, not random, even if the student does not articulate her goals. By “doing” Statistics a student “makes” it “artistic, practical, creative or routine”.

ACTIVITY THEORY: A PERSPECTIVE ON LEARNING

I review the literature interpreting Vygotsky’s approach and Leont’ev’s modifications of his ideas. I try to give a range of perspectives on their theories by including the views

of contemporary Russian theorists, such as V. V. Davydov and V. P. Zinchenko, and some well known Western scholars, such as J. V. Wertsch and J. Valsiner. To clarify my underlying theoretical position, I differentiate between theoretical approaches which are framed primarily in terms of Vygotsky's concepts and those which take the viewpoints of Leont'ev and his colleagues in the Khar'kov school.

Initially, I explain the background to my literature review and to my application of activity theory. I explain some important ideas of Vygotsky, particularly the cultural historical theory and other aspects of his framework associated with this theory. I develop the tenets of the activity theory approach: its historical emergence, its features and its principles and how these ideas are understood in contemporary research and education. I illustrate aspects of this theory with examples from my practice. These examples suggest that activity theory has important implications for education research and for understanding students' learning in practice. I comment on unsolved problems of the theory and some contemporary derivatives of the theories of Vygotsky and Leont'ev. I conclude by explaining the relevance of the theory to my investigation—how my approach builds on notions of activity theory and where it departs from these ideas.

BACKGROUND

Activity theory, in the view of Engeström “is one of the best kept secrets of academia”. Engeström speculates that this is partly due to language difficulties, in part because of the epistemological roots of the theory in German philosophy and dialectics and also due to the complex and “impressive” theorising behind the activity approach. Yet this theory is unquestionably evolving in Western thought as well as in its birthplace and has shown itself to have considerable implications for practice and research.

I develop the theoretical framework introduced. I review the principles of activity theory as they have been explored in the literature. Activity theory has been interpreted differently in different countries and at different historical periods. For instance, the translation of the original Russian scripts into various languages communicates different nuances of the theory. For example, the German word, “Tätigkeit”, evidently conveys a sense of serious purpose, such as work activity or occupation, which is lacking in the English word, activity.

Clearly, too, the perspectives of those who look back at a theory, which emerged in a period of tremendous turmoil and social change, will differ from the perspectives of those who wrote while they were embroiled in the events. Finally “theories in ‘use’ are always theories developing”. I have tried to piece together current interpretations of activity theory and its recent developments from the latest English publications available in Australia. The fact that a number of these scholarly works have been published since 1990 is an indication of the ongoing interest in activity theory and its relevance.

In Western education, the focus of the contemporary activity approach is on two major and related strands. One aspect pertains to the emphasis on goal driven actions as central to a student's learning. This aspect relates to Leont'ev's construct of activity.

Secondly, based on Vygotsky's expositions, which are more familiar to Western educationalists, current approaches stress the cultural, historical and social base of these actions as being germane to the learning that takes place.

My aim in this stage is to show how these cornerstones of the contemporary activity approach were developed in the work of Vygotsky and re-interpreted by Leont'ev and other of Vygotsky's colleagues and students. To begin with, I expand on my interpretation of students' learning of Statistics, taking aspects of activity theory as my starting points.

USING THE LENS OF ACTIVITY THEORY TO UNDERSTAND STUDENTS' LEARNING OF STATISTICS

I outlined my approach stemming from my belief that a student's learning proceeds through her active engagement with the task within a social and cultural setting. This draws on Leont'ev's notion of activity as "a system in the system of human relations". A focus on learning as activity which is socially and culturally situated is bound up with exploring what Varela et al term "histories that are lived". Leont'ev elaborating on Vygotsky's ideas, distinguished between three kinds of histories that are lived—evolutionary, societal and individual histories.

Firstly, evolutionary history has resulted in humans having bodily structures which enable us to interact with the world. This physical structure outlines the essentials of human existence. Secondly, from a societal perspective, the embodied person is spatially and temporally situated. The self speaks "not only as an individual voice but also as a collective voice", reflecting values and beliefs of communities and cultures which are significant to the individual. Finally, each person develops a personal story, a history of her own experiences which have left their mark. These three histories that are lived, act in tandem in human activity—such as the activity of learning Statistics. The actions a student takes to learn Statistics involve that student's cognitive abilities, which have developed over the history of the human race—her evolutionary history.

This history is inseparable from the development of tools, language and other cultural means of mediation. Secondly, a student's activity is related to her situation in an institutional and societal setting. Thirdly, and interwoven with these other histories, is the personal narrative of the student, both her biological and social development, which contributes to her ways of understanding Statistics and of interpreting the context in which she encounters it.

A key argument of activity theory is that personally meaningful goals have an "energising" function. Varela et al specify two aspects of goal directedness or intention.

These correspond to:

- What the system takes as its possibilities for action to be and to how the resulting situations fulfil or fail to fulfil these possibilities.

In my investigation, this means that actions taken to learn Statistics are directed by how the student construes her goals in context. In turn, how these goals become fulfilled, or fail to become fulfilled, is affected by the conditions surrounding her actions. Goals,

with accompanying evaluations, are interpreted within an institutional and broader cultural location. Hence the principle that learning is purposeful, or goal driven, links the two aspects of activity theory that I have delineated—the sociocultural situatedness of activity and the directed actions comprising the activity.

In summary, interpreting a student's learning from the activity perspective means attending to the "system of activities" or practices in which the student is engaged and understanding how the student is oriented or positioned with respect to these particular practices.

Evans and Tsatsaroni describe the positioning of a student engaged in a learning practice as:

- The resultant of the practice (s) in which all subjects in that situation *are positioned*, and of the practice (s) which the particular subject *calls up*.

This implies an operative interplay between participants and setting. This implies an operative interplay between participants and setting. Further, Evans and Tsatsaroni argue that there are critical aspects that make the practices, in which students are engaged, meaningful. These include: the students' goals; the resources available and the constraints confining the students; the social relations, including the exercise of power on which these relations are based and the "discourses which shape the practices themselves".

Their argument is based on suppositions which include both Vygotskian notions, such as the importance of speech and interpersonal relationships, and Leont'ev's emphasis on the goal directedness of all actions and on the conditions which determine the operations. Hence it is important for me to explain the roots of these ideas, and the others I have outlined, in Vygotsky's and Leont'ev's theories.

AN INTERPRETATION OF VYGOTSKY'S LEGACY

BACKGROUND

Vygotsky, as he would have been the first to admit, was a product of the culture and history of his time. Born in 1896 to a middle class Jewish family in a provincial Russian town and living only until 1934, he was an intellectual, moving comfortably from the realms of art and literature to science and psychology. In Russian culture at the time there were no sharp divisions of labour among science, art, philosophy, literature and even theology. Scholars such as Vygotsky were simultaneously "connoisseurs of these spheres of human activity".

Vygotsky was greatly influenced by philosophers, scientists, psychologists and writers: Spinoza, Hegel, Engels, Köhler, Freud, Piaget, Shakespeare and the poet Mandelstam, to mention but a few. The teachings of Marx and Engels had a profound effect on his thinking. Unlike many of his colleagues and followers, however, he did not reify Marxism, and ultimately did not accept that Marx' theory provided the explanatory principle of psychology. This led to his falling into disfavour with the authorities towards the end of his life. His works were banned, some posthumously, and much of his later

and most important writing was only published in Russia more than twenty years after his death. Vygotsky saw psychology as experiencing a crisis which was both deep and global. In his view this was a methodological crisis, stemming from the contradiction between the factual material of science and its theoretical premises.

He stated that:

- As long as we lack a generally accepted system incorporating all the available psychological knowledge, any important factual discovery inevitably leads to the creation of a new theory to fit the newly observed facts.

He tried to resolve this crisis by seeking a means of analysis specific to the field of psychology - a systematic way of organising the content, and delineating a framework for psychology.

To Vygotsky:

- ‘General psychology’ has its own laws, forms and structures.

In his, ultimately unsuccessful, search for the ideal analytic method, Vygotsky became inspired by Marx’ analysis of political economics. In the atmosphere of the early 1920’s in Russia, there was a struggle between psychology, which used introspective methods and focused on the inner world of the individual, and the study of physiological processes, such as reflexology.

According to Vygotsky:

- Reflexology, which has translated associationism into the language of physiology, sees the intellectual development of the child as a gradual accumulation of conditioned reflexes; and learning is viewed in exactly the same way.

Vygotsky was critical of this reduction of psychological processes to reflexes of various complexities. Reflexology and its opposing physiological theory, reactology, which was formulated by Kornilov and proclaimed the reaction as the basic unit of psychology, threatened to sweep away “subjective, empirical psychology”, as it was then in Russia. Marxism was seen by Vygotsky as the saviour of Russian psychology. However, he had his own perspective on Marxism. Vygotsky stressed that scientific truth concerning the mind was not a monopoly of Marx and developed his theories in dialogue with Marxism. This approach differed heretically from ideologically approved perspectives and publications of the time.

Vygotsky’s short life span, ill health and prodigious talent meant that, although he espoused many ideas, these did not represent a complete psychological system - they expressed an approach rather than a solution to the problems of psychology. Van der Veer and Valsiner point out that the very lack of conclusion of Vygotsky’s theories makes them useful for modern researchers who can develop them in their own settings. Today, there are many theoretical expositions which extend Vygotsky’s ideas.

For example, there is considerable research on how cognition develops in social practice, such as Scribner’s extensive work; research on situated learning in the form of legitimate peripheral participation; on the role of signs as mediators of cognition as well as “sociocultural” studies.

Indeed, Van der Veer & Valsiner contend that:

- The theoretical debate about the validity and fruitfulness of Vygotskian ideas is only at its very beginning.

In the West, the focus of recent research in education has been on a small selection of Vygotsky's concepts from the plethora of ideas he expressed. This is not surprising in our time of increased specialisation in areas of study. For example, Vygotsky is described by Wertsch et al as a "psychologist, semiotician and pedagogical theorist". These areas are usually considered quite distinct in contemporary Western disciplines. Also, some of the research work done by Vygotsky and his colleagues and students has been criticised as inappropriate for our political and temporal era. For example, one claim ascribed to the cultural historical school is that contemporary aboriginal peoples are similar in development to primitive peoples of the Prehistoric era.

This criticism refers to studies carried out on so-called primitive peoples in Central Asia by Vygotsky and his close associate, Luria, an important Soviet psychologist. Whether Vygotsky held that view in this simplistic way is, however, considered controversial. One interpretation is that Vygotsky considered the "huge wealth of vocabulary" of these rural people to be an impediment to their developing psychological tools. Such tools were, however, needed for the development of technology by cultural groups who had less abundant language. More importantly, such criticisms miss the point of Vygotsky's underlying thesis and its great strength - that psychological processes are not racially innate, or formed in the individual's head, in isolation, but develop in the cultural milieu of the individual.

Luria & Yudovich express this idea strongly in their promotion of Vygotsky's approach as follows:

- Only by understanding that the sources of all complex mental processes do not lie in the depths of the soul, but are to be found in complex forms of human social life and in the child's communication with people surrounding him, can we finally outgrow the prejudices which have been rooted for centuries in psychological science.

In contemporary education literature, Vygotsky's contribution concerning the importance of culture to formal or school learning is increasingly being recognised. In what follows, I will interpret some of Vygotsky's ideas, which are relevant to my approach. An important and fundamental aspect of Vygotsky's reasoning is his dialectical thinking. Based on Hegel's philosophy, dialectical thinking involves the resolution of disconnected or mutually discordant points of view—it is a rational method of synthesising conflicting ideas. To Vygotsky, opposing points of view meant that problems were being subsumed. These, in resolution, would lead to new insights.

Van der Veer & Valsiner argue that:

- The over-riding concern evident in Vygotsky's intellectual work is the quest for synthesis. Hence, though presented separately, Vygotsky's insights are interdependent.

SOME OF VYGOTSKY'S INSIGHTS

Human Evolution as Cultural-Historical Development

Perhaps the most well known contribution of Vygotsky to modern thinking in the West is his emphasis on the “historical development of humans”. This “cultural historical theory”, in summary, suggests that human higher processes developed in human history and have to be learnt anew by each child by means of social interactions. Vygotsky accepted Darwin’s idea that the evolution of man from animals was a continuous process. In marrying Darwin’s evolutionary theory and the Marxist concept that Man has control over his own economic destiny, Vygotsky interpreted the continuity of biological evolution and cultural human development in a dialectic way. That is, he believed that the qualitative differences between humans and animals—the development of higher mental abilities—arose in a complex interaction and antithesis of biologically produced, genetic factors and social interaction.

However, he assigned the major role in the development of particularly human traits to the mastering of culture, particularly the mastering of cultural artefacts such as sign systems and speech. For example, Vygotsky proposed that children’s intellectual growth was contingent on their mastering the social instrument of thought, speech—via inner speech.

He called socially developed thought: “verbal thought”, which he explained as follows:

- Verbal thought is not an innate, natural form of behaviour but is determined by a historical-cultural process and has specific properties and laws that cannot be found in the natural forms of thought and speech. Once we acknowledge the historical character of verbal thought, we must consider it subject to all the premises of historical materialism, which are valid for any historical phenomenon in human society. It is only to be expected that on this level the development of behaviour will be governed essentially by the general laws of the historical development of human society.

That is, complex processes such as thinking have their bases in the sociohistorical development of human society. To Vygotsky, following Engels’ writings, history is the history of cultural artefacts which allows humans to control their own environments. These cultural means, as they are mastered by the developing child, change not only the content of the child’s thinking but the very way of his thinking. This means that growing into a culture—appropriating its means—creates in the child a “second nature”. To Vygotsky and his colleague Luria, this adopting of cultural modes was not like putting on clothing, because cultural layers, once taken on, cannot be shed. For Vygotsky, who ignored learning at home as informal learning, the means by which this enculturation takes place is education. His empirical studies focused on children’s acquisition of “scientific concepts”, that is, abstract or generalised concepts and looked at “the interaction of development and instruction”. Vygotsky’s view of a rational man, one who is in control

of his intellect and behaviour, was that he is an educated man—where education is the process of mastering the means provided by the culture. Vygotsky's definition of culture was selective, emphasising those aspects of human life which enable people to control their environment, such as tools and communication systems—by means of which productive labour is possible. This was not because Vygotsky lacked interest in aspects of human evolution such as wisdom, music, or creativity.

Initially at least, he was inspired to frame his thinking in Marxist terms. Hence the idea of the tools of labour as the means by which people collectively master the physical world and gain economic power, influenced his thinking. The idea of mental tools developed from this notion of man using physical tools to conquer his own destiny. Vygotsky saw signs, for example, words and symbols as analogous to tools in mediating thinking. Van der Veer & Valsiner suggest, too, that Vygotsky would have seen little evidence for progress in areas of social evolution, such as wisdom or creativity. On the other hand, technological artefacts and communication systems—language, writing and symbols—were transforming the ways people functioned.

Hence a focus on these “hard” aspects of cultural development was consistent with Marx' and Engels' characterisation of human qualities as dependent on mastering tools. It was also consistent with Vygotsky's “sincere wish” to liberate and improve the quality of life of all citizens. By having access to education and the mediational means it provided, people could progress in their thinking and so, in Vygotsky's view, free themselves from the yokes of feudalism. It is possible that a nucleus of Vygotsky's theory of cultural transmission can also be found in his own, absorbed, religious heritage. Vygotsky's religion, Judaism, is rich in symbols and rituals. Further, each symbol, which is adopted by the community, and each ritual in which the community participates, contains the history of the Jewish culture. Also, each shared ritual and cultural symbol or artefact, is itself the seed of continuity—a means of passing on the history and culture from one year to the next, from generation to generation. This is precisely the point of Vygotsky's theory of cultural historical transmission.

Developmental Explanation of Human Mental Processes

Vygotsky's stated aim was to achieve a complete and dynamic analysis of higher psychological systems: their development, their structures and their functions.

He differentiated between the maturation of basic biological functions and the development of higher mental facilities as follows:

- *Within a general process of development, two qualitatively different lines of development, differing in origin, can be distinguished:* The elementary processes, which are of biological origin, on the one hand, and the higher psychological functions, of sociocultural origin, on the other. The history of child behaviour is born from the interweaving of these two lines. The history of the development of the higher psychological functions is impossible without a study of their prehistory, their biological roots, and their organic disposition.

So, in order to explain, rather than merely describe, mental processes, Vygotsky insisted that their origins must be examined: phylogenetic; ontogenetic; microgenetic.

Scribner asserts that to Vygotsky:

- The historical research of behaviour is not an additional or auxiliary aspect of theoretical study but forms the very basis of the latter.

Vygotsky tended to use the terms cultural and historical interchangeably and expressed a need to search for explanations of human behaviour, as opposed to that of animals, in history rather than biology. In his view, higher mental functions are collectively produced and have a sociohistorical, rather than biological, origin. So for phylogeny, the cultural development displaces the biological, that is, the “social” line takes over.

The biologically derived term “ontogeny” is used by Vygotsky as a generic term to refer to all processes of child development, so that in his work the concepts of the cultural and the biological development of the child are fused. According to the Vygotskian perspective, language is the main agency by which society changes biological development to social progress. Vygotsky’s understanding of history, both societal history and child development, was not a simple account of universal and linear progress. Rather, he recognised historical processes as changes that move humans in many directions and on many levels. Vygotsky proposed that the relationship between revolution and evolution, though not universally recognised, is understood in scientific thinking.

He expressed this as shown below:

- To the naive mind, revolution and evolution seem incompatible and historic development continues only so long as it follows a straight line. Where upheavals occur, where the historical fabric is ruptured, the naive mind sees only catastrophe, gaps, and discontinuity. History seems to stop dead, until it once again takes the direct, linear path of development.
- Scientific thought on the contrary, sees revolution and evolution as two forms of development that are mutually related and mutually presuppose each other.

Vygotsky’s understanding of the relation between history as change and history as unilateral progress is less clearly demonstrated in the work of Vygotsky’s students and followers. As outlined, Vygotsky’s theories are a powerful lens for viewing the development of the child. His basic claim was that there are two lines of development in the child. One is the biological growth and maturation of the child. The other, which guides the first, is the mastering of cultural artefacts, particularly speech. The child develops through the internalisation of activities carried out while interacting with others, usually adults. This theme has its roots in the work of Marx as well as taking account of the work of important psychologists of the time, including Piaget and Janet.

Origins of Higher Mental Functions in Social Life

The theme described above: That mental functions emerge from shared social experiences—is formulated by Vygotsky’s explanation that a child’s development appears

first on the social or interpsychological plane then on the personal or intrapsychological plane. That is, Vygotsky proposed that every higher mental function, as it develops in the individual, goes through an external stage in its development because it is initially a social function. In particular, adults' verbal interactions with the child underlie higher functions, such as voluntary attention or concept formation.

According to Vygotsky:

- We could therefore say that it is through others that we develop into ourselves and that this is true not only with regard to the individual but with regard to the history of every function.

Vygotsky described four major stages in the child's developing cognitive processes. These were based on his empirical studies of children carrying out various experimental tasks. In the first stage the child responds to the immediate stimuli around him and his own basic needs.

The beginnings of mental mediation arise in the second stage when the child, instead of solving a problem impulsively:

- Now solves it through an internally established connection between the stimulus and the corresponding auxiliary field.

At this stage, however, the child has not fully mastered the means of mediation. The third stage is reached when the child becomes aware of the role and utility of the mediator in cognitive activity. For example, in one of Vygotsky's experiments, children use pictures to decide which keys should be pressed to fulfil a given task. At this stage children can organise their environment but their regulation of their own behaviour is still dependent on the stimuli present in the external environment. The final stage in the development is characterised by the internalisation of the words or other signs and the relationships among the stimuli, mediator and responses. The operations can now be carried out without external stimuli. To me, the most important point about Vygotsky's account of this progression to self regulation is his emphasis that this is a social process. Díaz, Neal & Amaya-Williams specify four ways in which this is so. Firstly, the signs and tools that are brought to the setting, and are taken over by the child to control his or her behaviour, are social. They are features of the society or culture at large.

Vygotsky gives many examples of "psychological tools" used in this way, including:

- Language; various systems for counting; mnemonic techniques; algebraic symbol systems; works of art; writing; schemes; diagrams, maps, and mechanical drawings; all sorts of conventional signs.

Secondly, these signs are used by the child to influence others and interact with them - they have an immediate social role. Thirdly, in Vygotsky's view, speech is the most useful agency by which children master their environments. Speech is inherently social - it carries shared meanings with others in a particular setting. Speech not only mediates social interactions, but is also a tool for planning, organising and monitoring behaviour. The child's symbolic activity, especially private, internal speech allows the child independence from the external stimuli and enables mastery over her own responses.

Vygotsky expressed the evolution of “verbal thought” from simple generalisations to the most abstract concepts in this way:

- It is not merely the content of a word that changes, but the way in which reality is generalised and reflected in a word.

So the use of language and other semiotic systems transforms thought in ways that connects the child to her social environment. These transformations are not only in what the child thinks but also in the way she thinks. For example, the strategies people use to solve problems, the categories through which they organise their experiences and the ways people express themselves in language, depend on their cultural history and society.

One of the mechanisms by which Vygotsky explained how individuals realise their potential is his construct of the zone of proximal development. This relates to Vygotsky's aim of empowering people through instruction. It also draws on the Marxist notion that individuals can shape their worlds and develop their higher mental faculties through cooperating and interacting with others.

Referring mainly to children, Vygotsky defined this zone as the area between what a person can achieve on her own and what she can achieve with the help of a more capable person. This emphasises the child's potential to develop with strategic help. This support is called scaffolding by some researchers, following Bruner emphasising the support given to the learner. It is termed amplification by others which focuses on the child's own potential and current strengths.

The implication of the zone of proximal development is that the inexperienced learner will in the near, or proximate, future master a task for which she currently requires some help. Vygotsky chose the word “zone” to underscore the idea that development is a “continuum of behaviours” not a point on a scale. The zone of proximal development has a dynamic nature. As the child masters new and more difficult tasks, the zone of proximal development shifts, and the cycle of assisted learning followed by independent achievement is repeated.

This notion of Vygotsky's is well known and has been extensively researched in the West. It has important implications for mathematics and statistics education where, often, the focus is static - on what students can do unaided at a point in time - the lower boundary of the zone of proximal development. There may be little focus on the continuum of students' skills and abilities or on their potential to progress by means of structured interactions. Traditional assessment in mathematics and statistics reflects the view that a student “knows” or “does not know” a particular concept. Vygotsky's idea of a concept is dynamic. In his term a concept is a maturing process, rather than a mental acquisition

Vygotsky's perspective recognises the primacy of social contact as a basic human need and also rejects the dualism between the individual and society, insisting on the inseparability of the individual and the social realm in which she acts. The application of these ideas is well documented in the work of contemporary researchers such as

Lave & Wenger; Rogoff; Scribner and Walkerdine. Vygotsky and Leont'ev both stress the importance of the social role in human development and learning. However, while Vygotsky tended to emphasise immediate social relationships Leont'ev's focus was on the broader societal context.

Mediation of Thought

A major goal of Vygotsky was to develop a psychology able to account for the nature of human consciousness. Influenced by his knowledge and interest in art and literature, Vygotsky extended the Marxist concept of physical tools as mediating devices, to mental tools.

To Vygotsky:

- The most essential feature distinguishing the psychological tool from the technical tool is that it directs the mind and behaviour whereas the technical tool, which is also inserted as an intermediate link between human activity and the external object, is directed toward producing one or another set of changes in the object itself.

That is, psychological tools are aimed at mastery or regulation of oneself or others, rather than controlling the physical environment. In a dialectic way, these human transformations lead to the need for further mental tools which again give rise to new ways of conceptualising and acting in the world. Mental tools, for example, scientific theories, are cultural. They are invented and passed on to others through social processes and socially structured practices.

According to Wertsch et al they:

- Provide the link or bridge between the concrete actions carried out by individuals and groups, on the one hand, and cultural, institutional, and historical settings, on the other.

In Vygotsky's writing, the main mediators are words, symbols and other signs.

In this context, a sign has:

- A definite meaning that has evolved in the history of a culture.

A sign can serve as a system of reference, such as "notched sticks and knots", or writing. It can also make possible new, higher levels of thought and analysis, as do, for example, algebraic systems. Vygotsky was particularly interested in the role of the sign, especially the word, in the development of higher mental functions.

To Vygotsky:

All the higher psychic functions are mediated processes, and signs are the basic means used to master and direct them. The mediating sign is incorporated in their structure as an indispensable, indeed the central, part of the total process. In concept formation, that sign is the *word*, which at first plays the role of means in forming a concept and later becomes its symbol. So to Vygotsky, signs enable higher mental processes, such as concept formation, to develop and they also signify meanings. Vygotsky and his colleagues and students, including Luria and Leont'ev, carried out

many empirical studies on children and on rural peoples in order to understand how the processes of internalisation of speech and other semiotic systems allowed thinking to develop.

Luria & Yudovich posit that:

- The word, handing on the experience of generations as this is incorporated in language, locks a complex system of connections in the child's cortex and becomes a tremendous tool, introducing forms of analysis and synthesis into the child's perception which he would be unable to develop by himself.

That is, the internalisation of word meaning, which is socially developed, provides the structure for the development of cognition. Mental tools, in Vygotsky's thinking, extend the mind. They allow humans to do more than react to the environment as animals must—they enable us to change our environments. This implies that human behaviour is purposeful.

We use tools, and signs, especially language in order to carry out our plans, to cooperate with others and to regulate our own cognitive processes, such as focusing our attention or remembering, in considered, deliberate ways. In Vygotsky's thesis, cognitive functioning is shaped by the sociocultural setting. Mediation by means of cultural tools serves as the mechanism for this shaping.

DEVELOPMENT AND PRINCIPLES OF ACTIVITY THEORY

HISTORICAL BACKGROUND

Many of Vygotsky's conceptions greatly influenced and were further developed by his contemporaries, Luria, Leont'ev and other psychologists of the former Soviet Union. It is customary for Western authors to refer to Vygotsky, Luria and Leont'ev as a "troika" developing their theories in unity. Van der Veer & Valsiner dispel this unity as a myth, contending that different attitudes and thinking, as well as strong personal disagreements, marked the relationships among the three. Indeed Leont'ev did not co-author research publications with Vygotsky, although he did considerable empirical work in support of Vygotsky's theories. Leont'ev's work on psychological functions began under Chelpanov at the Moscow Institute of Experimental Psychology, formerly the Psychological Institute. In 1923 Chelpanov was expelled from the Institute, which he had himself established in 1912, in favour of his one time student and strident critic, Kornilov, a reactologist. Leont'ev, under the guidance of Luria and Vygotsky, worked at the Institute under Kornilov, who was rushing to re-organise it along the lines of the new Marxist psychology.

Luria is cited as describing, with tongue in cheek, his early days under Kornilov as follows:

- Our Institute was supposed to reform the whole psychological science by abandoning Chelpanov's idealistic theory and creating a new materialist one.

Meanwhile the reform of psychology was proceeding in two forms:

- First, by way of renaming things,
- Second, by way of moving furniture.

In the early 1930's Leont'ev moved from Moscow to Khar'kov where he created his own school of psychology. His approach came to be known as the "activity approach in psychology" or the "psychological theory of activity". For Vygotsky and others of the cultural historical school, the focus of research was the development of mind and consciousness and how these are mediated by psychological instruments, especially words and other signs. To Leont'ev, in accordance with Marxism, the unit of psychological analysis is activity, and physical tools and concrete objects mediate the connection of humans and the natural world.

Activity theory is, however, grounded in and uses Vygotsky's concepts. For example Leont'ev re-interpreted Vygotsky's analysis of speech in terms of speech activity. Wertsch et al argue that the "action orientation" of Vygotsky has been obscured by his distinction between speech and language. To Vygotsky, as to his Russian colleague Bakhtin, a linguistic philosopher, speech is a process, language is a semiotic means. That is, speech is a form of action. Language, while "it has the power to shape speaking and thinking" is not an action in Vygotsky's terms.

There is a difference between speech, as utterances, and language, as a system of communication. This difference is in the emphasis on different aspects—"the vocal and the semantic" which Vygotsky saw as related, but with different patterns of development. Vygotsky tended to focus on speech as opposed to language. However, he also paid attention to the symbolic properties of the mediator as central to the action of meaning making. Leont'ev, however, stressed the external active form of speech. While Vygotsky regarded word meaning as the appropriate unit of analysis, Leont'ev based his analysis on activity. In both cases these units are cultural. The title of Vygotsky's last volume *Thinking and Speech* highlights the "action dynamic" in Vygotsky's thesis. Indeed, the important part played by Vygotsky in the creation and development of the activity paradigm itself was acknowledged by Leont'ev in his obituary of Vygotsky.

Leont'ev's interpretation of Vygotsky's theories must be seen in the light of his attempts to rehabilitate Vygotsky's work after the latter's work was banned by the Russian authorities soon after his death in 1934. In the repressive climate of the former Soviet Union in the pre-Glasnost era, researchers were forced to demonstrate their loyalty to the prevailing ideology. Vygotsky came under increasing attack and his refusal to adhere to strictly Marxist principles regarding the primacy of material production was cited by Petr Zinchenko and other contemporaries of Vygotsky as a fundamental error.

Vladimir Zinchenko, the son of Petr Zinchenko, affirms that Vygotsky and any who supported him were in danger of losing their freedom, if not their lives. He notes that cultural-historical psychology emerged during the "dismemberment" of Russian culture, en route to communism, while the background to activity theory was the "unprecedented" enslavement of the country's peasants. Hence, the very emergence of these theories,

was, in V. P. Zinchenko's opinion, miraculous. Bodrova and Leong applaud the courage of those like Leont'ev who continued to expand and elaborate on Vygotsky's theories after the latter's death, despite the risks.

In their opinion, such people were then able to revive Vygotsky's ideas in the late fifties after the bans had been lifted. Other commentators paint a less rosy picture of the actions of Leont'ev and Luria, indicating that they did not always stand up for academic freedoms. Clearly Russian society at the time was a maelstrom and simplistic judgements cannot be made about those trying to steer their courses through the storms.

In V. P. Zinchenko's view, Vygotsky's students and colleagues, such as Leont'ev, did not simply camouflage their earlier views after Vygotsky's death but worked in a new direction.

Leont'ev's line of research was distinct from Vygotsky's and:

- *Today we are dealing with two scientific paradigms:* – Cultural-historical psychology
- The psychological theory of activity.

These two strands of research can be seen as mutually amplifying and enriching each other.

PRINCIPLES OF ACTIVITY THEORY

Davydov an "avowed adherent" of activity theory, and arguably one of its most well known Russian advocates in the contemporary Western world, explains that Marx took activity as a key concept to explain the development of economic life. This concept of activity was worked out in the context of German classical philosophy. Philosophers, such as Kant and Hegel formulated the groundwork for the concept of activity, its structure and its formation in the sociocultural realm. Marx emphasised the manufacture of artefacts as satisfying the basic needs of humans.

In his view, production, or labour, is the means by which man changes nature and at the same time, his own nature. This production depends on the use of tools. Mental abilities, too, arise in the process of material production and the inter-personal relationships surrounding it. According to Marx, human activity involving labour is the foundation for collective and individual consciousness. That is, activity is the basis of "the personal and social identity of the individual". In Russia, after the October 1917 Revolution, the concept of activity was developed in terms of disciplines such as philosophy and psychology.

There were many different versions of activity theory of which Leont'ev's became the most well known. Indeed, Gilgen & Gilgen comment that in scholarly publications written in English as well as in Russian, Leont'ev's account of Soviet psychology "predominated from the 1950's to the end of the Soviet regime in 1991". In part this is due to the prominent positions held by Leont'ev in Russia, which included Chairman of the Department of Psychology, Moscow State University. Lektorsky, a Russian philosopher, is cited by Cobb et al as an expert in activity theory.

Lektorsky explains cognition in terms of Marxist philosophy as follows:

- Marxist philosophy asserts that cognition is founded on practical activity and that the latter must be understood in its specifically human characteristics, to wit, as collective or joint activity, in which the individual enters upon definite relations with other persons, as mediated activity in which man places between himself and an external naturally emerging object other man made objects functioning as the implements of activity: and finally, as a historically developing activity carrying in itself its own history.

In this, somewhat dense, sentence are the basic tenets of activity theory. The first of these is that activity is the explanatory principle in psychological theory. Secondly, Lektorsky equates cognition as socialisation, as Vygotsky claimed, with cognition as the “interiorization” of external actions, in Leont’ev’s terms by describing all practical activity or labour as being collective. That is, all activity is of a social nature.

Thirdly, activity is mediated by cultural tools which shape the individual’s functioning. Finally there is the idea of the mutual formation of mind and the social world in history. This last idea is expressed cogently by Varela et al who propose that mind and the world co-emerge by structural coupling, much as the colours of flowers and the vision of bees co-evolve.

Davydov makes the point that the term “activity” does not have the same frame of reference everywhere, in research in the social sciences. In English, the word covers all kinds of overt and covert actions.

In Leont’ev’s activity theory the term:

- Refers exclusively to events in which actions are judged by their intention to change reality.

That is, activity is differentiated from other human actions by means of being goal driven.

Davydov defines activity in this framework in the following way:

- Activity is a specific species of human societal existence which aims at a goal directed change of physical and social reality.

So, individual activity is embedded in and transforms the social and cultural life of the community. At the level of the individual, activity brings together a goal, the means for realising this goal, the actual process of transformation and its effect or outcome. Each of these is organised within a societal setting. The goal of activity emerges first as a schematic model of the desired outcome of the actions. That is, practical actions start with mental plans. This has the effect of creating opportunities—of rising the limits as they exist and so increasing the options available.

The “system” of collective activity which relates people to each other, their cultural history and their realm of action is called an “activity system” by Engeström. Taken as a unit of analysis, it confers meaning on seemingly random and individual behaviour. The activity system of people acting in their historically formed and social world is in a process of change and contradiction.

The dialectics of the fusion of the subject and the external world are explained by Lektorsky by quoting Lenin as follows:

- The reflection of nature in man's thought must be understood not 'lifelessly', not 'abstractly,' *not devoid of movement, not without contradictions*, but in the eternal *process* of movement, the arising of contradictions and their solution.

It is these perturbations which lead to development of the system including the development of its participants. The following parts clarify the main principles underlying activity theory.

Activity is The Explanatory Principle of Psychology

Leont'ev, stressed that consciousness is "generated" by and inseparable from external activity. In his view, based on Marxism, activity must have associated external behaviour and mental images are direct products of contact with the world of objects. In this thinking, Leont'ev was influenced by others including, importantly, Rubinshtein and Vygotsky but his thinking differed from each of theirs.

Rubinshtein, like Leont'ev, proposed that consciousness and behaviour represent a unity. Rubinshtein defined individual consciousness as: the person's ability to regulate not only movements, acts and operations taking place at a particular time but also the entire life span. Rubinshtein's ideas accord with those of Piaget, in that he focused on the roles of individuals' independent explorations and interactions with the objective world as the sources of their reflections of it.

That is, Rubinshtein's position puts the subject at the centre of psychology. Vygotsky put interaction and communication among people at the centre of individual psychological development. As discussed earlier in this stage Vygotsky explained the development of higher mental faculties in terms of historically developed social interactions. In his theory, it is these internalised social interactions and communications that are the basis of cognitive development.

Diverging from both Vygotsky and Rubinshtein, Leont'ev proposed that activity is at the centre of a person's connection with others and the external world. Activity includes interaction with others and it also includes reflection, but Leont'ev's emphasis is on external actions. That is, according to Leont'ev individual psychological experiences are expressed through external, practical activity. Activity connects individuals to their communities and connects material objects to the psychological faculties that illuminate these objects.

This leads to the question of how objects are connected with the mind. Bakhurst, an American interpreter of activity theory, solves the problem by referring to the meaning of objects acquired by them in social practice. That is the "ideal" existence of objects is completely different from its physical nature - the stuff of which it is made. It is an artefact endowed with social significance. To illustrate this, Bakhurst describes a pen as a form which objectifies a social purpose. The meanings that a community give to an object transform it into a cultural artefact which has ideal properties. These properties

become accessible to our minds when we participate in the social practices for which the object was developed. Hence, writing with a pen gives it its “reality”. This idea of objects as mediators of sociohistorical experience is also applicable to symbols, referred to by Lektorsky as symbols-in-practice, rather than just symbols. For example, Lektorsky argues that when we read text and try to understand it, we focus on the content expressed with the help of letters, but ignore any characteristics of the letters that are “indifferent” to meaning making, such as their individual shapes.

Activity is Purposive

According to Leont'ev:

- There can be no activity without a motive. ‘Unmotivated’ activity is not activity devoid of a motive: It is activity with a motive that is subjectively and objectively concealed. Hence to Leont'ev, some motive, whether conscious or unconscious, always stands behind activity. Motives, needs and desires are translated into action by means of goals.

Motivation and goal formulation are inherently social. They connect psychological phenomena to the social world. Leont'ev proposes that when a desire meets the object of desire, it is filled “with content from the surrounding world”. That is, individual actions are embedded in larger systems of activity called “socially patterned systems of activities” by Scribner and are directed towards satisfying motives. These systems of activity are socially patterned in three ways: the goals they satisfy are socially generated, they are carried out through social means and they are connected to a system of collective activity which has developed over human history. The idea of a goal follows from the Marxist concept of labour.

This notion is interpreted by Davydov as follows:

- At the end of the labor process a result is obtained which existed in the person's conception even at the start of the process—that is, ideally.

That is, according to the theories of Marx and Engels, the human ability to formulate and carry out conscious goals provides the cognitive framework for humans to enter into production. In line with the Marxist paradigm, Leont'ev's theory postulates that human motives are determined by the division of labour in society, and that within this framework, specific, concrete actions are related to practical goals. Leont'ev's theory, the idea of individual potential emerges from his postulate that actions are consciously directed at a goal.

The intention of cognition, in the sense of directedness, implies an awareness of what possibilities for action there are, and an understanding of how the context constrains the actions or enables the fulfilment of these possibilities. Historically, goal formation was one of the major research topics in psychology in the former Soviet Union. Current research in the West has also provided empirical support for the importance of personal goals as directing the learning of university students. The research highlights the importance of goals in the self regulation of students' activities, the complexity of the

dynamics of goal formation and development in academic settings, and the ways goals are responsive or resistant to the environment. In contrast to Leont'ev's ideas, however, this research gives an account of students' goals that:

- Precludes any easy invocation of goals as causes of particular sets of actions.

That is, goal directedness is neither transparent nor unequivocal.

Activity Is Mediated

Activity can be thought of as having a tripartite structure consisting of the individual, the environment and the tool, which, as a cultural mediator, is both the agent and reflector of change. Leont'ev explains that the tool mediates activity and in this way connects humans with concrete objects and also with other people. He specifies that it is not the cultural tool, as an artefact, but the use of that tool that mediates or shapes mental processes.

That is:

- The formation of uniquely human functional systems takes place as a result of mastering tools and operations.

Moreover, Leont'ev contended that such mediators do not simply facilitate human thinking and agency. As the social structure and tools of production change, so does human thinking. This relates to Vygotsky's idea that as technical tools alter the operations of labour, so psychological tools transform "the entire flow and structure of mental functions".

This premise is most pertinent to our society with its rapid technological transformations. For example, the ways we use our memories and other mental abilities have changed dramatically with "extra-cerebral" tools like the Internet, and even calculators. These have taken over the function of storing vast amounts of information and of processing information by means of complex algorithms, thus allowing for increased expertise and more complex mental actions, such as problem solving.

Activity Theory Emphasises The Interdependence and Interaction of Sociocultural and Cognitive Processes

The view that cognitive abilities of the child or novice emerge from interaction with adults or experts was developed by Vygotsky who proposed that cognitive processes are carried out interpsychologically then intrapsychologically, or from the social to the individual plane. That is, they are first carried out initially with help from an adult or expert, then transformed to a self directed carrying out of the task, via the zone of proximal development. Speech is the major mediational means which directs this process.

In Leont'ev's modification of this theme, activity is the major source of cognitive development. Psychological functions, too, such as thinking, are actions in Leont'ev's terms. Leont'ev proposed that cognition can exist only in the presence of and embedded in social consciousness - what Leont'ev, echoing Vygotsky, called "co-knowing". It lives only in an environment of socially constituted meanings and is developed by the

appropriate use of cultural tools. Leont'ev proposed that inner mental processes originate from external activity which serves a practical, socially developed purpose. That is, thought is “not in the head”, in a private world of consciousness, but in the performance of social activities. External practical activity is internalised, that is, changed into processes that take place at the level of consciousness. Leont'ev expressed this as “... life underlies consciousness”. During the transition these external processes become generalised, verbalised, abbreviated and become the means for development beyond the bounds of only external activity. Leont'ev quotes Piaget as formulating that this conversion “leads from the sensory-motor plane to thought”.

A specific account of this essentially social process of transformation is provided by Davydov & Andronov. They call it a “test tube” example of the emergence of mathematical thinking. In their study of young children working on addition problems, the mental act of addition initially takes place as an overt action, a sweeping hand movement as the number is pronounced. The gesture, in turn, becoming a symbol for the number, is reduced, and, together with speech, becomes the means for organising the child's mathematical thinking during the problem solving.

Davydov explains that through instruction:

- Every individual appropriates to himself, converts into Forms of his own activity, the means and methods of thought that have been created by society at that historical epoch.

This idea stresses the sociohistorical roots of cognition. It also emphasises the engagement of the individual. The term “appropriate”, like the term “construct” highlights the active role of the learner. This is particularly relevant to students learning Statistics. Statistics is an artefact of our technological era with a history of development in society and it is perceived differently by each individual as he or she “appropriates” it, in Davydov's term.

Cobb et al sums up the different senses in which Vygotsky and Leont'ev view an individual's actions as being social as follows.

- Vygotsky views personal actions as involving direct interactions with others.
- Activity theorists view individual actions in terms of the broader sociocultural system.

The latter view is a stronger, more encompassing account of social.

Minick points out, as a key element of the activity framework, that:

- Psychological characteristics are conceptualized not as characteristics of the individual but as characteristics of the individual-in- (social) -action.

This stronger meaning of social has been applied in many studies of situated mathematical activity such as investigations into the delivery of milk crates or into supermarket shopping. In these studies, and in my own investigation, individuals participate in social activity which may, but does not always, include face to face interaction. Lektorsky argues that there is a continuity between Vygotsky's notion of social activity as the determinant of psychological development and Leont'ev's focus

on mental abilities being derived from labour or practical activity. According to Lektorsky, a person's cognition is shaped by interacting with others who teach him to use man-made things. Lektorsky combines Vygotsky's and Leont'ev's proposed mechanisms for the development of the psyche, by explaining that an individual's consciousness is founded in the direct links between practical activity, cognition and "the living communicative connections with other persons".

ILLUSTRATIONS OF THE FEATURES OF ACTIVITY THEORY: EXAMPLES FROM MY PRACTICE

So far, I have outlined a model for understanding learning suggested by the theory of activity. In summary, in explaining cognition, the emphases of activity theory are on its connected elements and mediated structure, how it functions by means of goal directed actions and on its cultural and historical development.

Engeström points out that activity theory does not offer ready made techniques and procedures.

Its conceptual tools and methodological principles:

- Have to be concretized according to the specific nature of the object under scrutiny.

In this part I present some of the elements of the theoretical framework which are of interest to me in defining the activity of learning Statistics. I shall attempt to illustrate these factors with examples drawn from my teaching practice. The part concludes with some implications of the activity theory approach for understanding learning and teaching.

DIALECTIC OF STUDENT, ACTIVITY AND SETTING

My perspective stresses that it is not fixed or pre-ordained features of the setting or of the individual learner that determine the learning. The student as a learner and the context of her learning are mutually constituted in social action.

I agree with Ratner's interpretation that:

- Social activity not only determines the content of psychological phenomena. Activity also conditions the particular fields of life in which a psychological phenomenon is employed.

Ratner points out that culture means that there is a specific, practical system operating. This determines the distribution of power, privileges and resources. It establishes the ways in which tasks are separated from one another, for example, school learning and vocational training. It influences how people act and relate to each other in different social practices, such as during work and at home. Ratner argues that, although culture is commonly recognised as the sharing of conceptual meanings and understandings, it also includes these practical aspects. These must be recognised for the important ways in which they affect people's psychological processes. Hofstede describes different

dimensions of culture, such as gender roles, individualism versus collectivism and distribution of power, that are important in the organisation of people's behaviour and thinking.

In Australia, higher education is undergoing massive changes, in part due to political and economic policies. These policies are transforming the ways in which educational processes are framed. For example, the ways that educators examine students, their priorities with respect to teaching and research, the efforts they make to reform the curriculum, the demands on their time and the resources available to them reflect the political and economic environment. These policies even transform perceptions of the past. For example, current "cut backs" in departmental budgets instantaneously transform the past into a golden era—a supposed time of plenty from which things have been cut back.

The educational context shapes the ways in which students learn. In turn, students' actions change the teaching situation which surrounds them, including the actions of educators. For example, if an educator teaches the same topic for two or more years, the teaching strategies and methods of communication which "worked" one year may be dramatically changed the following year. Such changes may reflect the social environment produced by the members of each new class.

The activity theory approach to education asserts that knowledge, and ways of gaining knowledge, have a social ontogeny. One implication of this is that students' previous experiences can be ameliorated. This fits with my own experience of teaching situations. For example, it was the case for Jane, an older student whom I helped with Statistics at the Mathematics Learning Centre. She felt that she had not succeeded with mathematical subjects before coming to university. She did not believe herself to be capable of doing so.

I interviewed her after she had achieved very good results in Statistics and she described her feelings as follows:

- When I've got my notes and my calculator I feel invincible.... I found that I actually knew more than some other people. Instead of being the one who knew the least. And that gave me an enormous sense of power - or a bit of power anyway - that I could actually know more than somebody else.

As in Jane's case, not only do students' perspectives and insights into a particular field of knowledge change, but also their very self concepts. Slotnick, Pelton, Fuller & Tabor propose that in answering a student's questions, an instructor is not only helping that person advance her skills in reasoning and logic and providing a role model for dealing with complex issues, but is also "validating" that person, that is, confirming her identity.

Changes in students have been empirically demonstrated in the case of mature students who attend university. These transformations are related to their experiences and to their goals, which unfold and are structured by the social and cultural meaning of the events. Further, this social meaning is embedded in specific, practical situations of

activity. Jane's activity took place in a particular university within the wider Australian culture, surrounded by the customs, norms and gender roles pertaining both to the institution and to our society. Each of these arenas - her immediate social environment, the institutional setting and the wider Australian society - are contextual orbits within which Jane's activity was organised.

By the word "orbits" I am emphasising that the structures operating on Jane were undergoing changes themselves, while seeming to be static - as the earth, surrounding us, appears static although we know it is moving. It is these contexts that influenced the way Jane thought about herself and structured the tasks in which she was engaged. They determined which of her actions were rewarded and how they were rewarded and what demands were placed on her.

VIGNETTES FROM PRACTICE

The theoretical approach offers ways of interpreting students' actions. In order to clarify the ideas I have introduced in this stage, I summarise my interpretation of key features of activity theory by means of examples. The following excerpts are from transcribed interviews with students whom I assisted with various statistics courses. These are snapshots, taken from different angles, of an activity system of students acting in their institutional worlds. They capture images of ongoing and dynamic processes.

Activities are Distinguished by their Energising Motives

Brenda, first year statistics student, age 23, said this:

- Like to understand what I'm doing. I don't just want to pass statistics, scrape through. I want to work. My whole attitude's totally different from when I was at school. I didn't care when I was at school. I became a Christian a couple of years ago. And so my whole attitude changed and now I guess I've got a goal and now I've got a reason for being alive basically. I never did well at anything at school and last year the course I did, out of 160 people I came first. That's amazing. That for me is amazing.

Brenda's quote illustrates the essence of activity as encompassing her total perspective on life. Her actions in the educational setting were guided and directed by her underlying *raison d'être*—in this case spiritual inspiration, and this resulted in a very favourable outcome.

Activity Theory Emphasises the Interdependence and Interaction of Learning Processes and Cultural Background

The following excerpt is from an interview with a first year statistics student, Huong, who recently entered Australia. Huong expressed a mismatch between her schooling in Vietnam and the ways she was expected to learn statistics in Australia. There is a big difference between the math I did at high school and this statistics course. Everything is new for me. From the beginning of this year I learn how to use a calculator and how to

work with the terms in math, how mathematics terms mean in English. But it is hard to learn to solve problems because in my country the way to learn is just to memorise by heart. We had no chance to practice to think for ourselves. Just to memorise what the teacher said to us. From the activity theory point of view, this excerpt is an invitation to explore the roles of memorisation in different cultures. For example, Biggs cites studies investigating how Asian and Australian students use memorising. These show, that in our culture, memorising with the intention of reproducing material may not lead to understanding, and is then inappropriate to the task of learning. For Chinese students in Hong Kong, it is more likely that memorising leads to understanding—the way that material is internalised is part of a deep approach to learning. Further exploration on the relation between memorisation and understanding for Chinese learners is provided by Marton et al.

Actions are Socially Formulated

Eve: Second year Psychology student describing her mainstream Statistics tutorials:

- Its much more like you imagine university maths to be nobody will ask questions. It doesn't seem to be the done thing.

Hence Eve's actions were prescribed by unwritten codes of social conduct in tutorials, as she perceived them.

Activity Involves Self Regulation

Student repeating Psychology II course:

- I can't use any of my own knowledge for Statistics. I can't use any of my analytical or philosophical skills or just like experiences in life—it's rote learning and its just so different.

This quote expresses the mutual structuring of this student's conceptions of learning Statistics and her regulation of her actions to learn it.

IMPLICATIONS FOR THE ANALYSIS OF LEARNING

The vignettes illustrate some aspects of my perspective drawing on activity theory. This perspective has implications for the ways in which we view the learning process.. These implications are not, of course, unique to this paradigm.

Indeed, viable research and teaching may follow from more than one paradigm.

- The activity framework implies a commitment to investigating the learning process as a dynamic process and one in which society and the individual interact and evolve rather than the student as a closed system or passive receiver of ready made information.
- The theory presents different levels at which learning can be analysed: the global level of the context of the learning, the level of the goal directed actions, that is, how the task was carried out and the level of the operations, which, according to Leont'ev, depend on the means by which an action is carried out.

- It emphasises the social dimension of learning: certain behaviour is acceptable in an arena of action, such as a tutorial session, other behaviour is not.
- It sensitises us to cultural effects. This is particularly pertinent for studying the learning of statistics as our current Western consumer society is immersed in statistics yet not all students are familiar or comfortable with this.
- It takes account of goals. The degree to which the goals set by the educator fit with those of the students will affect their involvement in the tasks set for them.
- A Vygotskian perspective recognises the importance of language and symbols. Statistics is a branch of mathematics which has a large verbal component. The language used in statistics, described by one European born student as “English acrobatics” is often the very aspect that confuses students.
- A further strength of the Vygotskian approach is its emphasis on underlying metacognitive skills, such as self regulation. Empirical research shows that these are important for the development of appropriate learning strategies.
- In sharp contrast to psychological theories which view cognitive processes as separate from the emotional domain, activity theory emphasises the inseparability of students’ feelings from their thinking. These cannot be considered as discrete, additive components. Leont’ev concurred with Vygotsky that motivation and the affective volitional sphere were the “hidden” plane of thinking.

According to Vygotsky, the separation of intellect and affect as subjects of study:

- Makes the thought process appear as an autonomous flow of ‘thoughts thinking themselves,’ segregated from the fullness of life, from the personal needs and interests, the inclinations and impulses, of the thinker. Rather than needs, feelings and emotions being treated as residing within the individual, influenced by the environment - a precondition of action, an activity theory approach regards these as unified aspects of human functioning.

ISSUES CONCERNING CONTEMPORARY INTERPRETATIONS OF ACTIVITY THEORY

PROBLEMS IN THE THEORETICAL ASSERTIONS OF ACTIVITY THEORY

A number of unsolved problems and issues pertaining to activity theory have been highlighted in recent research. These include the meaning of the construct “activity”; issues concerning learning by “internalization”; the nature of knowledge and of reality and methodological problems concerning activity as a unit of analysis. Davydov presents a number of problems pertaining to “activity”, itself.

Firstly, despite his own extensive work clarifying it, he regards the basic meaning of activity to be inadequately conceptualised. Activity, in his opinion, is insufficiently differentiated from other human actions, despite the emphasis on its goal orientation.

Davydov argues that each type of activity, such as art, politics or labour, must be understood in terms of its own cultural historical formation. For example, what types of instructional activity were privileged in specific historical periods? He also holds that each type of activity is embedded in processes of interpersonal relationships. These, too, must be explicated in order to understand the activity. In my project the activity of learning Statistics is interpreted in an institutional setting, which is culturally and historically formed and organised. It relates also to the interactions among students and academics.

One of the most widely criticised aspects of activity theory concerns the process of internalisation. This is a notion of Vygotsky which is also central to Leont'ev's theory. Two major concerns are highlighted by Bauersfeld. One pertains to how individual activity is formed on the basis of collective activity. The second aspect relates to the idea that what is internalised is ready made and singular.

The first issue was raised by Davydov who criticised Leont'ev for pointing out the similarities, but not the differences in structure, between collective activity and individual activity. To Davydov it is not clear how individual functioning, such as remembering, would be derived from a differently structured collective activity. One explanation that I offer in the case of remembering is that collective memories are individually accessed as metaphors. For example, children are told fairy stories about dangers which have their roots in "tribal" memories.

The chances of an Australian child encountering a wolf on route to her grandmother are remote for the vast majority. However, this "tribal" memory serves a useful purpose in the child's current behaviour, assisting her to cope with life by warning her of its dangers and also enabling her to access and express her emotions. Also, individual memories can be jogged and even amended by means of external aids such as photographs or the recollections of others.

The second concern relates to the idea that what is internalised is ready made and singular. This seems to suggest that there is a historically formed, objective reality "out there", which can be used to lead development. This aspect leads to problems explaining creativity and also suggests that knowledge is ready made.

To Rogoff the idea of internalisation suggests that:

- Something static is taken across a boundary from the external to the internal.

Rogoff like Davydov uses the term "appropriate" rather than "internalise" to emphasise the person's participation in an activity. Rogoff also proposes that "participatory appropriation" is a "process of transformation", whereby the boundary itself is questioned, since a person who is participating in an activity is part of that activity.

Rogoff comments that Vygotsky's notion of internalisation may be similar to her idea of appropriation in that transformation is inherent to his notion. To Leont'ev, too, internalisation is a dynamic, active process. However, Rogoff's term "participatory appropriation", in which the person and the social world are not separate, contrasts

with the “internalisation” of an event or knowledge which is external to the individual. The idea of historically formed, crystallised external knowledge can lead to a positivist and reductionist view of mathematics. Teaching mathematics from this perspective would mean guiding students to the “truth”, using direct instruction, powerful mediational means and explication of reflective strategies. In contrast, teaching mathematics is problematic from the constructivist perspective. A teacher cannot transmit knowledge or expose information.

Burton emphasises the:

- Interaction between individuals, society and knowledge out of which mathematical meaning is created.

This is a non-positivist position which fits with the metaphors of co-emergence of person and knowledge and of the interactionist perspective. My understanding agrees with the latter positions. Rather than mathematical knowledge as non-negotiable and absolute “truth”, it is constituted in the relationships among what is learned, those who learn or use it and the society within which it develops.

On a methodological level, Leont’ev has been criticised as viewing activity both as a unit of life which can be analysed by means of different levels and a means of investigating life itself. Kozulin considers this to be a consequence of the rejection by Leont’ev and his followers of Vygotsky’s notions concerning the role of semiotic systems, or cultural signs in general.

In his effort to adhere to Marxist principles, Leont’ev saw activity as providing not only the subject matter but also the methodology or explanatory principle for psychology. To Vygotsky, in contrast, higher mental functioning cannot be explained without reference to semiotic mediation. To avoid this circularity of explaining activity by means of activity, I will treat students’ reported activities in learning Statistics as the object of my investigation. Actions and operations will be regarded as components of their activity. Moreover, in my investigation I will confine my focus to what is accessible through students’ awareness. Leont’ev insists on all actions as goal directed. However, we do not always articulate, and are not necessarily even aware of, the motives for our actions. Activities that comprise of purposeful actions are, however, accessible to research through the awareness of the participants or actors.

TAKING ACTIVITY THEORY INTO THE FUTURE

The ideas of Vygotsky, Leont’ev and other theorists of the former Soviet Union have provided and continue to provide fruitful areas of research both in the East and in the West. Lektorsky comments that in the birthplace of activity theory, there is much criticism, in intellectual circles, of activity theory. Firstly, there is the understandable repudiation by these people of totalitarianism and philosophies based upon it.

As Davydov concurs, activity theory:

- Came to be fiercely criticised because it was connected with the idea of technological progress in which the ethical dimension is put between brackets.

Secondly, and linked to the previous critique, it is argued that the emphasis in activity theory on objectivity, as opposed to subjectivity, has led to the suppression of ideas about the creativity of individuals. That is, activity theory is criticised as representing humans not as creative beings but as interiorisers of ready made standards of behaviour and cognition.

While concurring with these criticisms, Lektorsky argues that Vygotsky's and Leont'ev's ideas should not all be rejected with Marxism. He also advocates that activity theory should be developed in new and fruitful ways. He argues that to understand activity we would do well to return to the ideas of Vygotsky who stressed not only interiorisation but exteriorisation—the ways people create new standards and rules. In this view, human beings are essentially creative beings who “determine themselves through objects that they create”.

Davydov argues that the concept of activity refers primarily to the:

- Openness and universality of human actions. Consequently, ‘activity’ should be interpreted as a form of creativity which is determined by the actual cultural-historical context.

To Davydov totalitarianism intruded on a concept of activity “torn loose” from its own cultural-historical roots in the work of Vygotsky. Davydov advocates the development of a multi-disciplinary activity theory, one that combines the differences of emphases of the semiotically oriented researchers with those taking as their paradigm Leont'ev's conceptualisation of the activity aspect of human existence. Indeed, in Davydov's opinion, a “broad spectrum” approach must be taken to solve the problems. That is, analysis must take place from a multi-disciplinary point of view.

Only in this way, he feels, can we:

- Shape our inheritance from the past in such a way that it can also serve us in the future.

Cole reiterates Davydov's view that there has not yet been close cooperation on an international scale between those who take activity theory as their point of departure for analysis and those who use some version of Vygotsky's ideas with the emphasis on language and semiotics as mediation. He also comments that Russian adherents of these approaches have a different perspective on the differences which are bound up with the historical and ongoing “wrenching political upheavals of the region”.

Wertsch et al distinguish between the terms “cultural-historical”, or “sociohistorical”, as used by Vygotsky and Leont'ev, and the heading “sociocultural” used by many Western researchers applying their approaches today. They feel that the first two terms are appropriate for referring to the heritage we have gained from the Russian psychologists, but that “sociocultural” expresses more closely the ideas as they have been appropriated in current Western research. For one thing, contemporary “sociocultural” scholars do not assume the “universalist” evolutionary claims for a linear progression of the development of the human psyche, which were common to Soviet thinking of the Vygotskian era, though not to Vygotsky himself. Also, a focus on consciousness and the

internal workings of the mind was regarded with deep suspicion as “bourgeois idealism” by Soviet officials in power at the time. This was one of the reasons why Vygotsky’s work was banned for so long in his country.

As I have explained, Soviet psychology, grounded in Marxism, has a very different basis to Western psychology. Certainly, too, culture is understood differently in Western and Eastern perspectives. However, a focus on action and on mediation, common to the thinking of Vygotsky, Leont’ev and other Soviet psychologists, is considered to be defining of contemporary sociocultural research. However, Wertsch et al acknowledge that the roles of action and of mediation in human functioning are problematic. For example, the cultural tools actually invoked in an activity, were not necessarily designed for that purpose.

Instead they may have been “dictated” by other sociocultural forces—and their benefits on mental functioning incidental. For example, technology developed for military purposes has had enormous benefits for the development of civilian systems of communication and has facilitated innovation in other fields such as engineering and medicine.

The goal of sociocultural research has been defined as being to:

- Explicate the relationships between human action, on the one hand, and the cultural institutional, and historical situation in which this action occurs, on the other.

Such relationships must include ambiguities, subject to interpretation and debate. This allows for multiple points of view and openness in investigation and analysis. These are essential if theory is to provide any tools for addressing human problems. My approach acknowledges these ambiguities and complexity of perspectives.

RESEARCH QUESTIONS, DESIGN, METHODOLOGY, TOOLS AND METHODS

My case study consists of two investigations. Study One is entitled “Exploring Mature Students’ Learning of Statistics”. In this study my aim is to tell the story of five students in a way that is generalisable to others—that is, I try to uncover recognisable truths from their narratives. Study Two is called “Understanding University Students Learning Statistics as a Service Course to Psychology”. This study is based mainly on a survey in which I tried to elicit students’ feelings about learning Statistics, their conceptions of the subject matter and their approaches to learning it. I also interviewed selected students and two academics involved with teaching Statistics.

In this stage, I describe my principal and sub-principal research questions, explain how I use a range of research tools to investigate them and describe some key features of my approach. These are specified separately for Study One and Study Two. I present some issues and controversies which are important to the current discourse on methodology in education research. I outline my methodology, that is, I explain the

rationale for my ways of collecting and analysing data. I summarise the procedures that I followed. I indicate how I have attempted to satisfy criteria for good research as advocated by researchers such as Guba & Lincoln; Kirk & Miller; Merriam; Miles & Huberman and Van Maanen.

I explain how I “triangulate” my results. I describe the modes of analysis I used in each of the two studies, drawing on Bruner’s two modes of thought. I then present an overview of the methods I used in each of the two studies undertaken, respectively. That is, I describe the research setting, methods of data collection and methods of analysis for each study.

In particular, I explain the assumptions underpinning the research specialisations, to use Marton’s term from which I developed my approaches. These research specialisations, or combinations of research orientations and research approaches, include naturalistic inquiry in Study One and phenomenography coupled with quantitative analysis in Study Two. I consider ethical issues that I encountered and how I responded to them.

RESEARCH QUESTIONS

The main research question which I investigate is:

- What are students’ orientations to learning Statistics?

This question is important because, from an activity theory perspective, the way students orient or position themselves to learn is critical to the quality of learning that unfolds. That is, the ways students feel about learning Statistics, their conceptions of Statistics and their approaches to learning it reflect their engagement with the learning task—their activities. These activities are inseparable from the setting in which the students’ learning is organised. They are shaped by and transform that setting in mutual and cyclic processes. The research question is specified as principal and sub-principal questions for each of the two studies.

Research Questions for Study One

In this study the research question stated translates to the following question:

- What can be learned about students’ orientations to learning Statistics by exploring the perceptions, actions and evaluations of five mature students studying it?

This question is divided into sub-principal research questions shown below:

- What were the students’ prior conceptions of mathematics and statistics based on their experiences of mathematical activities?
- How did the students’ perceptions of the context of the course mediate their goal directed actions?
- What were the students’ approaches to learning Statistics?
- How did the students regulate and evaluate their experiences of learning Statistics?

Research Questions for Study Two

In the context of this study, the research question outlined at the beginning of expanded to:

- What are students' feeling about learning Statistics, conceptions of it and approaches to learning it and what are the relationships of these factors to each other and to other variables including gender and performance on assessment tasks?

The survey, including the open-ended questions and Approaches to learning Statistics Questionnaire.

The principal research question is separated into sub-principal research questions as follows:

- What are the demographic or individual differences for the students participating in Study Two as indicated by the following variables: gender; age; fluency in English; degree for which enrolled; prior level of mathematics studied; prior level of statistics studied; intentions as to future study of Psychology and expected performance in Statistics and Psychology II.
- What are the affective factors reported in response to the question:
 - Would you study statistics if it were not a requirement of your psychology course?

Please give reasons for your answer.

- What categories of conceptions of Statistics emerge from students' written responses to the three open-ended questions in the survey, including the question:
 - What in your opinion is this statistics course about?
 - Please explain as fully as possible.
- What are the students' approaches to learning Statistics as indicated by a questionnaire—the Approaches to Learning Statistics Questionnaire?
- What are the assessment results of the surveyed students?
- What relationships can be found among the variables of interest?
- In particular, I am interested in the relationship between the categories of conceptions and students' scores on the ALSQ.
- More generally, I investigate the inter-relationships among the following variables:
 - Category of conception;
 - Willingness to study Statistics;
 - Scores on scales denoting surface and deep Approaches to learning Statistics on the ALSQ;
 - Performance in tests and examinations;
 - Previous level of mathematics studied;
 - Age;
 - Gender;
 - Reported fluency in English; i) previous level of statistics studied.

RESEARCH DESIGN AND RATIONALE

In this part, which embeds my methodology in the literature and discourse surrounding education research, I explain the issues I explored and the assumptions and rationale that guided my planning and investigations.

RESEARCH METHODOLOGIES

Education research is the systematic investigation of educational phenomena:

- A mode of thinking rather than a shortcut to answers.

Historically a spectrum of methodologies have been applied to education research. These include scientific experimentation, curriculum development, action research and ethnographic inquiries. At one extreme of this spectrum are methodologies based on what Nisbet terms the “agricultural model”—controlled experiments to improve the product. For a long time education research was dominated by this model which aspired to scientific precision in the behavioural sciences.

Kerlinger argues that science aims to “establish laws, systematic explanations or relations that apply generally”. This paradigm is based on a positivist view of knowledge—that the truth exists and can be uncovered empirically by research. Lincoln & Guba suggest that this paradigm belongs to a “positivist era” of faith in science and the scientific method. Although we are now in a “postpositivist era”, positivism has been “remarkably pervasive”. Over the last two decades, however, there has been growing disenchantment in education research with methodologies rooted in this paradigm.

One concern is that the very nature of the experimental method with its laboratory conditions and controls renders the educational situation artificial. This leaves the relevance of the results to educational practice open to challenge. Another issue concerns the uses and interpretations of inferential statistics, which underlie the experimental method in education research. For example, in recent editorials of the *Mathematics Education Research Journal*, Ellerton initiated debate and invited further discussion on statistical significance tests in education research. In this ongoing debate, the relevance of these statistical tests to the field of education is questioned. Finally, the problems of replicability and generalisability are considerably more complex for human situations than for agricultural products. The context of any educational phenomenon is unique and multi-faceted. For these and other reasons, there has been considerable movement in education research toward qualitative methodologies that involve open-ended inquiry—based on anthropological models or sociological models.

This paradigm is loosely described by Nisbet as:

- Go and live there and see what it is like.

Qualitative methodologies have interpretation and exploration as the rationale for data collection and analysis. Some major characteristics of qualitative methodology are described by Merriam as follows:

- The primary instrument is the researcher;
- The research involves natural settings or fieldwork;

- The results are qualitative description, that is “thick” rich text which gives depth to the experience;
- The inductive rather than deductive method is followed, that is abstractions are built from examples.

The methods by which qualitative research is expressed are diverse and may include interviewing, participant observation and content analysis as well as statistical analyses.

To Kirk & Miller:

- Qualitative research is an empirical, socially located phenomenon, defined by its own history, not simply a residual grab-bag comprising all things that are ‘not quantitative’.

To me the essence of the qualitative approach to research is the contextualising of the data.

As Van Maanen explains:

- The data developed by qualitative methods originate when a researcher figuratively puts brackets around a temporal and spatial domain of the social world.

This awareness of context implicitly signifies that the research framework is non-positivistic—the research cannot uncover laws about humans which apply at all times and places. This is particularly important for research inspired by Vygotsky’s insights where human processes are “historically and socially determined”. In my investigation, the context in which students learn Statistics is a focus of my analytic understanding of their learning processes.

Gill points out the high degree of contestation over different research methods and research methodologies. In particular there is rivalry between qualitative and quantitative approaches to research.

Rather than dialogue or debate, Gill likens the rivalry to staking a claim in goldmining days:

- Stake your claim of new ground and defend it by deriding all previous efforts.

In particular, Gill finds the objection to quantitative approaches, troubling and usually irrelevant to the researcher’s actual purpose of justifying an appropriate qualitative method.

I agree with Gill that what is significant in any research methodology is the relationship between the findings and the theorising, rather than what research methods were used. If the assumptions underlying the data collection and analysis are poorly articulated then this relationship may not be well developed. For example, misunderstanding of assumptions underlying statistical analysis can lead to a reliance on numerical outcomes and the reporting of such numbers as ends in themselves. Much of the debate about the role of statistical tests of significance, centres around the misapplication of statistical tests of significance. For example, these are sometimes misinterpreted as validating or invalidating the truth of a research hypothesis. However, poor research is no more inherent in quantitative methods than in qualitative methods. An uncritical acceptance

of “highly individualised” narratives can be just as implicated in obscuring the situation of the research in its broader social context as an uncritical acceptance of numbers. To Gill the adoption of a mixed methods approach is often more productive than a single line of inquiry.

Further, Gill suggests that:

- The schism in education research is a construction—and a construction that is fairly loose and somewhat insecure.

My position agrees with this. To me, different research methodologies have in common a search for meaning and understanding. The various research approaches are by no means mutually exclusive. Many education studies today use both controlled experimentation and qualitative exploration. Moreover, I believe that methodologies applying qualitative or quantitative methods are not even sharply separable. For example, the laws developed using quantitative methods are always statistical, that is concerned with variation—probabilistic. They are not absolute statements but conditional and uncertain. This is not only true of behavioural sciences, but also of the so called “exact” sciences, such as physics.

Precision and objectivity are always a matter of degree. “Fuzziness” is inherent in all processes, be they movements of an atom or a student’s performance in an examination. Further, experimental results cannot be divorced from the researcher’s carrying out the observations, nor from the context in which he or she does so. In short, all analysis involves interpretation. Mathematical statements may seem impersonal and indisputable but their interpretation is embedded in the rich soil of human experience. Moreover, while, positivist “laws” hypothesise and test relations at the collective and abstract level, they are useful in assisting understanding of the profile and social experience of the individual. The metaphor of a spectrum of methodologies conveys the idea of shedding light. That, in the end, is surely the aim of any education research.

MY METHODOLOGY

My primary aim is holistic and descriptive interpretation, aiming for depth of understanding. Consistent with a Vygotskian perspective, I consider the sociocultural context to be inseparable from individual actions. Hence my primary concern is to interpret the network of relationships between learner, subject matter and context. In attempting to make the data meaningful, I have drawn on various techniques of data collection and analysis. I have found it useful to combine qualitative and quantitative methods.

Overall, my mode of doing research is consistent with what Lincoln and Guba term the “naturalistic paradigm” in that I take as axiomatic that: realities are multiple; myself as researcher and my objects of research are inseparable; my working hypotheses are bound by time and context and all entities are in a state of mutual simultaneous shaping. Moreover, my inquiry is “value bound”. That is, my investigation is bound by my assumptions, theories and perspectives and is regulated by both cultural norms and my

individual beliefs. For example, my focus on the desirability of deep, rather than surface, approaches to learning and on personal development as part of the educational process reflects my commitment to these values.

The methodology I have used in this project is best described as case study research. Case study research does not claim any particular techniques for data collection or analysis. It focuses on a clearly defined area of investigation—what Smith calls an “instance of a more general class of events”.

The case study is appropriate when “interpretation in context” is essential to understanding the phenomenon under investigation. Yin defines the case study to be:

- An empirical inquiry in which the number of variables exceeds the number of data points.

This means that the phenomenon under investigation is complex and bound up with the context, so that the important and interacting variables are too composite and too numerous to isolate and measure. Miles & Huberman sometimes prefer the word “site” to “case” to emphasise that case studies take place in a specified social and physical settings. My design emphasises insight, discovery, description and interpretation within a bounded context. As with any style of research, reliability, validity and generalisability are prime concerns. However these terms have different meanings according to the position of the researcher. In a positivist framework there is an assumption that there are “true” scores which, given a good enough instrument, can be measured reliably and with validity. That is, the measuring instrument should capture accurately and with stability the true essence of what is being measured.

From a non-positivist perspective, however, there are no absolute truths or universal laws about human processes which can be captured. In such frameworks, the ways in which research findings are validated are socially construed. Constructs represent a consensus by the group concerned. This perspective suggests to me that the criteria for reliability and validity in education research depend on the recognition of the community of education researchers. These criteria are constituted by a convergence of social opinion about what is being investigated and how well this is done. My understanding of social consensus as the basis for the research criteria accords with the qualitative approach to education research. In particular, Merriam formulates questions about reliability, validity and generalisability as they relate to qualitative case study research. These questions suggest criteria which fit with my framework. In what follows I describe these criteria and give an overview of my attempts to satisfy them. The details of how these attempts were carried out will be described in the context of Study One and Study Two.

Merriam formulates the question of reliability for qualitative case study research as being:

- Are the results consistent?

I have tried to provide a basis for recognition of consistency by means of different modes of “triangulation”. This surveying term, now commonly used in education research, was first appropriated by Webb et al who spoke of subjecting findings “to the

onslaught of a series of imperfect measures”. This could suggest, however, that the perfect measure exists. In contrast, from my perspective, triangulation is about different ways of viewing a phenomenon, each of which is embedded in its own cultural and social context. Together, these ways provide a multi-faceted perspective on the phenomenon under investigation and better opportunities for understanding it than one method alone.

My modes of triangulation are listed below:

- I used triangulation “between methods”, that is different methods of data exploration. Both qualitative analyses and quantitative explorations were utilised to find congruencies and anomalies;
- I also applied triangulation “within methods”. For example, interview data was used to supplement, qualitatively, students’ written responses to open-ended survey questions and I used a range of statistical analyses to explore patterns in the data;
- My third mode of triangulation was triangulation “between investigators”. I checked findings with fellow researchers, informally in discussions and formally by independent classifications and peer reviews.

The prime question of validity for case study research suggested by Merriam is:

- How credible are my findings given the data I have? Can others recognise that my conclusions are supported by the data?

I have tried to achieve internal validity by:

- Making overt my assumptions and procedures and laying an “audit trail” so making it possible for other researchers to follow how I got from the data to my findings;
- Making explicit the context, my personal biases and the theoretical framework;
- Suggesting and checking alternative explanations for my findings;
- Most importantly, submitting preliminary analyses for publication, or presenting my findings at conferences, thereby ensuring peer examination of my research.

In order to satisfy user or reader generalisability that is to enable researchers to apply the findings to their own research, my aim is to make the data accessible—to create a virtual reality so that others can experience the phenomenon through my eyes. The passport to generalisability of my research is its “shock of recognition” by others.

I have tried to achieve this by:

- “Thick” description facilitated by my personal and deep relationships with some of the participants;
- Diverse sites for comparison, namely the Mathematics Learning Centre the Psychology II class and a General Statistical Methods class;
- Inclusion of both unusual and “typical” members of the group being studied;
- Sampling to capture diversity within the project;
- Submersion in the research situation;
- Making explicit the “transferable *theory*” from the investigation.

GROUNDING THEORY

Beginning with Glaser and Strauss there has been much discussion of “grounded research”, that is, the need for the research framework to evolve rather than to be imposed. In my case, a framework of beliefs, shaped by my teaching experience, previous research and interest in activity theory, resulted in my developing a loose theoretical framework early in the project. This was revised and changed at every stage of the research. Study One was explorative based on observations and intuitions gained by many years of teaching experience, in particular, teaching the Psychology students who attended the Mathematics Learning Centre for assistance in Statistics. The framework developed in Study One then served as the platform for developing the questions, hypotheses and procedures for the broader Study Two. This conceptual development is ongoing, informed by discussions and the act of writing this thesis and submitting research papers for review. Throughout the investigation, I have attempted to give a voice to the students. My analysis, though influenced by my experiences and by my interpretation of activity theory, is grounded in the perceptions expressed by the students.

ANALYSIS OF DATA

CRITERIA

Sieber suggests that good analysis usually involves the following:

Intertwining of analysis and data collection; Formulating classes of phenomena—that is, categorisation of concepts; identifying themes—linking concepts, noting regularities, patterns; provisional testing of hypotheses—looking for concomitant variation, ruling out confounding factors, identifying intervening variables. I attempted to meet these criteria in each of the studies.

TRIANGULATION USING QUALITATIVE AND QUANTITATIVE METHODS OF ANALYSIS

Qualitative and quantitative analyses each have assets and liabilities. The aim of triangulation is to exploit the assets of each method while minimising the deficiencies of each. To me, the singular features of quantitative analyses are the level of abstractness at which the findings can be described and the focus on pattern rather than detail. These generate a powerful way of communicating a complex and dynamic system. Quantitative data are useful for discerning distributions, trends, relationships and anomalies. The strengths of qualitative analyses are the very features lost in quantification—the richness and depth of description concerning the particular individual and context.

As Weis points out:

- Qualitative data are apt to be superior to quantitative data in density of information, vividness, and clarity of meaning.

My application of complementary methods of analyses produced opportunities for identifying consistencies within the data, thus enhancing my confidence and encouraging

better definitions and enrichment of the description. They also produced surprises and anomalies. These led to alternative and deeper analyses and more insightful interpretation and understanding of the data. Most importantly, the complementary methods allowed for my immersion in the data. This resulted in my improved closeness to the data.

Methods of Triangulation

As a researcher, my task is to gain insights into the data by systematically interpreting their meaning. Different methods of inquiry were used in Study One and in Study Two to investigate the network of relationships relating to the research problem. These include holistic description and statistical explorations. In both studies I have triangulated the analyses “between” and “within” methods by using a variety of approaches both qualitative and quantitative in order to better describe and understand the problems in their contexts. In Study One and in Study Two, “within methods” triangulation of the qualitative data included using material from interviews, surveys, informal observation and students’ written work. Quantitative indicators, such as performance on assessments and demographic data, added another dimension to the descriptive data in both these studies. In Study Two, my triangulation “within methods” also included using a range of statistical analyses such as cluster analysis, factor analysis and correlations. The different ways of exploring the data in this study produced patterns and trends which indicated consistency. For example, the qualitatively different categories of students’ conceptions of the subject matter, which emerged from students’ responses to open-ended questions, were confirmed as having links to students’ approaches to learning Statistics, analysed quantitatively, using scores on a Likert-type questionnaire. Strong relationships were found among the variables, such as students’ conceptions of Statistics and their willingness to study it.

Statistically significant trends converged with qualitatively significant differences between the categories. Further triangulation for both studies was provided by “between investigator” analysis and interpretation. Here, discussions with colleagues at each stage of the two studies provided valuable feedback on my research design, interpretations of findings and conclusions. Some independent classifications and checks were also carried out on my categorisation in Study Two.

For instance, students’ conceptions of Statistics as reported in their surveys, were categorised by myself and, independently, by another education researcher. Since this researcher is male, younger than me and has a different educational background, we brought very different perspectives to bear on the data. Discussions on the categorisation were held with a third researcher, allowing for further negotiation of different views. In addition, as explained earlier I consider it important to submit my work for publication as this enables me to experience peer reviews of each aspect of my project. A list of my relevant publications appears at the front of this thesis. This includes pilot studies and publications on preliminary findings, both for Study One and for Study Two as I carried them out. I will specify these in the context of each of the two studies.

MODES OF THOUGHT

I used a different genre for expressing my analysis in each of my two studies. In this I was influenced by Bruner's two modes of thought. Bruner calls these the "paradigmatic or logico-scientific mode" and the "narrative mode". The paradigmatic mode of thought is driven by "principled hypotheses" which relate not only to what is observed but also to possible, logically generated worlds.

This mode is the basis of science and is characterised by Bruner as:

- Attempts to fulfil the ideal of a formal, mathematical system of description and explanation. It employs categorization or conceptualization and the operations by which categories are established, instantiated, idealized, and related one to the other to form a system.

This is the mode of thought underlying my attempts to build up an "activity system" by a methodical and logical relation of parts to a whole, guided by empirical discovery and "reasoned" hypotheses. In Study Two my style of reporting is consistent with this mode of thought. However, in Study One and in my overall "story", I am also concerned with the narrative mode of thought. This mode is the basis of literature.

As an art form it:

- Deals in human or human-like intention and action and the vicissitudes and consequences that mark their course. It strives to put its timeless miracles into the particulars of experience, and to locate the experiences in time and place.

The essence of this mode is the attempt to create gripping drama and believable accounts. It relates events temporally as well as by themes. It provides a basis for understanding individual actions and events as part of a unified "plot". I have no interest in creating drama but I do have a concern with the unfolding of particular students' experiences and the insights these provide into human conditions. In short, these two modes are both cultural tools, forms of which I have employed in my research activity.

STUDY ONE: EXPLORING MATURE STUDENTS' LEARNING OF STATISTICS

RATIONALE FOR STUDY ONE

A focus of my inquiry is to interpret and extend the theory of activity by describing the learning of Statistics as purposeful and interactive activity that develops in a sociocultural context. As is the case for most students of the Mathematics Learning Centre, these students were at least 25 years old and had been out of the education system for a number of years. The rising number of older students in many universities means that research on their learning is becoming increasingly relevant and significant. In addition, observations made by mature adult students are insightful and supply valuable material for stimulating reflections on teaching. The five students who were invited to participate in Study One were selected purposefully. As will be explained in these students had

diverse life experiences, a range of academic backgrounds and singular personal attributes which led me to believe that their participation would be valuable to my research. In short, I believe that an understanding of the perceptions, actions and evaluations of the participants of Study One is helpful in shedding light on important issues concerning students who study statistics at university as a service course, rather than by choice. In keeping with a qualitative research design, my analysis of this data is interpretative rather than explanatory, acknowledging the complexity of the issues faced and aiming to indicate the dimensions of that complexity, as well as to stimulate reflection and dialogue.

MODEL FOR METHODOLOGY IN STUDY ONE

The design of this study reflects my consideration that the naturalistic paradigm has the best fit to my aims described. That is, in this study, with one exception, I have adopted the attributes of the naturalistic paradigm. The exception concerns my lack of adherence to grounded theory. What I studied was, to some extent, delineated in advance, informed by my understanding of activity theory. The attributes of the naturalistic paradigm are detailed with a description of how I complied with each in Study One.

- **Characteristic 1: Natural Setting**

- Study One was carried out in the Mathematics Learning Centre in the context of my teaching the five students during their second year of study of Psychology. All the students were regular and intensive users of the Centre and in some cases spent five or more hours per week working with me on their Statistics. Indeed the students frequently expressed the idea that they “lived in here” during their second year of study of Psychology. My observations of them for the purpose of the study are inseparable from my actions designed to assist them with their learning.

- **Characteristic 2: Human Instrument**

- I consider myself to be the primary data gathering instrument. That is, the evaluation of what the students said, did or wrote, is personal and subject to my own perceptions, beliefs and values.

- **Characteristic 3: Utilisation of Intuitive Knowledge**

- The study reflects my deep and warm relationship with the students. My analysis takes account of my experience of teaching them and others who have sought my assistance in learning Statistics.

- **Characteristic 4: Qualitative Methods**

- Qualitative methods of data collection and analysis were used, namely descriptive and exploratory techniques aiming to illuminate multiple truths in context.

- **Characteristic 5: Purposive Sampling**

- The students were not selected as being representative of any general group of students. Rather, it appeared to me that due to their singular characteristics

exploring these students' orientations to learning Statistics would be of value in understanding how students learn Statistics.

- **Characteristic 6: Inductive Data Analysis**

- I have tried to construct general themes from the particular examples provided by the students' accounts.

- **Characteristic 7: Grounded Theory**

- I did not wholly comply with the description by Lincoln and Guba of having “the substantive theory emerge from the data”, as I was building on activity theory. However, my theory evolved as the study unfolded.

- **Characteristic 8: Emergent Design**

- The design unfolded as I observed the students. It was influenced by my interactions with the students. My research questions evolved from informal observations and intuitions. They were incorporated into the questionnaires completed by the students during their study of Statistics and thence into the semi-structured interviews conducted with the students after they had completed second year Psychology. Themes emerged in these interviews and subsequent interactions with colleagues and experts in education. The design of Study One was only fully completed when I had written and revised a paper on the study for publication.

- **Characteristic 9: Negotiated Outcomes**

- My interpretations were discussed with the five participants during their interviews and re-evaluated together with peers and other researchers. I also tested my working hypotheses in other studies. I did not, however, ask the five participants to evaluate my interpretations of their interviews. I would have preferred to do so, but there were practical and ethical problems. By the time they were interviewed the students had already completed second year Psychology, and so had ended their association with me at the Mathematics Learning Centre. Not all of them were accessible to me afterwards. Also, I did not wish to exploit my personal relationship with the students by seeking their cooperation in my project beyond our negotiated agreement.

- **Characteristic 10: Case study reporting mode**

- I adopted this mode of reporting, also called the “narrative format” in order to constitute the stories of the students, as I perceived them. In this way of reporting I was influenced by Bruner's narrative mode of thought, as explained. This genre recognises the importance of description and context and is intended to make possible understanding on the part of the reader.

- **Characteristic 11: Idiographic Interpretation**

- I make no claims that the study fulfils any general human condition. The data are interpreted in a particular situation and this interpretation is dependent on my personal perspective of the context.

- **Characteristic 12: Tentative Application**

- While I hope that my research will be of value to other education researchers and to practitioners of teaching statistics, my findings are confined by temporal and physical boundaries - a snapshot of a situation. However, I have taken what steps I can to ensure that the extent of transferability and applicability by others can be judged by others.

- **Characteristic 13: Focus-Determined Boundaries**

- The boundaries of this study are determined on the basis of the theoretical concepts which frame the study, and by the particular activities undertaken by the students in the social setting.

- **Characteristic 14: Special Criteria for trustworthiness**

- I have tried to meet the criteria of reliability, validity and generalisability. The aim of meeting these criteria is to enable my conclusions to be recognised and used by others.

PREPARATORY RESEARCH FOR STUDY ONE

Two preliminary studies informed Study One. The first investigation formed the basis of the initial data collection on four of the participants in Study One. This investigation addressed students' attitudes to and beliefs about mathematics and statistics. The students were surveyed during a bridging course in statistics.

The course was advertised as a preparation for those students lacking the basic mathematical skills needed for their study of statistics in second year Psychology or for postgraduate courses in Public Health. The participants in the bridging course were self selecting: each of the 60 students enrolled felt poorly prepared for the statistics course which was a compulsory component of his or her chosen course. One interesting finding of this preliminary study related to differences in students' perceptions of statistics as compared to mathematics. In response to free association with the words "statistics" and "mathematics", students reported that statistics is "useful" and statistics is "misleading". Disraeli's saying: "Lies, damned lies and statistics!" evidently rings true for many people.

Mathematics, on the other hand, was seen by these students as being either true or false. I took these perceptions into consideration in my analyses of Study One and of Study Two. That is, I paid attention to differences in the students' usage of the words "statistics" and "mathematics" in their written responses to open-ended questions and in interview data. The second investigation looked at the attributes and perceptions of Psychology II students who attended the Mathematics Learning Centre: their motivation for studying Statistics, their reasons for using the Centre, their background in mathematics and their evaluations of the learning environment. Correlations between the students' examination grades and their ages, prior levels of mathematics studied and performances in other areas of Psychology were addressed in this study. I discovered relevant questions to be addressed, for example: What were students' expectations about themselves as

learners of Statistics? I also identified methodological issues from this investigation. For instance, I applied the “action research spiral” to this investigation. At the same time as building up the pragmatic and methodological foundations for Study One, I was building up a theoretical framework. This was based on the ideas of Vygotsky and Leont’ev as I have related. I applied their theories to the context of the Mathematics Learning Centre by illustrating the major features of the theory of activity with examples taken from my teaching practice.

I also developed a model for teaching these students based on my own, adult, experiences of learning to ice skate. In order to view the activity of learning Statistics from the students’ perspectives, I related my own deficiencies in the three C’s needed for ice skating, to the students’ feelings of inadequacies about themselves as learners of Statistics. I compared my ways of operating in the zone of proximal development with the support of the skating instructor, to the ways students learned to overcome their difficulties with my support. In these ways—by developing theoretical concepts, intuitive knowledge and metaphors for my teaching and learning, and submitting publications on these to peer review, I developed my themes and working hypotheses for Study One.

METHOD FOR STUDY ONE

Participants

Each of the five participants of Study One had exceptional attributes in some respect. The names I use to refer to the students, are fictitious but indicate gender.

Norman, Sandra and Alice were extremely successful students of second year Psychology:

- Norman and Sandra both achieved High Distinction grades, a rank attained by only four percent of their Psychology II class of 473 students, while Alice gained a Distinction for the course. Alice and Hettie had the unusual characteristic of not having studied mathematics at secondary school level at all. In fact Hettie’s school education ended six months into secondary school, making singular her later academic achievements. Both succeeded in passing Statistics after a considerable struggle although both had high grades in other areas of the course. Four of these students were pursuing a study of psychology for vocational reasons. Ernest, on the other hand, was of an age where vocational interests are not relevant and was studying psychology to satisfy his intellectual curiosity. This resulted in his having a different attitude to his studies.

Data Collection

My data collection for Study One included informal observations made while participating in the students’ learning, audio taped interviews, short surveys and questionnaires relating to the students’ attitudes to and strategies for learning Statistics,

the students' written evaluations of the teaching and environment of the Mathematics Learning Centre, demographic information and assessment results. As is customary with qualitative research my data consists of descriptions, direct quotations and excerpts, obtained by close psychological contact with those being studied. The major sources of these data were interviews conducted after the students had completed the course. The students trusted me and were uninhibited in the interview situation. Some of the students' quotations which I report were obtained from questionnaires completed by them at various stages of the academic year and from their written notebooks.

My initial data collection consisted of survey responses of students taking a bridging course in statistics. Four of the participants in Study One took part in this bridging course. My aim was to elicit the students' own understandings and perspectives instead of imposing my own ideas. Accordingly, at the start of the first session, without more than a brief personal introduction, the students were asked to respond freely with their associations to "mathematics" and "statistics".

These open questions were followed immediately by a more structured questionnaire. From this investigation, I obtained an initial idea of the issues pertinent to the four students' views on mathematics and statistics. Demographic information about the five participants and their reasons for seeking help in Statistics were obtained from their enrolment forms for the Mathematics Learning Centre. This was completed by the five students early in the first semester.

Throughout the year I observed the students informally, while I was engaged in teaching them individually, or while they were working in small groups in the Centre. As each of the students spent considerable and regular time with me according to their own needs, I had many opportunities to view their written work and to observe their strategies for learning. I spent considerable time interacting with them, both informally and in formal instruction. A survey consisting of an open-ended questionnaire and Likert-type items was completed by the five students in semester two. Information on their evaluations of the Mathematics Learning Centre, as the setting for learning Statistics, was obtained from Mathematics Learning Centre Evaluation forms for the students, all of whom opted to sign their names on this survey.

After their final examinations in second year, at the conclusion of the academic year, I invited each of the students to come into the Centre for a long interview. According to Patton the purpose of the interview is to enter into the other person's perspective. To be successful, the interviewer must be neutral, that is non judgemental, and have rapport with the subject. The protection of the respondent must be ensured and ethical issues taken into account. The logistics of the interview must be worked out.

I based my interview questions on the activity theory framework and on my data analysis of the students' responses undertaken so far. The interview questions were designed to gain an in-depth knowledge of the students' goals, perceptions, attitudes and beliefs, relating to the Statistics course which they had recently completed. The interviews were semi-structured but neither the exact wording nor the order of the

questions was precisely determined in advance. I allowed for feedback, interactive reassessment and probing. Furthermore, I took advantage of the students' relaxed frames of mind and the interpersonal trust we had established, to follow up any unseen issues that arose in the course of the interview. I was also able to explore new openings suggested by the students' responses. In short, I acknowledged the sub texts of the conversations. The interview questions were divided into four broad areas, as advocated by Patton.

These were about:

- *Behaviour or Experience, for Example:* Did any changes in your learning pattern occur during the year?
- *Opinions and Values, Such as:* How useful do you think what you have learned in Statistics will be to you?
- *Emotional Responses and Feelings:* How did you feel about mathematics then?
- *Background Demography, for Example:* Have you done any mathematics or statistics since school, apart from your psychology course?

The ways in which the questions were framed included four modes, described by Strauss, Schatzman, Bucher, and Sabshin.

These are:

- *The Hypothetical Mode:* For example—If Statistics had not been assessed, but nothing else changed, how would you have approached learning it?
- *The Devil's Advocate Mode:* For example—Some people would say, that if you can't cope with statistics you have no business doing psychology. Would you comment on that?
- *The Mode of Posing the Ideal:* For example—What to you would be the ideal statistics course?
- *The Interpretative Mode:* What do you mean by?/Could you explain?

I used this mode for probing. The interviews were audio taped and I transcribed them myself, as a way of getting close to the data. I also collected and read the students' notebooks in which they had made their own summaries of Statistics. This added further insights into each individual's ways of working and feeling.

For example, Alice prefaced her written summary of a part she found difficult with:

- Analysis of Variance G-d help me!

The notebooks were returned directly and promptly to the students but I retained confidential photocopies of illuminating parts.

Data Analysis

The aim of the data analysis is to tell the story of the five students in a way that could be recognised by other researchers and educators. As advocated by Sieber each stage of the data collection was analysed and suggested further data collections and analysis in an ongoing cycle. In the first level of analysis I arrived at categories of description, namely nodes or clusters of similar objects or characteristics, by reading systematically through the case records and finding patterns. In particular, the act of personally

transcribing the audio taped interviews suggested categories of similar phenomena. Then I found units of information: for example, a sentence or notion that stood by itself, and I classified each into a category. In this way the categories were clarified and refined to ensure that they were, as proposed by Holsti and Merriam, relevant, exhaustive, mutually exclusive, independent and formed from a single classification principle. The categories I arrived at were: demographic information; students' goals with respect to learning Statistics; their perceptions of school mathematics; perceptions of "real life" statistics; perceptions of context in the activity of learning Statistics and approaches taken to learn Statistics. The categories were entered on a database as a precursor to my writing an integrated and interpretative description.

That is, from the earliest stage of analysis, my database was a narrative, intended to be read by an audience of educators. Each stage of the analysis corresponded to a refinement of this narrative—from disjointed elements, through motifs to a unified whole. The second stage of the analysis was to identify themes which connected the categories. The themes were derived from what the students said—common threads in the interview data. These were analysed in terms of Leont'ev's breakdown of activity into three levels: the milieu or social setting of the activities, the goals directing the actions and the operations or automatic components of the actions. One major theme which emerged, relates to students' evaluations and self regulation of the task of learning Statistics. The next step was to formulate working hypotheses.

These were outlined are reproduced below for the convenience of the reader.

- The students identified two different sorts of statistics or mathematics—school mathematics and "life" statistics or mathematics.
- Actions were goal directed.
- The students' approaches to learning Statistics were related to how they positioned themselves with respect to the learning task.
- The students' metacognitive processes of monitoring and evaluation were paramount in defining their activities.

The last stage of analysis involved integrating my interpretations and explanations into a coherent description. The results of this study served as the starting points for Study Two.

STUDY TWO: UNDERSTANDING UNIVERSITY STUDENTS LEARNING STATISTICS AS A SERVICE COURSE TO PSYCHOLOGY

SETTING FOR STUDY TWO

The setting for Study Two is the class of students studying Statistics as a compulsory component of Psychology II at the University of Sydney in 1995. All the participants had studied a short introductory course in statistics in Psychology I. The students

completed the survey during a Statistics lecture, and had by this time written two Statistics class tests. The excerpts reported are written expressions from this survey and vignettes from interviews conducted after the survey, later that year.

GROUNDING FOR STUDY TWO

Study Two is grounded in Study One. The results of Study One were instrumental in determining the questions to be answered for the whole cohort of Psychology II students. The methodology owes much to my previous research projects, in particular, a pilot study and collaborative research on the learning experiences of mainstream mathematics students. A pilot study was conducted by surveying 52 students entering second year Psychology and 59 Arts students starting a first year general statistics course. In this survey, I asked students to report on their reasons for studying statistics, their expectations of their courses and their usual approaches to studying mathematics. These responses were elicited by three open-ended questions. Question 1 differed for the Psychology students and the General Statistical Methods students because statistics was an elective for the latter group.

The Psychology students were asked to respond to the following question:

- Would you study statistics if it were not a requirement of your psychology course?
- Please give reasons for your answer.

General Statistical Methods students were asked the following question:

- Why did you decide to enrol for this course in statistics?

Both groups were asked to respond to the following two questions:

- What do you expect this statistics course at university to be about? How do you usually go about learning some maths?

The students surveyed in this pilot study were starting their statistics courses and many had little or no experience of studying statistics, as distinct from mathematics. Hence I did not ask about their usual methods of studying statistics. I identified categories of description for the students' responses to each of the questions. These were discussed with a colleague experienced in education research. This researcher also, independently, classified each response. We then discussed our classifications and reached agreement on all classifications. The results showed that those who chose to study statistics were motivated primarily by perceptions of the relevance of the knowledge. In contrast, personal and negative evaluations of mathematics dominated the responses of those unwillingly studying a compulsory course in statistics. About half the students surveyed expected their statistics course to provide a tool; most of the others focused on statistical processes. Disturbingly, almost 80% of them reported surface approaches to learning mathematics. The three open-ended questions from the pilot study were modified in Study Two. In this study I tried to ascertain students' conceptions of Statistics during their study of Psychology II, rather than investigating the expectations of students entering Psychology II about Statistics, as was the case in the pilot study.

METHOD FOR STUDY TWO

All second year Psychology students attending a Statistics lecture were asked to complete a questionnaire on their conceptions of Statistics and their attitudes and approaches to learning it. Interviews with selected students were carried out during the following months while analysis of the survey was ongoing, in order to flesh out and clarify the findings emerging from the participants' written responses. I presented my preliminary findings at an international conference, the 20th Conference of the International Group for the Psychology of Mathematics Education.. Feedback from this presentation was used to refine the analysis. I gained further feedback on my ongoing analysis at an Australian symposium on research methodologies and at the 20th Annual Conference of the Mathematics Education Research Group of Australasia.

Participants

The survey was completed by 279 students. The majority of these participants were female. Most of the students were 19 years old (48%) or 20 years old (18%) .

Data Collection

Written Data

The survey was completed during twenty minutes of a Statistics lecture near the end of the first semester. The survey includes questions relating to demographic variables such as age, gender and prior level of mathematics studied, three open-ended questions and a Likert type questionnaire—the Learning Statistics Questionnaire. The Approaches to learning Statistics Questionnaire consists of items from the Learning Statistics Questionnaire. I personally explained the research to the students and asked them to assist by participating in the research. I then presented the survey papers to the students, with the help of three assistants. Students were asked to detach and keep the information sheet. To ensure confidentiality, all surveys were given a code number. The last sheet, containing the names and telephone numbers of students who were willing to be interviewed, was coded with the same number, detached and kept in a separate and secure place. Test and examination results were publicly available from the departments concerned. These identify students by their Student Identification Number only, that is, no names are supplied on the lists that are publicly available. Hence I could, and did, only access the marks of students who had given their permission for this to be done by supplying their Student Identification Number on the survey.

Interviews—Vignettes

In Study One, interviews were the primary sources of data. However, in Study Two, interviews with students served only to illustrate and clarify aspects of students' written survey responses. Hence it was my aim to interview few but carefully selected students. Approximately one third of the surveyed students supplied telephone numbers indicating

their willingness to be interviewed. This meant that I had a large pool from which to select prospective interviewees. It also indicates a high degree of engagement with the research by the participants so enhancing the richness of their written responses.

Seven students of Statistics were interviewed. These were selected because they gave diverse survey responses both in terms of what they wrote and how engaged with the issues they seemed. In particular, I interviewed students with different attitudes to learning Statistics, conceptions of Statistics and approaches to learning it—as far as I could judge at the time from my ongoing analysis of the three open-ended survey questions. I also ensured a range of values on variables such as age, gender and background in mathematics. I interviewed some who wrote a lot, or expressed themselves passionately, and some who did not articulate much of their thinking. During the period in which I conducted interviews no assessment marks were available to me, nor had I commenced analysis of the Approaches to Learning Statistics Questionnaire.

Extracts from six interviews are quoted. These flesh out the written responses of these two students. Short vignettes from four other interviews are included to illustrate four different student profiles identified. The interview data from the seventh student is not included as it did not provide any additional clarity nor shed new light on the findings. I also interviewed two academics. For the sake of maintaining confidentiality, I will refer to them only as teachers of Statistics.

However, these show “seed” questions only. The interviews were unstructured. I attended carefully to what the individual interviewees were saying and followed up on the issues which seemed important to that person or provided me with new insights.

Summary of Procedures for Data Analysis

I outline the procedures I followed to analyse students’ responses to the survey.

The parts fill out this overview by specifying the details for each portion of these analyses:

- For each variable relating to personal differences, including sex, age and prior level of mathematics studied, I defined categories and then coded these numerically.

For example, I identified six hierarchical levels of prior mathematics studied by the participants:

- Less Than Year 12,
- Unit Mathematics In Society,
- Unit Mathematics,
- Unit Mathematics,
- Unit Mathematics,
- University Mathematics.

The middle four of these relate to the Higher School Certificate in the Australian state of New South Wales. Some of these categories were further divided into sub-categories.

- I analysed students' willingness or reluctance to study Statistics from their responses to Question 1 of the survey. Initially I defined a dichotomous variable CHOICE: YES or NO. Then I classified students' reported reasons for their choice.
- I categorised students' conceptions of Statistics using a phenomenographic approach. Five main categories were obtained from their responses to the open-ended questions.
- I analysed students' approaches to learning Statistics as indicated by their scores on two scales, the Deep Scale and Surface Scale of the Approaches to Learning Statistics Questionnaire. The ALSQ was derived from Biggs and modified by my colleagues and myself in previous research. My construction of these scales and checks on their reliability coefficients.

Higher Levels of Analysis: Developing Relationships and Themes

I analysed students' survey responses to explore the inter-relationships among the variables. That is, I explored the relationships among students' conceptions of Statistics their scores on the Deep and Surface Scales of the Approaches to Learning Statistics Questionnaire, their expressed willingness to study Statistics, their attainment in tests and examinations on Statistics, and various demographic and individual difference variables, such as gender and prior level of mathematics studied. Strong patterns and relationships were found. The data collection and analysis was not a linear process. I was interviewing students at the same time as I was analysing the survey responses. Each stage of the analysis and data collection resulted in my reviewing the earlier stages and, in many cases, rethinking my conclusions and proposing new hypotheses. For example, my analysis of data on students' marks which became available at the end of the year guided me to new insights about the students' experiences. My exploration of students' responses also suggested questions about teachers' perceptions and led to my interviewing two teachers of Statistics. Themes emerged—such as the relation between students' "personal sense" of Statistics and the meaning of Statistics as collectively understood. I synthesised the themes arising from Study Two with those of Study One.

ANALYSIS OF STUDENTS' APPRAISALS OF LEARNING STATISTICS

This part outlines the method used to analyse the first open-ended survey question:

- Would you study statistics if it were not a requirement of your psychology course? Please give reasons for your answer.

As I specified the first part of the question was simply analysed in terms of whether the student answered "Yes" or "No".

The reasons that students gave for their choice were categorised in the following way:

- I started with the categories that had been identified in my pilot study. However, I expected that differences would emerge in Study Two. The pilot study had

been carried out three years earlier. A different lecturer taught Statistics then and the pilot study was conducted at the beginning of the academic year, so that students had less experience of Statistics than the participants in Study Two. Moreover my pilot study sampled not only Psychology II students but also students of General Statistical Methods. Nevertheless, many of the same categories were found to be useful for Study Two.

- Most students gave several reasons for their decisions. For example, some students who reported that mathematics was boring also believed that it was hard. I classified each portion of the student's response into one of the specified categories. If the student gave a reason which indicated a negative view, I assigned her a score of -1 for the appropriate category. I assigned a score of 1 for favourable observations. If the student made no response in a particular category, a score of zero for that category was allocated. In this way students could have non zero scores in a number of categories.
- Each student scored either -1 , 0 or 1 for each of the categories. Hence the total number of favourable responses and unfavourable responses in any one category was equal to the number of students whose reported reasons were classified in that category.
- If the student's response did not fit into any of the categories identified so far, a new category was specified.
- When all 279 surveys had been classified, some of the original categories were found to be empty and dropped.

ANALYSIS OF STUDENTS' CONCEPTIONS OF STATISTICS

I based my analysis of students' conceptions of Statistics on the modifications of the phenomenographic method that were developed in earlier research by my colleagues and myself. As was the case in those investigations, the categories of description and their patterns of distribution are considered to be major results of Study Two. The phenomenographic analysis of the data on students' conceptions involved three stages. In the first stage the categories of conception were identified.

The second stage consisted of classifying the 279 surveys into these categories. In this process the categories were clarified and refined. In the third and later stage I checked each survey response to pick out any that I felt had been misclassified and submitted these to further examination and discussion.

Stage One: Identifying The Categories of Conception

The first stage in the analysis of the data was to identify a set of qualitatively different categories of description of students' conceptions of Statistics.

This involved the following procedure:

- An initial set of categories was identified by myself and another education researcher, Mr Peter Fletcher, experienced in phenomenographic research. This

was achieved by our independently reading and classifying the entire set of 279 written responses to the open-ended question below:

- What in your opinion is this statistics course about? Please explain as fully as possible.
- We two researchers then compared and discussed our initial categories and agreed on a draft set of categories and sub-categories.
- Together with a third researcher, we independently classified 30 selected responses in terms of this draft set of categories.
- The three of us compared and discussed the individual classifications for the 30 responses and agreed on a final set of clear statements for each category.

The categories were arrived at by negotiation and discussion. We looked at commonalities in the students' responses and differences between them.

Stage Two: Classifying the Surveys

All 279 responses were then classified into the identified categories by myself and my co-researcher, working independently. This classification was done in the following way.

- For each student the response to Question 3 was read in isolation to the rest of her survey.
 - What in your opinion is this statistics course about? Please explain as fully as possible.

An initial decision was made as to the category the response best fitted. I was mindful of the fact that, although I had phrased the question in order to elicit the student's own perception of the subject matter, some students were likely to interpret the question as asking about their perceptions of the teacher's view of Statistics, rather than their own awareness.

- Each of us re-appraised our initial decisions in the light of students' responses to the other open-ended questions, namely:
 - Question 1 Would you study statistics if it were not a requirement of your psychology course? Please give reasons for your answer.
 - Question 2 Think about the statistics you've done so far this year.
 - a. How do you go about learning it?
 - b. What are you trying to achieve?

In this way we amplified our understanding of each student's awareness and perceptions of Statistics and gained further insight into her way of experiencing the knowledge. We individually recorded a final category for each student's conception of Statistics by taking account of the student's overall responses to all three open-ended questions. The initial category that had been identified from Question 3 alone played no further part in the analysis, but was found to be very useful in the ensuing discussions between myself and fellow researchers in making explicit the reasons for the final categorisation.

- We then discussed all classifications and reached agreement on any that did not match. In this process of classification and discussion the categories themselves were further clarified.

That is, as Marton explains, the analysis:

- Is dialectical in the sense that meanings are developed in the process of bringing quotes together and comparing them.

Not only core meanings are important in specifying the categories, but, as Marton points out, “borderline cases” play an important part in delineating the categories. In accord with Marton’s phenomenographic method, the categories transcended the boundaries between students’ responses to different questions and also transcended the boundaries between individuals. That is, the students’ separate responses to different questions and their individual ways of expressing themselves were subsumed into general categories.

4.5.5.3 Stage Three: Triangulation and Checking

Triangulation on the analysis of students’ conceptions of Statistics was achieved by:

- Interpersonal agreement between at least two researchers;
- Interviewing selected subjects;
- Peer review. I presented my ongoing analysis at three research conferences, conducted approximately six months apart. In this way I obtained feedback on my ideas, methodology and findings as my study progressed.

After a period of some months I read through all the responses, one category group at a time. On the basis of the better understanding of the data now available to me, for example through interviews, I discovered a few responses that appeared to me to have been misinterpreted in earlier discussions. These were further discussed with my colleague and reclassified in some cases.

DEVELOPMENT OF THE APPROACHES TO LEARNING STATISTICS QUESTIONNAIRE

The Approaches to Learning Mathematics Questionnaire and the Approaches to Learning Statistics Questionnaire were derived from the two scales of the Study Process Questionnaire which denote surface and deep approaches to learning. While the Biggs’ scales purport to investigate approaches to studying in general, the Approaches to Learning Mathematics Questionnaire investigates students’ approaches to learning mathematics, as a generic field of study. That is, in the Approaches to Learning Mathematics Questionnaire, the wording of Biggs’ deep and surface scales was changed to refer to approaches to learning mathematics, rather than to studying in general. My Learning Statistics Questionnaire from which the items of the ALSQ were selected, was modified from our Approaches to Learning Mathematics Questionnaire by changing “mathematics” to “statistics” in the items. The Learning Statistics Questionnaire refers to students’ approaches to learning Statistics—as a particular subject

studied for Psychology II. Hence the three questionnaires: the Study Process Questionnaire; the Approaches to Learning Mathematics Questionnaire and the ALSQ represent a progression—each focusing and particularising the constructs of interest in previous investigations.

The Learning Statistics Questionnaire completed by the Psychology II students contains 28 items on approaches to learning Statistics. The 14 odd numbered items were derived from items in the previous questionnaires indicating a surface approach to learning and the 14 even numbered items were derived from the scales denoting deep approaches to learning. For each item a choice of responses numbered 1 to 5 was provided. The lower end indicates that the student “only rarely” adopts this approach when studying Statistics, while the upper end indicates that the student “almost always” does so. Hence, a student’s score on each subscale signifies how usual it was for her to adopt the specific approach to learning Statistics.

I carried out item factor analyses and scale reliability analyses in order to determine the structure and internal consistency of the items. As a result of these analyses the ALSQ was defined as consisting of 18 items, twelve making up the Deep Scale and 6 items constituting the Surface Scale. Scale validity of the ALSQ is indicated by relationships with other variables such as willingness to learn Statistics.

ETHICS

This study was approved by the University of Sydney Human Research Ethics Committee. In order to ensure that I fulfilled my ethical obligations to the students and academics participating in my research I addressed the following issues.

Voluntary Participation

Survey

In the case of the survey, I invited students to participate both verbally and in writing. I explained the purpose of the research and how their responses would be used. Students who did not wish to complete a portion of the survey, or indeed the whole survey, were at liberty to desist. A few students omitted to give their Student Identity Number and one person refused to complete the survey. No pressure was brought to bear on them.

Interviews

Only students who gave their written consent to be interviewed were contacted by telephone. All interviews took place on a face to face basis and were audio taped. Hence the participants were able to see when the tape recorder was running, and I invited them to switch the recorder off at any time if they so wished. I asked students to sign a consent form. I also signed this form assuring the student of confidentiality and explaining again the purpose of the research. During the interview each participant was given the opportunity to ask any questions and discuss any difficulties relating to

Statistics. Some students took this opportunity to ask questions relating to the course content, the assessments or my research. Students were offered full editorial rights over the transcript of their own interviews. None opted to take advantage of this offer.

Disclosure of my Aims

I informed all participants of my aims: in writing, on page one of the survey, on the interview consent form, and verbally. Teaching staff were invited to discuss the research with me at various stages of preparation for the project and analysis.

Guarantee of Confidentiality

All students and staff participating in my research were guaranteed confidentiality. False names are used throughout. As recognition of students by staff members close to them is impossible to prevent, no excerpts from students' interviews were published during the year the students were in Psychology II.

Opportunity to Complain of any Ethical Misconduct

Students were given written information as to whom any complaints or reservations about any aspect of their participation in this research could be addressed.

Taking Account of Power Structures

As a staff member of the Mathematics Learning Centre, I have no input whatsoever over students' grading and am not involved with tests and examinations in any way. Similarly I have no authority over other academics. My job is to assist students at their own behest with any subject at The University of Sydney which involves elementary level mathematics and statistics. Students and teachers were invited to assist me by participating in this research.

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